

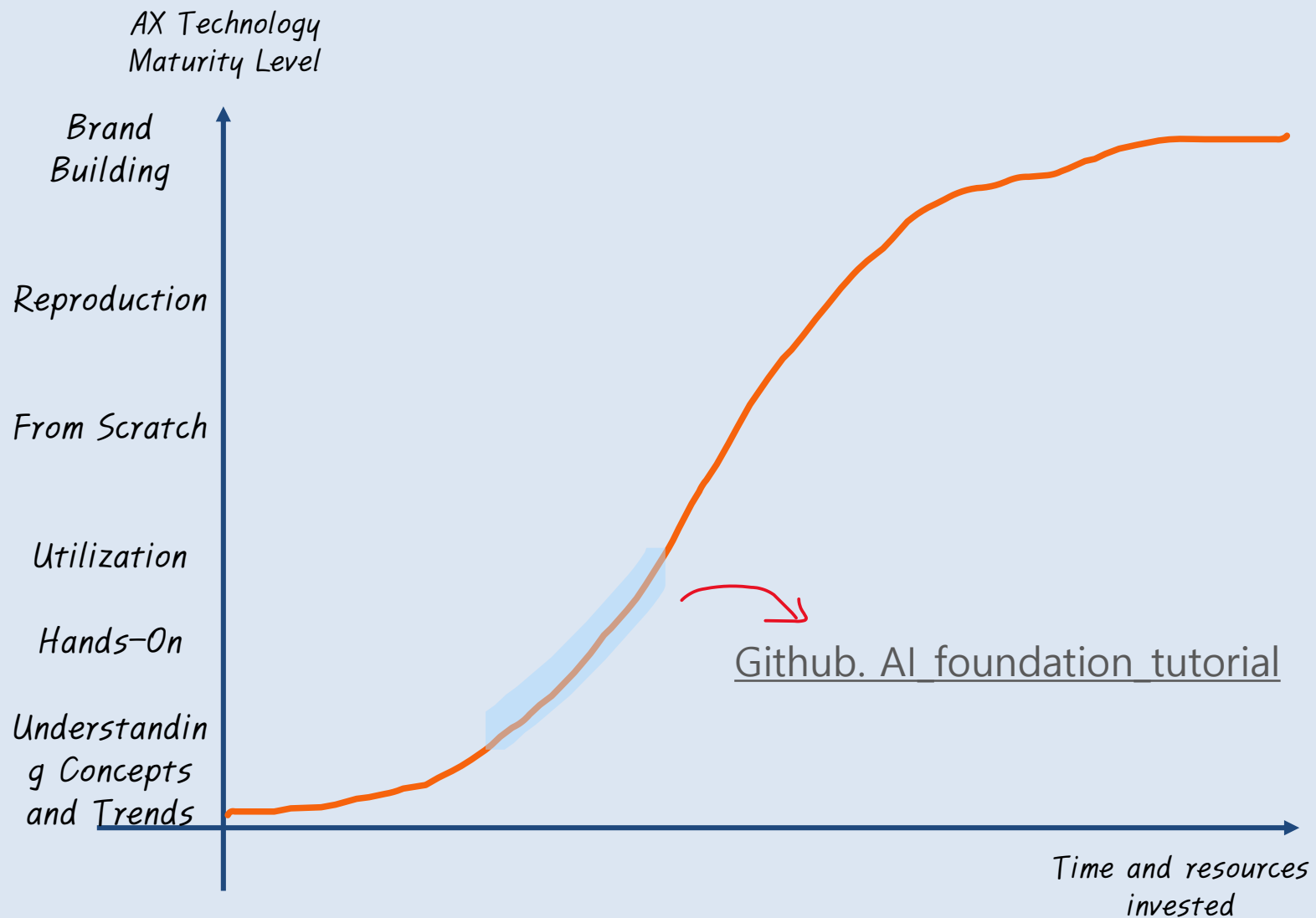
AX foundation and Trend 2025

AI
Transformation

Ph.D Kang Taewook
laputa99999@gmail.com
Github. AI foundation tutorial
Daddynkidsmakers.blogspot.com



2025/7



Content

AX Trends and Services
Basic Machine Learning Concepts
Structured Understanding of Deep Learning
Natural Language Processing
Transformer and the Latest AI Models
LLM and AI Agents
Overview of AI Model Service-Based Solutions
AX Strategy

AX Trends and Services

AX Trends

AI vs. Traditional Software Architecture Comparison
AI Service Development Flow

AX trend

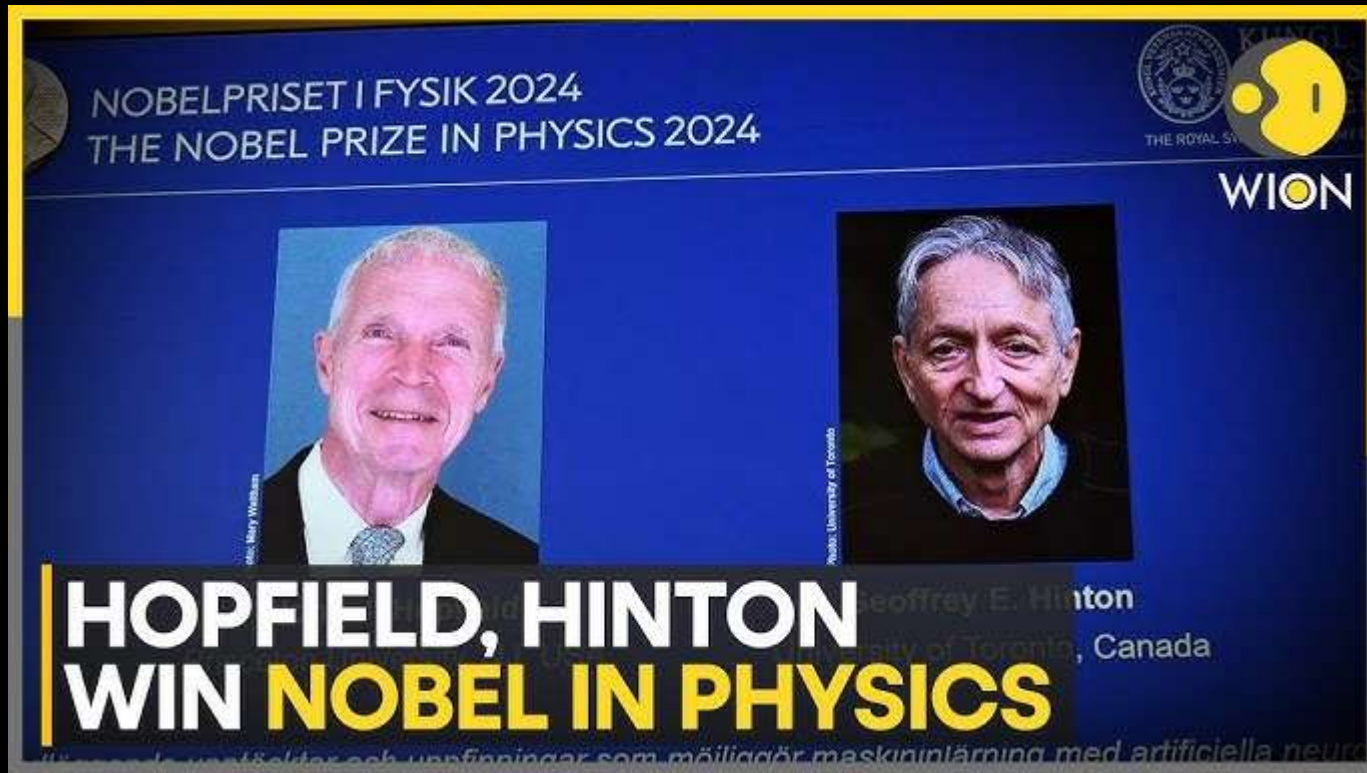
Key Question

What impact will the AX revolution have on the industrial economy?

What can and cannot AI do?

What is the future direction of AI agents?

AX revolution



Members of the Nobel Committee for Chemistry at the Royal Swedish Academy of Sciences explain the work of 2024 Nobel Prize in Chemistry winners David Baker, Demis Hassabis and John M. Jumper. JONATHAN NACKSTRAND/AFP via Getty Images

AI Pioneers Geoffrey Hinton And John Hopfield Win Nobel Prize For Physics | Latest News | WION

AX revolution

THE BUSINESS TIMES

Google and OpenAI's AI models win milestone gold at global math competition

Of the 630 students participating in the 66th IMO on the Sunshine Coast in Queensland, Australia, 67 contestants, or about 11 per cent, achieved gold-medal scores

▶ Listen to this article



Google and OpenAI's AI models win milestone gold at global math competition - The Business Times

AX revolution Multi-modal Agents



AX revolution

Multi-modal Agents

We've been testing the capabilities of Gemini, our new multimodal AI model.

We've been capturing footage to test it on a wide range of challenges, showing it a series of images, and asking it to reason about what it sees.

Large Language Models as General Pattern Machines

Suvir Mirchandani Fei Xia Pete Florence Brian Ichter
Danny Driess Montserrat Gonzalez Arenas Kanishka Rao
Dorsa Sadigh Andy Zeng

general-pattern-machines.github.io



Stanford

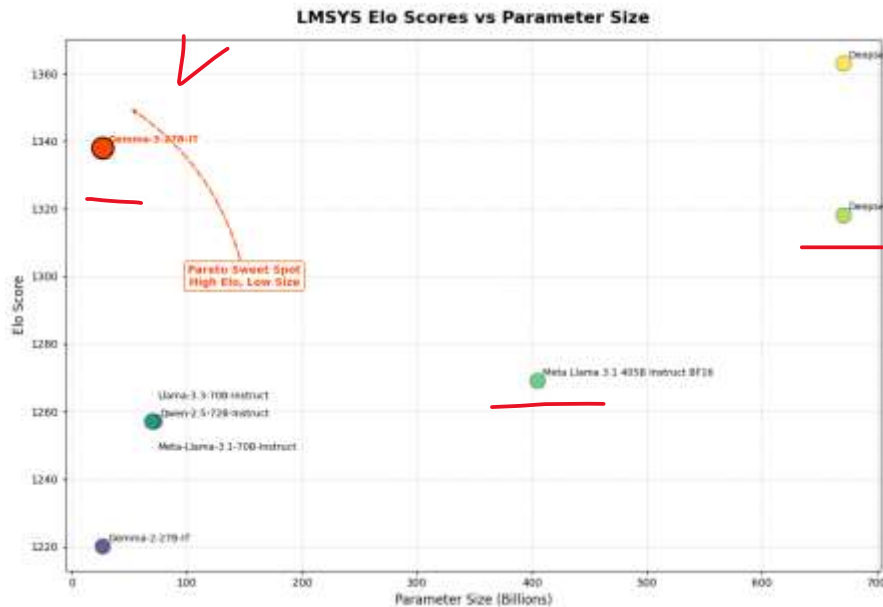


TITLE	CITED BY	YEAR
Large language models as general pattern machines S Mirchandani, F Xia, P Florence, B Ichter, D Driess, MG Arenas, K Rao, ... arXiv preprint arXiv:2307.04721	103	2023

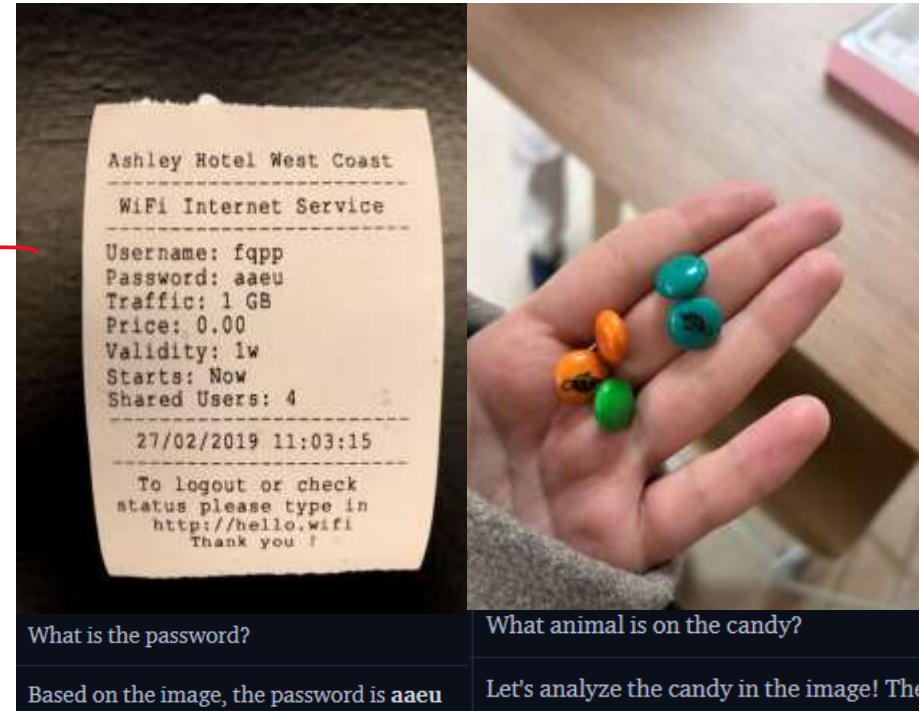
AX revolution open source basd sLLM

Gemma 3

LLM (1B, 4B, 12B, 27B), developed by Google and released on March 10, 2025, is a next-generation lightweight, open multimodal AI model that supports simultaneous text and image processing. It supports context sizes of 32k (1B) and 128k (4B-27B).



Gemma 3 Release - a google Collection



Link: [Gemma3 기반 Ollama 활용 AI 에이전트 개발 핵심 Function Call 구현해 보기](#)

AX revolution Rapid growth of AI services

Generative AI

The image shows a web browser window displaying the KREA AI interface. The browser address bar shows <https://www.krea.ai/chat>. The interface includes a chat area with a text input field and a button to send a message. Below the chat area, there are several buttons: "More", "Change style", "Cinematic", "Change background", "16:9", and "9:16". A red arrow points from the chat area to a terminal window on the right.

The terminal window displays the command to run the Llama 3.2 vision model:

```
python -m llama_run llama3.2-vision
```

The terminal output shows the following information:

- Environment Variables: OLLAMA_HOST (IP Address for the ollama server (default 127.0.0.1:11434)), OLLAMA_NOHISTORY (Do not preserve readline history).
- Added image: /zaha_ddp.jpg
- The image depicts a modern, white building with a sleek and minimalist design. The purpose of the image is to showcase the architectural features of the building.
- A large, open space in a building.
- The space is well-lit, with natural light pouring in through the windows.
- The floor is made of polished concrete or marble, reflecting the light and adding to the sense of openness.
- There are no visible walls or partitions, creating a sense of freedom and flexibility.
- A staircase leading up to the second floor.
- The staircase is curved and sleek, with a minimalist handrail that adds to the overall aesthetic.
- The steps are made of the same material as the floor, creating a sense of continuity.
- The staircase appears to be the main entrance to the building, drawing attention upwards.
- A glass wall on the left side.
- The glass wall is floor-to-ceiling, providing an unobstructed view of the surrounding area.
- The wall is framed by a thin metal strip, adding a touch of sophistication.
- The glass wall appears to be a design feature, rather than a functional one, as it does not seem to serve any practical purpose.

Overall, the image showcases a modern and sleek building with an emphasis on openness and minimalism. The use of natural light, polished floors, and curved lines creates a sense of fluidity and freedom.

The KREA AI interface also displays a description of the image:

Image of a modern, all-white building with curved lines and a large open space. The building features a sleek and minimalist design, with a staircase leading up to the second floor and a glass wall on the left side. The interior is well-lit, with natural light pouring in through the windows.

Below the description, there are three small images showing different views of the building. A red arrow points from the description to the terminal window.

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Generative AI



Krea | Video

AX revolution

Generative AI



“Make this look like a cozy summer evening in a Ghibli film — soft golden light, fireflies glowing, and a gentle breeze moving the grass.”

“Turn the background into a Ghibli-style village with tiny tea shops, lanterns, and warm, inviting streets.”

AX revolution

AX-ization of research and de

AI researcher



Sources:

- Camuffo, A., et al. (2020). *A Scientific Approach to Entrepreneurial Decision Making: Evidence from a Randomized Control Trial*. *Management Science*, 66(2). (RCT on lean startup methods) [OVERVIEW](#) [OVERVIEW](#)
- Lee, S.R. & Kim, J.D. (2024). *When do startups scale? Large-scale evidence from job postings*. *Strategic Management Journal*, 1–37. (Scaling timing and success/failure study) [STRATEGICMANAGEMENTJOURNAL](#) [STRATEGICMANAGEMENTJOURNAL](#)
- Startup Genome Project (Marmer et al., 2011). *Why Startups Fail: A deep dive into premature scaling*. (Report on 3,200+ tech startups) [MIDWINTERJOURNAL](#) [MIDWINTERJOURNAL](#)
- Picken, J. (2017). *From startup to scalable enterprise: Laying the foundation*. *Business Horizons*, 60(2), 175–185. (Challenges in the transition stage) [NEWSJOURNALJOURNAL](#) [NEWSJOURNALJOURNAL](#)
- March, J.G. (1991). *Exploration and exploitation in organizational learning*. *Organization Science*, 2(1), 71–87. (Foundational theory of exploration vs exploitation) [ESSAYJOURNAL](#)
- Blank, S. (2013). *Why the Lean Startup Changes Everything*. *Harvard Business Review*, 91(5), 63–72. (Conceptual piece on search vs execution in startups) [STRATEGICMANAGEMENTJOURNAL](#) (Referenced for context).

🔊 📄 👍 🔄

 **Strategic Management Society**

STRATEGIC INSIGHTS PUBLICATIONS & RESOURCES COPY PAR

Lee and Kim's new study, published in the *Strategic Management Journal*, explored when startups should begin scaling. They define scaling as the process in which startups primarily focus on acquiring and committing new resources to implement their core business idea and grow their customer base. To test the question of whether startups should scale early, Lee and Kim had to find a way to empirically measure when startups start scaling their business, because companies don't publicly announce such information.

Their solution: job postings. The researchers analyzed when these companies planned to hire their first manager or sales personnel, which they deemed to be the common markers of scaling. They created a dataset of 6.3 million job postings for more than 38,000 startups founded in the U.S. after 2010, which included information on the date and the occupational family of each job posting.

They found that, most notably, startups that scale early are less likely to engage in experimentation through A/B testing. They also found that startups operating in a new market (i.e., no competitors before entry) tend to scale later than those in a more established market – despite the popular belief that they should scale early to reduce the risk of imitation (the popular examples being Facebook and Uber). Importantly, they found that early scalers are associated with a higher likelihood of failure than their peers that scale later.

"What was surprising to us wa:

The New York Times

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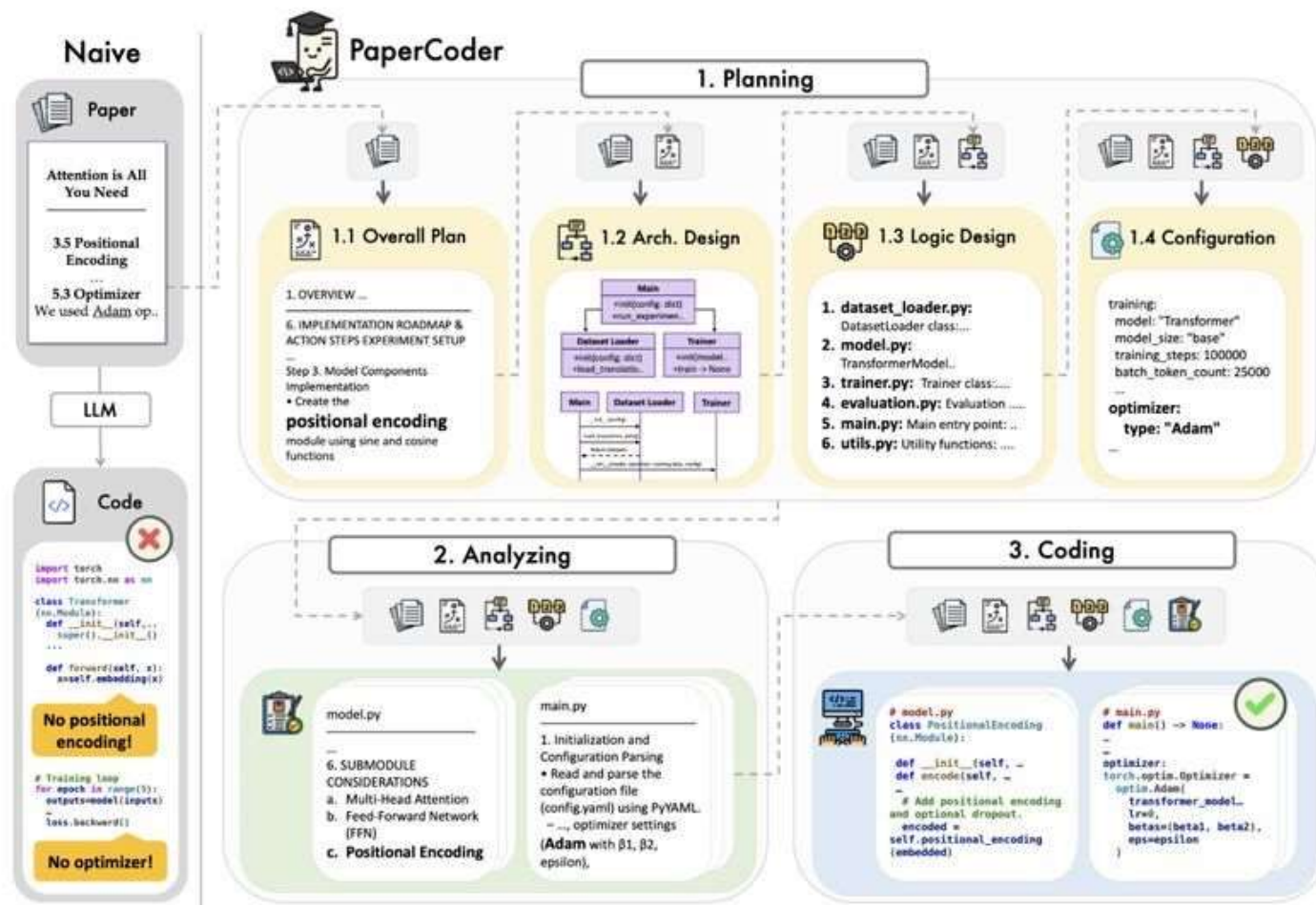
454 Hints That a Chatbot Wrote Part of a Biomedical Researcher's Paper

Scientists show that the frequency of a set of words seems to have increased in published study abstracts since ChatGPT was released into the world.

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AI developer

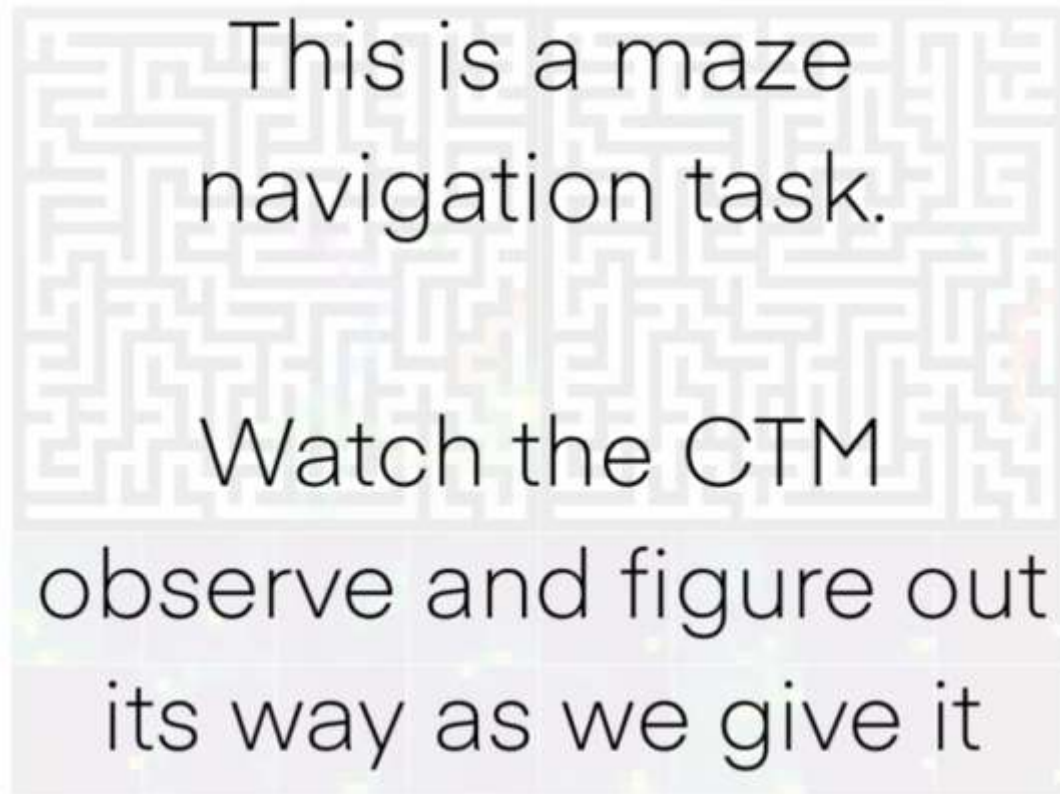
Paper2Code: Automating Code Generation from Scientific Papers in Machine Learning



AX revolution 연구 개발 전문 분야의 AX화

AI scientist

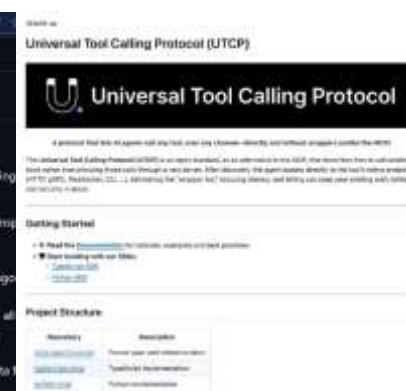
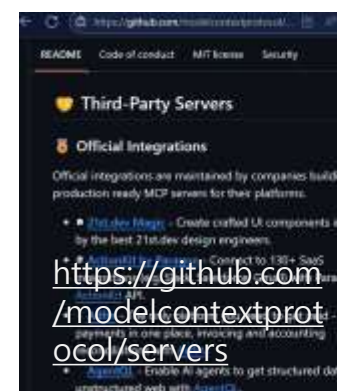
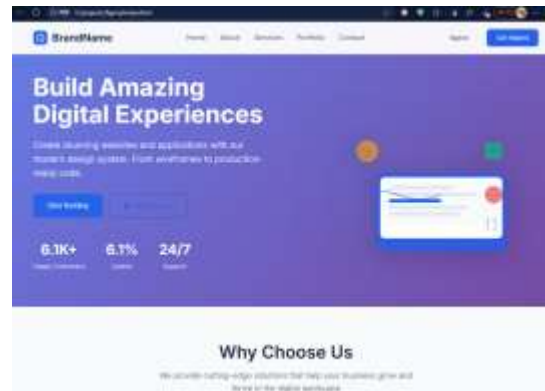
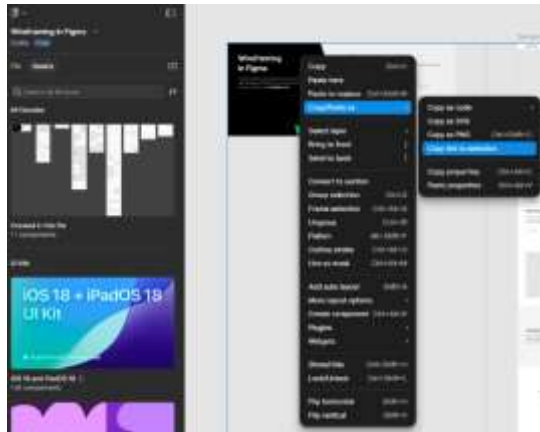
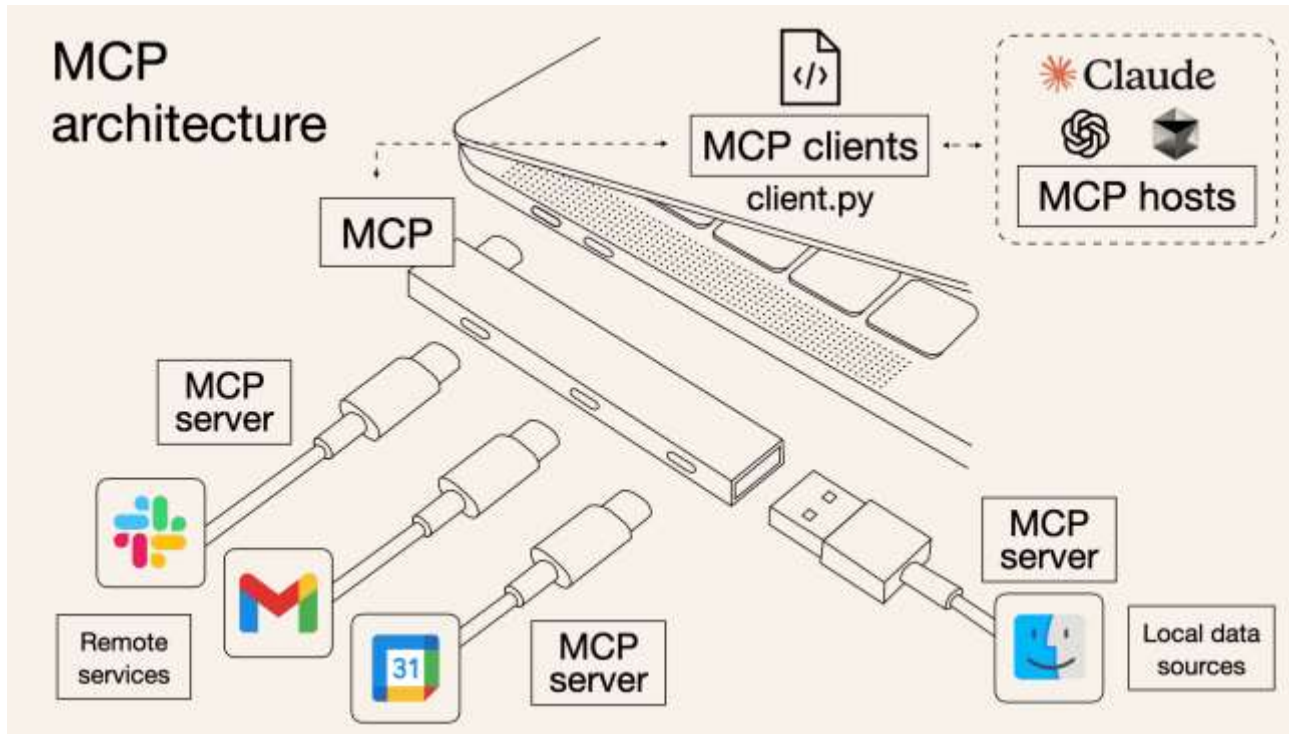
[Sakana AI](#) Scientist is a system that automatically performs the entire scientific research process.



Time to Think

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MCP (Model Context Protocol)



Link: 인공지능 AI 에이전트 표준 프로토콜 MCP 개념, 사용, 개발 및 동작 구조 분석하기

A2A (Agent2Agent Protocol)



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Vibe Coding



TechCrunch Latest Startups Venture Apple Security AI Apps Events Podcasts Newsletters

Vibe Coding: The Future of Software Development or Just a Trend? - Lovable Blog

Lovable becomes a unicorn with \$200M Series A just 8 months after launch

Anna Heim · 5:11 AM PDT · July 17, 2025

This Game Created by AI 'Vibe Coding' Makes \$50,000 a Month. Yours Probably Won't

flyplanet.com was totally made in just 30 minutes with AI tools and is now generating thousands of dollars a month. The future of AI-assisted game development will not be that simple.



A game created with AI in just 30 minutes is generating more than \$50,000 a month. Its creator claims—just could be a peak of how AI can, and can't, change game design in the future.

WIKI, LARRY TUBES

WIX ACQUIRES SIX-MONTH-OLD AI "VIBE CODING" STARTUP BASE44 FOR \$80M CASH

Wix Business Capital Markets News Style My Startup Company June 20, 2025

Published on July 20, 2025 · 11:11 AM GMT+2

Cognizant's Vibe Coding Lesson for Indian IT

Cognizant aims to generate 50% of its code using AI within a year and states it has already achieved 20% of that target.



BASE44

Build Apps with AI in Minutes | Base44

This Game Created by AI 'Vibe Coding' Makes \$50,000 a Month. Yours Probably Won't, Wix Acquires Six-month-old AI "Vibe Coding" Startup Base44 for \$80M Cash, Cognizant's Vibe Coding Lesson for Indian IT, Vibe Coded a Website With My Daughter Using an AI Tool Called Bolt - Business Insider

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Vibe Coding

OpenAI Codex: Transforming Software Development with AI Agents

OpenAI Codex: Transforming Software Development with AI Agents

[OpenAI Codex: Transforming Software Development with AI Agents - DevOps.com](#)

BY: **TOM SMITH** ON MAY 18, 2025

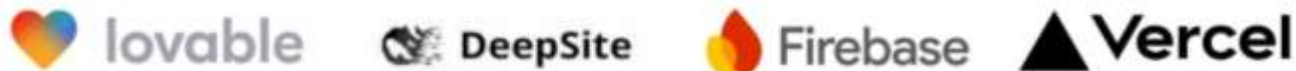
Code Generation



Smart IDEs



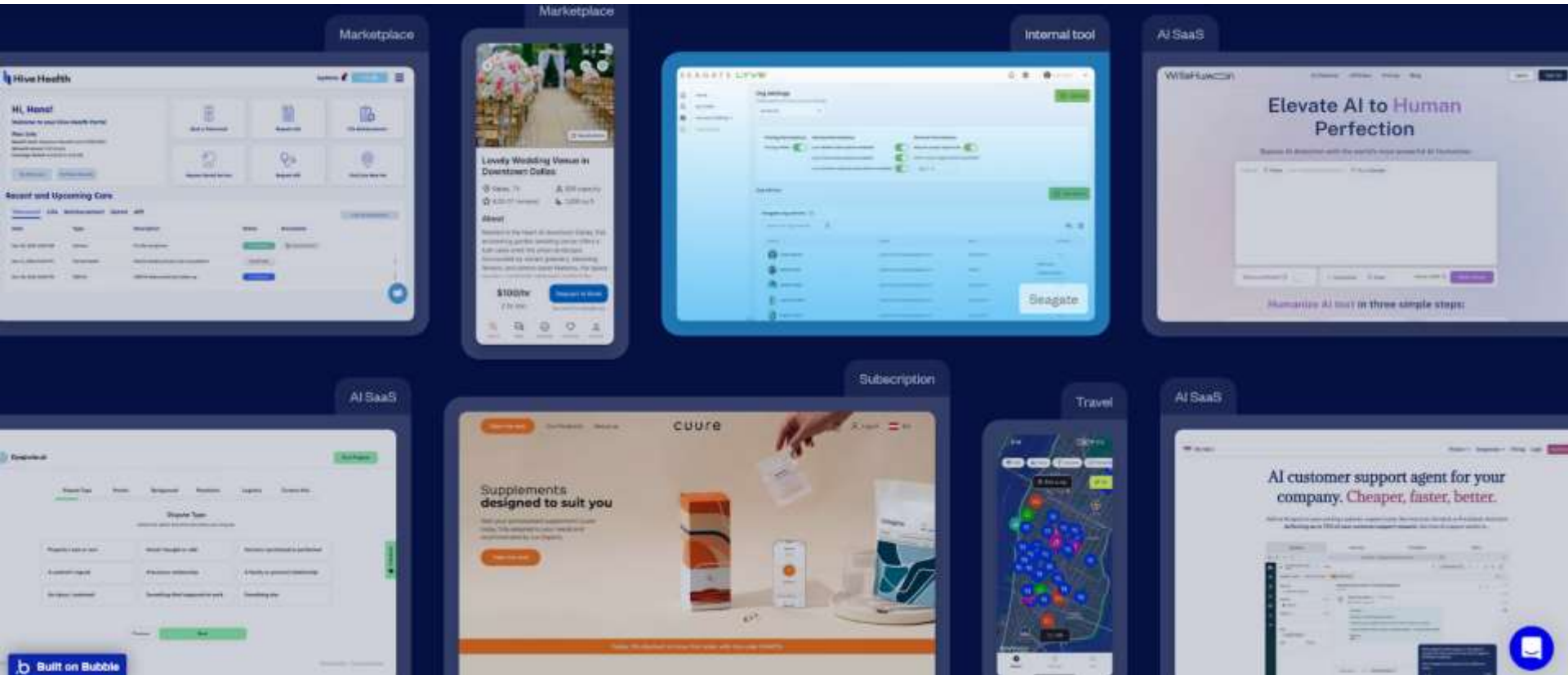
Frontend + Deployment



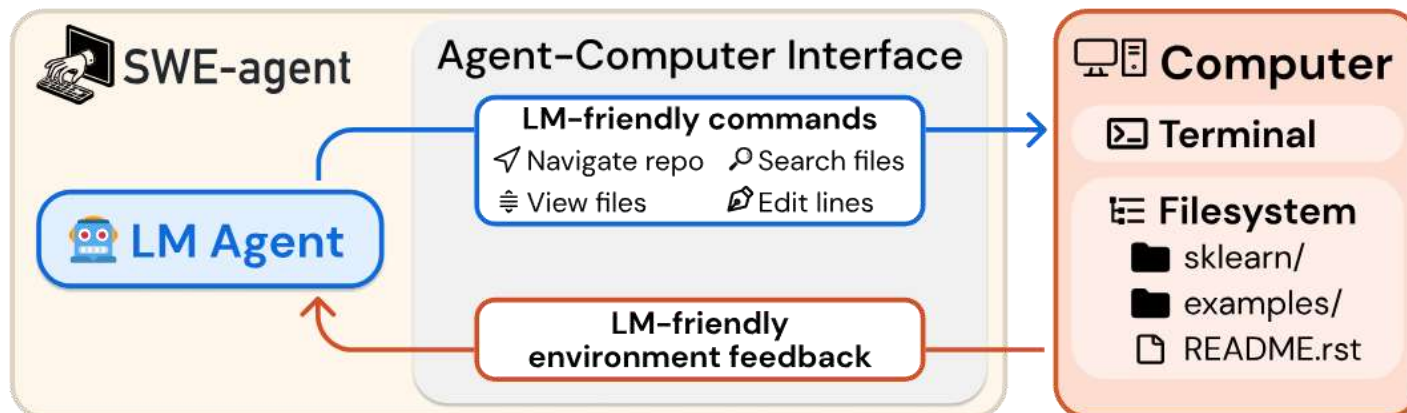
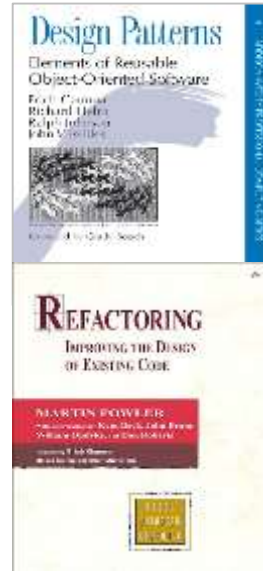
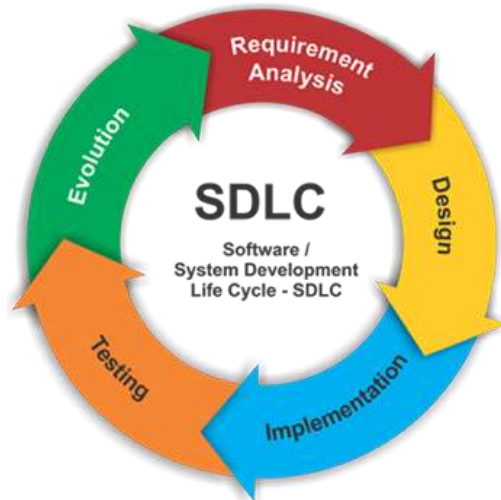
[Gemini CLI vs Claude Code vs Cursor – Which is the best option for coding? – Bind AI IDE](#)
[Is Vibe Coding the Future of Software Development? A Deep Dive into AI's Role](#)

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No-code spread



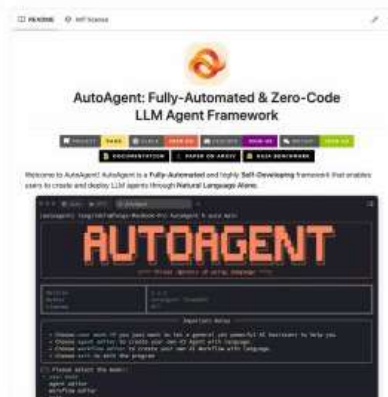
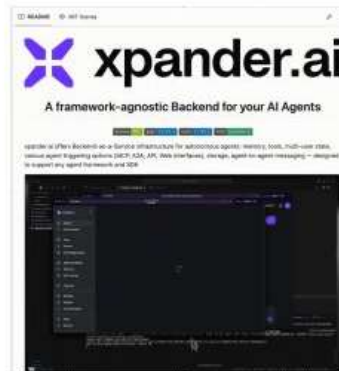
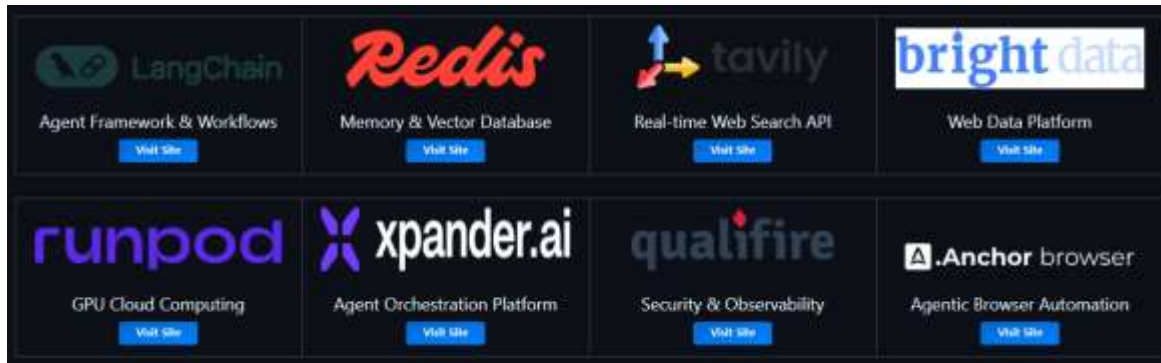
Advances in AI Agent Technology in Software Engineering



Popular AI Agents for Devs: Chatdev, SWE-Agent & Devin [Example Project]
Software Development Life Cycle Models and Methodologies - Mohamed Sami

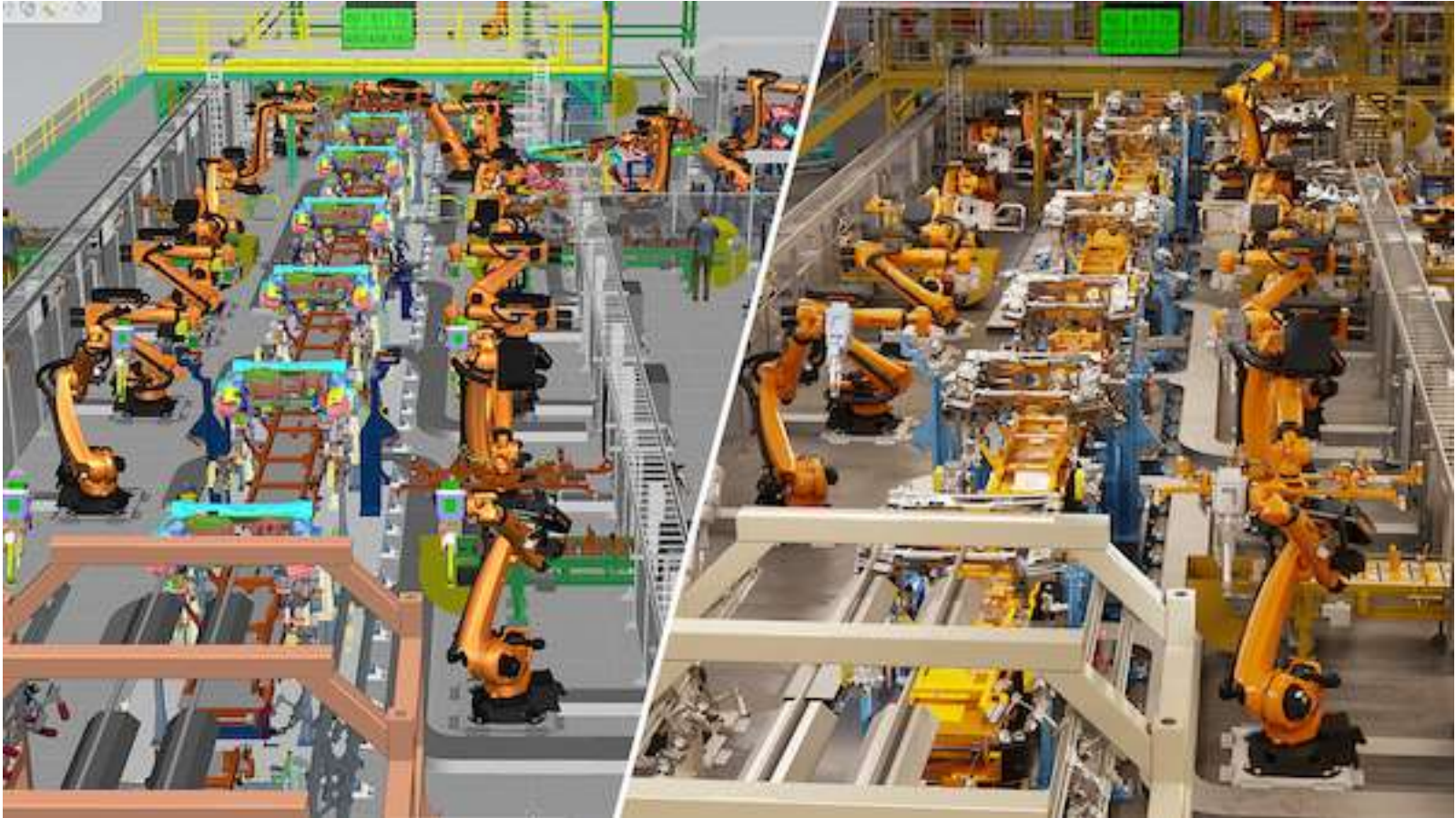
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Multi-AI Agent Technological Advancements



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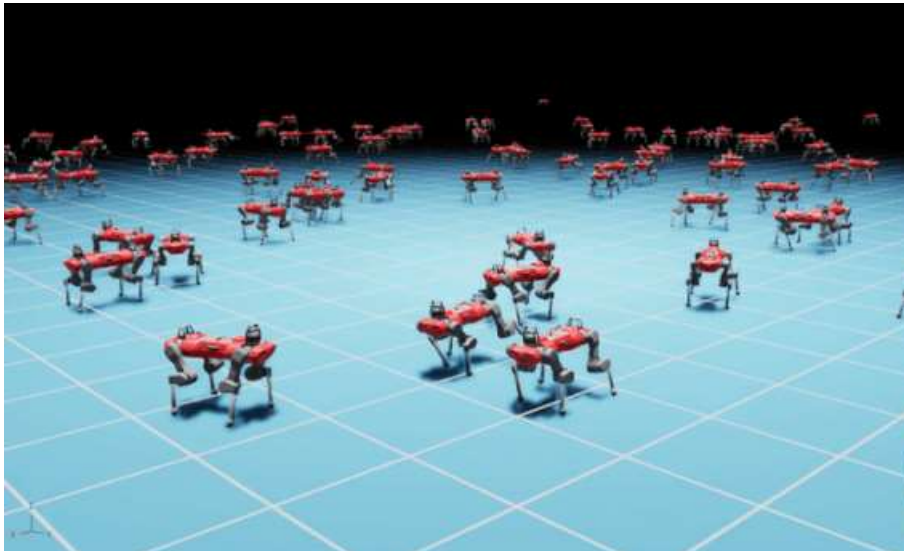
Digital Twin + AX + Robotics



Siemens Process Simulate (left) connects to NVIDIA Omniverse (right) to enable a full-design-fidelity, photorealistic, real-time digital twin.
https://www.robotics247.com/article/siemens_xcelerator_nvidia_omniverse_accelerate_digital_twins_manufacturing

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Digital Twin + AX + Robotics



Unitree R1
Price from \$5,900

Ultra-lightweight, Fully Customizable



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China Initiative



[Link: 딥시크\(deep seek\) open source 코드 및 구조 분석하기](#)

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 머니투데이

"추격조"에 GPU 몰아주면 韓 딥시크 10개 나와"

"글로벌 AI 추격조를 만들어 '3년간 한국 데이터를 다 가져다 써라' 해야 합니다. 정부가 연내 GPU를 1만개를 확보해 상하반기 각 5개 기업에 2000..."

Feb 6, 2025



 한국일보

[현장] "GPU·데이터 몰아주면 한국에서도 딥시크 10개 나온다"

6일 과학기술정보통신부가 개최한 중국 인공지능(AI) 모델 '딥시크'의 등장 이후 국내 AI 산업의 경쟁력을 점검하는 회의에서 거대언어모델(LLM)을...

Feb 7, 2025



 중앙일보

韓 AI 회사 "지금이 오히려 기회...딥시크 10개 만들 방법 있다" [팩플]

중국 생성 인공지능(AI) 모델 딥시크의 등장은 한국 AI 산업에 위기일까 기회일까. 국내 AI 기업들은 'AI 추격조'를 만드는 등 정부의 전폭적 지원이...

Feb 6, 2025



 노컷뉴스

[Q&A]AI 강국되려면 GPU 잡아라..."품귀현상에도 구할 수 있다"

"더 많은 인프라를 가지고 양질의 데이터를 확보해서 AI 모델을 만드는 것이 중요하다" (배경훈 LG AI연구원장) "GPU 1만 개 확보해서 10개 회사에..."

1 month ago



AX revolution

China Initiative

Structured 3D Latents for Scalable and Versatile 3D Generation

Jianfeng Xiang^{1,3*} Zelong Lv^{2,3*} Sicheng Xu³ Yu Deng³ Ruicheng Wang^{2,3*}
Bowen Zhang^{2,3*} Dong Chen³ Xin Tong³ Jiaolong Yang^{3†}
¹Tsinghua University ²USTC ³Microsoft Research
<https://trellis3d.github.io>

2 Dec 2024



I am currently a Ph.D student majoring in Computer Science at Tsinghua University (THU) under the supervision of Prof. Baining Guo. Besides, I have been a research intern at Microsoft Research Asia (MSRA) since Jul 2021, working closely with Sicheng Xu, Jiaolong Yang and Xin Tong.

Before that, in Jun 2022, I received my B.E. in Electronic Information Engineering at University of Science and Technology of China (USTC) from the School of the Gifted Young and the School of Information Science and Technology.

[Jianfeng Xiang | Home](#)



[Baining Guo at Microsoft Research](#)

Link: [open source 기반 2D Image to 3D model 소개](#)

AX revolution AX service explosion

Rapid advancements in AI model engineering technology



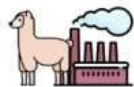
Finetune Qwen3, Llama 4, Gemma 3, Phi-4 & Mistral 2x faster with 80% less VRAM!

Finetune for Free

Notebooks are beginner friendly. Read our [guide](#). Add your dataset, click "Run AI!", and export your finetuned model to GGUF, Ollama, vLLM or Hugging Face.

Unsloth supports	Free Notebooks	Performance	Memory use
Qwen3 (14B)	Start for free	2x faster	70% less
Qwen3 (4B): GRPO	Start for free	2x faster	80% less
Gemma 3 (4B)	Start for free	1.6x faster	60% less
Llama 3.2 (3B)	Start for free	2x faster	70% less
Phi-4 (14B)	Start for free	2x faster	70% less
Llama 3.2 Vision (11B)	Start for free	2x faster	50% less
Llama 3.1 (8B)	Start for free	2x faster	70% less
Mistral v0.3 (7B)	Start for free	2.2x faster	75% less
Sesame-CSM (1B)	Start for free	1.5x faster	50% less

- See all our notebooks for [SageMaker](#), [GRPO](#), [ITP](#) & [Vision](#)
- See all our models and our [Synthetic Dataset notebooks](#) in collaboration with Meta
- See detailed documentation for Unsloth [here](#)



LLaMA-Factory
Easy and Efficient LLM Fine-Tuning



Used by Amazon, NVIDIA, Aliyun, etc.

Supporters ❤️



Warp, the agentic terminal for developers
[Available for MacOS, Linux, & Windows](#)



Overview

Axolotl is a tool designed to streamline post-training for various AI models.

Features:

- **Multiple Model Support:** Train various models like LLaMA, Mistral, Mistral, Pythia, and more. We are compatible with HuggingFace transformers causal language models.
- **Training Methods:** Full fine-tuning, LoRA, QLoRA, GPTQ, QAT, Preference Tuning (DPQ, IPQ, KTO, ORPO), RL (GRPO), Multimodal, and Reward Modelling (RM) / Process Reward Modelling (PRM).
- **Easy Configuration:** Re-use a single YAML file between dataset preprocess, training, evaluation, quantization, and inference.
- **Performance Optimizations:** Multipacking, Flash Attention, Xformers, Flash Attention, Liger Kernel, Cut Cross Entropy, Sequence Parallelism (SP), LoRA optimizations, Multi-GPU training (PDP, PDP2, DeepSpeed), Multi-node training (Torchrun, Ray), and many more!
- **Flexible Dataset Handling:** Load from local, Huggingface, and cloud (S3, Azure, GCP, OCI) datasets
- **Cloud Ready:** We ship [Docker images](#) and also [PyPi packages](#) for use on cloud platforms and local hardware.

Quick Start

Requirements:

- NVIDIA GPU (Ampere or newer for 32B and Flash Attention) or AMD GPU
- Python 3.11
- PyTorch 2.5.1



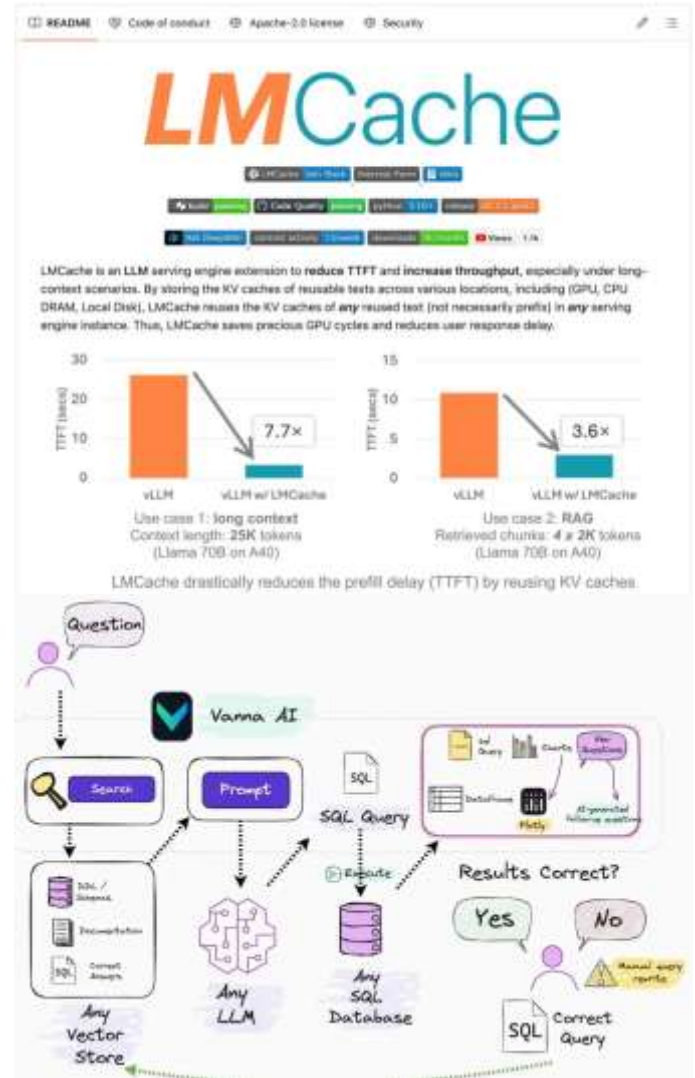
Extreme Speed and Scale for DL Training and Inference

DeepSpeed enabled the world's most powerful language models (at the time of this writing) such as [MT-530B](#) and [BLOOM](#). It is an easy-to-use deep learning optimization software suite that powers unprecedented scale and speed for both training and inference. With DeepSpeed you can:

- Train/inference dense or sparse models with billions or trillions of parameters
- Achieve excellent system throughput and efficiently scale to thousands of GPUs
- Train/inference on resource constrained GPU systems
- Achieve unprecedented low latency and high throughput for inference
- Achieve extreme compression for an unparalleled inference latency and model size reduction with low costs

DeepSpeed's four innovation pillars

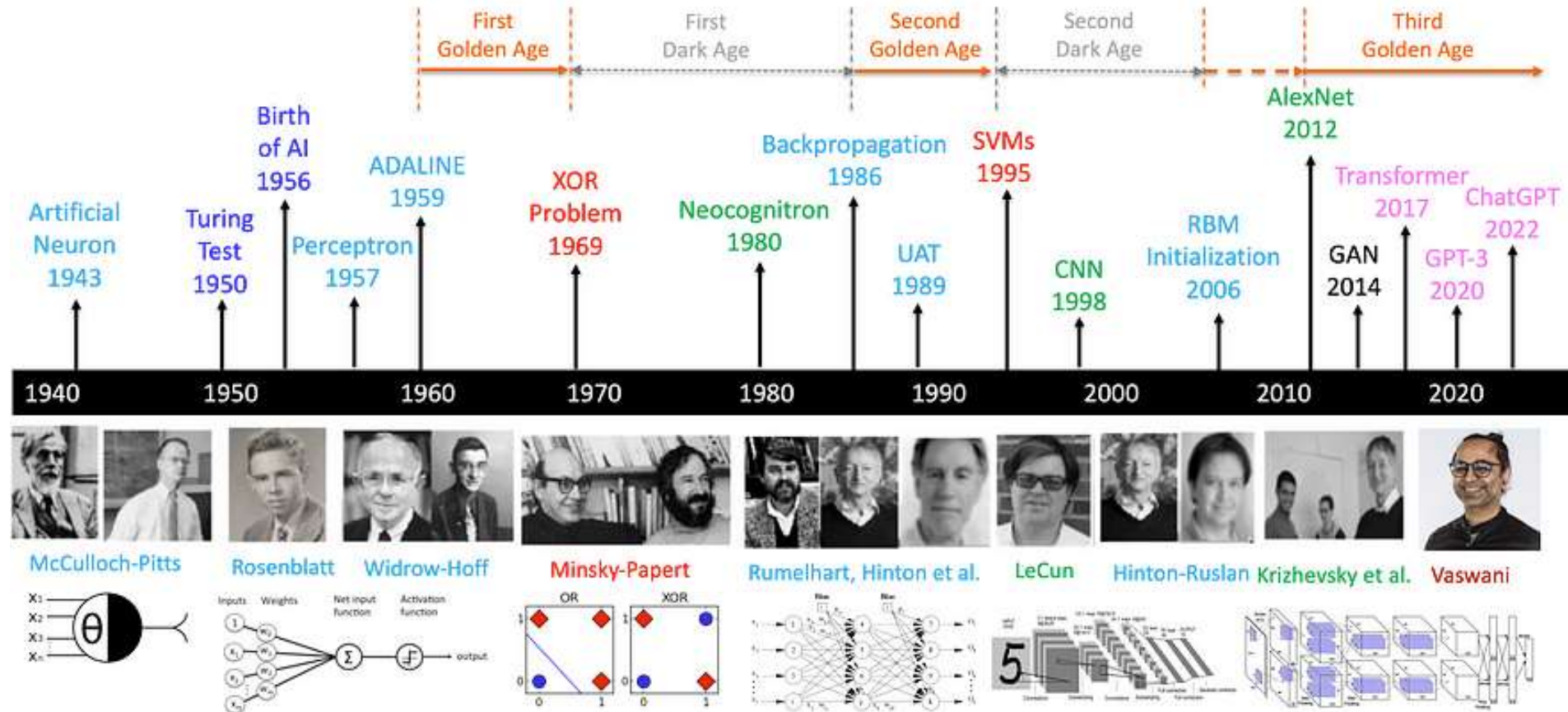
Training	Inference	Compression	Science
• Speed • Scale • Democratization • MoE models • Long sequence • RLHF	• Large models • Latency • Serving cost • Agility	• Latency • Composability • Runnable on client devices	• Speed • Scale • Capability • Diversity • Discovery



AI service AI vs Traditional software

History

A Brief History of AI with Deep Learning



AI service

AI vs Traditional software

History

Convolutional neural networks running on GPUs (2012) Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton, Advances in Neural Information Processing Systems 2012



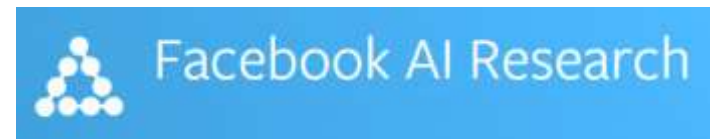
Deep CNN, ReLU, Dropout



2013. Google acquired DNN

Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
<i>SIFT + FVs [7]</i>	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	16.4%
1 CNN*	39.0%	16.6%	—
7 CNNs*	36.7%	15.4%	15.3%

[ImageNet Classification with Deep Convolutional Neural Networks](#)



Yann LeCun.
Chief AI Scientist for Facebook AI Research (FAIR)

AI service

AI vs Traditional software

History

Computer Vision & Learning Group



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Prof. Dr. Björn Ommer

[Contact](#)

University of Munich

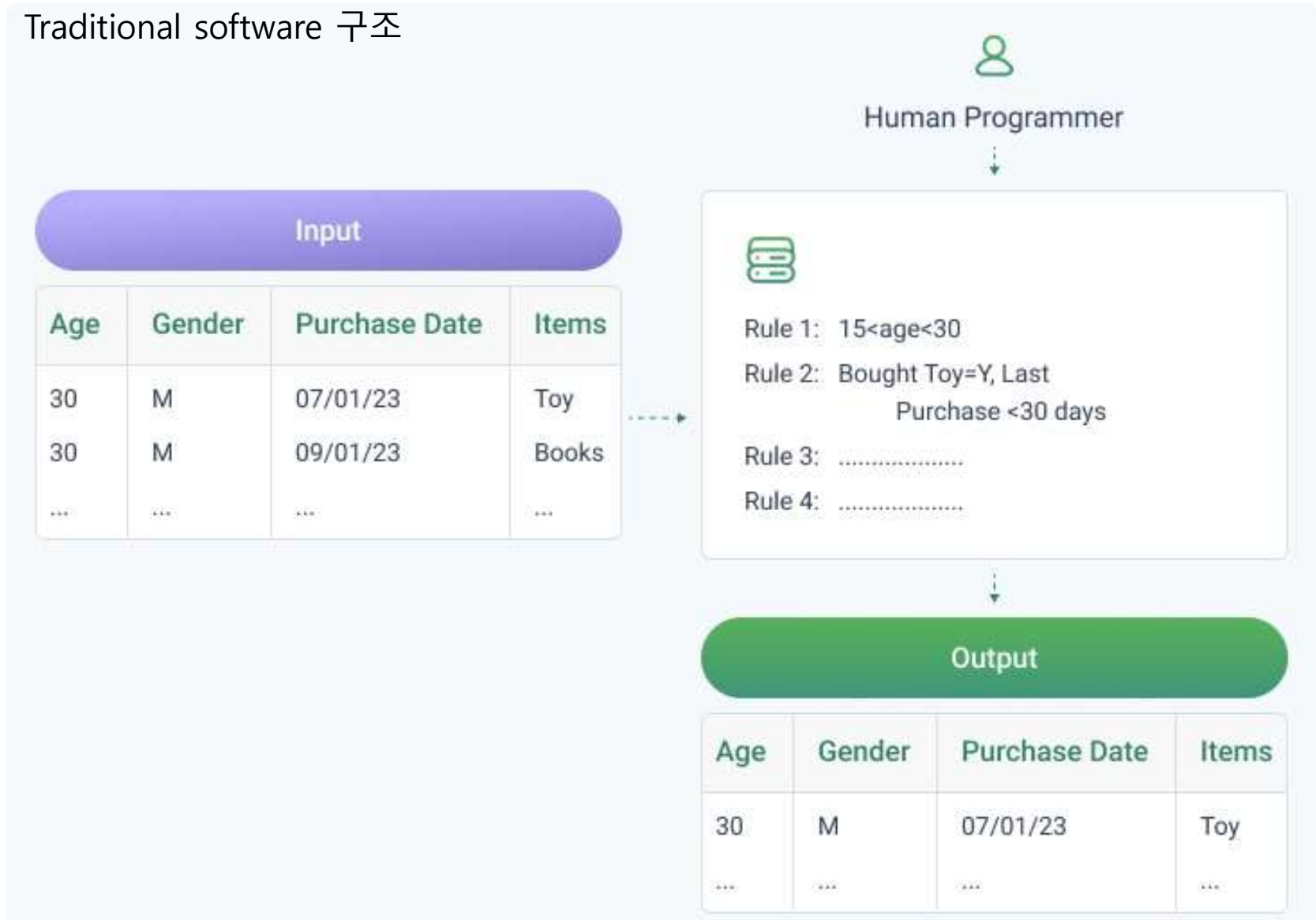


Stable Diffusion Developer [Computer Vision & Learning Group \(ommer-lab.com\)](https://ommer-lab.com)

AI service

AI vs Traditional software

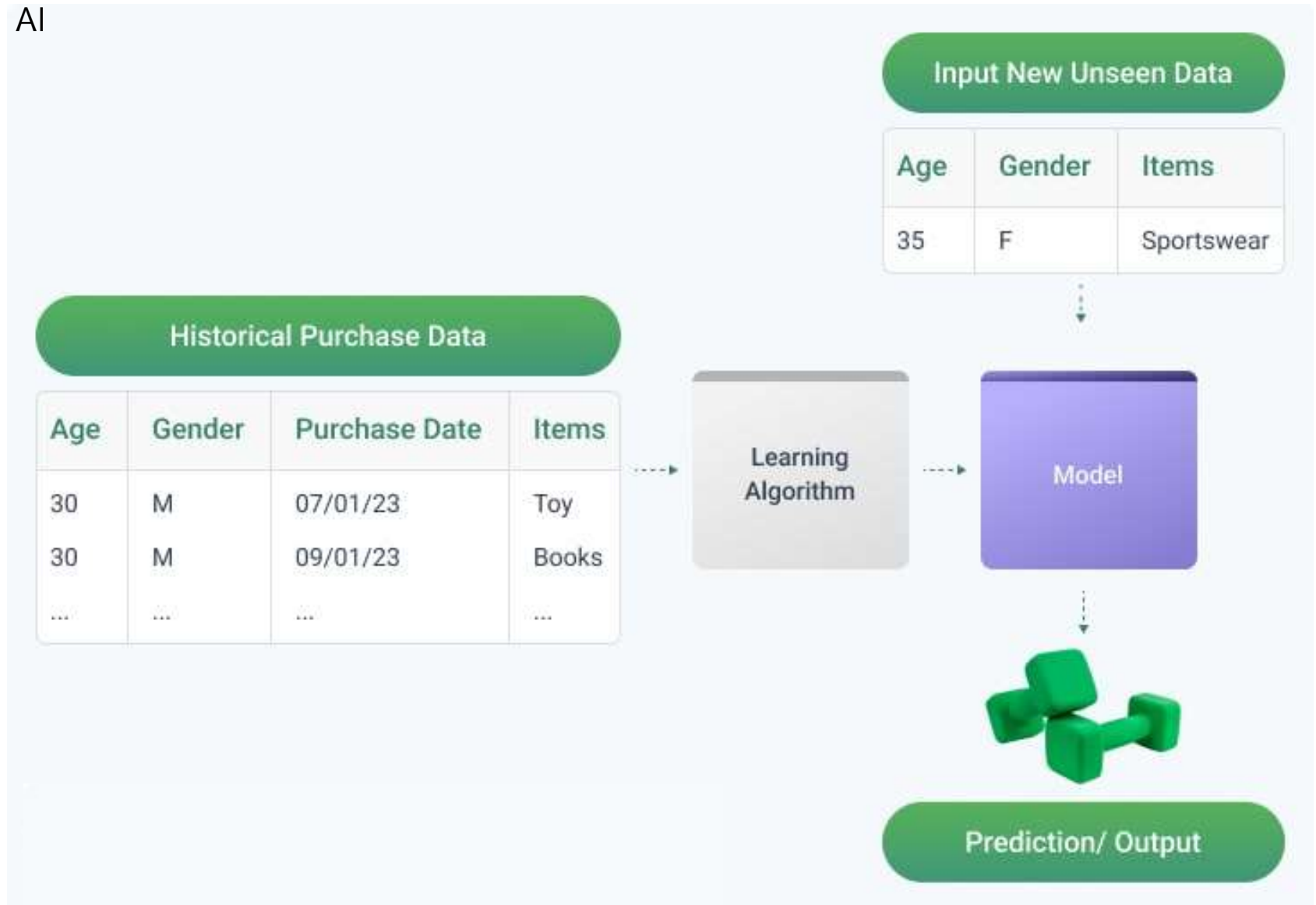
Traditional software 구조



AI service

AI vs Traditional software

AI



AI service

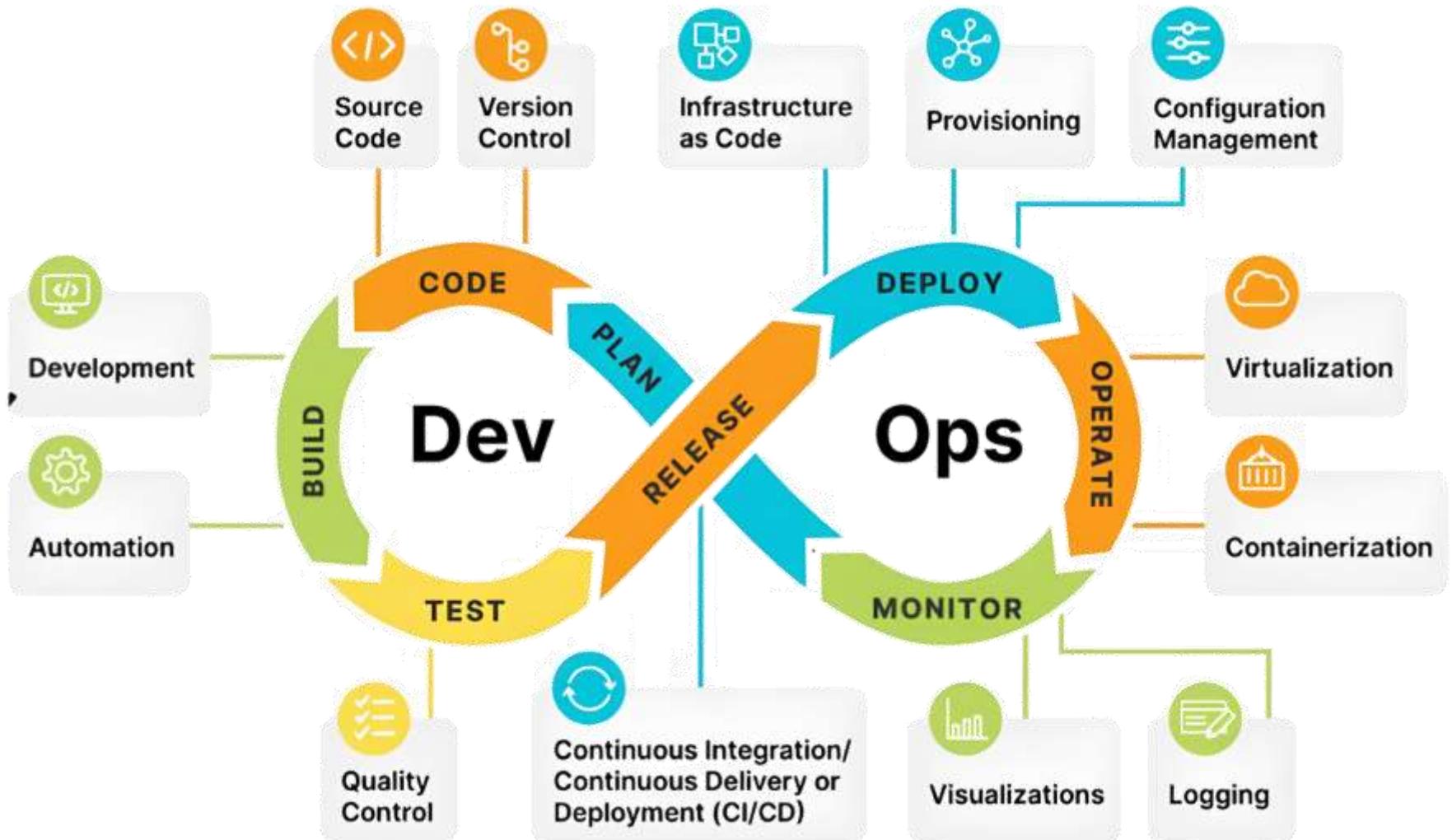
AI vs Traditional software

division	Traditional software	AI service
core logic	Explicit rules and algorithms coded directly by developers	Patterns learned from data (models)
How it works	Deterministic Always output the same result for the same input	Probabilistic & Adaptive Results may vary depending on training data and model status, and performance can be continuously improved with new data.
development process	Requirements analysis → Design → Coding (implementation) → test	Data collection/processing → model training and evaluation → serving → monitoring and retraining
data role	Inputs that the system must process	Key ingredients for training and validating models
maintenance	Code updates including bug fixes and feature additions	Monitoring performance degradation (Drift), retraining models, and managing data bias
merit	- High predictability and reliability - Easy to debug and verify because the logic is explicit - If the requirements are clear, the development period and cost can be predicted.	- Solving complex, probabilistic, and irregular problems - Self-adapting/learning/improving performance based on data - Pattern recognition requires human intuition. Personalization strengths.
disadvantage	- Difficulty responding to unexpected data or changes - Problems where rules cannot be defined (e.g. image recognition) is impossible to solve - As the logic becomes more complex, development and maintenance become more difficult. becomes exponentially more difficult	- The 'black box' problem: Explaining why a decision was made difficult to command - Data dependency: A large amount of high-quality data is required. Numerical and vulnerable to data bias - High computing costs and specialized personnel for learning and operation power is needed - Hallucinations. 100% accuracy cannot be guaranteed.

AI service

AI service Development Flow: Data → Model → Servicing

DevOps



AI service

AI service Development Flow: Data → Model → Servicing

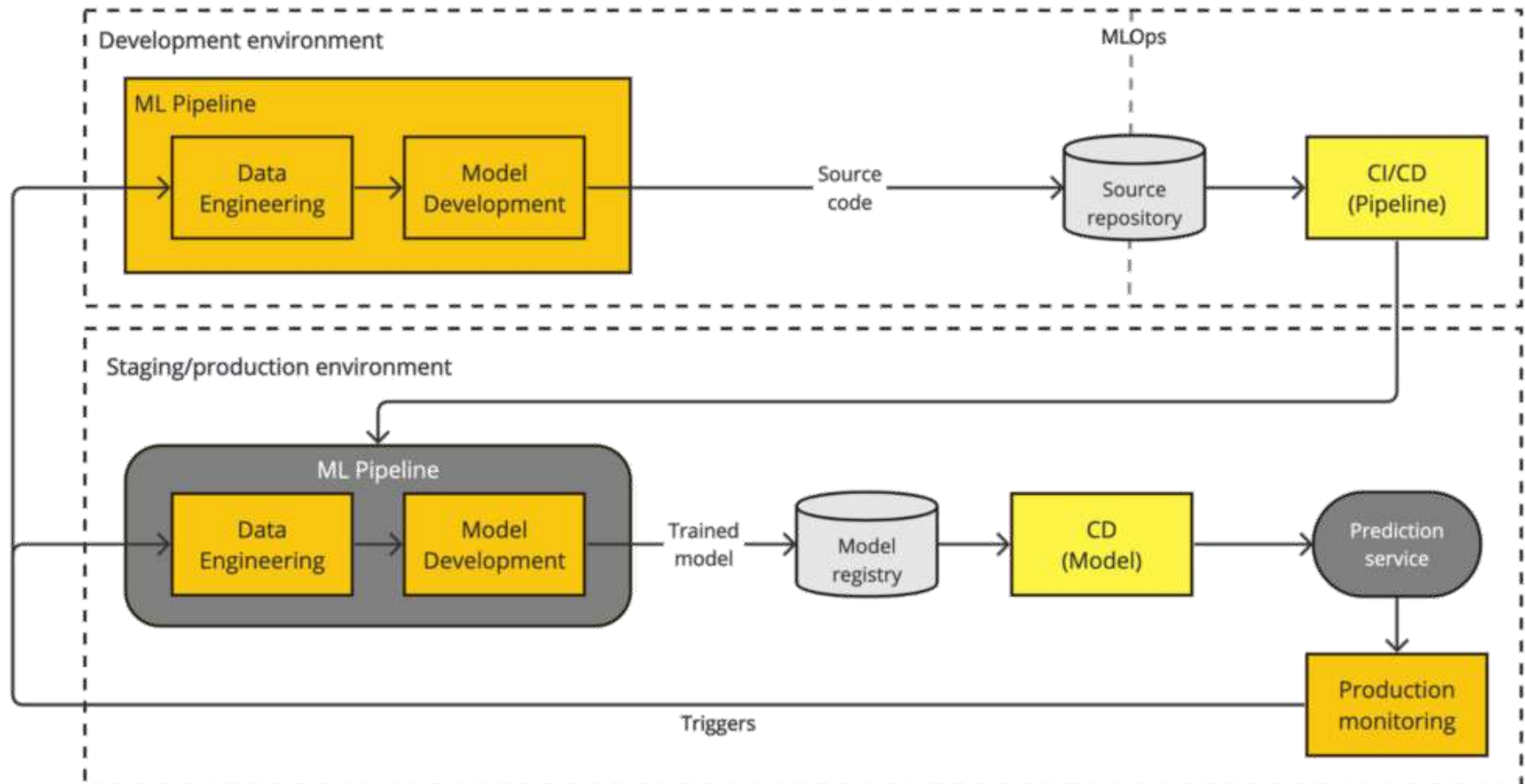
MLOps

Technology and infrastructure that effectively accelerate AI solution development productivity and maintain data quality

Infrastructure that supports machine learning engineers to accelerate model development and deployment

Continuous monitoring and validation of ML models is essential

Completing the continuous integration/deployment/training (CI/CD/CT) cycle is crucial



AI service

AI service Development Flow: Data → Model → Servicing

MLOps

step	Key Activities	Detailed description
Problem Framing	Understand business goals, define success metrics, and assess data availability.	The initial planning stage where business problems are converted into machine learning problems and the project's success criteria and direction are established.
Data Engineering	We carry out data collection, cleaning, labeling, feature engineering , and data version management.	Collect the data required for model training, refine it to improve quality, and process it into features that the model can understand.
Model Engineering	Performs algorithm selection, model code writing, hyperparameter tuning, model performance validation, experiment management, etc.	A key step in finding the optimal model by training a model with prepared data and objectively evaluating its performance through various indicators.
Model Deployment	Package the trained model and prepare it for deployment to serving infrastructure (API server, cloud environment, etc.).	Transferring a verified model to an operational environment accessible to actual users.
Serving & Inference	Provides prediction results in real-time or batch form through deployed models, and connects with other services through APIs.	The model makes predictions on actual user data and provides the core functionality of the AI service.
Model Monitoring	Continuously track and record model performance degradation (Drift), data bias (Bias), system resource usage, etc.	Monitor in real time whether the operating model maintains consistent predictive performance and whether there are any unexpected errors.
Retraining	Based on the monitoring results, the model is retrained with new data and a new pipeline is run when necessary.	The process of continuously maintaining and improving the overall performance and accuracy of a service by allowing the model to adapt to changing data patterns over time.

AI service

AI service development flow: Data → Model → Servicing

MLOps

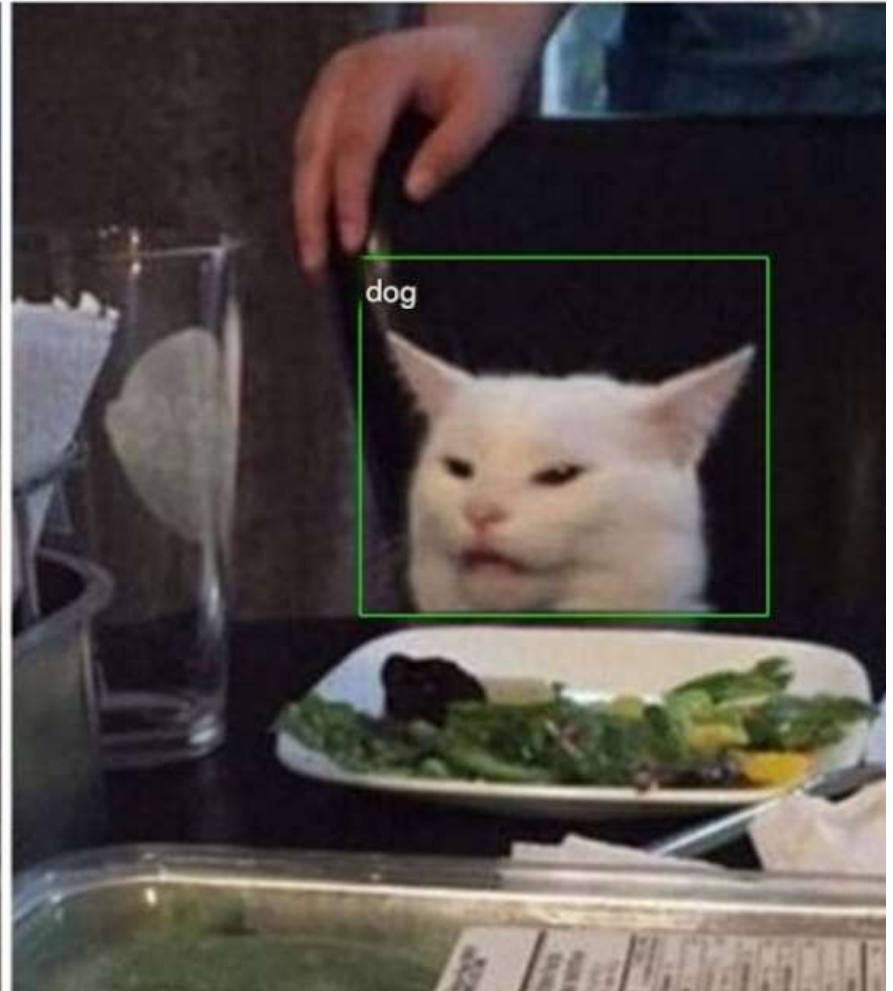
단계 (Phase)	솔루션 Example
Problem Framing	Our own analytics tools, spreadsheets, Nocode, and Vibe coding services
Data Engineering	PyTorch, Kafka, Spark, Pandas, Scikit-learn, NumPy
Model Engineering	TensorFlow, PyTorch, Keras, Scikit-learn, MLflow, Weights & Bias
Model Deployment	ONNX((Open Neural Network Exchange), Docker, Kubernetes, Tens orFlow Serving, TorchServe, FastAPI, Flask
Serving & Inference	TensorFlow Serving, TorchServe, NVIDIA Inference Server, AWS Sag eMaker Inference, Google AI Platform Prediction, Azure Machine Lear ning Endpoints
Model Monitoring	Prometheus, Grafana
Retraining	Kubeflow Pipelines, MLflow Projects, AWS SageMaker Pipelines, Goo gle Cloud AI Pipelines, Azure Machine Learning Pipelines

AI will rule the world

AI Misunderstandings:



AI I created:



Basic Machine Learning Concepts

AI Learning and Development Environment

Differences between Classification and Regression

Concepts of Overfitting, Regularization, and Validation Sets

Overview of Major Algorithms (Decision Trees, KNN, etc.)

Introduction to Pipeline Examples Using Scikit-learn

ML basic

Key Question

How do we classify and analyze data using machine learning?
Why do we need pipelines in machine learning?

Machine Learning Basics

AI development environment

Python (Version 3.11 or higher recommended for library compatibility. As of July 2025)

Download Python

Mac Users: Python Installation Guide on Mac

After installation, check if the installation works properly with `python --version` in Terminal.

NVIDIA Driver (for NVIDIA GPUs)

Download NVIDIA Driver

After installation, check with `nvidia-smi` in Terminal

CUDA Toolkit (for NVIDIA GPUs)

Download CUDA Toolkit

Check GPU and driver compatibility during installation

Add the CUDA path to the environment variable.

Install Github Tools

Install the tools from the following link:

<https://docs.github.com/ko/desktop/installing-and-authenticating-to-github-desktop/installing-github-desktop>

Anaconda (Version 24.0 or higher recommended)

Install the tools from the following link: [Download Anaconda](#)

Install the PyTorch library

Visit the following link and install it from the terminal, depending on your CPU or GPU driver

version: <https://pytorch.org/get-started/locally/>

For CPUs, enter the following command in the terminal to install.



Machine Learning Basics

AI development environment

Install Docker (Optional)

Docker installation is required for virtual image-based tasks.

Visit <https://www.docker.com/get-started/> to install

Install Ollama

To use the local LLM AI tool, you need to install Ollama.

Visit <https://www.ollama.com/> to install

Install a Code Editor and IDE

Install <https://www.sublimetext.com/>

Install <https://code.visualstudio.com/download>. For detailed installation instructions,

Install Claude Desktop

Install <https://claude.ai/download>

<https://docs.github.com/ko/desktop/installing-and-authenticating-to-github-desktop/installing-github-desktop>

Machine Learning Basics

AI development environment

GitHub: AI foundation tutorial

README

MIT license

AI foundation and trend seminar tutorial

This repository contains materials for an [AI Foundation seminar](#), covering fundamental concepts of AI, Machine Learning, Deep Learning, Natural Language Processing, Transformers, and Large Language Models (LLMs), including agent-based approaches and related services. It is designed to provide hands-on experience, primarily utilizing Jupyter Notebooks. In reference, you can learn [How to develop AI agent with LLM](#), [computer vision with deep learning](#) and [AI for media art](#).



Repository Structure

The repository is organized into several folders, each focusing on a specific area of AI, along with supplementary documents:

- 1_AK_trend : AI Transformation trends.
- 2_ML_basic : Basic Machine Learning concepts.
- 3_DL_foundation : Deep Learning foundations.
- 4_NLP : Natural Language Processing.
- 5_transformer : Transformer models.
- 6_LLM_agent_vibe : LLM Agent concepts and vibe coding.
- 7_service : AI services related topics.
- 8_AK_reference : AI Transformation references.
- AI_foundation_and_trend.pdf : A PDF slide document possible.
- AI_foundation_syllabus.docx : The syllabus for the AI Foundation.
- LICENSE : Contains the MIT License information.
- LLM-lesson-plan.docx : A lesson plan related to LLMs.
- README.md : This README file.

Getting Started: Development Environment Setup

This section outlines the [prerequisites and installation steps](#) to prepare your working environment for a smooth hands-on experience. All materials can be downloaded from this repository.

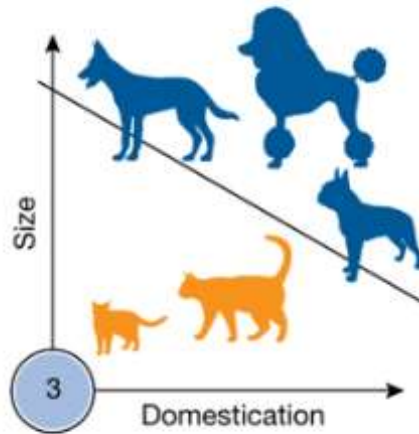
- **LICENSE** : Contains the MIT License information.
- **LLM-lesson-plan.docx** : A lesson plan related to LLMs.
- **README.md** : This README file.

Getting Started: Development Environment Setup

This section outlines the [prerequisites and installation steps](#) to prepare your working environment for a smooth hands-on experience. All materials can be downloaded from this repository.

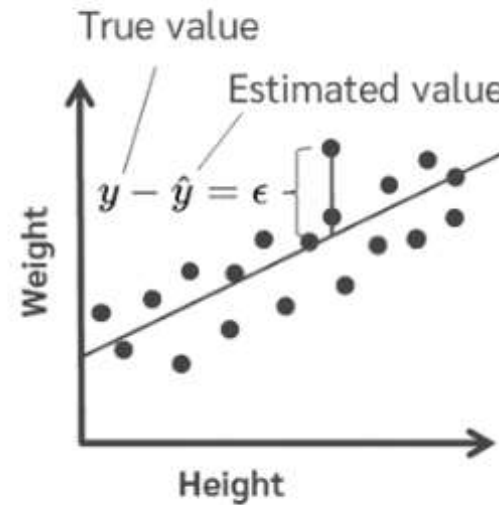
Machine Learning Basics classification and regression

Classification



Classification is a machine learning technique that assigns input data to one of a number of predefined categories (classes).

Regression analysis



Regression analysis learns the relationship between input values and target values with the minimum error. It is a model that can estimate the result y for the input.

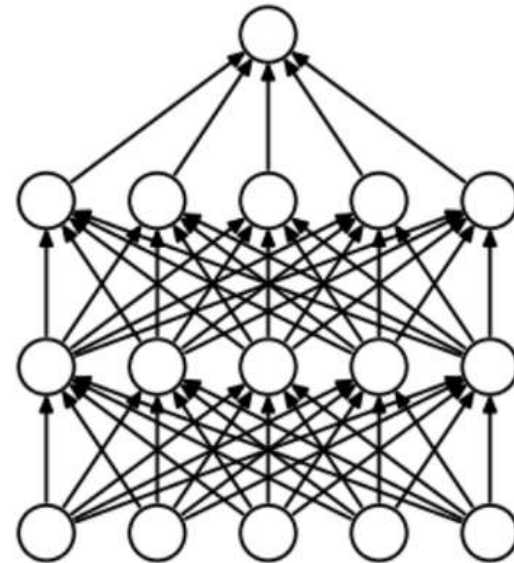
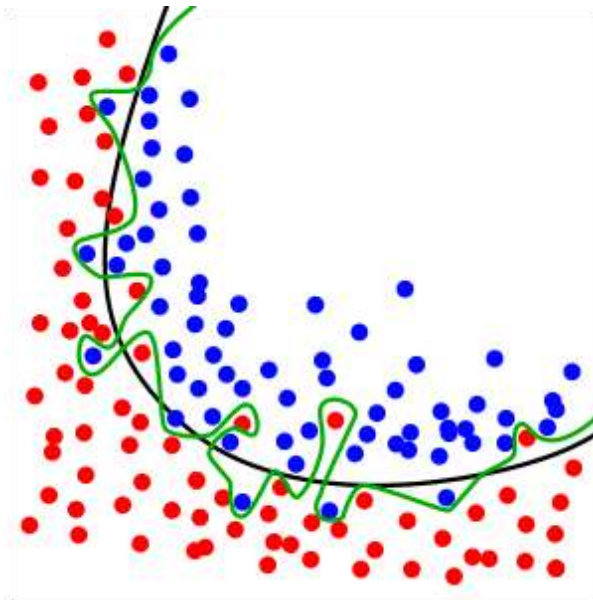
characteristic	Classification	Regression analysis
target	Classify/assign data into predefined categories	Predict/estimate continuous numeric values
output type	Discrete and categorical labels (e.g., "Yes/No", "A/B/C")	Continuous, numeric values (e.g., 1.23, 100,000)
Example problem	Spam detection, disease diagnosis, image recognition	House price forecasts, temperature forecasts, and sales forecasts

Machine Learning Basics

Overfitting, Normalization, validation

Overfitting

A phenomenon in which the prediction performance for new data deteriorates due to overtraining on the training data.



Machine Learning Basics

Overfitting, Normalization, validation

Korean Society of Magnetic Resonance in Medicine, 2019,
Deep Learning in MR Image Processing

Overfitting and 검증셋



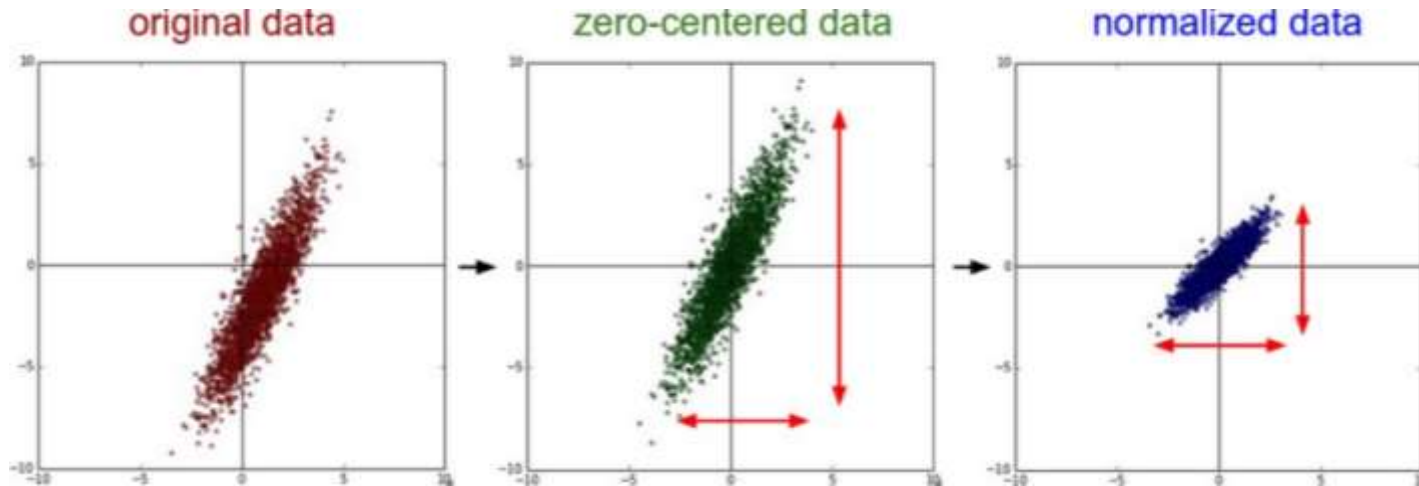
characteristic	Training Set	Validation Set	Test Set
purpose	Model training and parameter optimization	Hyperparameter tuning and model selection	Evaluate the final performance of the model and measure its generalization ability.
When to use	Initial and iterative learning phases of model development	Intermediate learning and hyperparameter tuning stages	once after the use model development is complete Only
data exposure	Data that the model directly learns from	The model is used to evaluate performance during training, but is not used for direct training.	completely unseen by the model new data
Avoid overfitting	It may overfit the training data.	Helps monitor and prevent overfitting	Objectively measure the true generalization performance of a model.

Machine Learning Basics

Overfitting, Normalization, validation

Normalization

Regularization is a general term for techniques to prevent machine learning models from overfitting to training data and to improve generalization performance for new data.



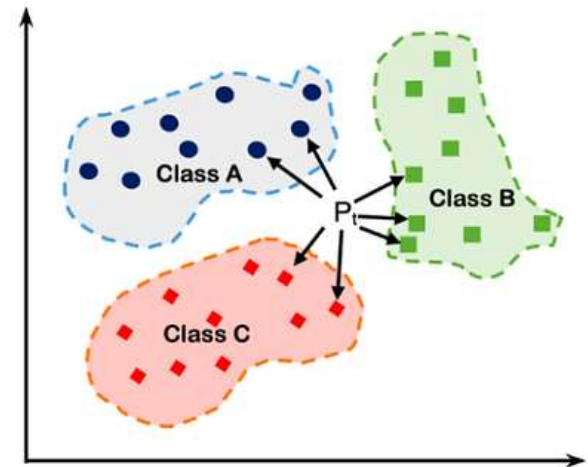
Practical Aspects
of Deep Learning -
2 | Ashish Jaiswal

techniques	principle	main effect
L1 regularization	Preventing bias towards specific weights	Mitigating overfitting
L2 regularization		Model stabilization and overfitting mitigation
dropout	Randomly set some neuron outputs to 0 during training	Inhibition of excessive interneuron dependence, generalization
Early termination	Stop training if there is no improvement in performance on validation data.	Performance improvements
data augmentation	Augmenting data through transformations such as rotation, translation, flipping, and adding noise.	Prevent overfitting and save training time
		Expanding data diversity, making models robust to deformation, and improving generalization performance

Machine Learning Basics algorithm

K-Nearest Neighbors (KNN)

Neighborhood-based learning (Instance-based learning) finds the K nearest training data points for a given new data point and performs clustering. Used for data classification.



[K Nearest Neighbours — Introduction to Machine Learning Algorithms | by Sachin Soni](#)

Function KNN_Predict(training_data, training_labels, new_data_point, K):

 distance_list = empty list

 # Calculate distance to all training data points

 For each training_data_point in training_data:

 distance = Calculate_Distance(new_data_point, training_data_point) # Euclidean Distance

 Add (distance, training_data_point_label) pair to distance_list

 # Sort by distance and select the K nearest

 Sort distance_list in ascending order by distance

 k_nearest_neighbors = top K items from distance_list

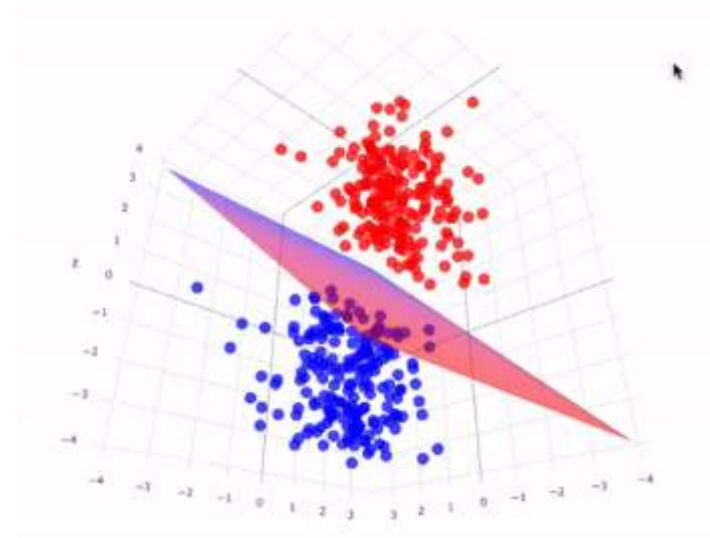
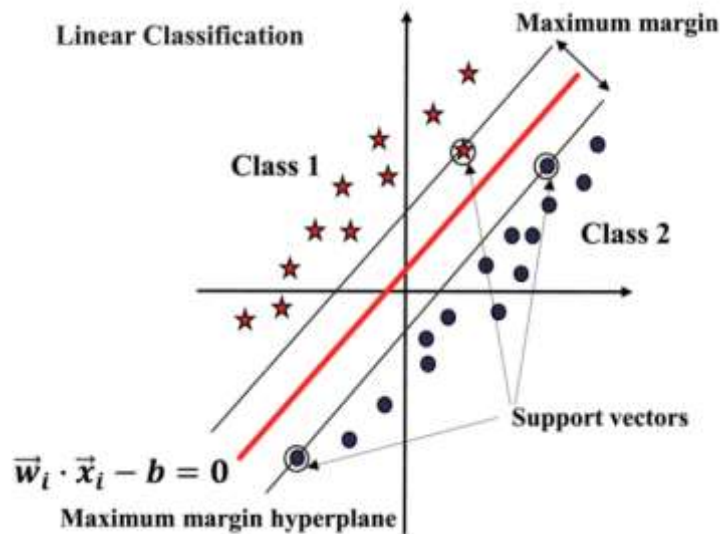
 # Make Prediction

 Return the most frequent label among k_nearest_neighbors_labels

Machine Learning Basics algorithm

Support Vector Machine (SVM)

Calculate the optimal decision boundary that divides the given data into different classes.
This is used to classify the dataset.



Machine Learning Basics algorithm

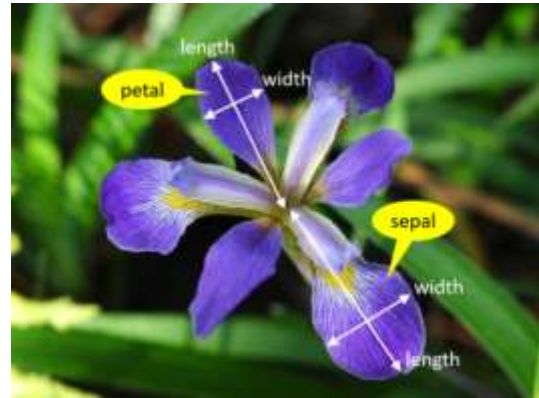
Decision Tree

An algorithm that represents a series of rules that continuously divide data according to specific criteria in a tree structure.

Root Node: The starting point of the tree and contains the entire data.

Internal Node: Represents a "question" or "condition" regarding a specific feature. Data is divided into subnodes based on the question results.

Leaf Node: The end point of the tree, where no further division occurs and contains the final decision (class label or predicted value).



Dataset Order	Sepal length	Sepal width
1	5.1	3.5
2	4.9	3.0
101	6.3	3.3
102	5.8	2.7

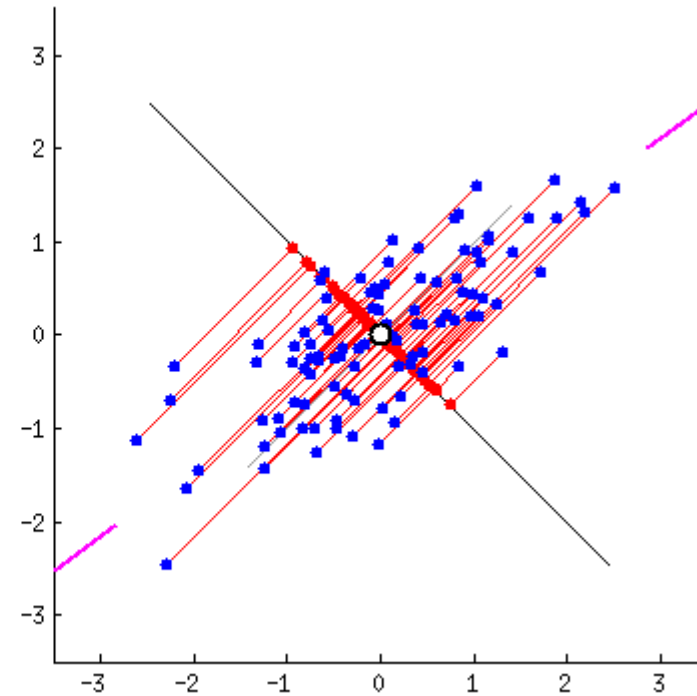
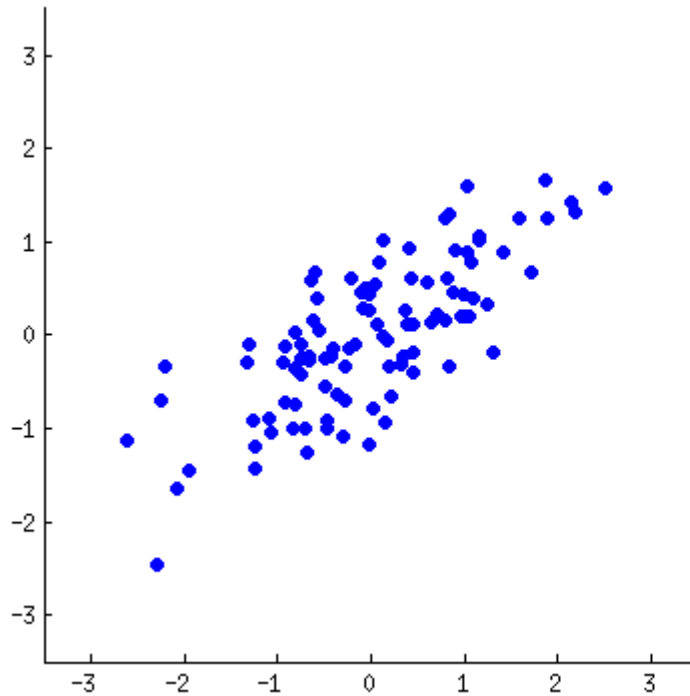
Petal length	Petal width	Species
1.4	0.2	<i>I. setosa</i>
1.4	0.2	<i>I. setosa</i>
6.0	2.5	<i>I. virginica</i>
5.1	1.9	<i>I. virginica</i>



Machine Learning Basics algorithm

Principal Component Analysis (PCA)

Transforms high-dimensional data into a low-dimensional space and calculates orthogonal axes that best describe the variance of the original data. Used to reduce data dimensionality.



Machine Learning Basics algorithm

algorithm	implementation principle	application
K-Nearest Neighbors (KNN)	Use the K closest neighbor values for prediction of new data.	Recommendation systems, image classification, etc.
Linear Regression	Find the optimal linear relationship between input and target to predict	Price prediction, statistical inference, etc.
Logistic Regression	Apply sigmoid to linear combination to output probability and classify binary.	Spam classification, disease diagnosis, etc.
Support Vector Machine (SVM)	Compute the optimal surface boundary with the maximum margin to divide the data.	Image classification, handwriting recognition, etc.
Decision Tree	Predict by dividing data into a tree structure according to rules	Create decision rules, segment customers, etc.
Random Forest	Ensemble learning, which generates multiple decision trees and synthesizes predictions.	Fraud detection, medical diagnosis, etc.
Principal Component Analysis (PCA)	Reduce high-dimensional data to low dimensions while preserving its distribution.	Image compression, feature extraction, etc.

Machine Learning Basics

Pipeline using Scikit-learn

Scikit-learn

An open-source machine learning library. Provides supervised learning (classification, regression) and unsupervised learning (clustering, dimensionality reduction) algorithms, data preprocessing, and model evaluation tools.

Algorithms: Provides numerous algorithms, including linear regression, logistic regression, decision trees, random forests, K-NN, and K-Means.

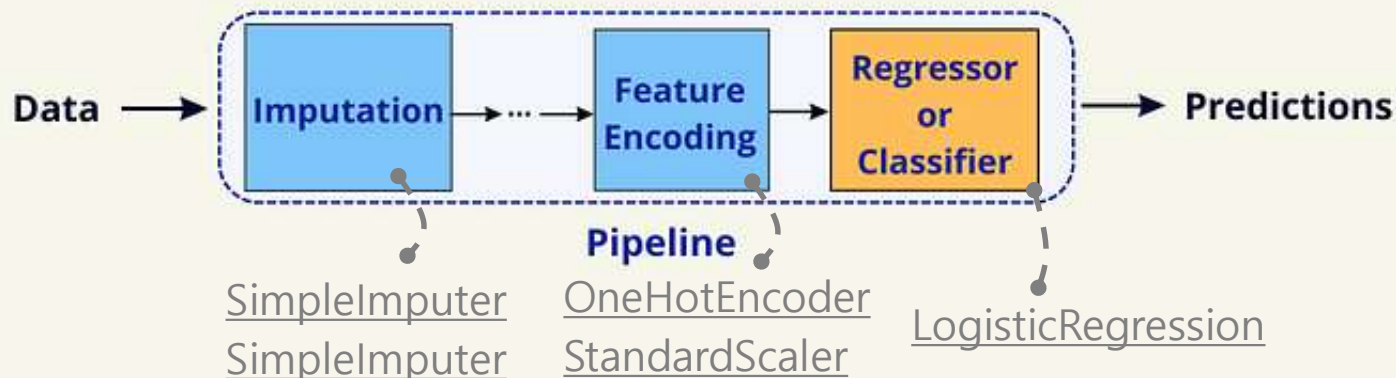
Data preprocessing: Data cleaning and transformation tools, including scaling (StandardScaler, MinMaxScaler), encoding (OneHotEncoder), and feature transformation (PolynomialFeatures).

Model selection and evaluation: Provides cross-validation (cross_val_score), hyperparameter tuning (GridSearchCV, RandomizedSearchCV), and performance metrics (accuracy, classification).

Consistent API: All estimators use consistent methods, such as fit(), transform(), and predict().

Pipeline is a class that sequentially connects multiple transformers and a final estimator, acting as a single combined estimator.

Simplify Machine Learning Workflow With Scikit-Learn Pipelines



[Scikit-learn Pipelines Explained: Streamline and Optimize Your Machine Learning Processes | by Sahin Ahmed, Data Scientist | Medium](#)

Machine Learning Basics

Pipeline using Scikit-learn

```
import numpy as np, pandas as pd
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline, make_pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.model_selection import train_test_split
```

```
data = {
    "state": ["CA", "WA", "CA", np.nan, "NV", "WA"],
    "gender": ["male", "female", "female", "male", np.nan, "female"],
    "age": [34, 29, 22, 44, 55, np.nan],
    "weight": [122, 150, 130, np.nan, 140, 175],
    "target": [0, 1, 0, 1, 0, 1]
}
df = pd.DataFrame(data)
X = df.drop("target", axis=1)
print("Input DataFrame:")
print(X)
```

```
y = df["target"]
print("Target Series:")
print(y)
```

	state	gender	age	weight	
0	CA	male	34.0	122.0	0
1	WA	female	29.0	150.0	1
2	CA	female	22.0	130.0	0
3	NaN	male	44.0	NaN	1
4	NV	NaN	55.0	140.0	0
5	WA	female	NaN	175.0	1

Machine Learning Basics

Pipeline using Scikit-learn

```
categorical_preprocessor = Pipeline(  
    steps=[ ("imputation_constant", SimpleImputer(fill_value="missing", strategy="constant")),  
            ("onehot", OneHotEncoder(handle_unknown="ignore")) ]  
)
```

CA: [1, 0, 0, 0]
WA: [0, 1, 0, 0]
NV: [0, 0, 1, 0]
missing: [0, 0, 0, 1]

```
numeric_preprocessor = Pipeline(  
    steps=[ ("imputation_mean", SimpleImputer(missing_values=np.nan, strategy="mean")),  
            ("scaler", StandardScaler()) ]  
)
```

```
preprocessor = ColumnTransformer(  
    [  
        ("categorical", categorical_preprocessor, ["state", "gender"]),  
        ("numerical", numeric_preprocessor, ["age", "weight"]),  
    ]  
)
```

	state	gender	age	weight	
0	CA	male	34.0	122.0	0
1	WA	female	29.0	150.0	1
2	CA	female	22.0	130.0	0
3	NaN	male	44.0	NaN	1
4	NV	NaN	55.0	140.0	0
5	WA	female	NaN	175.0	1

```
pipe = make_pipeline(preprocessor, LogisticRegression(max_iter=500))  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)  
pipe.fit(X_train, y_train)
```

```
print(f"Input test set: {X_test}")  
predictions = pipe.predict(X_test)  
print(f"Predictions on the test set: {predictions}")
```

	state	gender	age	weight
0	CA	male	34.0	122.0
1	WA	female	29.0	150.0

Predictions on the test set:
[0 1]

Understanding the Structure of Deep Learning

Basic Neural Network Structure (MLP)

Meanings of Optimizer, Learning Rate, and Batch Size

Conceptual Differences Between CNN, RNN, and LSTM

Comparison of TensorFlow and PyTorch

Key Question

How does deep learning mimic neural networks for learning?

What is the principle of backpropagation, the core of learning?

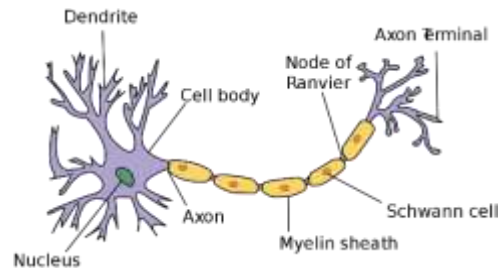
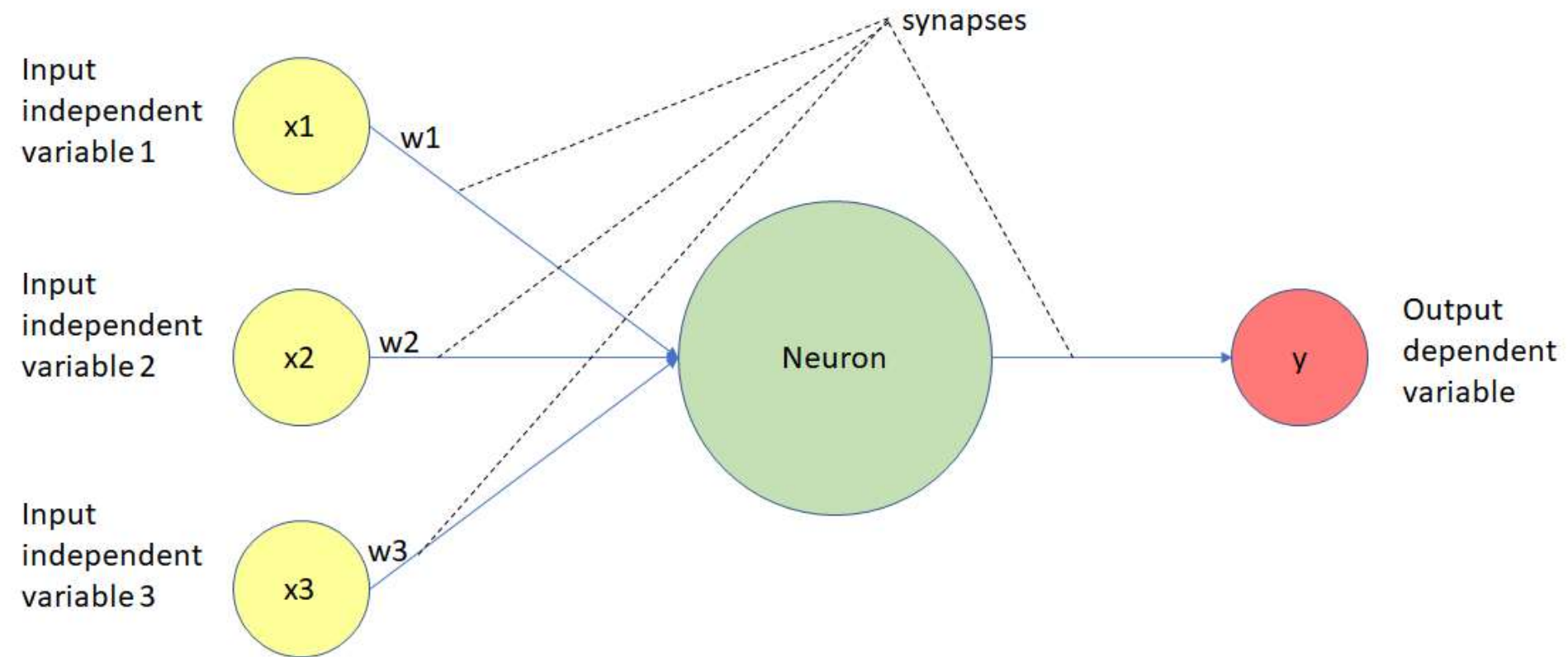
What does the output from each neural network layer mean?

What is important when designing a neural network?

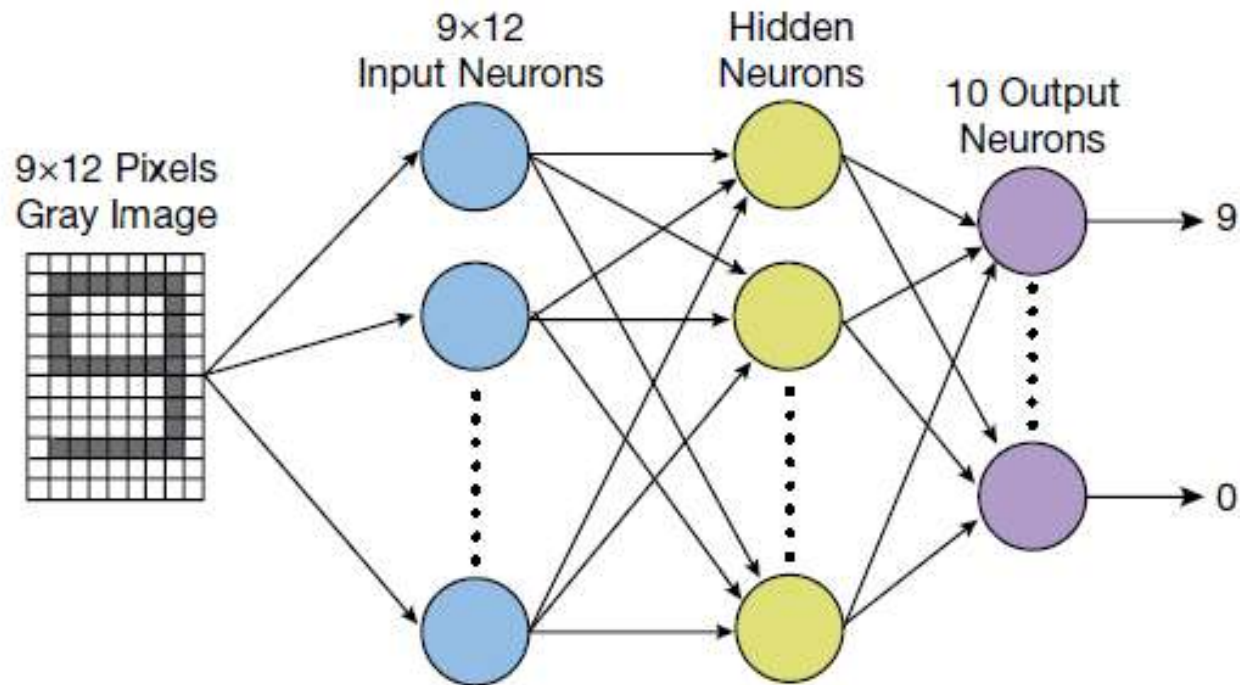
How can we effectively extract and train data features to achieve our goals?

What is the purpose of the mathematical formulas used to compute neural networks?

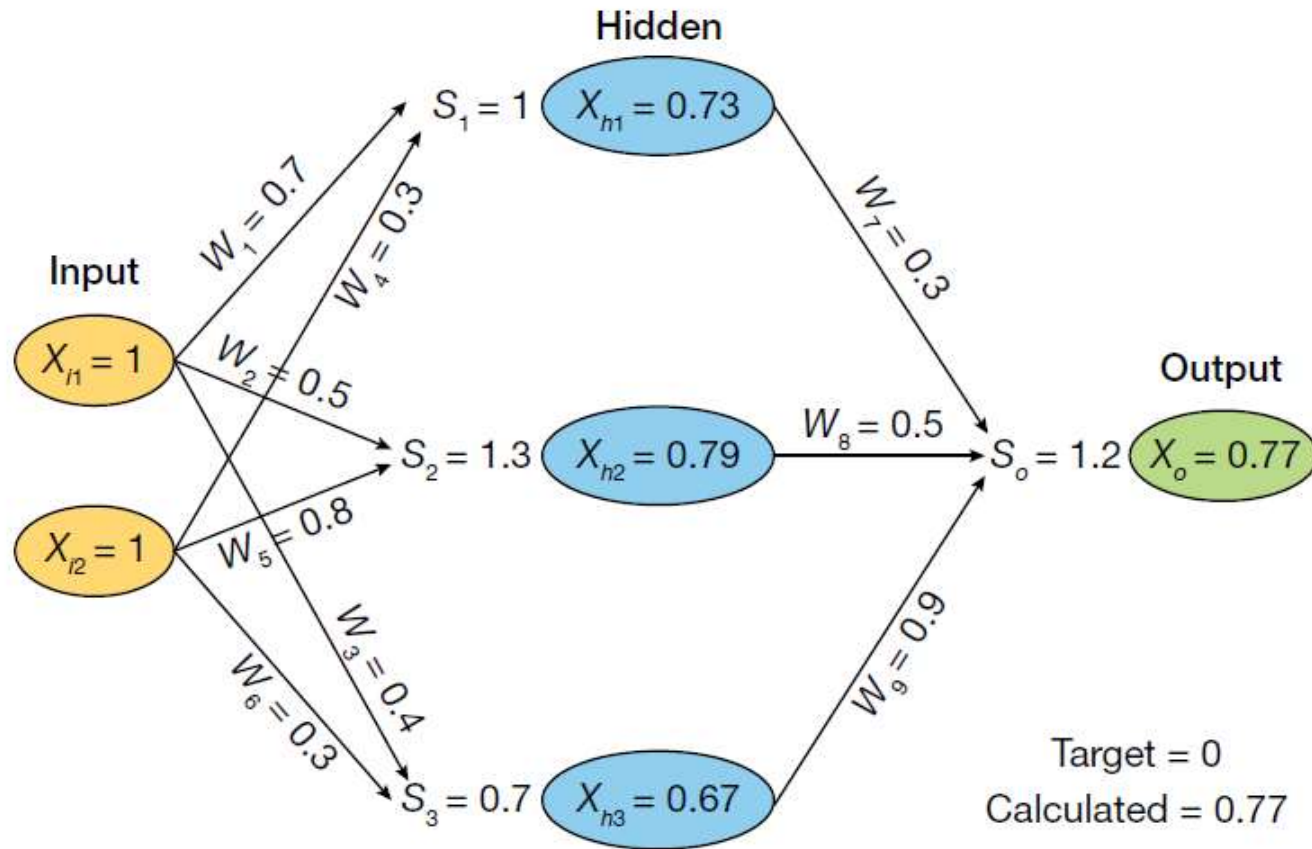
Structural Understanding neural networks (MLP)



Structural Understanding neural networks (MLP)

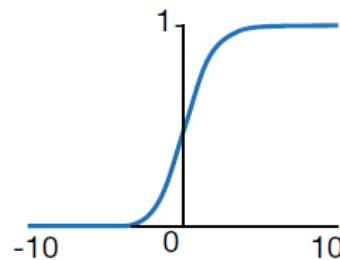


Structural Understanding neural networks (MLP)

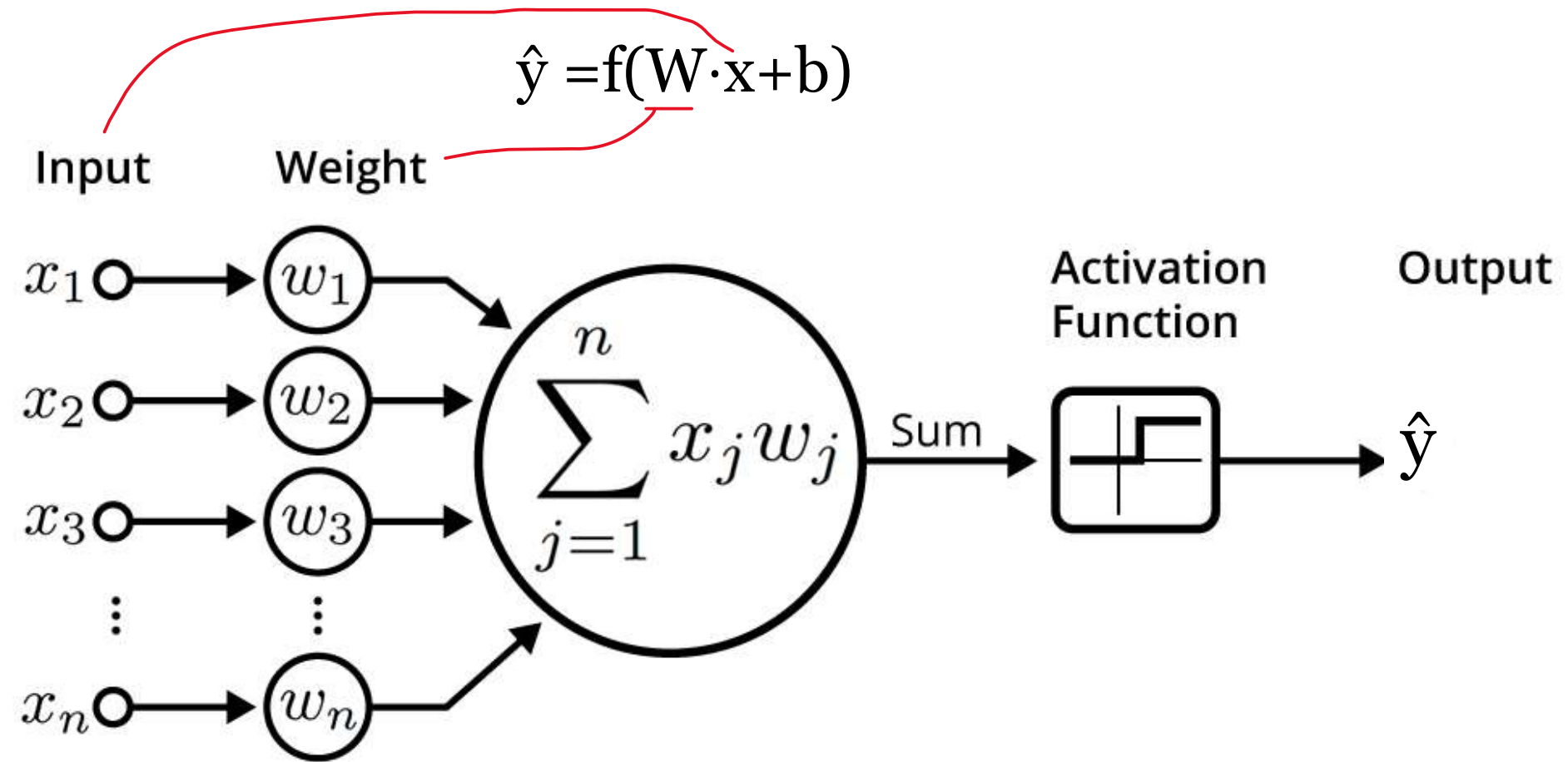


Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



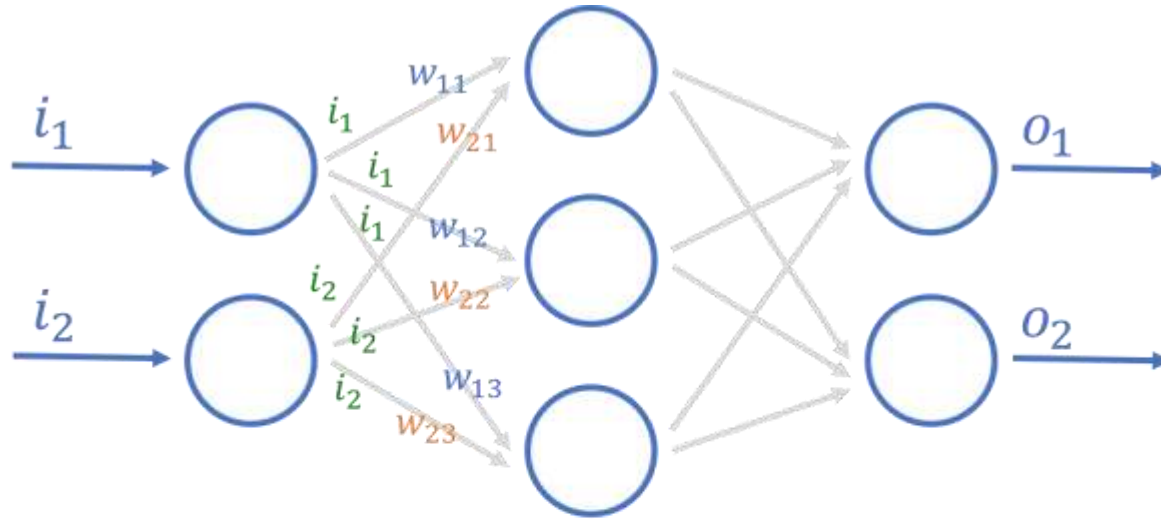
Structural Understanding neural networks (MLP)



$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Mean
Squared
Error

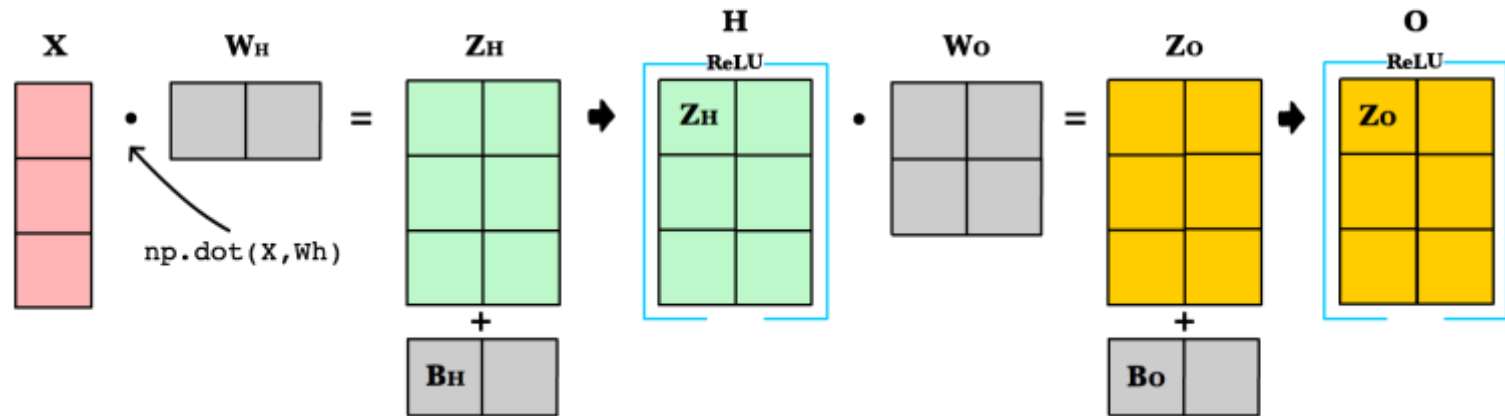
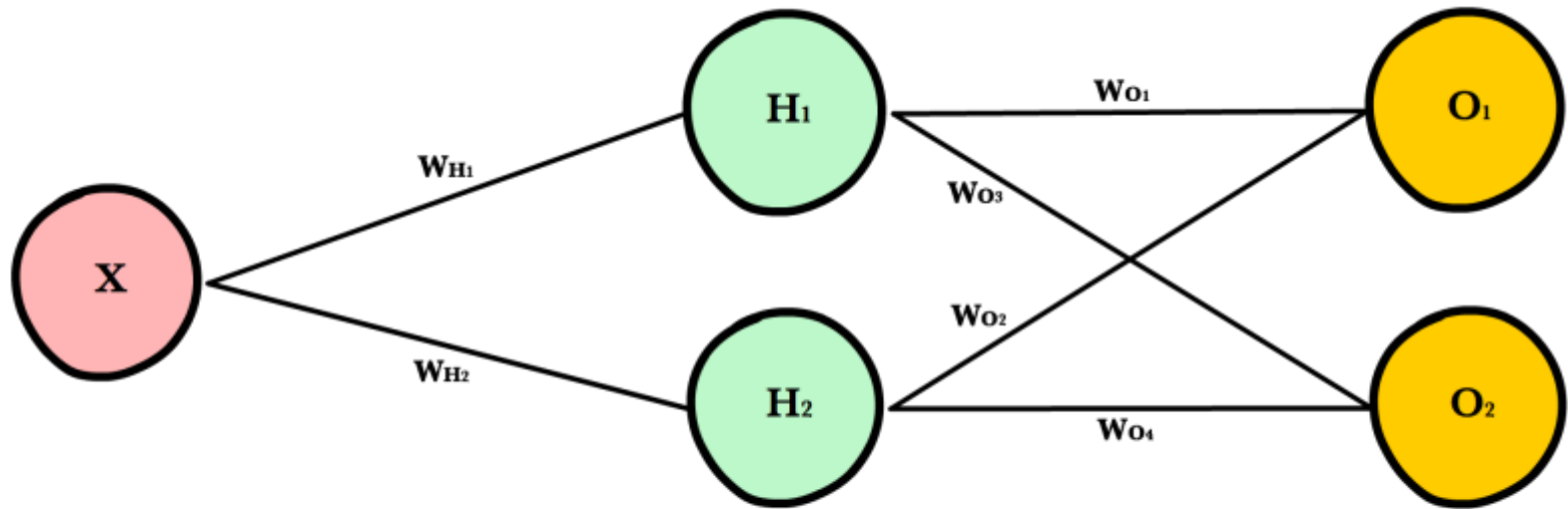
Structural Understanding neural networks (MLP)



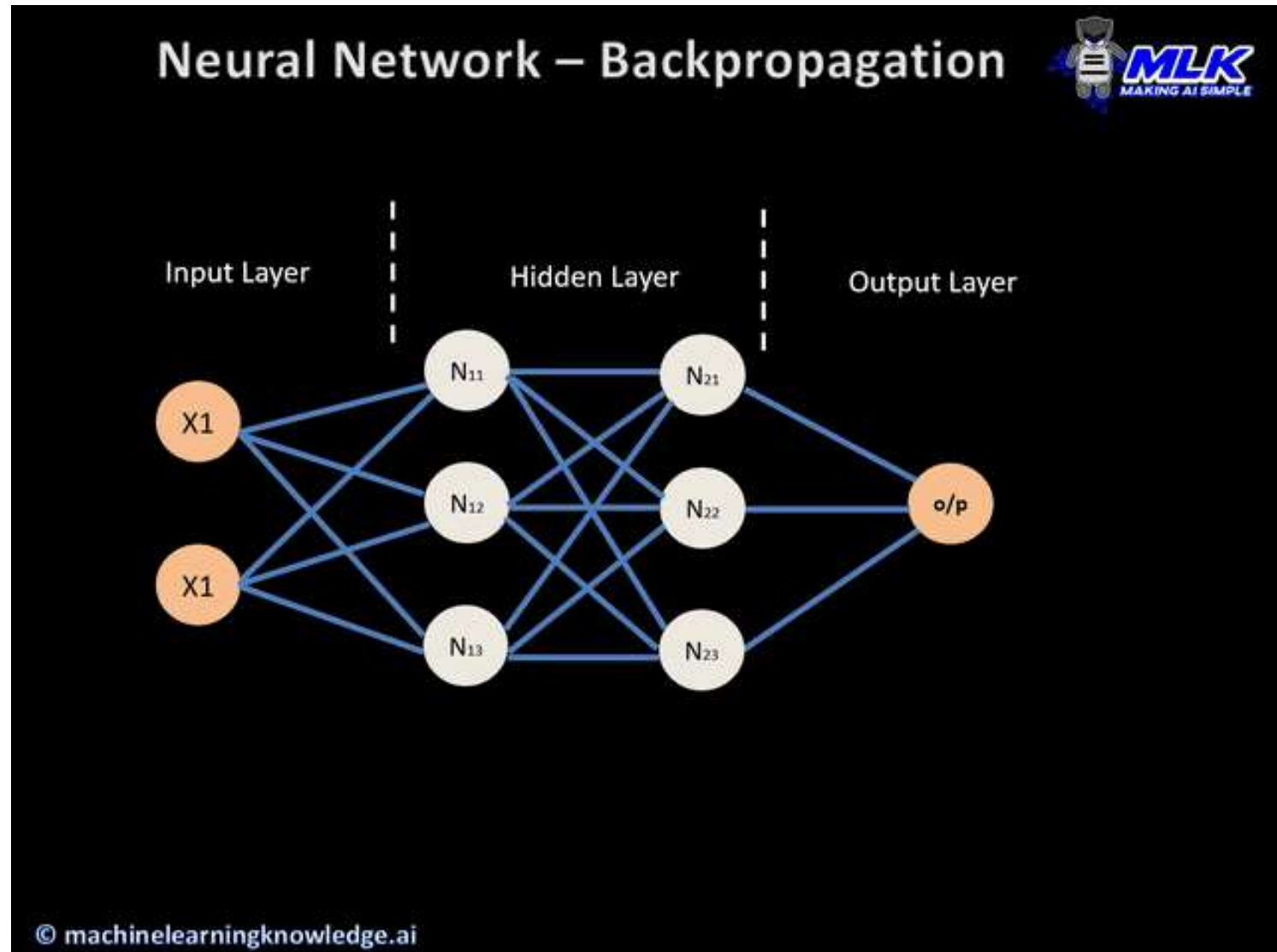
$$f(\mathbf{W} \cdot \mathbf{x} + \mathbf{b}) = \hat{\mathbf{y}}$$

$$\begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} (w_{11} \times i_1) + (w_{21} \times i_2) \\ (w_{12} \times i_1) + (w_{22} \times i_2) \\ (w_{13} \times i_1) + (w_{23} \times i_2) \end{bmatrix}$$

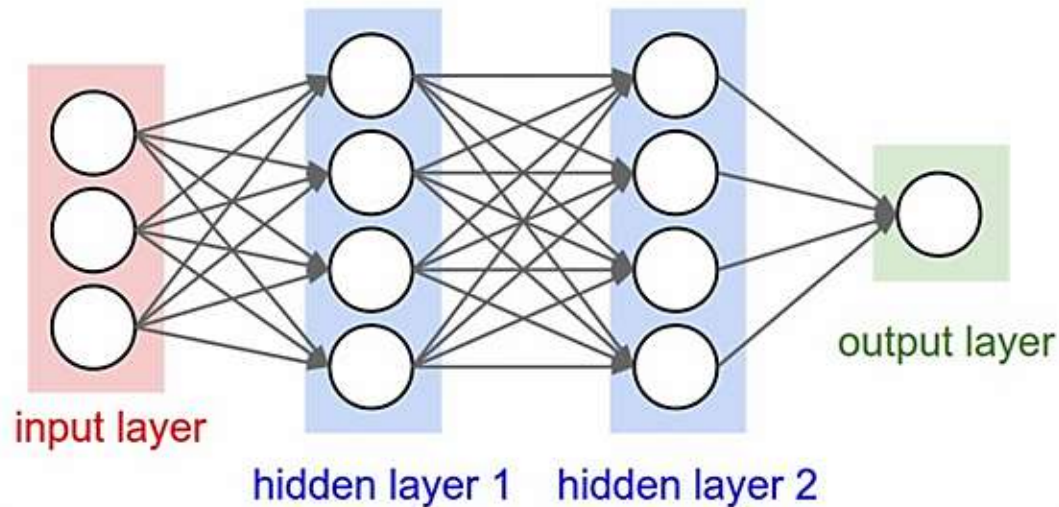
Structural Understanding neural networks (MLP)



Structural Understanding neural networks (MLP)



Structural Understanding neural networks (MLP)



```
# forward-pass of a 3-layer neural network:  
f = lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)  
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)  
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)  
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)  
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

Structural Understanding neural networks (MLP)

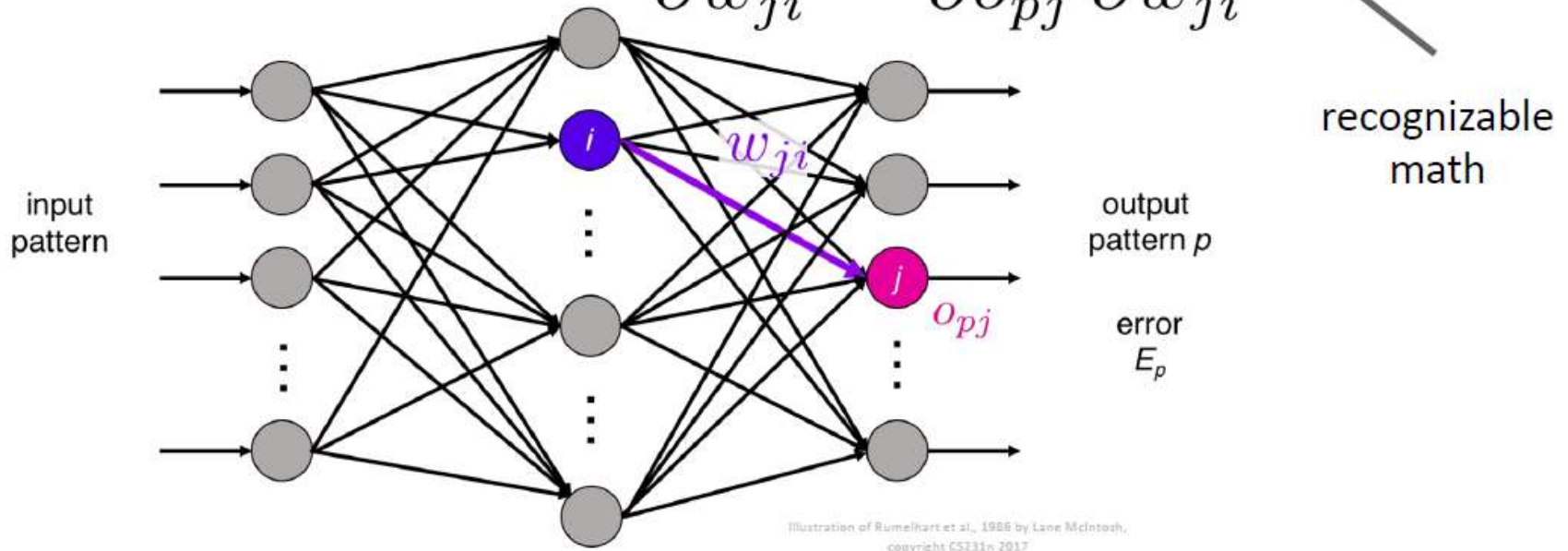
역전파(Backpropagation)

An algorithm that trains a model by efficiently updating artificial neural network weights. It uses the chain rule to calculate the gradient of the loss function and uses this gradient to update the weights using an algorithm like gradient descent.

$$\Delta w_i = -\alpha \frac{dE}{dw_i}$$

$$w_{next} = w + \Delta w$$

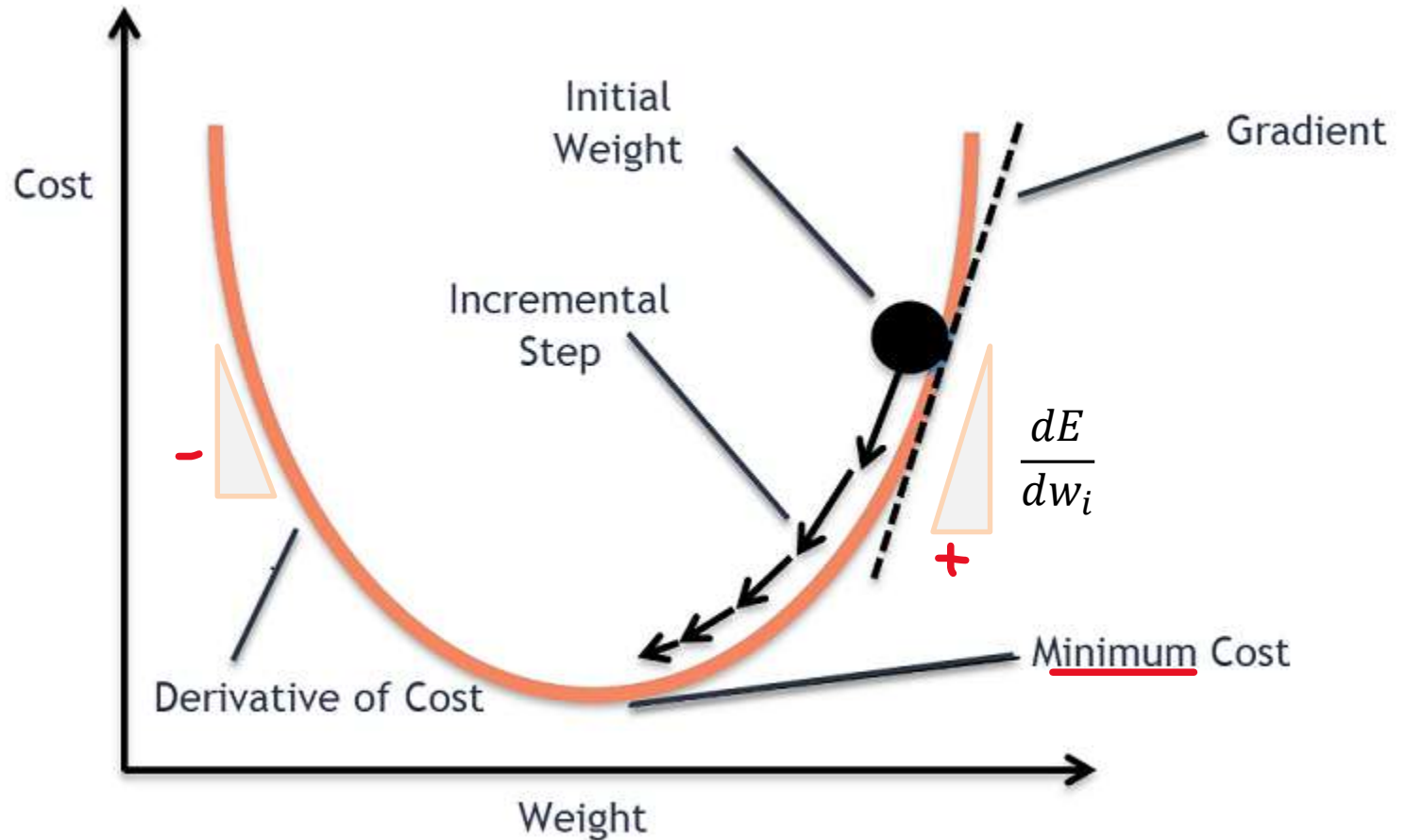
$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial o_{pj}} \frac{\partial o_{pj}}{\partial w_{ji}}$$



Structural Understanding neural networks (MLP)

$$\Delta w_i = -\alpha \frac{dE}{dw_i}$$

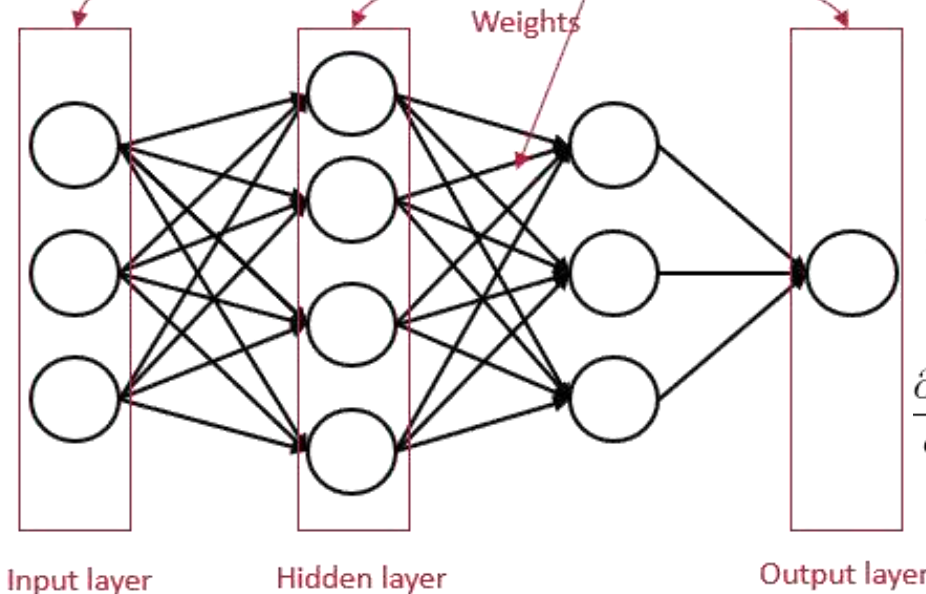
$$w_{next} = w + \Delta w$$



Structural Understanding neural networks (MLP)

A neural network layer is not a single function of a single parameter, but a composite function of multiple parameters. The chain rule from calculus is used to calculate the change in error due to small changes in weights.

$$f(x) = g \circ f_K \circ \dots \circ f_2 \circ f_1(x) \quad z_i = w_i a_{i-1} + b_i \quad a_i = \sigma(z_i)$$



$$\frac{\partial w_i}{\partial L} = \frac{\partial o}{\partial L} \times \frac{\partial z}{\partial o} \times \frac{\partial w_i}{\partial z}$$

$$\frac{\partial o}{\partial L} = (o - y) \quad (\text{derivative of the loss function})$$

$$\frac{\partial z}{\partial o} = o(1 - o) \quad (\text{derivative of the sigmoid function})$$

$$\frac{\partial w_i}{\partial z} = x_i \quad (\text{derivative of the weighted sum})$$

Structural Understanding neural networks (MLP)

A neural network layer is not a single function of a single parameter, but a composite function of multiple parameters. The chain rule from calculus is used to calculate the change in error due to small changes in weights.

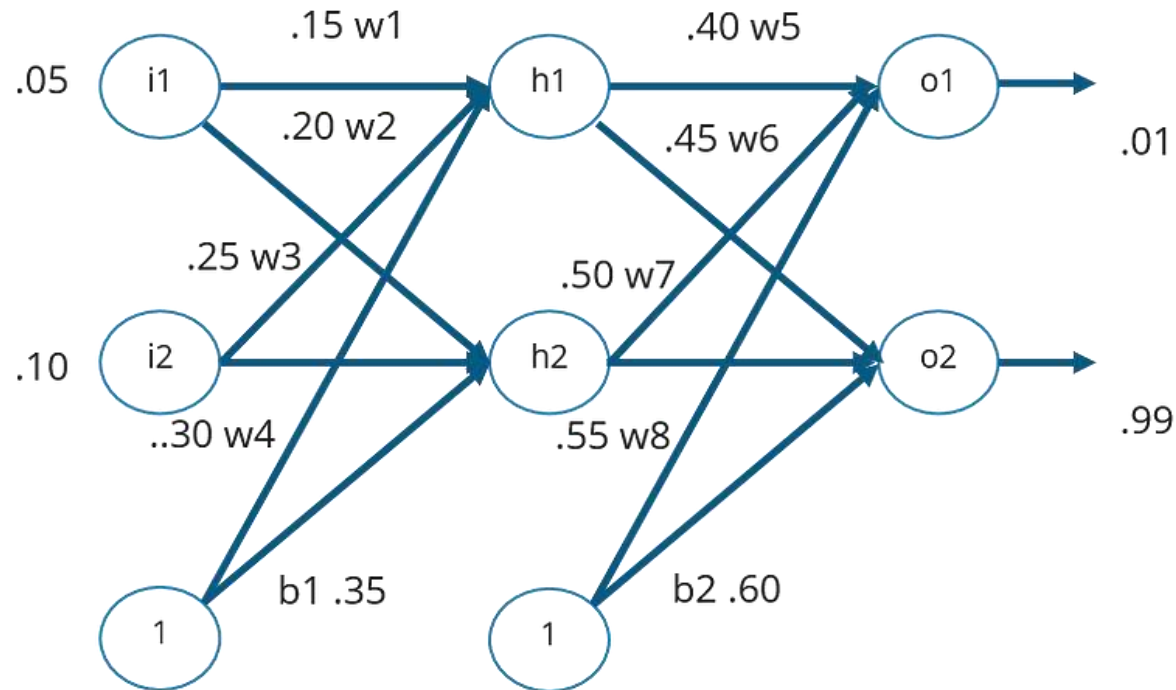
Rule	$f(x)$	Scalar derivative notation with respect to x	Example
Constant	c	0	$\frac{d}{dx}99 = 0$
Multiplication by constant	cf	$c\frac{df}{dx}$	$\frac{d}{dx}3x = 3$
Power Rule	x^n	nx^{n-1}	$\frac{d}{dx}x^3 = 3x^2$
Sum Rule	$f + g$	$\frac{df}{dx} + \frac{dg}{dx}$	$\frac{d}{dx}(x^2 + 3x) = 2x + 3$
Difference Rule	$f - g$	$\frac{df}{dx} - \frac{dg}{dx}$	$\frac{d}{dx}(x^2 - 3x) = 2x - 3$
Product Rule	fg	$f\frac{dg}{dx} + \frac{df}{dx}g$	$\frac{d}{dx}x^2x = x^2 + x2x = 3x^2$
Chain Rule	$f(g(x))$	$\frac{df(u)}{du}\frac{du}{dx}$, let $u = g(x)$	$\frac{d}{dx}\ln(x^2) = \frac{1}{x^2}2x = \frac{2}{x}$

$$y=f(g(x))=\sin(x^2) \quad \left. \begin{array}{l} y = \sin \\ \uparrow \\ u = \text{square} \\ \uparrow \\ x \end{array} \right\} \frac{dy}{du} \frac{du}{dx}$$

$$u=x^2, y=\sin(u) \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} \quad \frac{\frac{du}{dx}}{\frac{dy}{du}} = \frac{2x}{\cos(u)}$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \cos(u)2x = 2x\cos(x^2)$$

Structural Understanding neural networks (MLP)



Net Input For h_1 :

$$\text{net } h_1 = w_1 \cdot i_1 + w_2 \cdot i_2 + b_1 \cdot 1$$

$$\text{net } h_1 = 0.15 \cdot 0.05 + 0.2 \cdot 0.1 + 0.35 \cdot 1 = 0.3775$$

Output Of h_1 :

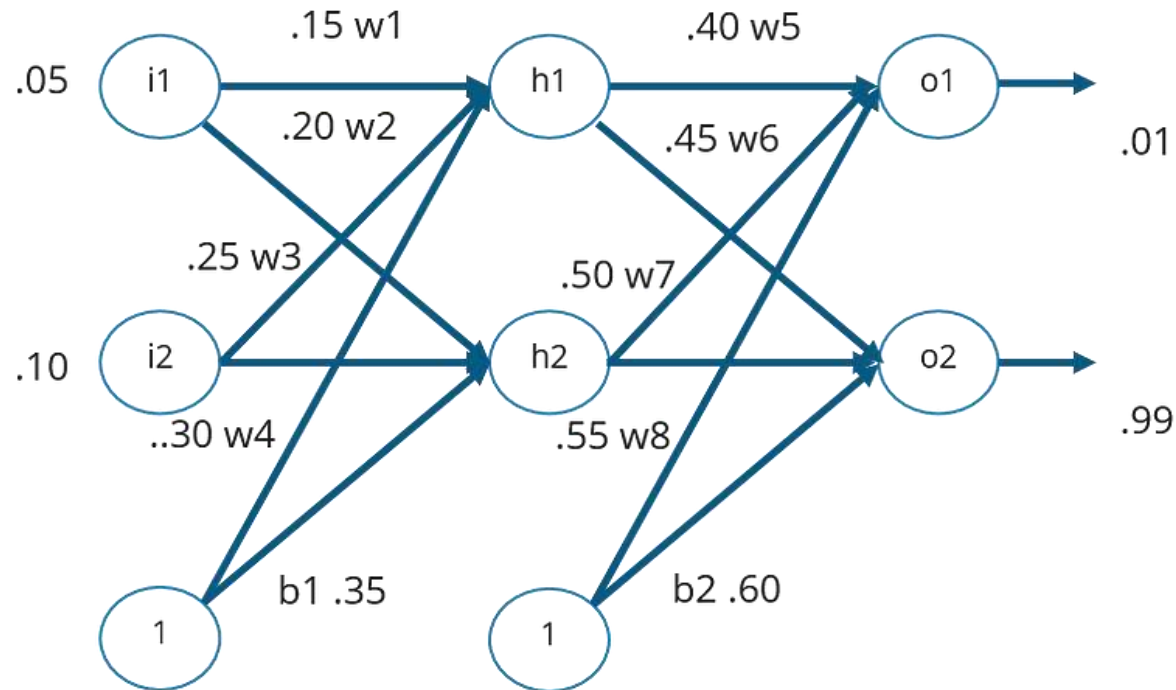
$$\text{out } h_1 = \frac{1}{1 + e^{-\text{net } h_1}}$$

$$\frac{1}{1 + e^{-0.3775}} = 0.593269992$$

Output Of h_2 :

$$\text{out } h_2 = 0.596884378$$

Structural Understanding neural networks (MLP)



Output For o1:

$$\text{net } o1 = w5 \cdot \text{out } h1 + w6 \cdot \text{out } h2 + b2 \cdot 1$$

$$0.4 \cdot 0.593269992 + 0.45 \cdot 0.596884378 + 0.6 \cdot 1 = 1.105905967$$

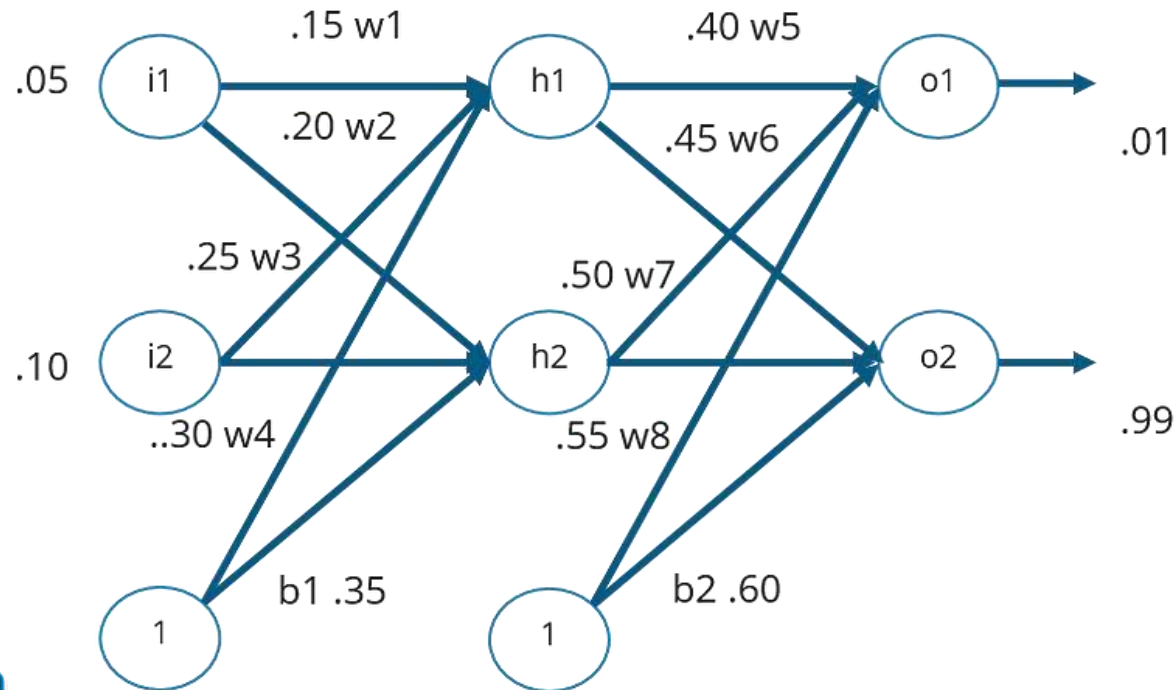
$$\text{Out } o1 = 1 / (1 + e^{-\text{net } o1})$$

$$1 / (1 + e^{-1.105905967}) = 0.75136507$$

Output For o2:

$$\text{Out } o2 = 0.772928465$$

Structural Understanding neural networks (MLP)



Error For o_1 :

$$E_{o1} = \sum 1/2(\text{target} - \text{output})^2 \rightarrow \frac{1}{2} (0.01 - 0.75136507)^2 = 0.274811083$$

Error For o_2 :

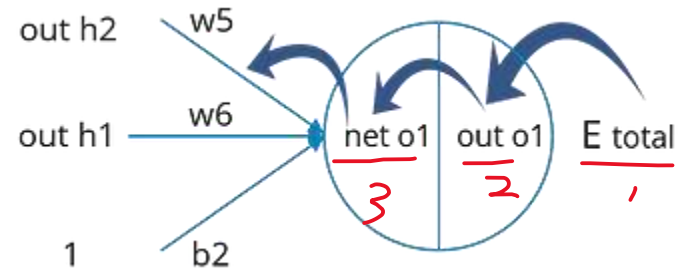
$$E_{o2} = 0.023560026$$

Total Error:

$$E_{\text{total}} = E_{o1} + E_{o2} \rightarrow 0.274811083 + 0.023560026 = 0.298371109$$

Structural Understanding neural networks (MLP)

$$\frac{\delta E_{total}}{\delta w_5} = \frac{\delta E_{total}}{\delta out\ o1} * \frac{\delta out\ o1}{\delta net\ o1} * \frac{\delta net\ o1}{\delta w_5}$$



$$E_{total} = 1/2(target\ o1 - out\ o1)^2 + 1/2(target\ o2 - out\ o2)^2$$

$$\frac{\delta E_{total}}{\delta out\ o1} = -(target\ o1 - out\ o1) = -(0.01 - 0.75136507) = 0.74136507$$

$$out\ o1 = 1/1+e^{-net\ o1}$$

$$\frac{\delta out\ o1}{\delta net\ o1} = out\ o1 (1 - out\ o1) = 0.75136507 (1 - 0.75136507) = 0.186815602$$

$$net\ o1 = w_5 * out\ h1 + w_6 * out\ h2 + b_2 * 1$$

$$\frac{\delta net\ o1}{\delta w_5} = 1 * out\ h1 = 0.593269992$$

$$\frac{\delta E_{total}}{\delta w_5} = \frac{\delta E_{total}}{\delta out\ o1} * \frac{\delta out\ o1}{\delta net\ o1} * \frac{\delta net\ o1}{\delta w_5} = 0.082167041$$

$$w_5^+ = w_5 - \eta \frac{\delta E_{total}}{\delta w_5} \quad w_5^+ = 0.4 - 0.5 * 0.082167041$$

Updated w5

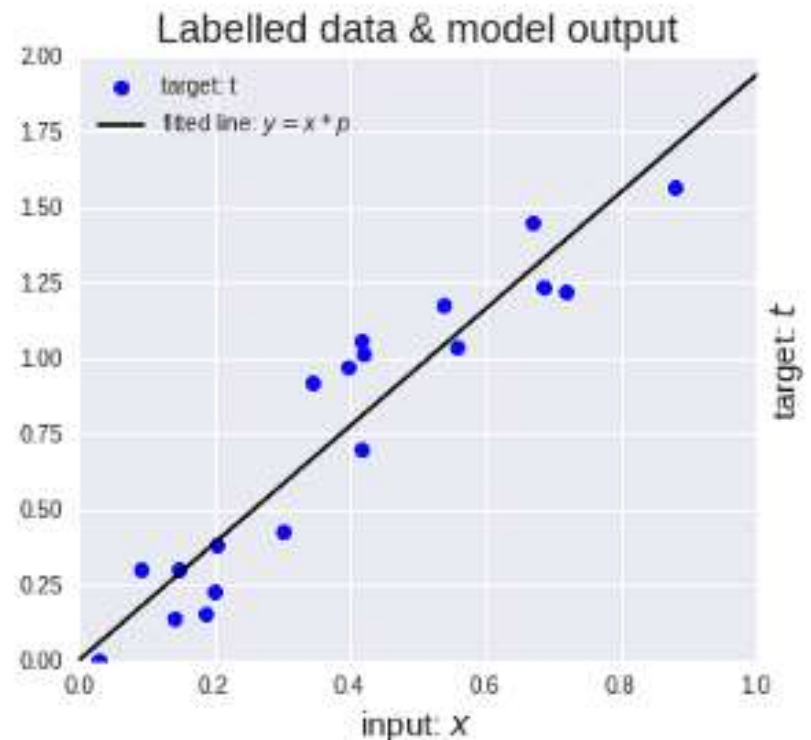
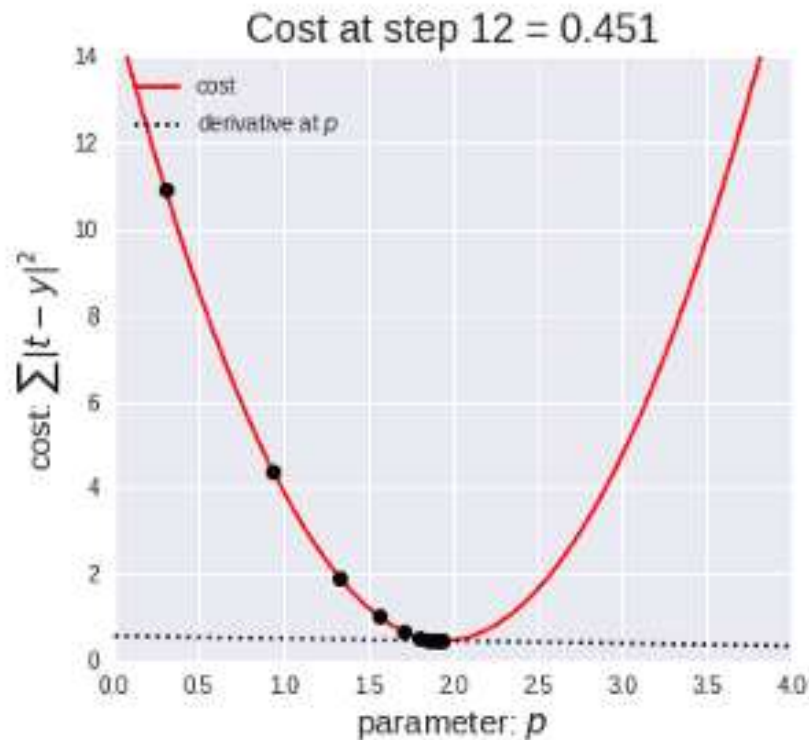
0.35891648

Structural Understanding neural networks (MLP)

$$\hat{y} = f(W \cdot x + b)$$

$$\text{target} = \min_{\text{LOSS}}(y - \hat{y})$$

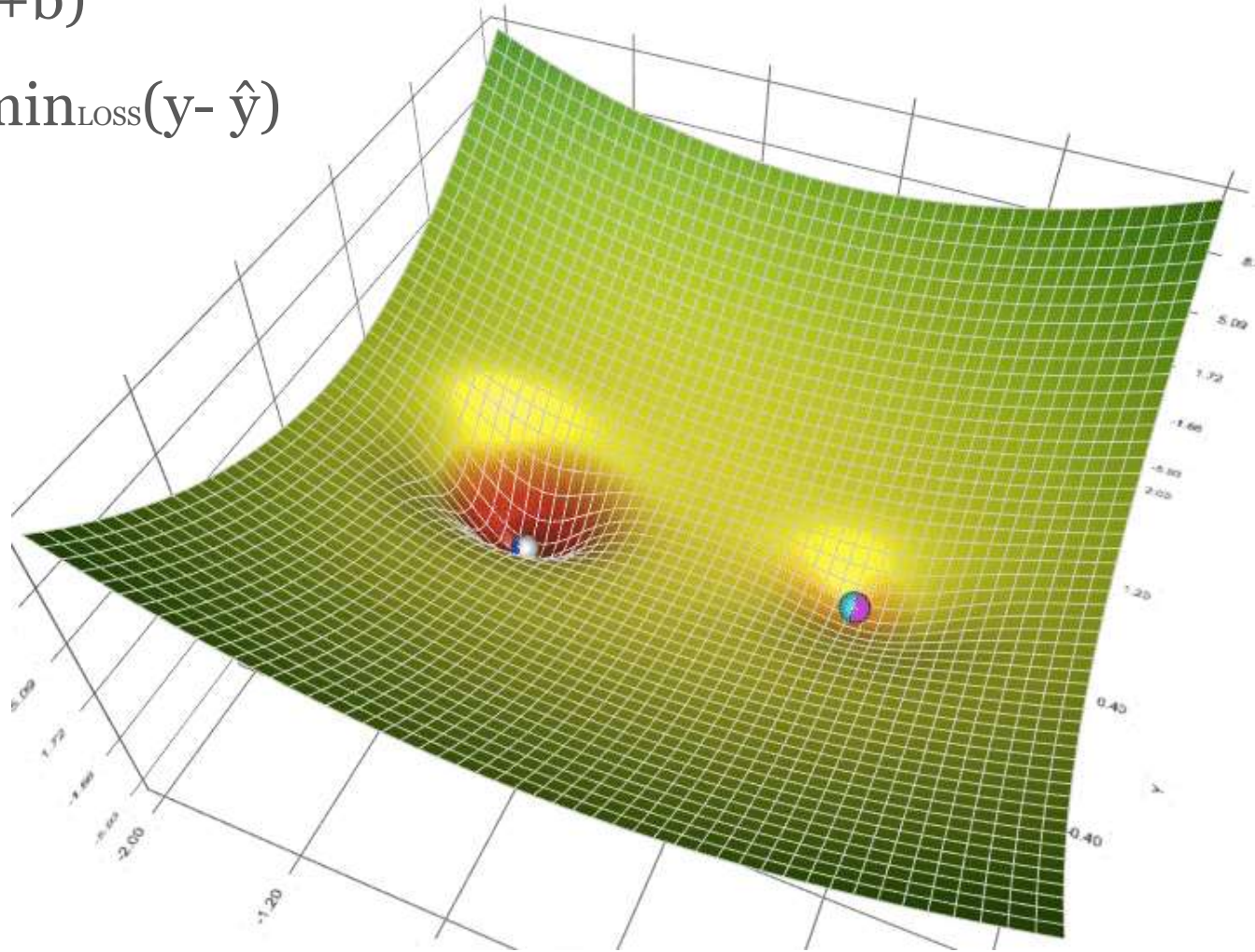
$$w_{\text{new}} = w_{\text{old}} + \eta \Delta w$$



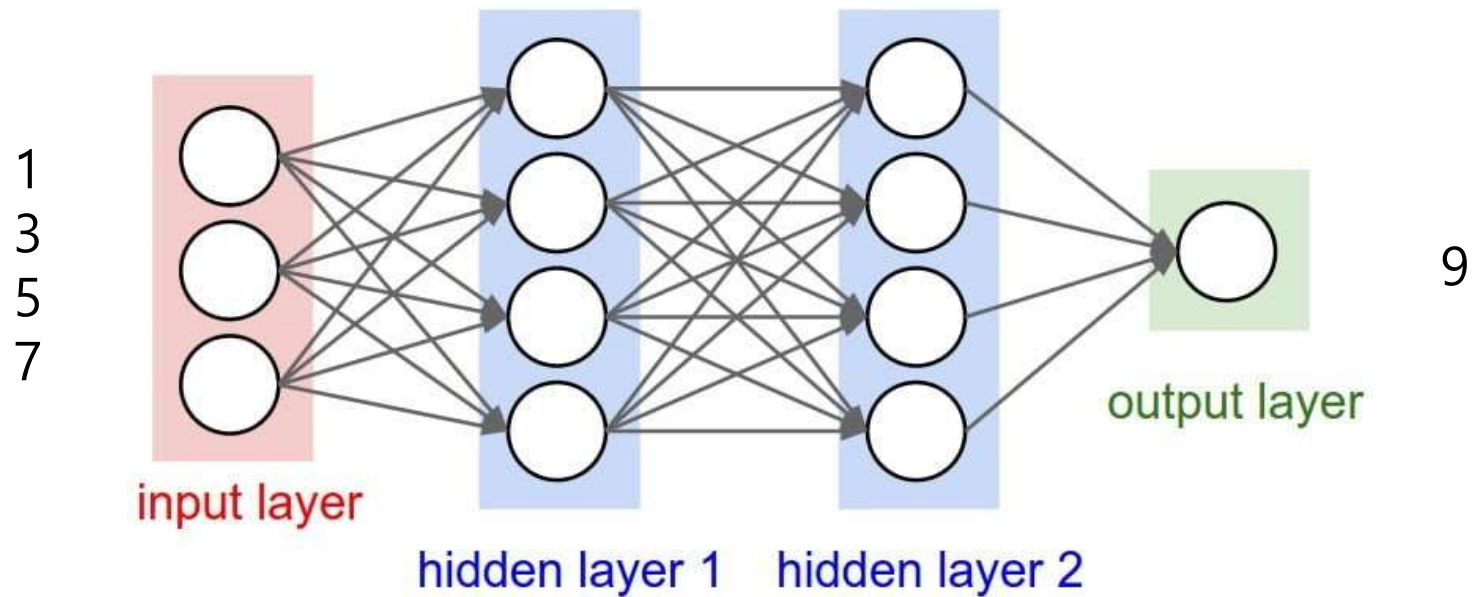
Structural Understanding neural networks (MLP)

$$\hat{y}=f(W\cdot x+b)$$

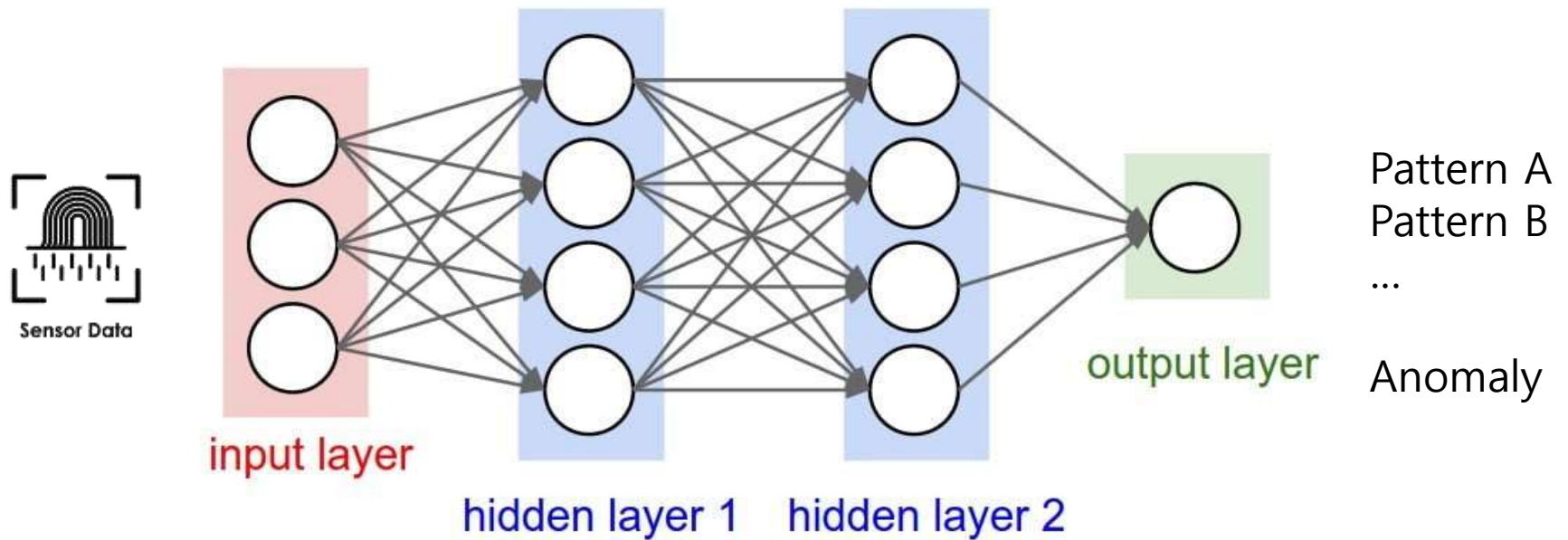
$$\text{target}=\min_{\text{LOSS}}(y-\hat{y})$$



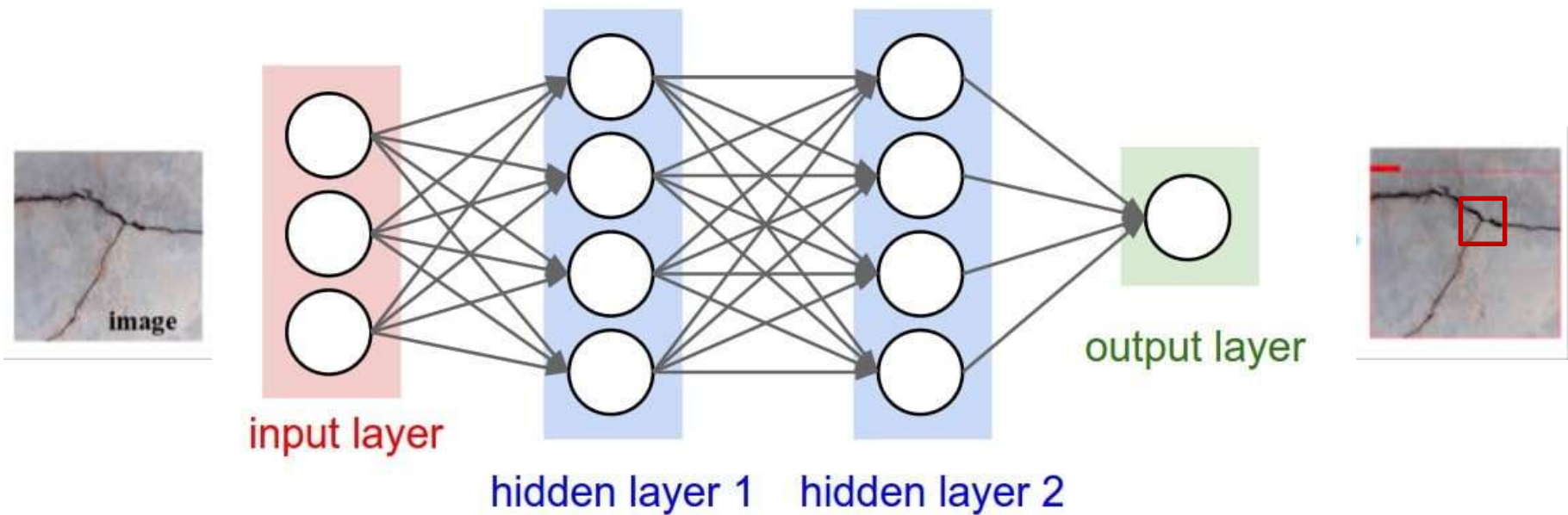
Structural Understanding neural networks (MLP)



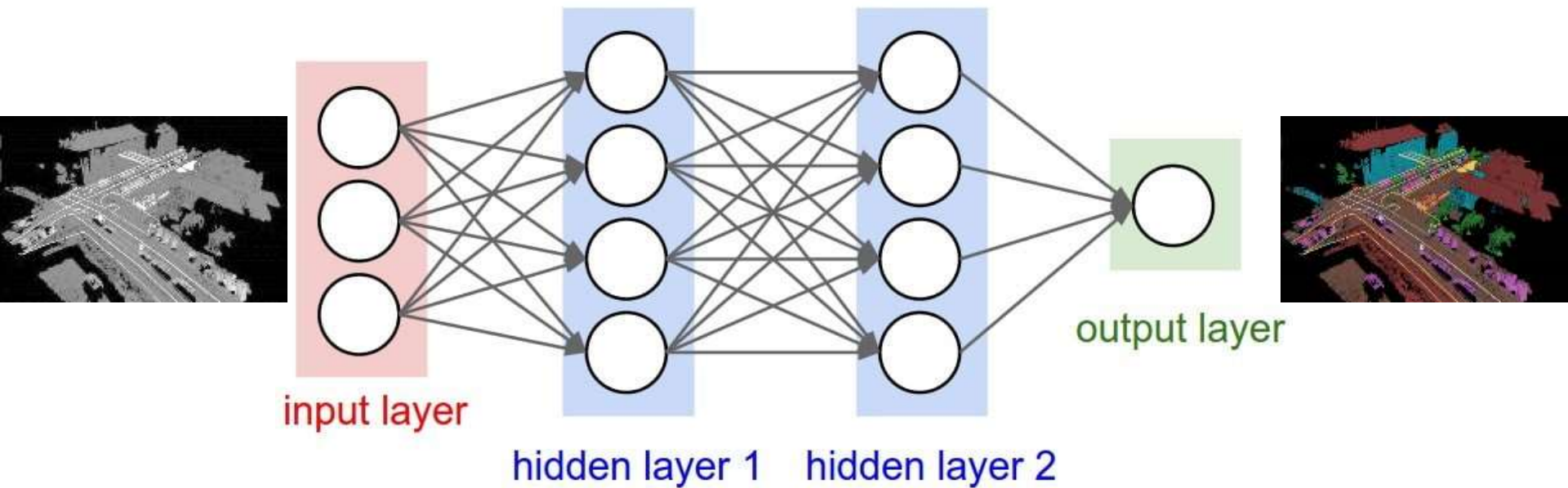
Structural Understanding neural networks (MLP)



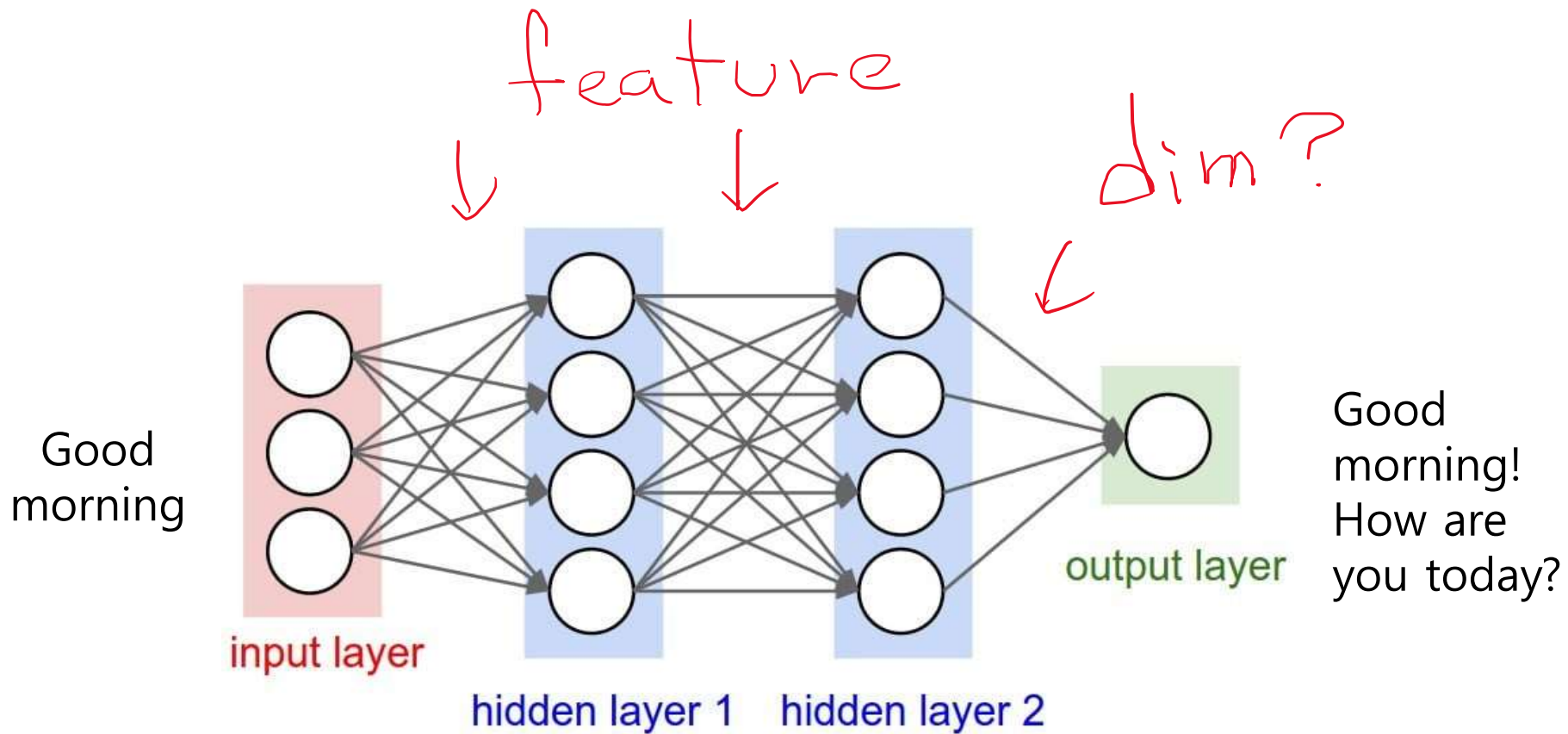
Structural Understanding neural networks (MLP)



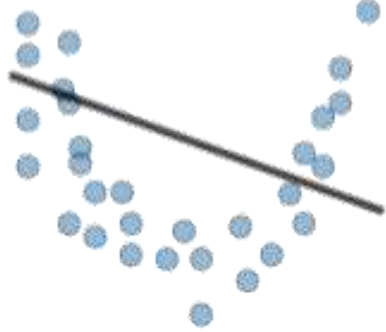
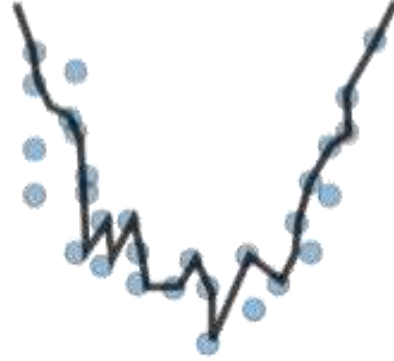
Structural Understanding neural networks (MLP)



Structural Understanding neural networks (MLP)



Structural Understanding neural networks (MLP)

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Regression illustration			
Deep learning illustration			
Possible remedies	• Complexify model • Add more features • Train longer		• Perform regularization • Get more data

Abhishek Shrivastava, 2020, Underfitting Vs Just right Vs Overfitting in Machine learning, Kaggle(www.kaggle.com/getting-started/166897)

Structural Understanding

Optimizer, Learning Rate, Batch Size

Epoch

The process of continuously updating a model's weights and biases to minimize the loss function. This update is typically performed by feeding data to the model in small batches.

An epoch is when the entire training dataset is used as input to the model, predictions are calculated through forward propagation, and the weights and biases are updated through backpropagation.

Batch gradient descent (BGD)

A type of gradient descent, an optimization algorithm that calculates the gradient using the entire training dataset at once when updating the model's weights.

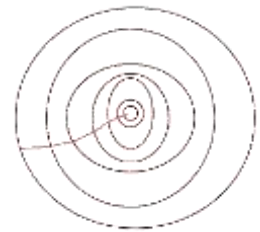
Batch size



1 Epoch

Dataset number = Sample = 10

```
for index in range(epoch):  
     $\hat{y}$  = Tensor(All dataset) * NN in GPU  
    Loss = Sum ( $\hat{y} - y$ ) / N  
    Back propagation (Loss)
```



Structural Understanding

Optimizer, Learning Rate, Batch Size

Mini-batch gradient descent (MB-GD. batch size or subdivision)

Minibatch gradient descent divides data into several small batches (minibatches) during model training, training them sequentially. At each update step, it calculates the average gradient of the loss function for one minibatch and uses this gradient to update the model's parameters. Processing all minibatches completes one epoch.



Mini-batch size = 2

Dataset = 10, Batch size = 2
(GD), Steps of 1 Epoch = 5

```
for index in range(epoch):
```

```
    for mini_batch in (Batch / subdivision):
```

```
         $\hat{y}$  = Tensor(mini_batch.dataset) * NN in GPU
```

```
        Loss = Sum ( $\hat{y} - y$ ) / N
```

```
        Back propagation (Loss) to update NN weights using Gradient Descent (GD)
```

Structural Understanding

Size

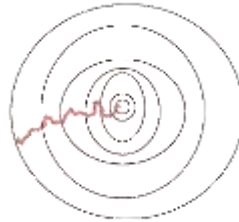
Optimizer, Learning Rate, Batch

Stochastic gradient descent (SDG)

Unlike BGD or MB-GD, SGD learns computationally from a single, randomly selected sample at each update.



Dataset = 10, Sample size = 1



For each epoch:

- Shuffle the training dataset

- For each single sample in the dataset:

 - $\hat{y} = \text{Neural_Network}(\text{sample_features})$

 - $\text{Loss} = (\hat{y} - y)^2$

 - Backpropagate(Loss)

 - Update_NN_Weights()

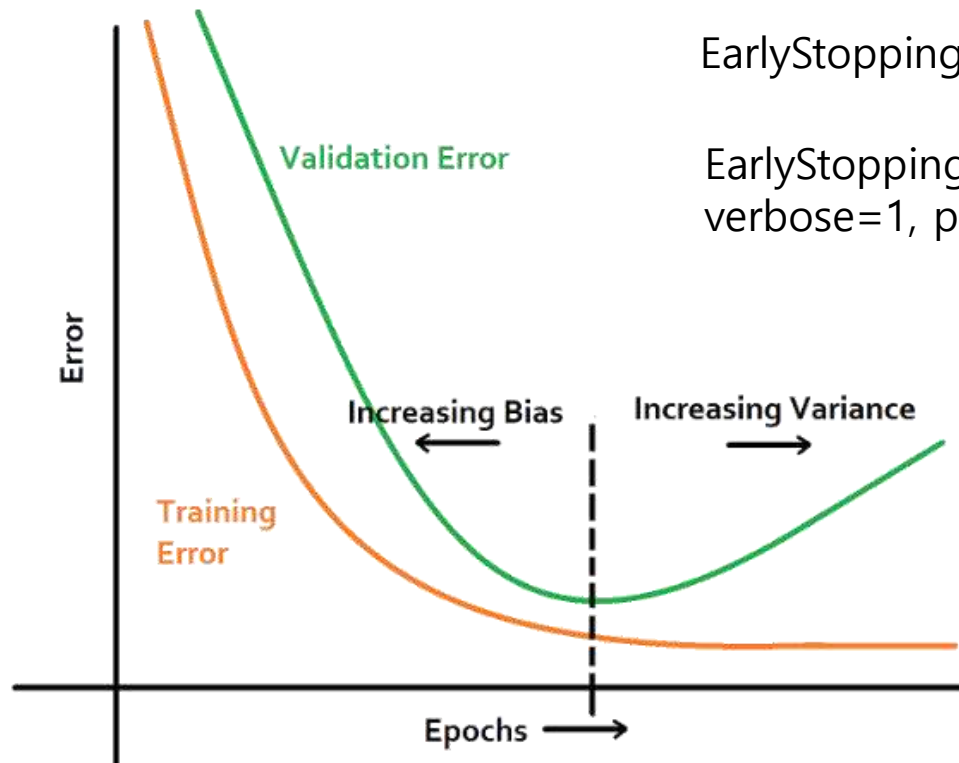
Structural Understanding Optimizer, Learning Rate, Batch

Regularization Techniques

Various strategies are available to prevent models from overfitting to the training data and improve generalization performance on new data.

Early stopping

Early stopping is a regularization technique used during iterative model training to prevent overfitting. Even if performance on the training set continues to improve, training is halted when performance on a separate validation set begins to deteriorate.



`EarlyStopping(monitor='val_loss', mode='min')`

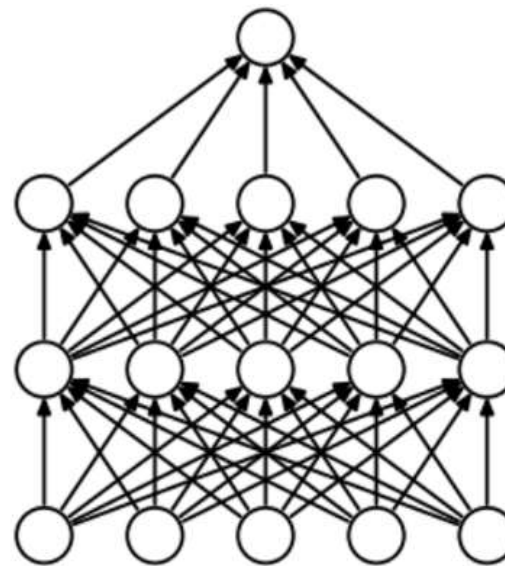
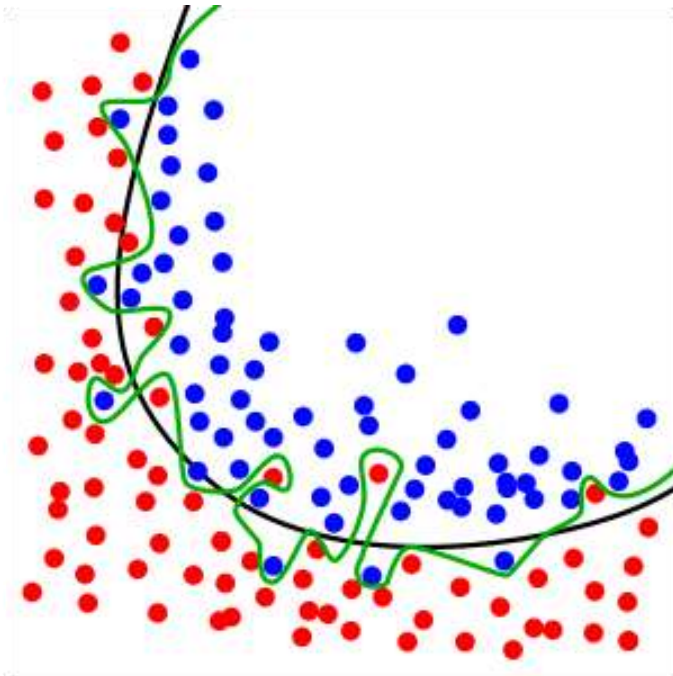
`EarlyStopping(monitor='val_loss', mode='min',
verbose=1, patience=50)`

Structural Understanding Optimizer, Learning Rate, Batch

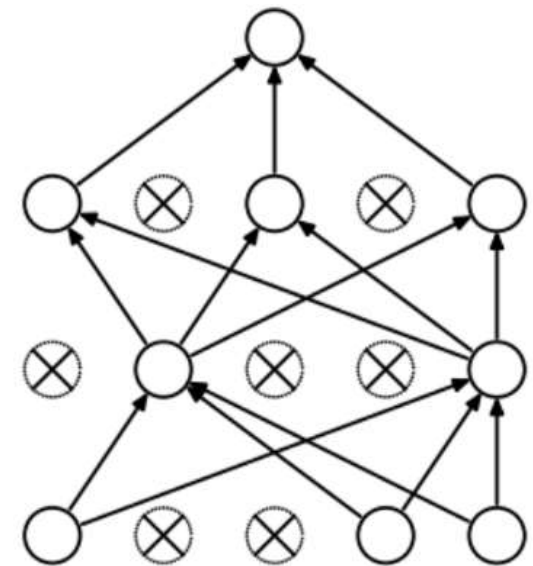
Dropout

A regularization technique to prevent overfitting in neural networks.

During training, a subset of neurons in each layer of the neural network is randomly selected and temporarily disabled.



(a) Standard Neural Net



(b) After applying dropout.

```
tf.model.add(tf.keras.layers.Dropout(drop_rate))
```

Structural Understanding

Optimizer, Learning Rate, Batch

L1 Norm (Lasso)

Regulates the weights by adding the sum of the absolute values of the weights to the loss function. This prevents the feature weights of specific neurons from becoming too large. By adding a penalty proportional to the sum of the absolute values of the weights, some weights are set to zero, resulting in a feature selection effect.

$$Cost = \frac{1}{n} \sum_{i=1}^n \{L(y_i, \hat{y}_i) + \frac{\lambda}{2} |w|\}$$

regularization
strength

```
Dense(64, activation='relu',  
kernel_regularizer=regularizers.l1(0.01))
```

L2 Norm (Ridge)

Add a penalty proportional to the sum of the squares of the weights. This reduces model complexity and prevents overfitting by ensuring that weights are uniformly smaller.

$$Cost = \frac{1}{n} \sum_{i=1}^n \{L(y_i, \hat{y}_i) + \frac{\lambda}{2} |w|^2\}$$

```
Dense(64, activation='relu',  
kernel_regularizer=regularizers.l2(0.01))
```

Structural Understanding Optimizer, Learning Rate, Batch

Optimization algorithm

How to adjust the parameters (weights and biases) of a machine learning model to improve its performance.

Gradient Descent with Momentum

Conventional gradient descent moves from the current position in the direction of steepest decrease in the loss function (negative slope). This method can cause zigzag behavior.

When the loss function surface is a long, narrow valley, it cannot descend in a straight line toward the minimum, but rather oscillates violently left and right.

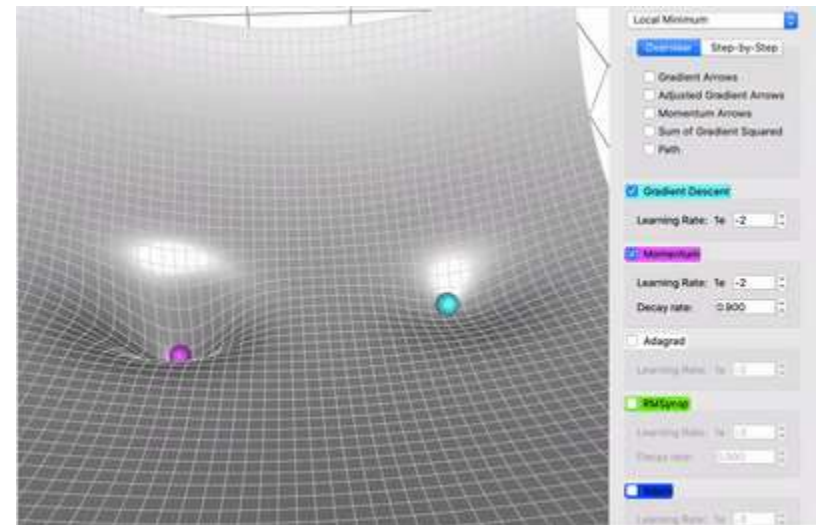
Use the inertia of the previous parameter update value

$$\begin{aligned} V(t) &= m * V(t-1) - \alpha \frac{\partial}{\partial w} Cost(w) \\ W(t+1) &= W(t) + V(t) \end{aligned}$$

m (momentum coefficient): A value between 0 and 1

$V(t-1)$: Velocity (momentum vector) at the previous time step $t-1$

If $t=1$, then $V=0$



Structural Understanding Optimizer, Learning Rate, Batch

ADAM (Adaptive Moment Estimation)

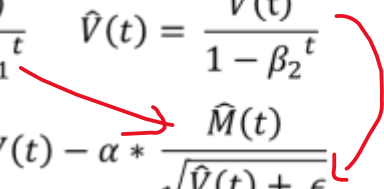
An optimization algorithm based on gradient descent. It efficiently updates weights and biases during model training.

Momentum: Determines the direction of weight updates by calculating an exponentially weighted moving average of previous gradients. This reduces unnecessary oscillations during the optimization process. It provides inertia to overcome local minima.

RMSProp (Root Mean Square Propagation): Calculates an exponentially weighted moving average of the squared gradients of parameters and uses this to adjust the learning rate. For parameters with large gradients, the learning rate is reduced, while for parameters with small gradients, the learning rate is increased, promoting stable learning.

Bias Correction: When calculating the first moment m and second moment v , $m = 0$ and $v = 0$ at $t = 0$. This initial value of 0 induces a strong bias, stabilizing unstable weight updates.

$$M(t) = \beta_1 M(t-1) + (1 - \beta_1) \frac{\partial}{\partial w(t)} \text{Cost}(w(t))$$
$$V(t) = \beta_2 V(t-1) + (1 - \beta_2) \left(\frac{\partial}{\partial w(i)} \text{Cost}(w(i)) \right)^2$$

$$\hat{M}(t) = \frac{M(t)}{1 - \beta_1^t} \quad \hat{V}(t) = \frac{V(t)}{1 - \beta_2^t}$$
$$W(t+1) = W(t) - \alpha * \frac{\hat{M}(t)}{\sqrt{\hat{V}(t) + \epsilon}}$$


$$m_1 = \beta_1 m_0 + (1 - \beta_1) g_1$$

$$m_1 = 0.9 \cdot 0 + (1 - 0.9) \cdot 5$$

$$m_1 = 0 + 0.1 \cdot 5 = \mathbf{0.5}$$

$$v_1 = \beta_2 v_0 + (1 - \beta_2) g_1^2$$

$$v_1 = 0.999 \cdot 0 + (1 - 0.999) \cdot 5^2$$

$$v_1 = 0 + 0.001 \cdot 25 = \mathbf{0.025}$$

$$\hat{m}_1 = \frac{m_1}{1 - \beta_1^t} = \frac{0.5}{1 - 0.9^1} = \frac{0.5}{1 - 0.9} = \frac{0.5}{0.1} = \mathbf{5.0}$$

$$\hat{v}_1 = \frac{v_1}{1 - \beta_2^t} = \frac{0.025}{1 - 0.999^1} = \frac{0.025}{1 - 0.999} = \frac{0.025}{0.001} = \mathbf{25.0}$$

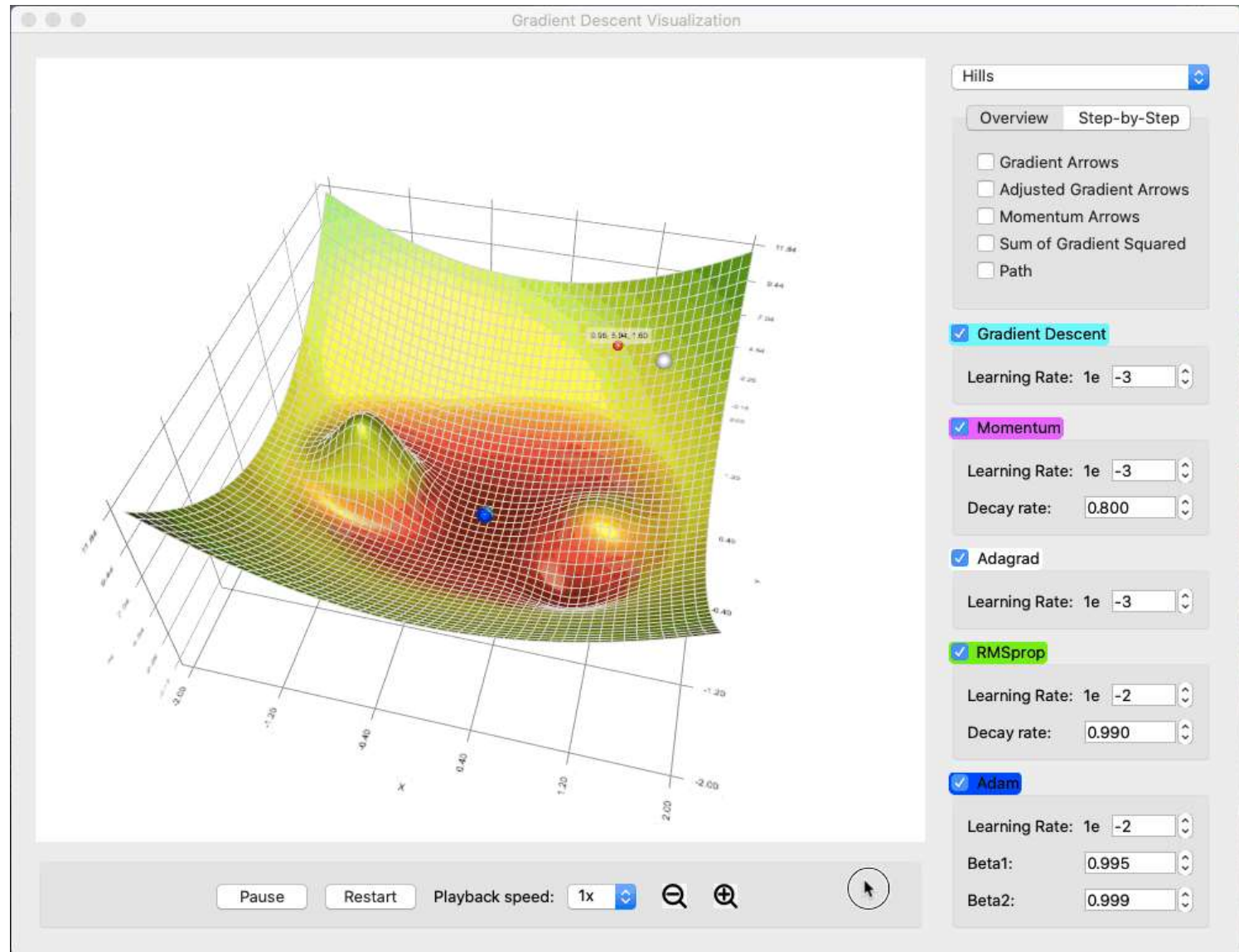
$$W_{1+1} = 10 - 0.01 \cdot \frac{5.0}{\sqrt{25.0 + 10^{-8}}}$$

$$W_2 = 10 - 0.01 \cdot \frac{5.0}{5.0 + 10^{-8}}$$

$$W_2 \approx 10 - 0.01 \cdot 1.0 = \mathbf{9.99}$$

Structural Understanding Optimizer, Learning Rate, Batch

Gradient Descent Method Simulation



interactive visualization of gradient descent methods with step-by-step illustration and hyperparameter tuning UI

Structural Understanding Optimizer, Learning Rate, Batch

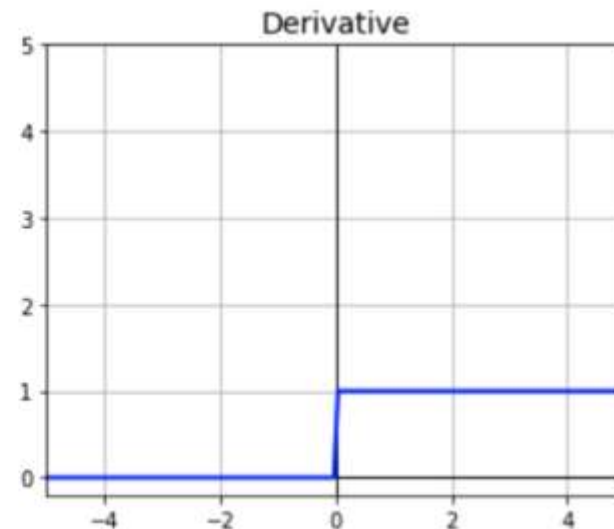
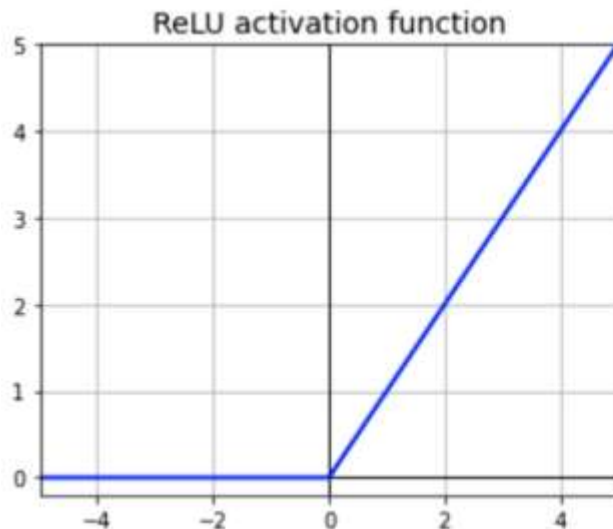
ReLU (Rectified Linear Unit)

Easy to calculate and fast. The gradient in the positive region is always 1, preventing the vanishing gradient problem, which causes learning to halt when the gradient becomes 0 during backpropagation. Negative inputs output 0, disabling the neuron and resulting in the dropout effect. However, the dying ReLU problem also occurs, preventing the weights associated with these neurons from being updated.

$$f(x) = \max(0, x)$$

$$\begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$$

```
def relu(x):  
    if x < 0:  
        return 0  
    else:  
        return x
```

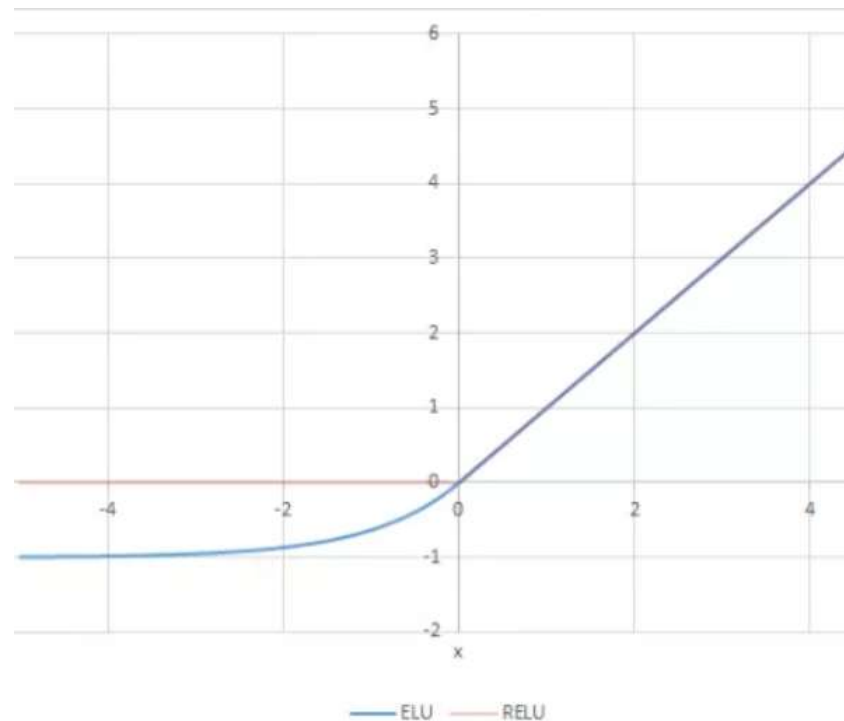


Structural Understanding Optimizer, Learning Rate, Batch

Exponential Linear Unit (ELU)

Outputs small negative values even for negative inputs, preventing neuron deactivation issues like Dying ReLU.

$$R(z) = \begin{cases} z & z > 0 \\ \alpha \cdot (e^z - 1) & z \leq 0 \end{cases}$$



```
def elu(z,alpha):  
    return z if z >= 0 else alpha*(e^z -1)
```

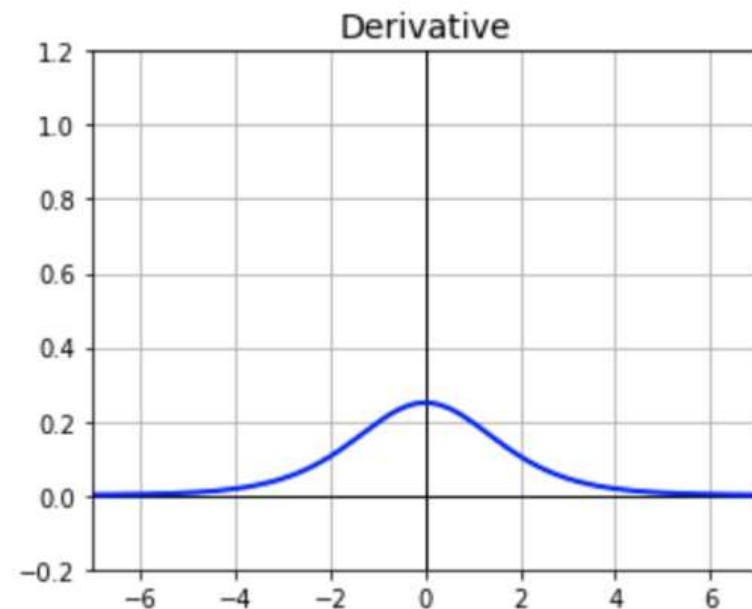
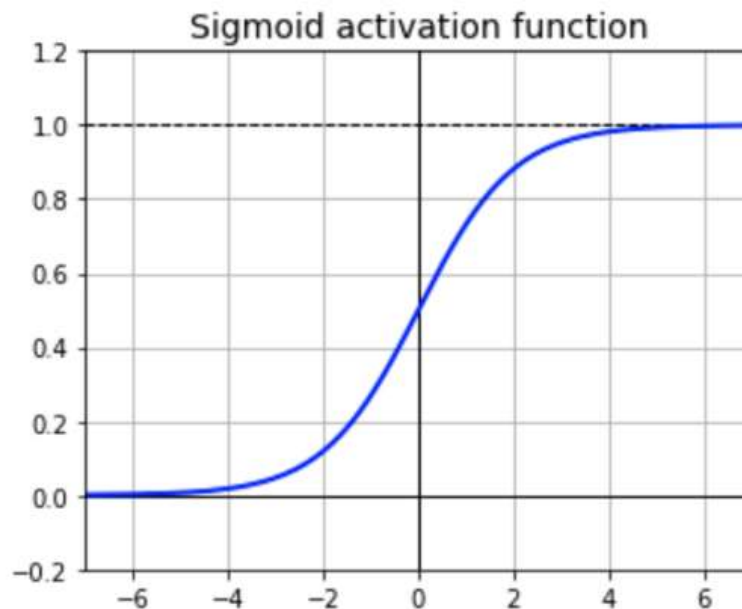
Structural Understanding Optimizer, Learning Rate, Batch

Sigmoid

Normalizes output between 0 and 1 and is differentiable. High computational cost. Beneficial for binary classification problems. However, when the input value is very large or small, the differential gradient value approaches 0, causing a gradient vanishing problem. Since the output average is not 0, biased weight updates occur, resulting in slow learning convergence.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

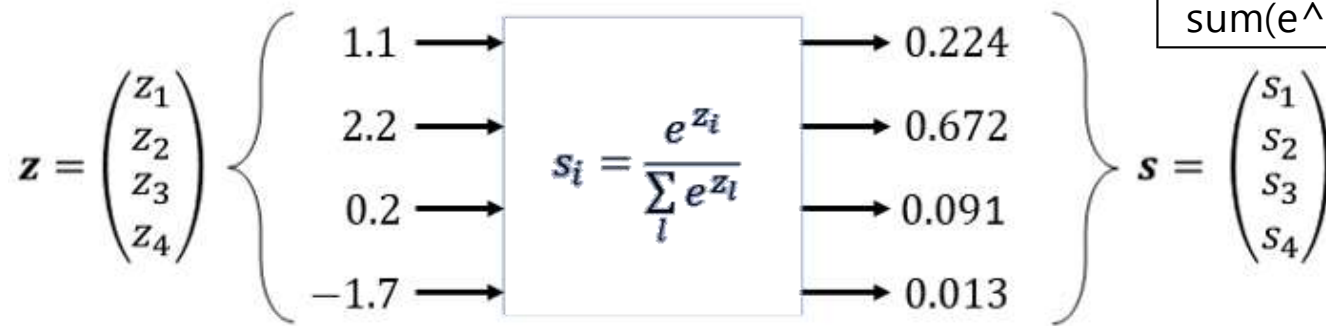
Vanishing gradient
Computationally expensive
The output is not zero centered



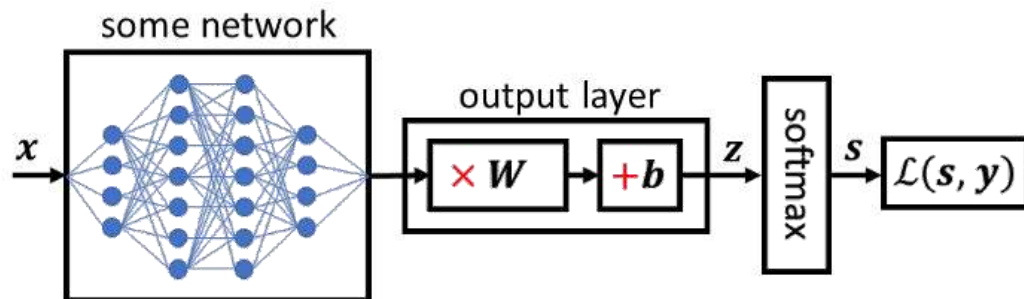
Structural Understanding Optimizer, Learning Rate, Batch

Softmax

$$s_i = \frac{e^{z_i}}{\sum_{l=1}^n e^{z_l}}$$



z_i	e^{z_i}	s_i
1.1	3.00417	0.224
2.2	9.02501	0.672
0.2	1.22140	0.091
-1.7	0.18268	0.014
sum(e^{z_i})	13.43	



$$-1.7 < 0.2 < 1.1 < 2.2 \rightarrow 0.013 < 0.091 < 0.224 < 0.672$$

Structural Understanding Loss function

loss function	merit	drawback
MSE(Mean Square Error) $MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$	1. Differentiable and easy to optimize 2. Penalty for large errors: Square the error to apply a larger penalty to large errors.	Sensitive to outliers: Vulnerable to outliers due to large penalties for large errors.
MAE(Mean Absolute Error) $MAE = \frac{\sum_{i=1}^n y_i - \hat{y}_i }{n}$	1. Robust to outliers: Less sensitive to outliers by taking the absolute value of the error. 2. Intuitive interpretation: The units of the loss value are the same as the actual/predicted units, so the interpretation is intuitive.	1. Existence of non-differentiable points: Differentiation is not possible at points with an error of 0. Optimization is difficult. 2. Slow convergence: The gradient magnitude is constant regardless of the error size, and convergence may be slow near the minimum.
RMSE(Root Mean Square Error) $RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}}$	1. Easy to interpret units: Unlike MSE, it is easy to interpret because the units are the same as the actual/predicted units by taking the square root. 2. Maintaining the penalty for large errors: Maintaining the large error penalty characteristic of MSE.	Sensitive to outliers: Since it is derived from MSE, it is still sensitive to outliers.

Structural Understanding Loss function

Loss function

BCE(Binary Cross Entropy Loss)

$$L(y, \hat{y}) = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))$$

$$L = -(1 \cdot \log(0.9) + (1 - 1) \cdot \log(1 - 0.9))$$

$$L = -(1 \cdot \log(0.9) + 0 \cdot \log(0.1))$$

$$L = -\log(0.9)$$

$$L \approx -(-0.1053)$$

$$L \approx \underline{\mathbf{0.1053}}$$

$$L(y, \hat{y}) = -\sum_{k=1}^C y_k \log(\hat{y}_k) \text{ py)}$$

C : 총 클래스 수

y_k : 실제 정답을 원-핫 인코딩한 값 (정답 클래스면 1, 아니면 0)

$\hat{y}_k$: 모델이 예측한 k 번째 클래스의 확률

[1, 0, 0] (고양이)

[0.9, 0.05, 0.05] (고양이 90%, 강아지 5%, 새 5%)

$$L = -(y_0 \log(\hat{y}_0) + y_1 \log(\hat{y}_1) + y_2 \log(\hat{y}_2))$$

$$L = -(1 \cdot \log(0.9) + 0 \cdot \log(0.05) + 0 \cdot \log(0.05))$$

$$L = -\log(0.9)$$

$$L \approx -(-0.1053)$$

$$L \approx \mathbf{0.1053}$$

merit

1. Suitable for measuring the difference in probability distributions: It is mathematically suitable for measuring how well the predicted probability matches the actual correct answer distribution.
2. Convenient for convergence: The loss increases steeply as the model makes incorrect predictions, allowing the model to learn quickly and converge in the early stages.
3. Differentiable: It is differentiable at all points, and when used with a sigmoid (0~1), the vanishing gradient problem is alleviated, making learning efficient.

1. Multi-class classification: Effective classification of three or more mutually exclusive classes
2. Probabilistic interpretation: Provides probability distribution of belonging to each class along with Softmax output
3. Differentiable and easy to optimize: The loss function is differentiable, making it suitable for gradient descent-based optimization

disadvantage

1. Vulnerable to class imbalance: When the positive/negative class ratio is imbalanced, the model is biased toward majority class predictions.
2. Unintuitive interpretation: Intuitive interpretations of "mean of error" metrics like MSE and MAE are difficult.

1. Vulnerable to class imbalance: When a specific class's data is overwhelmingly abundant, the majority class is biased, resulting in poor performance for the minority class.
2. One-hot encoding is essential: Softmax processing outputs are required for the number of classes. The correct answer labels must be prepared in one-hot encoding format.

Structural Understanding

Loss function

Loss function

use

MSE
(Mean Squared Error)

Regression problems: Used to predict continuous numerical values (e.g., predicting house prices or temperature). Suitable when the size of the error is important and larger errors are penalized more heavily.

MAE
(Mean Absolute Error)

Regression problems: Used to predict continuous numerical values (e.g., predicting house prices or temperature). Suitable when you want to be less sensitive to outliers.

RMSE
(Root Mean Squared Error)

Regression problems: Similar to MSE, this is used to predict continuous numeric values. It's useful when you want to improve interpretability by matching the MSE units to the actual predicted units.

BCE
(Binary Cross-Entropy)

Binary classification problem: Used when classifying into one of two mutually exclusive classes (e.g., spam/regular email classification, disease prediction). The output layer uses the **sigmoid activation function**.

CCE
(Categorical Cross-Entropy)

Multi-class classification problems: Used when classifying objects into one of three or more mutually exclusive classes (e.g., handwritten digit recognition, object classification in images). The output layer **uses the softmax activation function**.

Structural Understanding Optimizer, Learning Rate, Batch

Learning Rate Decay

A strategy that gradually reduces the learning rate value as training progresses. Used to improve the stability and performance of model training.

Time-based decay

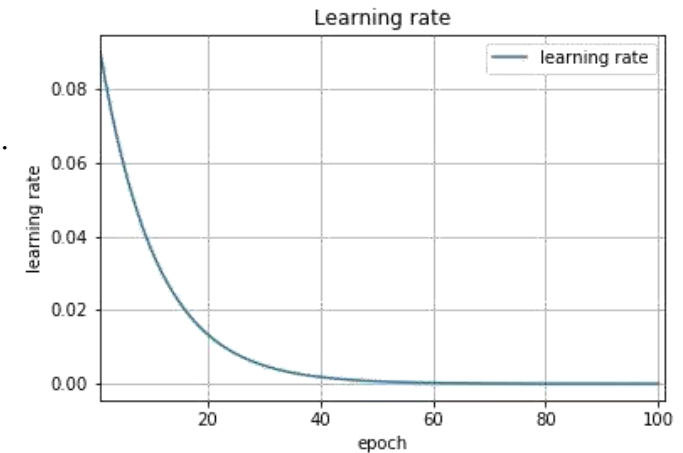
$$\alpha_t = \frac{\alpha_0}{1+k \cdot t}$$

learning_rate = 0.1

decay_rate = learning_rate / epochs

momentum = 0.8

optimization = SGD(lr=learning_rate, momentum=momentum, decay=decay_rate)



Step decay

$$\alpha(t) = \alpha_0 \cdot \gamma^{\lfloor \frac{t}{s} \rfloor}$$

$\lfloor \cdot \rfloor$ = floor(), s = drop num, t = epoch

```
def step_decay(epoch):
```

```
    initial_lrate = 0.1
```

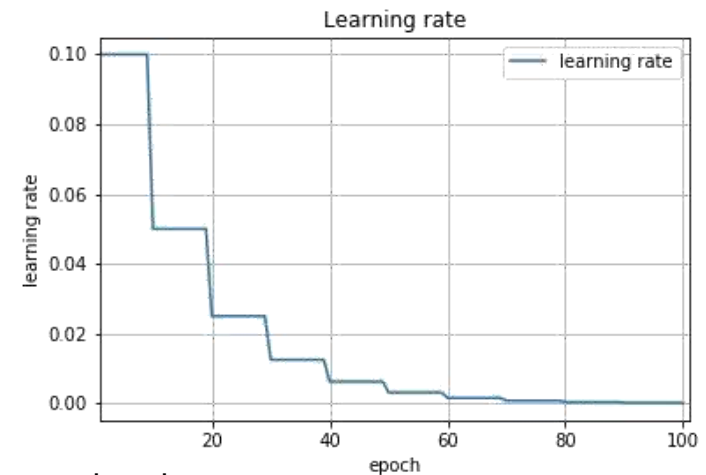
```
    drop = 0.5
```

```
    epochs_drop = 10.0
```

```
    lrate = initial_lrate * math.pow(drop, math.floor((1+epoch)/epochs_drop))
```

```
    return lrate
```

```
lrate = LearningRateScheduler(step_decay)
```



Structural Understanding Optimizer, Learning Rate, Batch

Batch normalization

Batch normalization (batchnorm) improves neural network performance accuracy. Batch normalization occurs batch by batch.

Instead of normalizing the inputs to the network, it normalizes the inputs to each hidden layer within the network.

During training, incoming mini-batch data is normalized to a mean of 0 and a variance of 1, then adjusted using learnable scale (gamma) and shift (beta) variables.

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1...m}\}$;

Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$

```
from keras.models import Sequential
from keras.layers import Dense, Dropout
from keras.layers import BatchNormalization
...
```

```
y = to_categorical(y)
n_train = 800
trainX, testX = X[:n_train, :], X[n_train:, :]
trainy, testy = y[:n_train], y[n_train:]
```

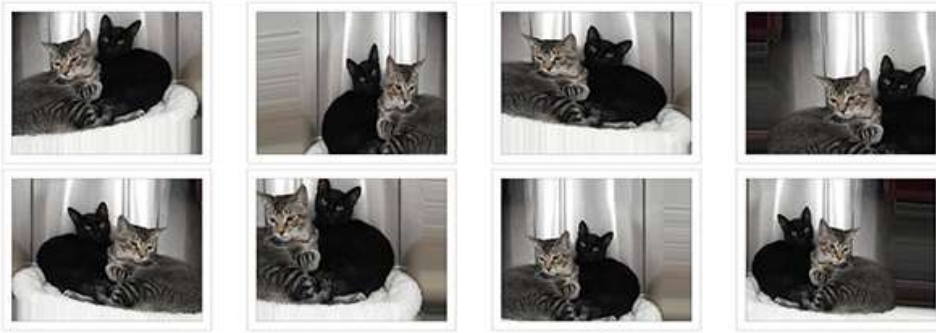
```
model = Sequential()
model.add(Dense(25, input_dim=2, activation='relu'))
model.add(BatchNormalization())
model.add(Dropout(0.2))
model.add(Dense(8, activation='relu'))
model.add(BatchNormalization())
model.add(Dropout(0.2))
model.add(Dense(4, activation='softmax'))
model.compile(loss='categorical_crossentropy',
              optimizer='adam', metrics=['accuracy'])
```

```
model.summary()
```

Structural Understanding Optimizer, Learning Rate, Batch

Data Augmentation

Enhancing the model's generalization ability by securing learning data diversity.



```
from keras.preprocessing.image import ImageDataGenerator
from keras.utils import array_to_img, img_to_array, load_img
```

```
datagen = ImageDataGenerator(
    rotation_range=40, width_shift_range=0.2, height_shift_range=0.2, shear_range=0.2, zoom_range=0.2,
    horizontal_flip=True, fill_mode='nearest')
```

```
img = load_img('./cat0.jpg')          # this is a PIL image
x = img_to_array(img)                 # this is a Numpy array with shape (3, 150, 150)
x = x.reshape((1,) + x.shape)         # this is a Numpy array with shape (1, 3, 150, 150)
```

```
i = 0
for batch in datagen.flow(x, batch_size=1, save_to_dir='preview', save_prefix='cat', save_format='jpeg'):
    i += 1
    if i > 20:
        break # otherwise the generator would loop indefinitely
```


Structural Understanding Optimizer, Learning Rate, Batch



Link

Tinker With a Neural Network
<http://playground.tensorflow.org/#activation=relu®ularization=L2&batchSize=10&dataset=spiral®Dataset=reg-plane&learningRate=0.01®ularizationRate=0.001&noise=0&networkShape=8,6,3,2&seed=0.88358&showTestData=false&discretize=false&percTrainData=80&x=true&y=true&xTimesY=false&xSquared=false&ySquared=false&cosX=false&sinX=false&cosY=false&sinY=false&collectStats=false&problem=classification&initZero=false&hideText=false>

Structural Understanding CNN/RNN/LSTM's concepts

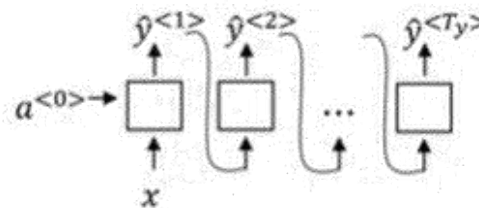
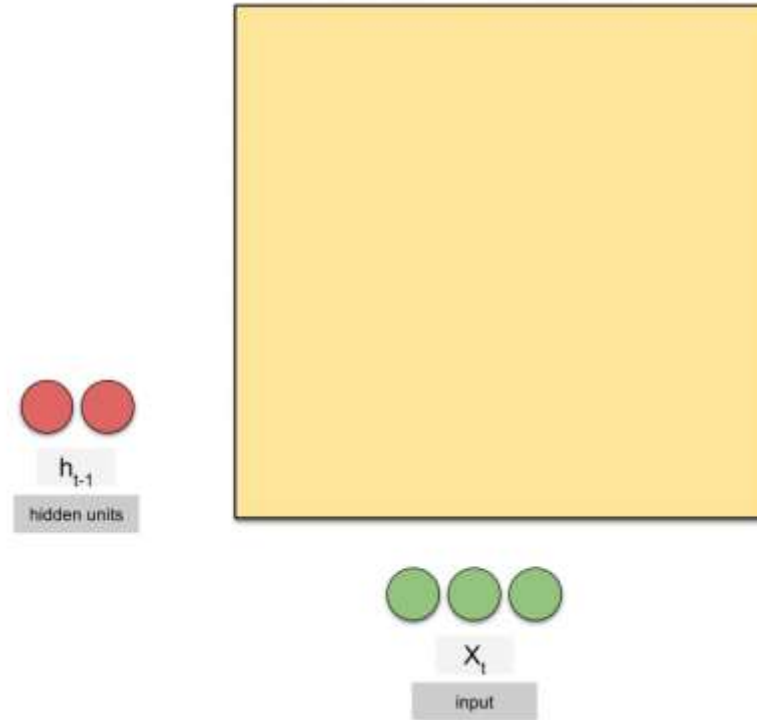
RNN(Recurrent Neural Network)

Recurrent neural networks (RNNs) have been used in Apple's Siri and Google's voice search. RNNs are a generalized form of feedforward neural networks with internal memory.

RNNs perform the same function for all data inputs, while the output of the current input depends on a single past computation, making them inherently repetitive.

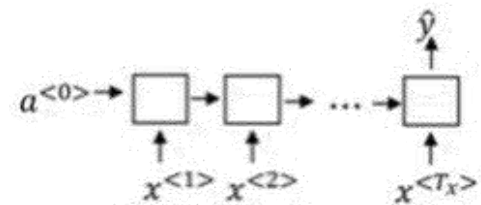
After generating the output, it is copied and fed back into the recurrent network. The final output considers the current input and the output learned from previous inputs. RNNs have a memory function, allowing them to remember previous data. They use the current input and previous states to compute the current state.

RNNs can suffer from the vanishing gradient problem.



One to many

음악 Generative



Many to one

감정 분석

[Animated RNN, LSTM and GRU. Recurrent neural network cells in GIFs](#)
| by Raimi Karim | [TDS Archive](#) | [Medium](#)

[Understanding Recurrent Neural Network \(RNN\) and Long Short Term Memory\(LSTM\)](#) | by Vijay Choubey | [Analytics Vidhya](#) | [Medium](#)

Structural Understanding CNN/RNN/LSTM's concepts

LSTM (Long Short Term Memory)

When new information is added to an RNN, the RNN faces the problem of significantly modifying existing information. LSTMs use input, output, and forget gates to determine which data is important and potentially useful in the future, and which data should be discarded.

input gate i_t

Decide whether to store new information in the cell state from the current input x_t and the previous hidden h_{t-1} . Calculate a new candidate cell, multiply it, and output it.

oblivion gate f_t

Decide whether to keep or forget information from the previous cell state. The output sigmoid (0~1) value is multiplied by the previous C_{t-1} .

Cell State Update C_t

Add input and forget gate information, and calculate a new C_t

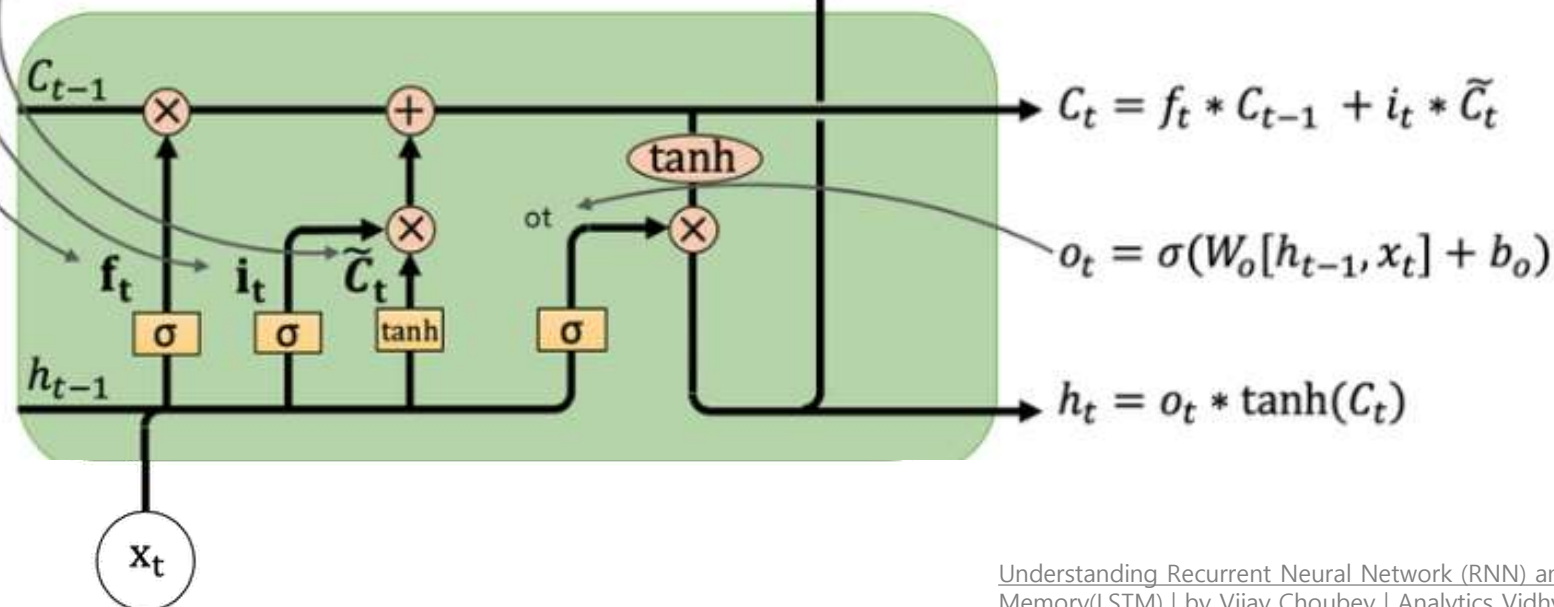
Output gate O_t

Select and print important information from the current cell

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

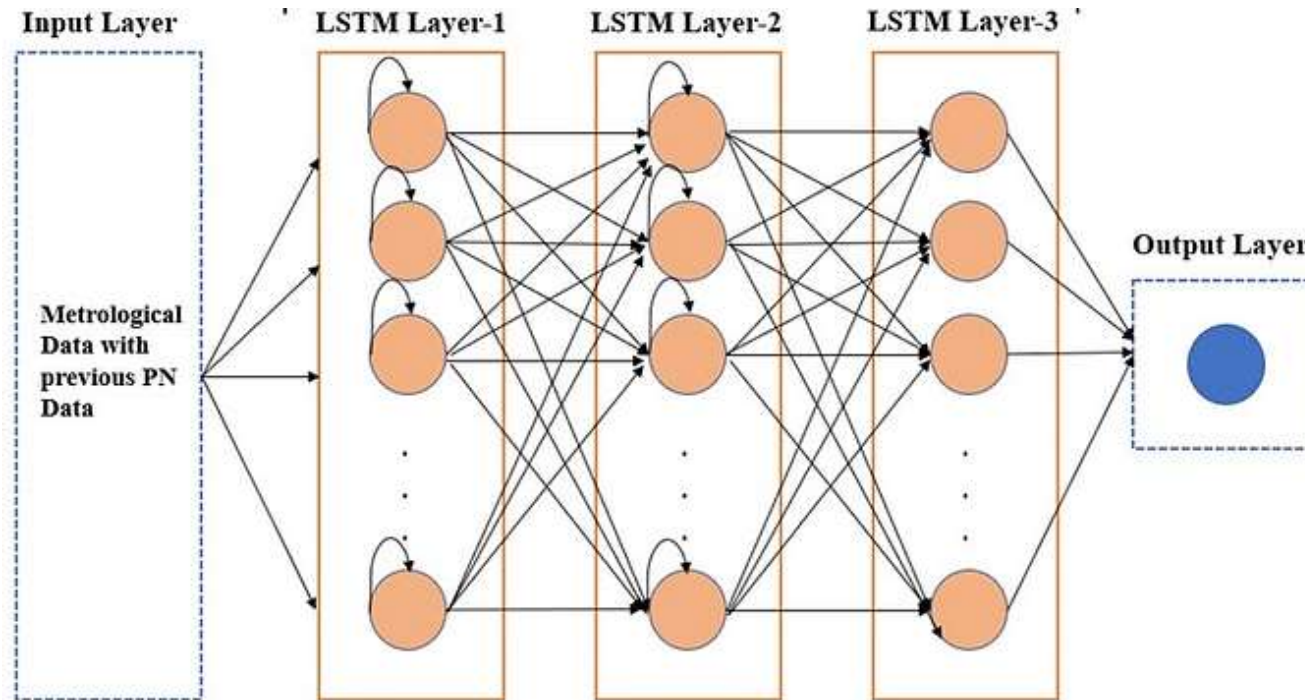
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$



Structural Understanding CNN/RNN/LSTM's concepts

LSTM (Long Short Term Memory)



```
from keras.models import Sequential
from keras.layers import LSTM, Dense, Embedding, SpatialDropout1D
```

```
model = Sequential()
```

```
...
```

```
model.add(LSTM(100, dropout=0.2, recurrent_dropout=0.2))
```

```
model.add(Dense(1, activation='sigmoid'))
```

Structural Understanding CNN/RNN/LSTM's concepts

ImageNet Classification with Deep Convolutional Neural Networks

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Ilya Sutskever
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Abstract

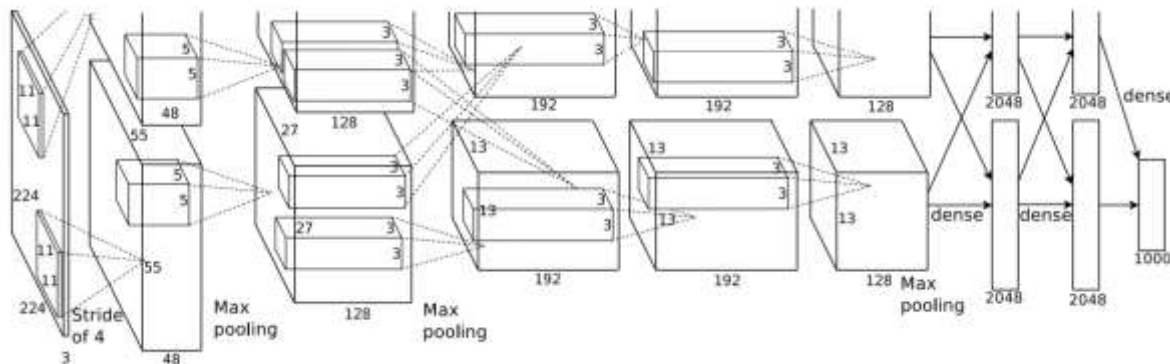


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Structural Understanding CNN/RNN/LSTM's concepts

CNN (Convolutional Neural Network)

Convolutional neural networks (ConvNets or CNNs) are one of the main categories that perform image recognition and image classification.

The CNN model passes each input image through a series of convolutional layers with filters (Kernels), pooling, and fully connected layers (FC), and then applies the Softmax function to classify objects with probability values between 0 and 1.

Filter kernel

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

5 x 5 – Image Matrix



1	0	1
0	1	0
1	0	1

3 x 3 – Filter Matrix

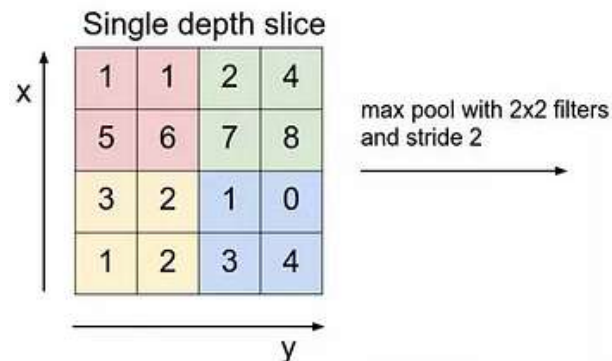
1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Image

4		

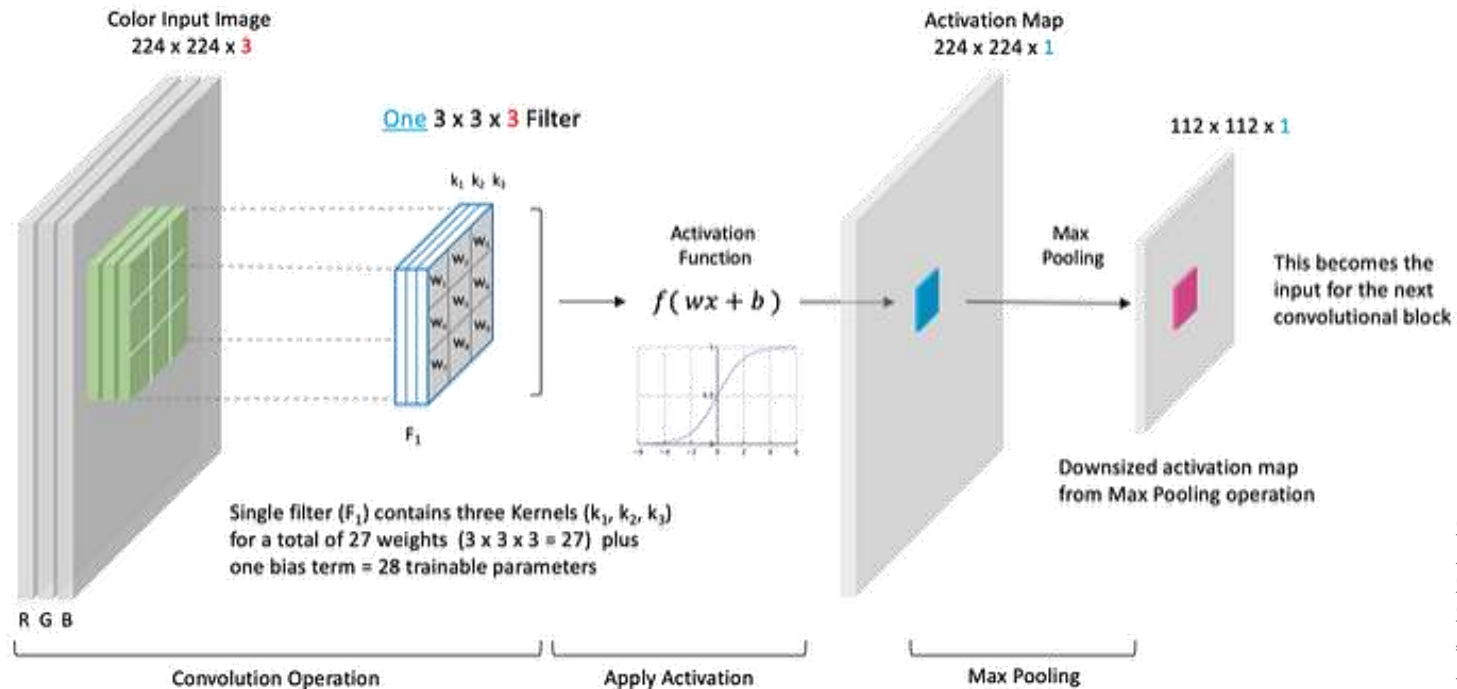
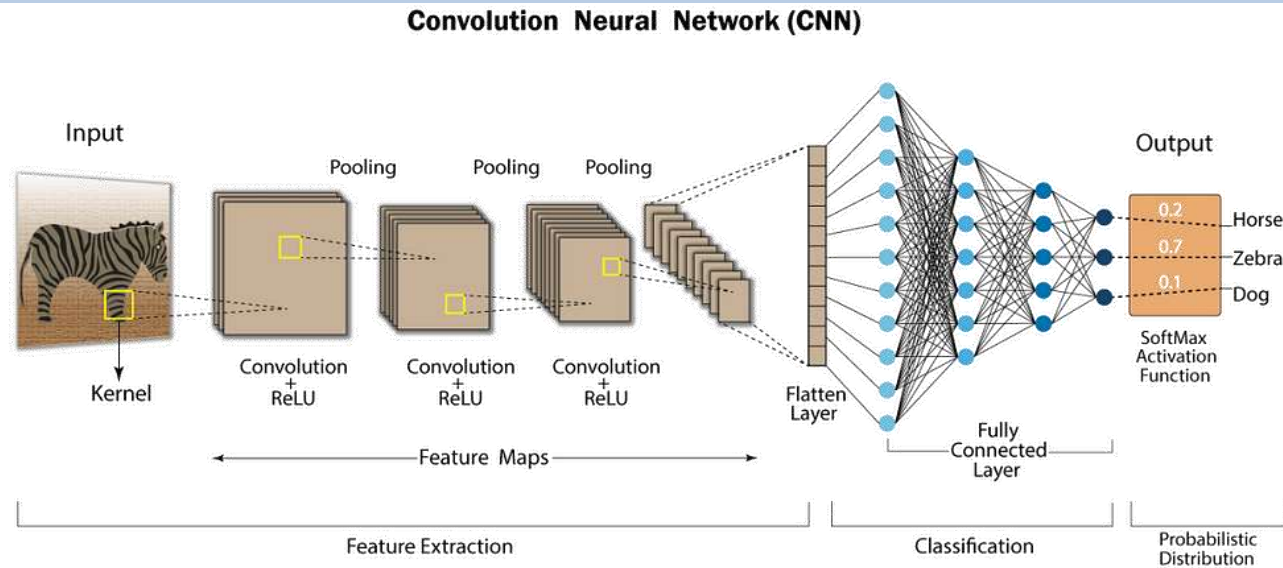
Convolved
Feature

Max pooling



Structural Understanding CNN/RNN/LSTM's concepts

CNN



Structural Understanding CNN/RNN/LSTM's concepts

0	0	0	0	0	0	...
0	156	155	156	158	158	...
0	153	154	157	159	159	...
0	149	151	155	158	159	...
0	146	146	149	153	158	...
0	145	143	143	148	158	...
...	...	...	...	...	...	...

Input Channel #1 (Red)

0	0	0	0	0	0	...
0	167	166	167	169	169	...
0	164	165	168	170	170	...
0	160	162	166	169	170	...
0	156	156	159	163	168	...
0	155	153	153	158	168	...
...	...	...	...	...	...	...

Input Channel #2 (Green)

0	0	0	0	0	0	...
0	163	162	163	165	165	...
0	160	161	164	166	166	...
0	156	158	162	165	166	...
0	155	155	158	162	167	...
0	154	152	152	157	167	...
...	...	...	...	...	...	...

Input Channel #3 (Blue)

-1	-1	1
0	1	-1
0	1	1

Kernel Channel #1



308

1	0	0
1	-1	-1
1	0	-1

Kernel Channel #2



-498

0	1	1
0	1	0
1	-1	1

Kernel Channel #3



164

+

+

+ 1 = -25

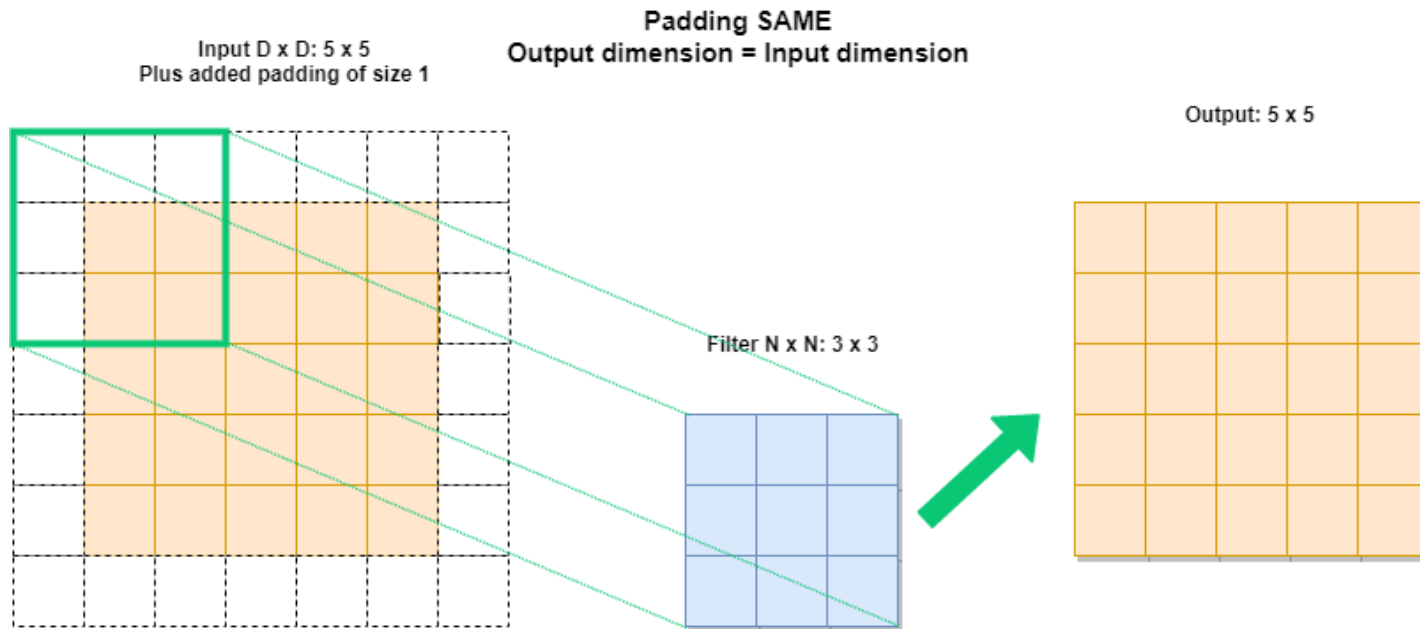
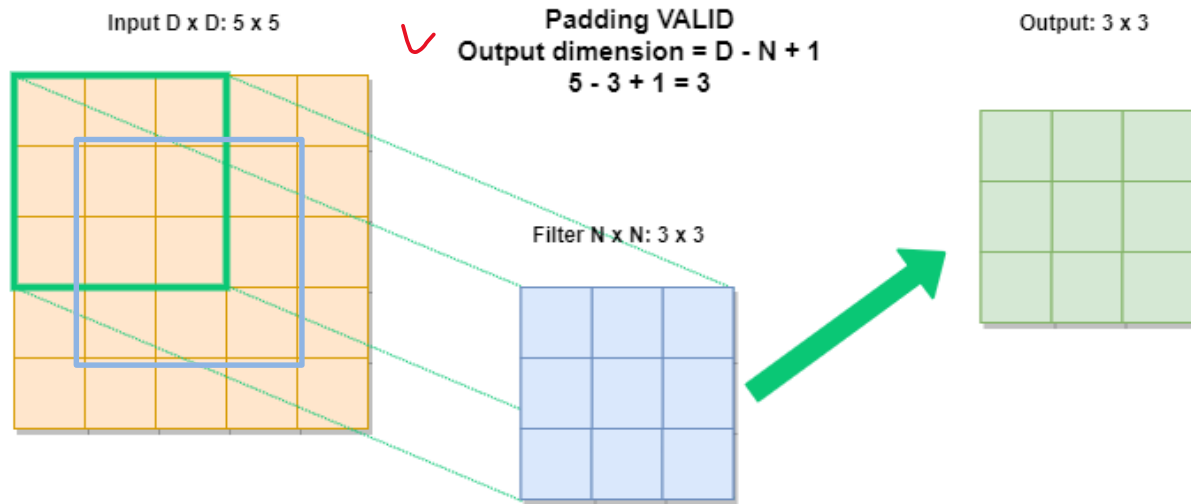


Bias = 1

Output

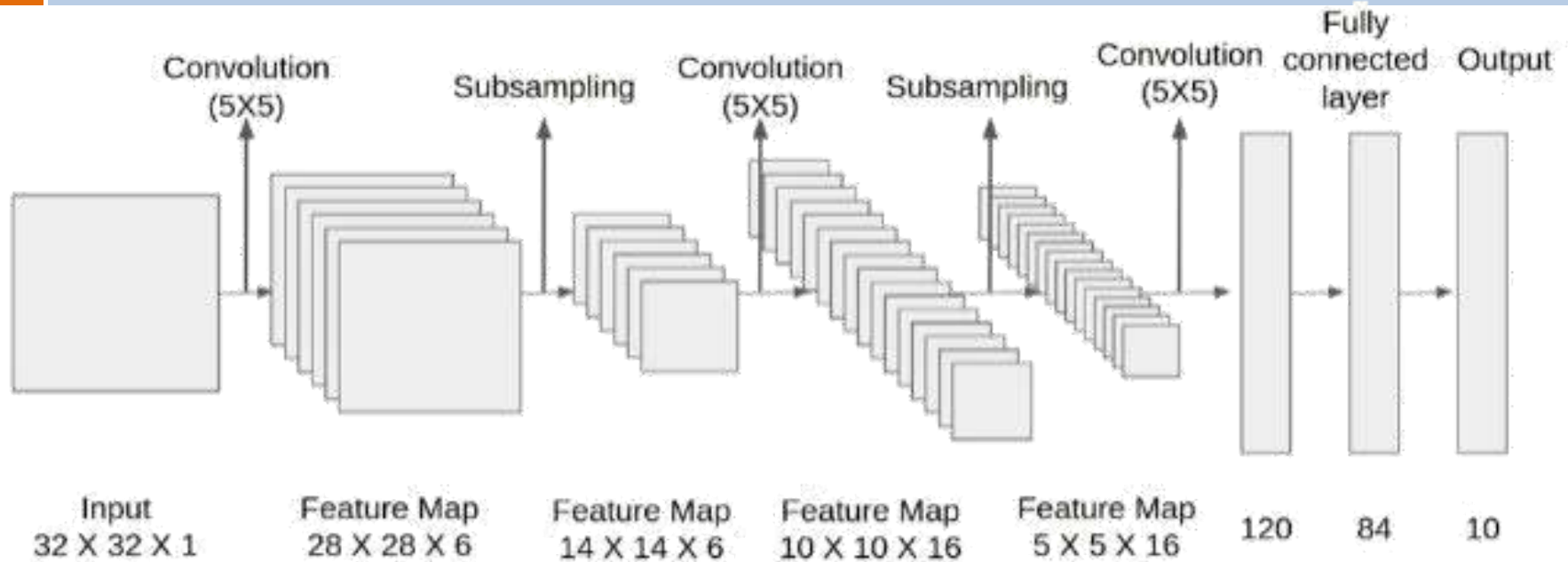
-25				...
				...
				...
				...
...	...	...	...	...

Structural Understanding CNN/RNN/LSTM's concepts



AI Geek Programmer, 2019,
Convolutional neural
network 2: architecture,
<https://aigeekprogrammer.com/convolutional-neural-network-image-recognition-part-2/>

Structural Understanding CNN/RNN/LSTM's concepts



Layer	# filters / neurons	Filter size	Stride	Size of feature map	Activation function
Input	-	-	-	32 X 32 X 1	
Conv 1	6	5 * 5	1	28 X 28 X 6	tanh
Avg. pooling 1		2 * 2	2	14 X 14 X 6	
Conv 2	16	5 * 5	1	10 X 10 X 16	tanh
Avg. pooling 2		2 * 2	2	5 X 5 X 16	
Conv 3	120	5 * 5	1	120	tanh
Fully Connected 1	-	-	-	84	tanh
Fully Connected 2	-	-	-	10	Softmax

$$O = \frac{I - F + 2P}{S} + 1$$

$$O = \frac{I - F}{S} + 1$$

Shipra Saxena, 2021, The Architecture of Lenet-5, analyticsvidhya
www.analyticsvidhya.com/blog/2021/03/the-architecture-of-lenet-5/

Structural Understanding CNN/RNN/LSTM's concepts

```
#Instantiate an empty model
```

```
model = Sequential()
```

```
# C1 Convolutional Layer
```

```
model.add(Conv2D(6, kernel_size=(5, 5), strides=(1, 1), activation='tanh', input_shape=input_shape, padding='same'))
```

```
# S2 Pooling Layer
```

```
model.add(AveragePooling2D(pool_size=(2, 2), strides=2, padding='valid'))
```

```
# C3 Convolutional Layer
```

```
model.add(Conv2D(16, kernel_size=(5, 5), strides=(1, 1), activation='tanh', padding='valid'))
```

```
# S4 Pooling Layer
```

```
model.add(AveragePooling2D(pool_size=(2, 2), strides=2, padding='valid'))
```

```
# C5 Fully Connected Convolutional Layer
```

```
model.add(Conv2D(120, kernel_size=(5, 5), strides=(1, 1), activation='tanh', padding='valid'))
```

```
#Flatten the CNN output so that we can connect it with fully connected layers
```

```
model.add(Flatten())
```

```
# FC6 Fully Connected Layer
```

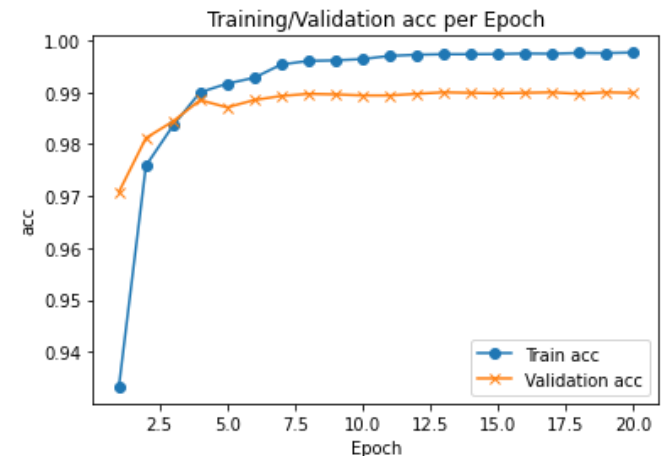
```
model.add(Dense(84, activation='tanh'))
```

```
# Output Layer with softmax activation
```

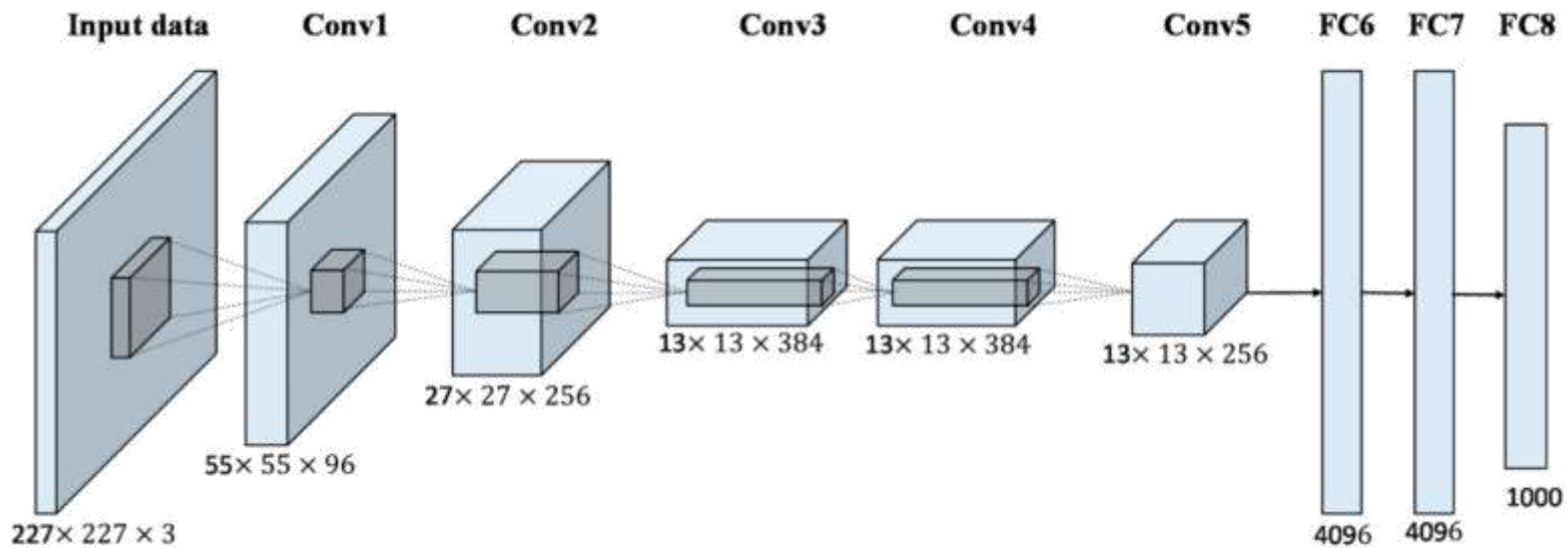
```
model.add(Dense(10, activation='softmax'))
```

```
# print the model summary
```

```
model.summary()
```



Structural Understanding CNN/RNN/LSTM's concepts



```
model = Sequential(name="Alexnet")
```

```
# 1st layer (conv + pool + batchnorm)
```

```
model.add(Conv2D(filters= 96, kernel_size= (11,11), strides=(4,4), padding='valid', kernel_regularizer  
=l2(0.0005),
```

```
input_shape = (227,227,3)))
```

```
model.add(Activation('relu'))
```

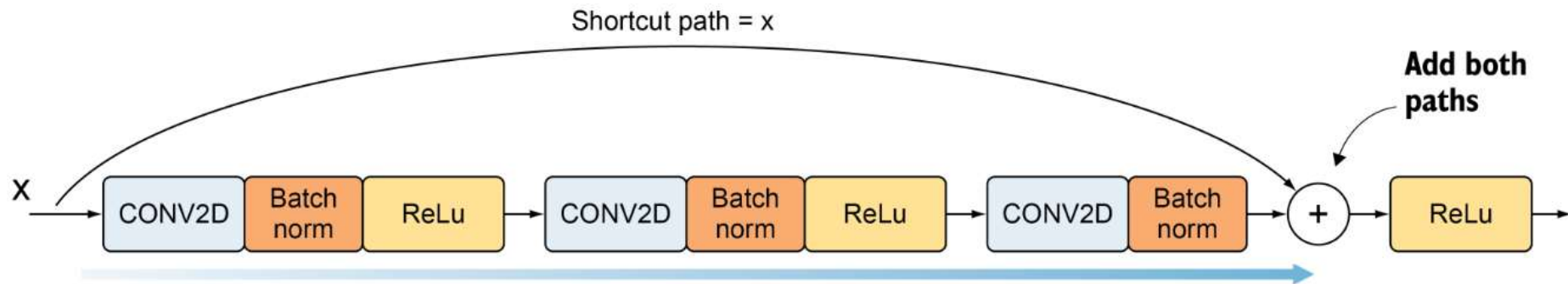
```
model.add(MaxPool2D(pool_size=(3,3), strides= (2,2), padding='valid'))
```

```
model.add(BatchNormalization())
```


Structural Understanding CNN/RNN/LSTM's concepts

Residual Connections (Skip Connections)

A method used in architectures such as ResNet, where the input of a particular layer is directly added to the output of that layer.



$X_{\text{shortcut}} = X$

$F = 32$

$X = \text{Conv2D}(\text{filters} = F, \text{kernel_size} = (3, 3), \text{strides} = (1, 1))(X)$

$X = \text{Activation}(\text{'relu'})(X)$

$X = \text{Conv2D}(\text{filters} = F, \text{kernel_size} = (3, 3), \text{strides} = (1, 1))(X)$

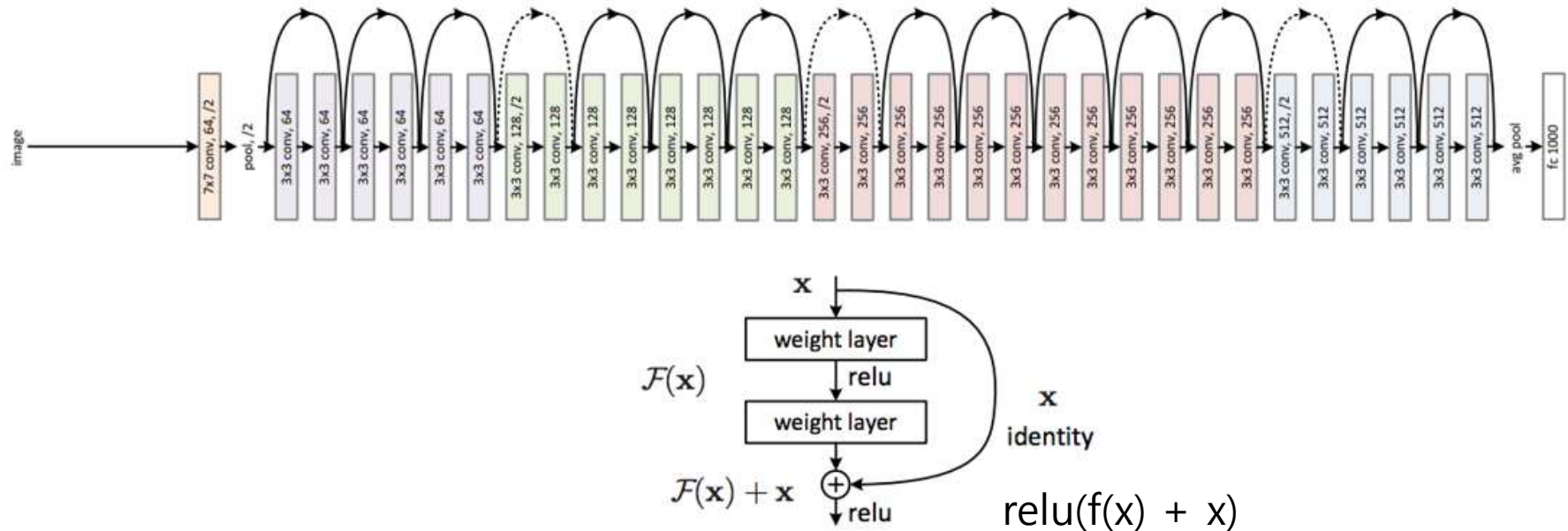
$X = \text{Add}()([X, X_{\text{shortcut}}])$

$X = \text{Activation}(\text{'relu'})(X)$

Structural Understanding CNN/RNN/LSTM's concepts

Residual Connections (Skip Connections)

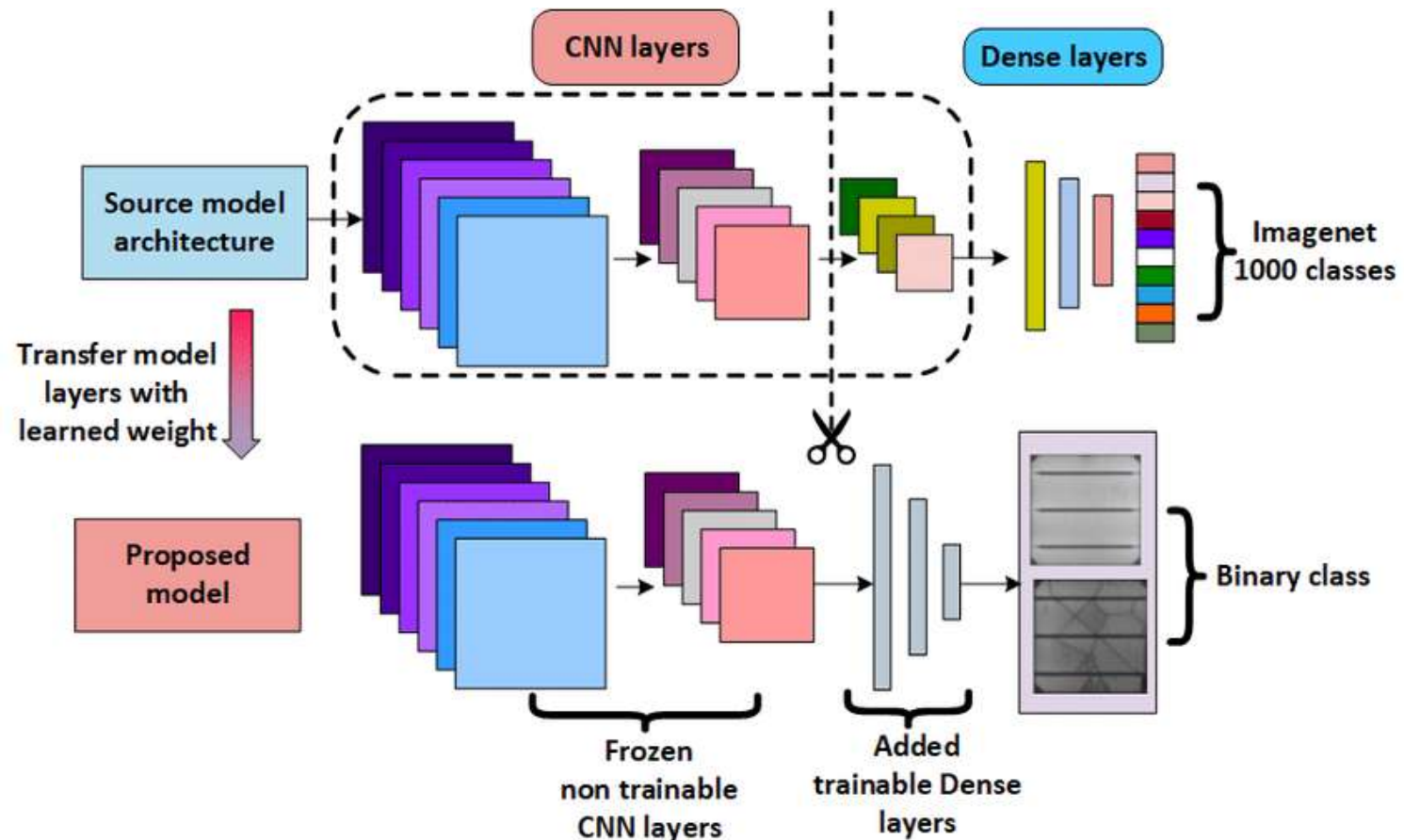
Residual Networks (ResNet50)



Structural Understanding CNN/RNN/LSTM's concepts

Transfer learning

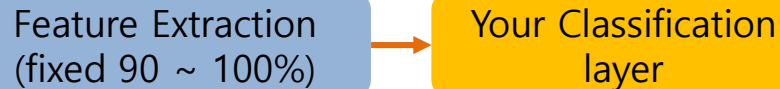
A machine learning technique that reuses the knowledge and features of a pre-trained model for new related tasks.



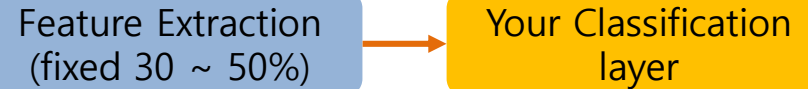
Structural Understanding CNN/RNN/LSTM's concepts

Transfer learning

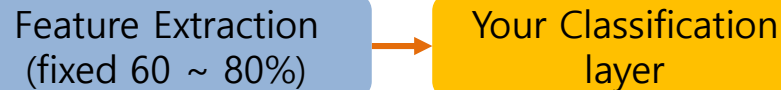
SS. Small target dataset size, Similar domains



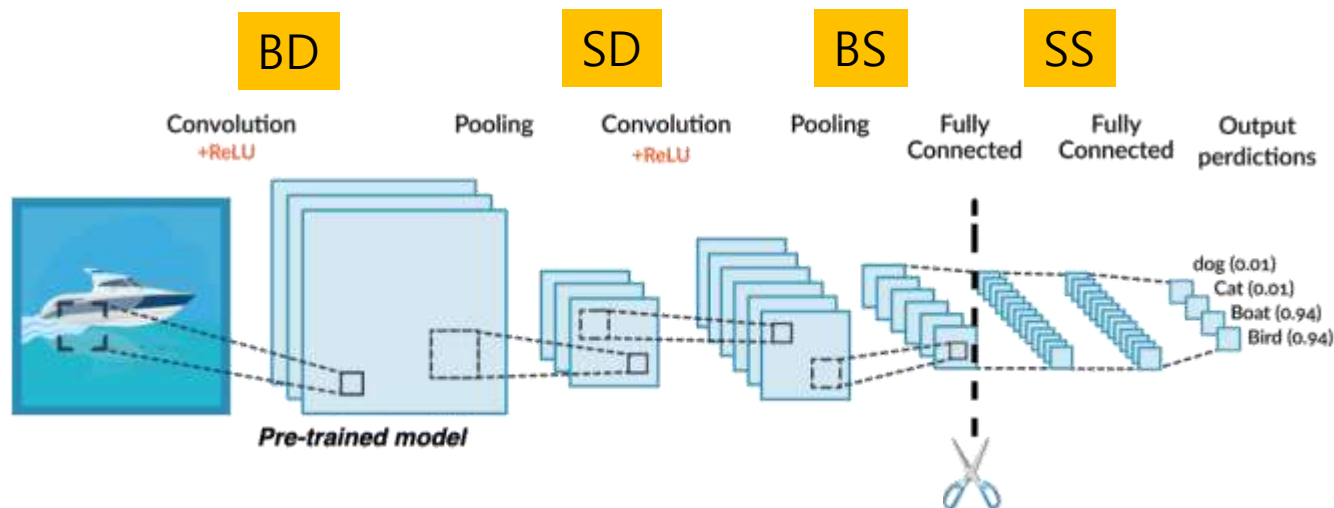
SD. Small target dataset size, Different domains



BS. Big target dataset size, Similar domains



BD. Big target dataset size, Different domains



Structural Understanding

Tensorflow vs Pytorch Diff

Differences Between Deep Learning Frameworks

TensorFlow, PyTorch, and Keras have structural differences in how they compute and update gradients in deep learning models. Specifically, they differ in how they implement graph data structures that manage partial differential operator information.

Deep learning models calculate the gradient of each parameter relative to the loss function to optimize the parameters. To efficiently execute the backpropagation algorithm, deep learning frameworks incorporate automatic differentiation. Automatic differentiation can be implemented in two main ways:

Static Graph: Predefining the entire computational graph before model execution. Applied to TensorFlow.

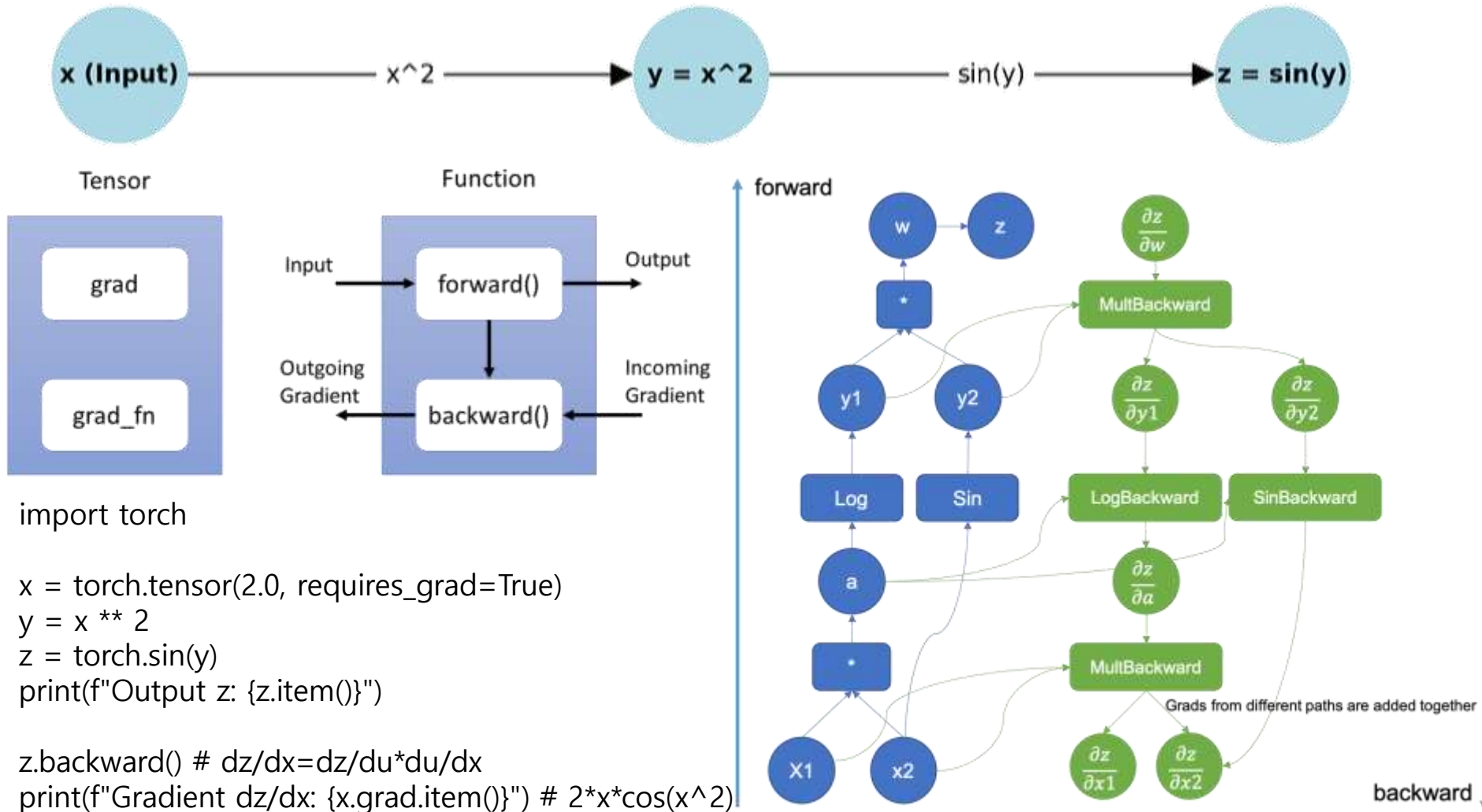
Dynamic Graph: Creating the computational graph on the fly each time the code is executed. Applied to PyTorch.

Characteristic	Static (as of TensorFlow 1.x, Keras)	Dynamic (PyTorch)
graph method	Predefine the entire computational graph before running the model.	Performs calculations each time the code is executed, generating a graph at the same time.
merit	Once compiled, the graph is optimized for fast execution, making it ideal for distributed learning and deployment on mobile/embedded devices.	Checking intermediate variable values like regular Python code, debugging is suitable for cases where the model structure changes dynamically (variable length, sequence, etc.)
disadvantage	It is difficult to debug variable values until the graph is executed. It is vulnerable to dynamic changes in the model structure.	Additional overhead at runtime compared to static graphs
Gradient operator connection method	Operation nodes are connected within <code>tf.Graph</code> , and <code>tf.GradientTape</code> records the operations. calculate slope	Operations are connected through the <code>grad_fn</code> property of <code>torch.Tensor</code> to form a computational graph for backpropagation.

Structural Understanding Tensorflow vs Pytorch Diff

automatic differentiation engine

Computational Graph: $y = x^2$, $z = \sin(y)$



Structural Understanding

Tensorflow vs Pytorch Diff

tensorflow
검색어

keras
검색어

pytorch
검색어

+ 비교 추가

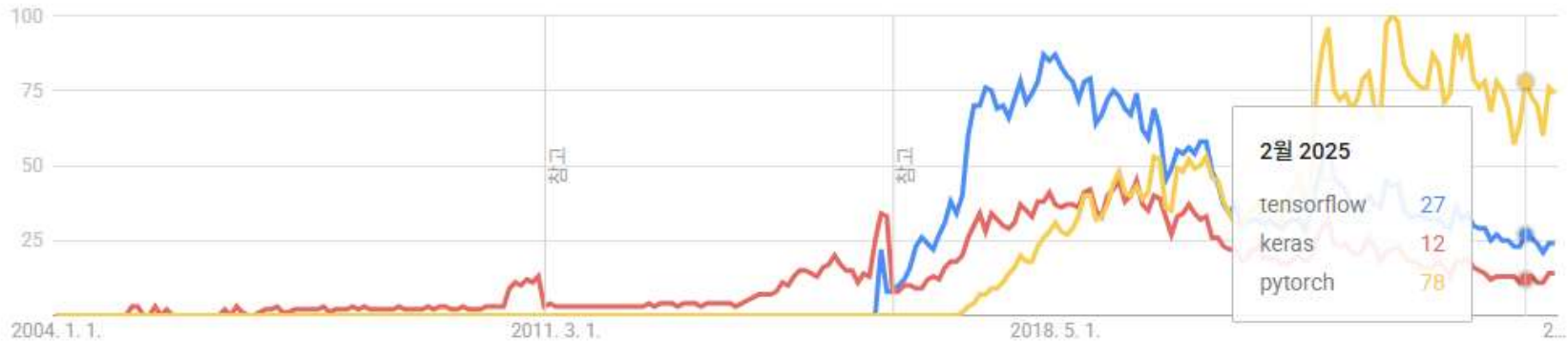
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2004 - 현재 ▼

모든 카테고리 ▼

웹 검색 ▼

시간 흐름에 따른 관심도 변화 ②



Structural Understanding

Tensorflow vs Pytorch Diff

PyTorch

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1.12.1+cu102

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QUICKSTART

This section runs through the API for common tasks in machine learning. Refer to the links in each section to dive deeper.

Working with data

PyTorch has two primitives to work with data: `torch.utils.data.DataLoader` and `torch.utils.data.Dataset`. `Dataset` stores the samples and their corresponding labels, and `DataLoader` wraps an iterable around the `Dataset`.

```
import torch
from torch import nn
from torch.utils.data import DataLoader
from torchvision import datasets
from torchvision.transforms import ToTensor
```

Structural Understanding

Tensorflow vs PyTorch Diff

Tensorflow

```
import tensorflow as tf

x_data = [1, 2, 3, 4, 5]
y_data = [2, 4, 5, 4, 5]

model = tf.keras.Sequential([
    tf.keras.layers.Dense(1, input_dim=1)
])

model.compile(optimizer='sgd',
              loss='mean_squared_error')

model.fit(x_data, y_data, epochs=1000,
          verbose=0)

print(model.predict([10]))
```

PyTorch

```
import torch, torch.nn as nn, torch.optim as optim

x_data = torch.tensor([[1.], [2.], [3.], [4.], [5.]])
y_data = torch.tensor([[2.], [4.], [5.], [4.], [5.]])

class LinearRegressionModel(nn.Module):
    def __init__(self):
        super(LinearRegressionModel, self).__init__()
        self.linear = nn.Linear(1, 1)

    def forward(self, x):
        return self.linear(x)

model = LinearRegressionModel()
optimizer = optim.SGD(model.parameters(), lr=0.01)

for epoch in range(1000):
    optimizer.zero_grad()
    outputs = model(x_data)
    loss = criterion(outputs, y_data)
    loss.backward()
    optimizer.step()

print(model(torch.tensor([[10.]])
```

Structural Understanding Pytorch vs Keras Diff

Keras

```
from keras.models import Sequential
from keras.layers import Dense, Dropout,
Conv2D, MaxPool2D, Flatten
from keras.layers import
BatchNormalization

model = Sequential()
model.add(Conv2D(32, (3, 3),
activation='relu', input_shape=(32, 32, 3)))
model.add(MaxPool2D())
model.add(Conv2D(16, (3, 3),
activation='relu'))
model.add(MaxPool2D())
model.add(Flatten())
model.add(Dense(10, activation='softmax'))

model.summary()
```

PyTorch

```
import torch

nn = torch.nn
F = torch.nn.functional

class Net(nn.Module):
    def __init__(self):
        super(Net, self).__init__()
        self.conv1 = nn.Conv2d(3, 32, 3)
        self.conv2 = nn.Conv2d(32, 16, 3)
        self.fc1 = nn.Linear(16 * 6 * 6, 10)
        self.pool = nn.MaxPool2d(2, 2)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = x.view(-1, 16 * 6 * 6)
        x = F.log_softmax(self.fc1(x), dim=-1)
        return x

model = Net()
print(model)
```

Structural Understanding Tensorflow vs Pytorch Diff

1 Load data

2 Define model

3 Train model

4 Evaluate model

General

PyTorch is a open source machine learning framework. It uses **torch.Tensor** – multi-dimensional matrices – to process. A core feature of neural networks in PyTorch is the autograd package, which provides automatic derivative calculations for all operations on tensors.

<code>import torch</code>	Root package	<code>torch.randn(*size)</code>	Create random tensor
<code>import torch.nn as nn</code>	Neural networks	<code>torch.Tensor(L)</code>	Create tensor from list
<code>from torchvision import datasets, models, transforms</code>	Popular image datasets, architectures & transforms	<code>tnsr.view(a,b, ...)</code>	Reshape tensor to size (a, b, ...)
<code>import torch.nn.functional as F</code>	Collection of layers, activations & more	<code>requires_grad=True</code>	tracks computation history for derivative calculations

Layers

 nn.Linear(m, n): Fully Connected layer (or dense layer) from m to n neurons	 nn.ConvXd(m, n, s): X-dimensional convolutional layer from m to n channels with kernel size s; $X \in \{1, 2, 3\}$
 nn.Flatten(): Flattens a contiguous range of dimensions into a tensor	 nn.MaxPoolXd(s): X-dimensional pooling layer with kernel size s; $X \in \{1, 2, 3\}$
 nn.Dropout(p=0.5): Randomly sets input elements to zero during training to prevent overfitting	 nn.BatchNormXd(n): Normalizes a X-dimensional input batch with n features; $X \in \{1, 2, 3\}$
 nn.Embedding(m, n): Lookup table to map dictionary of size m to embedding vector of size n	 nn.RNN/LSTM/GRU: Recurrent networks connect neurons of one layer with neurons of the same or a previous layer

torch.nn offers a bunch of other building blocks.

A list of state-of-the-art architectures can be found at <https://paperswithcode.com/sota>.

[PyTorch Tutorial for Reshape, Squeeze, Unsqueeze, Flatten and View - MLK - Machine Learning Knowledge](#)

Link - <https://www.stefan-nseeger.de/media/pytorch-cheatsheet-EN.pdf>

Structural Understanding

Example
ML Basic

Natural Language Processing

Text Preprocessing

Token and Word Embedding

Introducing the Hugging Face Model Example

NLP

Key Question

How to digitize words for neural network training?

How to differentiate between words?

What can we do with embedding vectors?

Can we predict the next word based solely on the preceding words?

Can we differentiate between a given sentence and a predicted sentence?

Can we classify the type of sentence?

AI model

Transformer Structure (Self-Attention, Positional Encoding)

Deep learning models cannot understand text directly.

Sentence → Tokenizing → Token ID → Embedding Vector

"I am a student" → ["I", "am", "a", "student"]

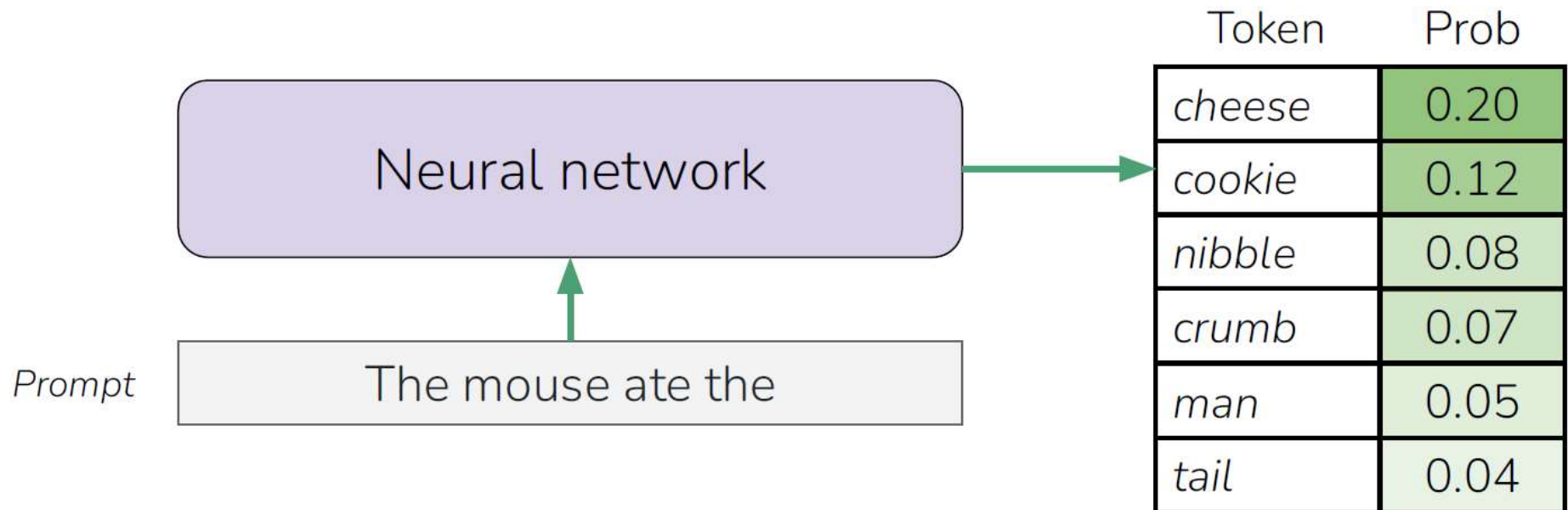
["I", "am", "a", "student"] → [101, 2023, 1037, 3076]

```
embedding = nn.Embedding(vocab_size, d_model)
token_vec = embedding(token_ids)
```

[-0.1, 0.3, ..., 0.05]

Natural Language Processing

Tokenization

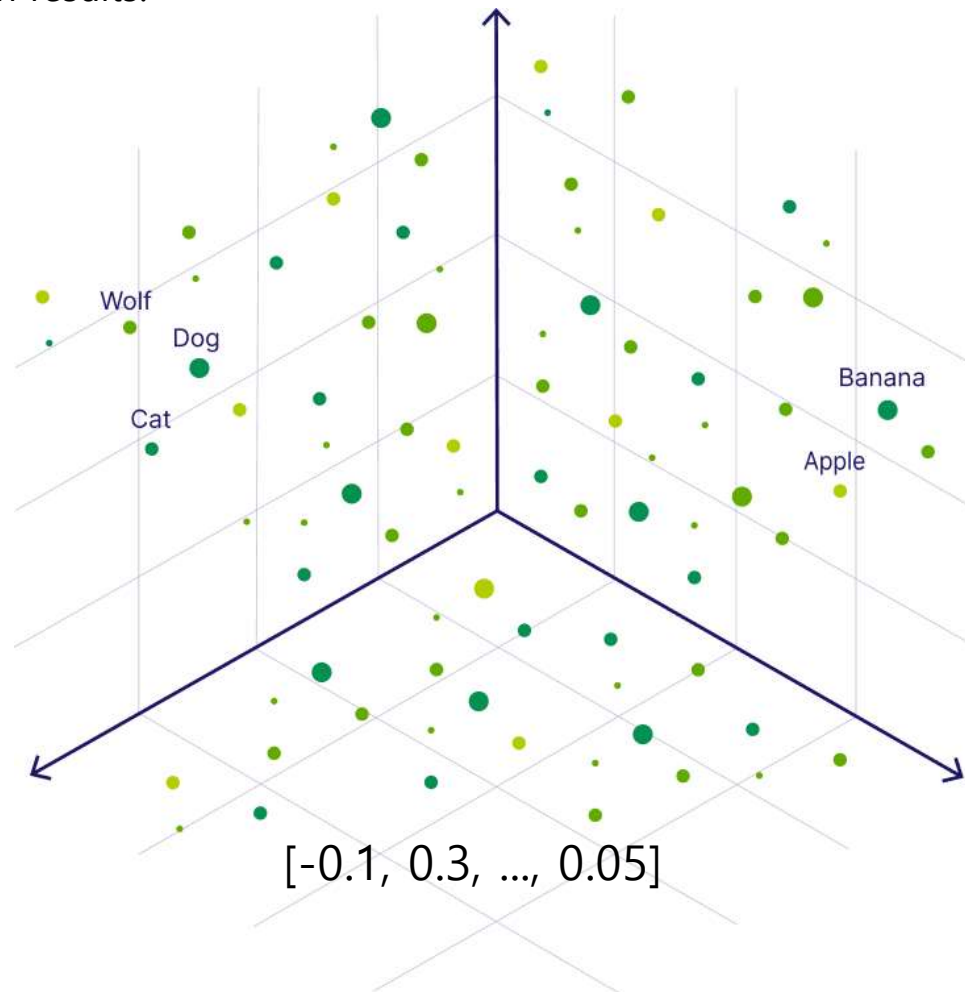


AI model Transformer Structure (Self-Attention, Positional Encoding)

Embedding vectors

A compressed representation of high-dimensional, sparse data, such as words, sentences, images, and users, into a low-dimensional, dense vector space.

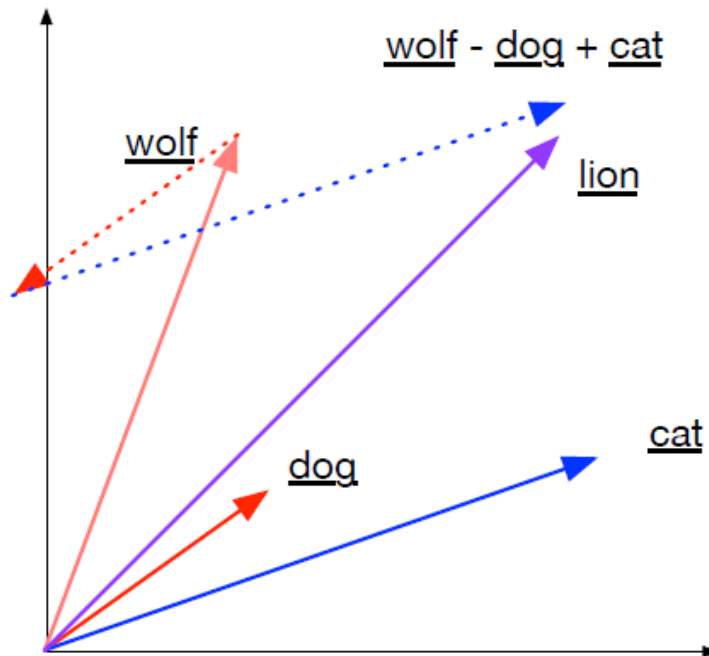
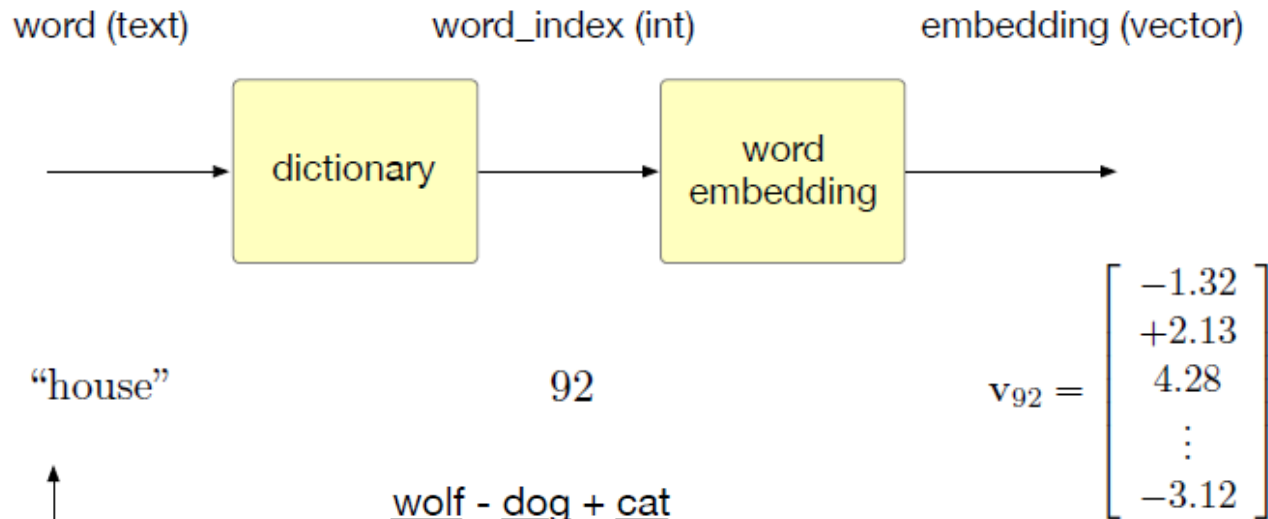
If the token dictionary and embedding model are different, it is difficult to obtain proper model training, prediction, and pattern calculation results.



Arrange words in space in a way that is "easy for the model to predict"

Natural Language Processing Tokenization

embedding vector



Natural Language Processing embedding (Word2Vec)

Word2Vec

An embedding method proposed by Google in 2013, it utilizes two learning structures: Skip-gram and CBOW (Continuous Bag of Words).

Skip-gram predicts surrounding words by taking a central word as input. CBOW predicts the central word using surrounding words.

In the sentence "The quick brown fox jumps," if the window size for the central word "fox" is 2, the surrounding words would be "brown" and "jumps."

Word2Vec is trained by minimizing the loss for predicting surrounding words. Embedding models are typically trained using unsupervised learning on large-scale text data.

```
# Word2Vec Skip-gram
for center_word, context_words in dataset:
    center_vec = word_embedding(center_word)
    for context in context_words:
        context_vec = word_embedding(context)
        loss += negative_sampling_loss(center_vec, context_vec)
    loss.backward()
    optimizer.step()
```

Natural Language Processing Tokenization

Cosine Similarity

A measure of how similar the directions of two vectors are.

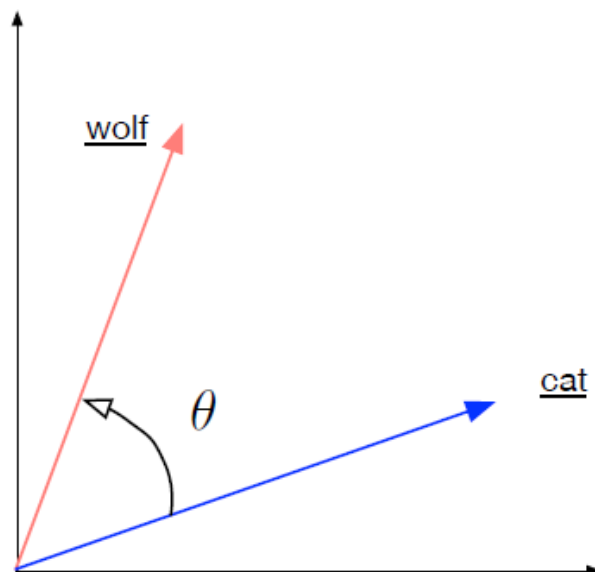
In word embeddings, words with similar meanings, such as "cat" and "leopard," are trained to point in close directions in the vector space. This similarity can be measured using cosine similarity.

$$\cos(\theta) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|}$$

1: When the two vectors point in exactly the same direction (most similar)

0: When the two vectors are orthogonal (perpendicular)

-1: When the two vectors point in completely opposite directions



Natural Language Processing 정규식

Regular Expressions

A string search tool for finding, replacing, or separating specific patterns in strings.

First introduced in the 1950s by American mathematician Stephen Kleene.

grammar	explanation	example	explanation
.	one random character	a.c	Matches 'abc', 'axc'
^	start of string	^a	Matches 'apple', X for 'banana'
\$	end of string	e\$	Matches 'apple' and 'title'
[]	character set	[credit]	matching to vowels
[^]	Excluded Character Set	[^0-9]	non-numeric characters
*	0 or more repetitions	a*	11 "", 'a', 'aa' all match
+	1 or more repetitions	a+	'a', 'aa' match, "" is X
?	0 or 1	a?	"", 'a' matches
{n}	repeat exactly n times	a{3}	Matches only 'aaa'
{n,}	Repeat n or more times	a{2,}	Matches 'aa', 'aaa', ...
{n,m}	Repeat n~m times	a{1,3}	Matches 'a', 'aa', 'aaa'
\d	numeric characters	\d+	Matches '123', '5', etc.
\w	word letters	\w+	Matches 'abc123', '_', etc.
\s	space character	\s+	Matches "", '\t', '\n', etc.
()	grouping	(abc)+	Matches 'abc', 'abcabc'

Natural Language Processing Tokenization

WordPiece

Tokenizes subwords using a tokenization process.

Byte Pair Encoding (BPE)

A type of subword tokenizer that compresses data by replacing the most common consecutive byte pairs in the data with bytes that do not occur in the data.

BPE was initially developed as a text compression algorithm and is then used for tokenization during GPT model pretraining.

Efficiently encodes large vocabularies by splitting words into more manageable subwords or symbols.

Significantly reduces vocabulary size and improves the model's ability to handle rare or out-of-vocabulary (OOV) terms.

```
tokenizer.encode('123234234')  
[10163, 24409, 24409]  
tokenizer.encode('123')  
[10163]  
tokenizer.encode('234')  
[24409]  
tokenizer.encode('234')  
[24409]
```

Natural Language Processing Tokenization

```
from transformers import DistilBertTokenizerFast, DistilBertModel
```

```
tokenizer = DistilBertTokenizerFast.from_pretrained("distilbert-base-uncased")
```

```
tokens = tokenizer.encode('This is a lfcBuilding.', return_tensors='pt')
```

```
print("These are tokens!", tokens)
```

```
for token in tokens[0]:
```

```
    print("This are decoded tokens!", tokenizer.decode([token]))
```

```
model = DistilBertModel.from_pretrained("distilbert-base-uncased")
```

```
print(model.embeddings.word_embeddings(tokens))
```

```
for e in model.embeddings.word_embeddings(tokens)[0]:
```

```
    print("This is an embedding!", e)
```

```
These are tokens! tensor([[ 101,  2023,  2003,  1037,  2065, 27421, 19231,  4667,  1012,  102]])
This are decoded tokens! [CLS]
This are decoded tokens! this
This are decoded tokens! is
This are decoded tokens! a
This are decoded tokens! if
This are decoded tokens! ##cb
This are decoded tokens! ##uil
This are decoded tokens! ##ding
This are decoded tokens! .
This are decoded tokens! [SEP]
```

```
This is an embedding! tensor([ 3.8952e-02, -1.2318e-02, -2.0844e-02, -5.2684e-04, -1.9758e-02,
                               3.8324e-02, -2.0617e-02,  3.3877e-03, -2.2452e-02, -4.3990e-02,
                               1.2990e-02, -1.6670e-02,  9.7562e-03, -1.2669e-02, -4.5170e-02,
                               2.5090e-02,  4.4694e-02,  5.9726e-02, -5.0432e-03, -3.6367e-02,
```


Natural Language Processing Tokenization

```
from transformers import BertTokenizerFast, BertModel
import torch
from torch import nn
```

```
# BERT 토크나이저 사전학습모델 로딩
```

```
tokenizer = BertTokenizerFast.from_pretrained('bert-base-uncased')
print(tokenizer.tokenize("[CLS] Hello world, how are you?"))
```

```
print(tokenizer.tokenize("[newtoken] Hello world, how are you?"))
tokenizer.add_tokens(['[newtoken]'])
```

```
[['[',
 'newt',
 '##oke',
 '##n',
 ']',
 'hello',
 'world',
 ',',
 'how',
 'are',
 'you',
 '?']]
```

Natural Language Processing Tokenization

```
# Add tokens and tokenize again.
```

```
tokenizer.add_tokens(['[newtoken]'])
```

```
tokenizer.tokenize("[newtoken] Hello world, how are you?")
```

```
# Tokenization is done correctly.
```

```
['[newtoken]', 'hello', 'world', ',', 'how', 'are', 'you', '?']
```

```
# Check the token values.
```

```
tokenized = tokenizer("[newtoken] Hello world, how are you?", add_special_tokens=False,  
return_tensors="pt")
```

```
print(tokenized['input_ids'])
```

```
tkn = tokenized['input_ids'][0, 0]
```

```
print("First token:", tkn)
```

```
print("Decoded:", tokenizer.decode(tkn))
```

```
# As shown below, the token values are correctly assigned. tensor([[30522, 7592, 2088, 1010,  
2129, 2024, 2017, 1029]])
```

```
First token: tensor(30522)
```

```
Decoded: [newtoken]
```

Natural Language Processing

Named Entity Recognition (NER)

Named Entity Recognition (NER)

A technology that identifies specific entities in a sentence and classifies them into categories such as person, organization, or location. It can be used for sentiment analysis and news classification.

In the sentence "Apple Inc. was founded by Steve Jobs," "Apple Inc." is classified as an organization and "Steve Jobs" as a person.

B- (Begin): Beginning of an entity.

B-PER: Beginning of a person's name.

B-ORG: Beginning of an organization.

B-LOC: Beginning of a location.

N-gram

A group of N consecutive words or character units. It is mainly used for contextual modeling in natural language processing (NLP).

Unigram (1-gram): Word unit

["I", "rice", "ate"]

Bigram (2-gram): Two consecutive words

[("I", "rice"), ("rice", "ate")]

Trigram (3-gram): Three consecutive words

[("I", "rice", "ate")]

Natural Language Processing BLEU

BLEU(Bilingual Evaluation Understudy)

A metric for evaluating machine translation results. It calculates the degree of n-gram agreement between the translated sentence (candidate) and the reference sentence.

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

BP is the Brevity Penalty applied when a translation is shorter than the reference translation. w_n is the weight for each n-gram ($N=4$, $w_n = 0.25$).

If the base sentence is "the cat is on the mat" and the machine translation is "the cat is on mat", the 1-gram precision is 0.8, since 4 out of 5 matches

Candidate sentence: "the dog is on the mat"
Reference sentence: "the cat is on the mat"

Candidate sentence 1-grams: "the", "dog", "is", "on", "the", "mat"

Reference sentence 1-gram: "the", "cat", "is", "on", "the", "mat"

Number of 1-gram matches per:

"the": 2 candidates, maximum 2 references $\rightarrow \min(2,2)=2$

"dog": 1 candidate, at most 0 references $\rightarrow \min(1,0)=0$

"is": 1 candidate, at most 1 reference $\rightarrow \min(1,1)=1$

"on": 1 candidate, at most 1 reference $\rightarrow \min(1,1)=1$

"mat": 1 candidate, at most 1 reference $\rightarrow \min(1,1)=1$

"cat": 0 candidates, at most 1 reference (only in references)

Sum of matching 1-grams = $2+0+1+1+1=5$ Total number of 1-grams in candidate sentences = 6

$P1 = 5/6 \approx 0.8333$

Candidate sentences 2-gram: "the dog", "dog is", "is on", "on the", "the mat"

2-grams of the reference sentence: "the cat", "cat is", "is on", "on the", "the mat"

Number of matches for each 2-gram:

"the dog": $\min(1,0)=0$

"dog is": $\min(1,0)=0$

"is on": $\min(1,1)=1$

"on the": $\min(1,1)=1$

"the mat": $\min(1,1)=1$

Matching 2-gram sum = $0+0+1+1+1=3$

Total number of 2-gram candidate sentences = 5. $P2 = 3/5 = 0.6$

$$BLEU = BP \cdot \exp(w_1 \log p_1 + w_2 \log p_2)$$

$$BLEU = 1.0 \cdot \exp(0.5 \cdot \log(0.8333) + 0.5 \cdot \log(0.6))$$

$$BLEU \approx 0.7071$$

Natural Language Processing Hugging Face Example

Hugging Face

Known for providing a standardized interface for easily accessing the latest deep learning models, Hugging Face offers excellent LLM and transformer libraries, training datasets, and model execution space functionality.

 **Hugging Face**

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[google-bert/bert-base-uncased](#) · Hugging Face

[Papers](#) [arxiv:1810.04805](#)

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Published on Oct 11, 2018

Authors: [Jacob Devlin](#), [Ming-Wei Chang](#), [Kenton Lee](#), [Kristina Toutanova](#)

Abstract

BERT is a bidirectional transformer-based model that pre-trains on unlabeled text and fine-tunes for various NLP tasks, achieving state-of-the-art results across multiple benchmarks.

AI-generated summary

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications. BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

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 **google-bert/bert-base-chinese**

[Fill-Mask](#) • [0.1B](#) • Updated 20 days ago • [↓ 3.35M](#) • [♥ 1.27k](#)

 **dslim/bert-base-NER**

[Token Classification](#) • [0.1B](#) • Updated Oct 8, 2024 • [↓ 2.25M](#) • [♥ 623](#)

 **google-bert/bert-base-multilingual-cased**

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 **Babelscape/wikineural**

[Viewer](#) • Updated Nov 13, 2022 • [1.03M](#) • [↓ 3.31k](#) • [♥ 33](#)

Natural Language Processing Hugging Face Example

Hugging Face Model and Transfer Learning

Hugging Face provides pre-trained models, allowing for reuse and fine-tuning for new tasks.

[finetune-bert-cot-train.py](#)

The screenshot displays the Hugging Face homepage with a dark theme. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Community, Docs, Enterprise, and Pricing. The main content area is divided into sections for Tasks, Parameters, Libraries, and Apps. The Models section is prominently featured, showing a grid of popular models with their names, tasks, update dates, and statistics.

Tasks: Text Generation, Any-to-Any, Image-Text-to-Text, Image-to-Text, Image-to-Image, Text-to-Image, Text-to-Video, Text-to-Speech, +42

Parameters: < 1B, 6B, 12B, 32B, 128B, > 500B

Libraries: PyTorch, TensorFlow, JAX, Transformers, Diffusers, Safetensors, ONNX, GGUF, Transformers.js, MLX, Keras, +40

Apps: vLLM, TGI, llama.cpp, MLX LLM, LM Studio, Ollama, Jan, +12

Models (1,871,163):

- deepseek-ai/DeepSeek-R1**: Text Generation • Updated Mar 27 • ± 844k • + • 12.5k
- black-forest-labs/FLUX.1-dev**: Text-to-Image • Updated 20 days ago • ± 1.44M • + • 10.9k
- CompVis/stable-diffusion-v1-4**: Text-to-Image • Updated Aug 24, 2023 • ± 3.48M • 6.87k
- stabilityai/stable-diffusion-xl-base-1.0**: Text-to-Image • Updated Oct 31, 2023 • ± 2.4M • + • 6.73k
- meta-llama/Meta-Llama-3-8B**: Text Generation • Updated Sep 28, 2024 • ± 314k • + • 6.25k
- bigscience/bloom**: Text Generation • Updated Jul 29, 2023 • ± 3.82k • 4.92k
- stabilityai/stable-diffusion-3-medium**: Text-to-Image • Updated Aug 12, 2024 • ± 13.4k • + • 4.8k
- hexgrad/Kokoro-82M**: Text-to-Speech • Updated Apr 11 • ± 1.5M • + • 4.67k
- openai/whisper-large-v3**: Automatic Speech Recognition • Updated Apr 11 • ± 3.05M • + • 4.65k
- meta-llama/Llama-2-7b-chat-hf**: Text Generation • Updated Apr 17, 2024 • ± 1.04M • 4.5k
- mistralai/Mixtral-8x7B-Instruct-v0.1**: Text Generation • Updated Aug 19, 2024 • ± 388k • + • 4.49k
- meta-llama/llama-2-7b**: Text Generation • Updated Apr 17, 2024 • 4.37k
- meta-llama/Llama-3.1-8B-Instruct**: Text Generation • Updated Sep 26, 2024 • ± 6.25M • + • 4.31k
- black-forest-labs/FLUX.1-schnell**: Text-to-Image • Updated Aug 16, 2024 • ± 651k • + • 4.09k

Natural Language Processing

Hugging Face Example

[google/gemma-2b-it · Hugging Face](#)

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Text Generation



Transformers



Safetensors



Korean

Llama



Inference Endpoints



Model card



Files and versions



Community **1**



Gated model You have been granted access to this model



Natural Language Processing

Hugging Face Example

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Gated model You have been granted access to this model

Gemma Model Card

Model Page: [Gemma](#)

This model card corresponds to the 7B base version of the Gemma model. You can also visit the model card of the [2B base model](#), [7B instruct model](#), and [7B instruct model](#).

Resources and Technical Documentation:

- [Gemma Technical Report](#)
- [Responsible Generative AI Toolkit](#)
- [Gemma on Kaggle](#)
- [Gemma on Vertex Model Garden](#)

Downloads last month
73,221

Safetensors Model size: 8.54B params Tensor type: BF16 Files info

Inference Providers [NEW](#)

Text Generation

This model isn't deployed by any Inference Provider. [Ask for provider support](#)

Model tree for google/gemma-7b

Adapters	9182 models
Finetunes	360 models
Merges	7 models
Quantizations	23 models

[google/gemma-7b](#) · Hugging Face

Natural Language Processing Hugging Face Example

Fine-tuning examples

1 [finetuning/finetune-gemma.py](#)

You can find fine-tuning notebooks under the [examples/ directory](#). We provide:

- A script to perform Supervised Fine-Tuning (SFT) on UltraChat dataset using [QLoRA](#)
- A script to perform SFT using FSDP on TPU devices
- A notebook that you can run on a free-tier Google Colab instance to perform SFT on English quotes dataset. You can also find the copy of the notebook [here](#).

Running the model on a CPU

```
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("google/gemma-7b")
model = AutoModelForCausalLM.from_pretrained("google/gemma-7b")

input_text = "Write me a poem about Machine Learning."
input_ids = tokenizer(input_text, return_tensors="pt")

outputs = model.generate(**input_ids)
print(tokenizer.decode(outputs[0]))
```

Running the model on a single / multi GPU

```
# pip install accelerate
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("google/gemma-7b")
model = AutoModelForCausalLM.from_pretrained("google/gemma-7b", device_map="auto")

input_text = "Write me a poem about Machine Learning."
input_ids = tokenizer(input_text, return_tensors="pt").to("cuda")
```

Quantized Versions through `bitsandbytes`

- Using 8-bit precision (int8)

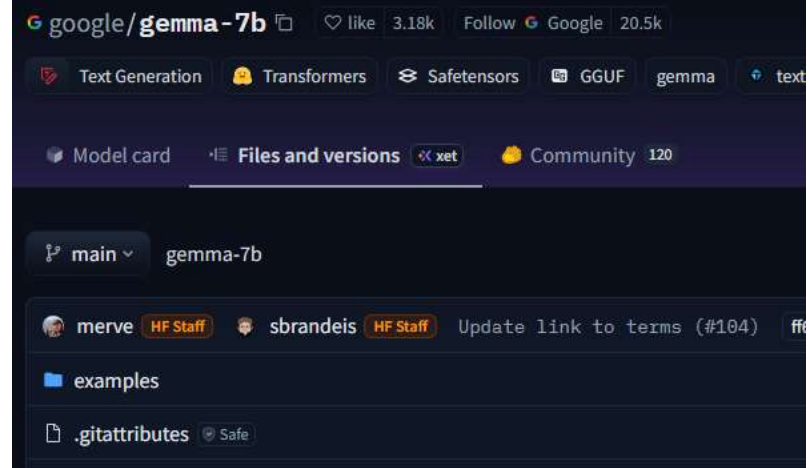
```
# pip install bitsandbytes accelerate
from transformers import AutoTokenizer, AutoModelForCausalLM

quantization_config = BitsAndBytesConfig(load_in_8bit=True)

tokenizer = AutoTokenizer.from_pretrained("google/gemma-7b")
model = AutoModelForCausalLM.from_pretrained("google/gemma-7b", quantization_config=quantization_config)

input_text = "Write me a poem about Machine Learning."
input_ids = tokenizer(input_text, return_tensors="pt")

outputs = model.generate(**input_ids)
print(tokenizer.decode(outputs[0]))
```



Natural Language Processing

Hugging Face Example

Datasets: sanjay920/gemma-function-calling like 8 Dataset		
Split (1) train · 112k rows		
Search this dataset		
messages list · lengths 306 67.9%	tools string · lengths 4760948 40.9%	text string · lengths 39202.01k 65.7%
[{ "content": "You are a helpful assistant, with no access to external functions.", "role": "system" }, {-	null	<start_of_turn>user You are a helpful assistant, with no access to external functions.<end_of_turn>-
[{ "content": "You are a helpful assistant with access to the following functions. Use them if required - \n[\n {\n \"type\": \"function\", \n \"function\": { \"name\": \"send_email\", \"description\": \"Send an email to a recipient\", \"parameters\": { \"type\": \"object\", \"properties\": { \"recipient\": { \"type\": \"string\", \"description\": \"The email address of the recipient\", \"type\": \"string\", \"description\": \"The email address of the recipient\" }, \"subject\": { \"type\": \"string\", \"description\": \"The subject of the email\" }, \"message\": { \"type\": \"string\", \"description\": \"The body of the email message\" } } }, \"required\": [\"recipient\", \"subject\", \"message\"] }], \"role\": \"system\" }, { \"content\": \"I need to send an email to my boss. Can you help me with that?\", \"role\": \"user\" }, { \"content\": \"Of course, I can help you with that. Could you please provide me with the recipient's email address, the subject of the email, and the	[{ "type": "function", "function": { "name": "send_email", "description": "Send an email to a recipient", "parameters": { "type": "object", "properties": { "recipient": { "type": "string", "description": "The email address of the recipient" }, "subject": { "type": "string", "description": "The subject of the email" }, "message": { "type": "string", "description": "The body of the email message" } }, "required": ["recipient", "subject", "message"] }]	<start_of_turn>user You are a helpful assistant with access to the following functions. Use them if required - [{ "type": "function", "function": { "name": "send_email", "description": "Send an email to a recipient", "parameters": { "type": "object", "properties": { "recipient": { "type": "string", "description": "The email address of the recipient" }, "subject": { "type": "string", "description": "The subject of the email" }, "message": { "type": "string", "description": "The body of the email message" } }, "required": ["recipient", "subject",

Natural Language Processing

Hugging Face Example

 Hugging Face

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mesh

Browse [ZeroGPU Spaces](#)

Full-text search

Sort: Trending



sudo-ai

Aug 23



sudo-ai

Aug 20



MarcusLozen

Jun 8



Yiwen-ntu

Aug 7



dawood

Jul 13



Yiwen-ntu

Aug 8



Create a new Space

Spaces are Git repositories that host application code for Machine Learning demos. You can build Spaces with Python (Streamlit, Gradio, or Colab) or using Docker images.

Owner

meow99

Space name

New Space name

Short description

Short description

License

License

Select the Space SDK

You can choose between Streamlit, Gradio, and Docker for your Space. Or you can use the first way when you click on the SDKs below.

Streamlit

Gradio

Docker

Static

Space hardware

Mikro K205 small - 4 vCPU - 15 GB - \$3.00 per hour

Sleep after: 3 hour / off inactivity / Sleep time is not billed

Natural Language Processing Hugging Face Example

 Spaces

shariqfarooq / ZoeDepth 

 like 685

Running on **A10G**



Depth Prediction | **Image to 3D** | 360 Panorama to 3D

Image to 3D mesh

Convert a single 2D image to a 3D mesh





☐ Keep occlusion edges

Submit



Natural Language Processing Hugging Face Example

 **Spaces**

 **shariqfarooq / ZoeDepth**

 like 685

Running on **A10G**



Depth Prediction

Image to 3D

360 Panorama to 3D

Panorama to 3D mesh

Convert a 360 spherical panorama to a 3D mesh

ZoeDepth was not trained on panoramic images. It doesn't know anything about panoramas or spherical projection. Here, we just treat the estimated depth as radius and some projection errors are expected. Nonetheless, ZoeDepth still works surprisingly well on 360 reconstruction.

Input Image



3d mesh reconstruction



Natural Language Processing

Hugging Face Example

Example
NLP

Transformer and the Latest AI Models

Transformer Structure

How BERT, GPT, and Others Work

Why Has Transformer Become the Standard?

[transformer](#)

Key Question

How can we predict the words that follow the preceding words?

What is context?

How do we calculate the similarity between sentences?

How can we mathematically represent the context within a sentence sequence?

Which module in a Transformer computes the context as a feature vector?

Which module computes the probability of the next word using the context feature vector?

How does cross-attention translate using self-attention context?

Why is the attention module computed separately using QKV to predict the next word?

AI model Transformer Structure (Self-Attention, Positional Encoding)

트랜스포머(Transformers)

In the realm of text generation, transformer-based models like the Generative Pre-trained Transformer (GPT) learn patterns from massive amounts of text data through attention calculations. This enables the learning of consistent and contextually relevant token sequences. They are used to generate conditioned images.

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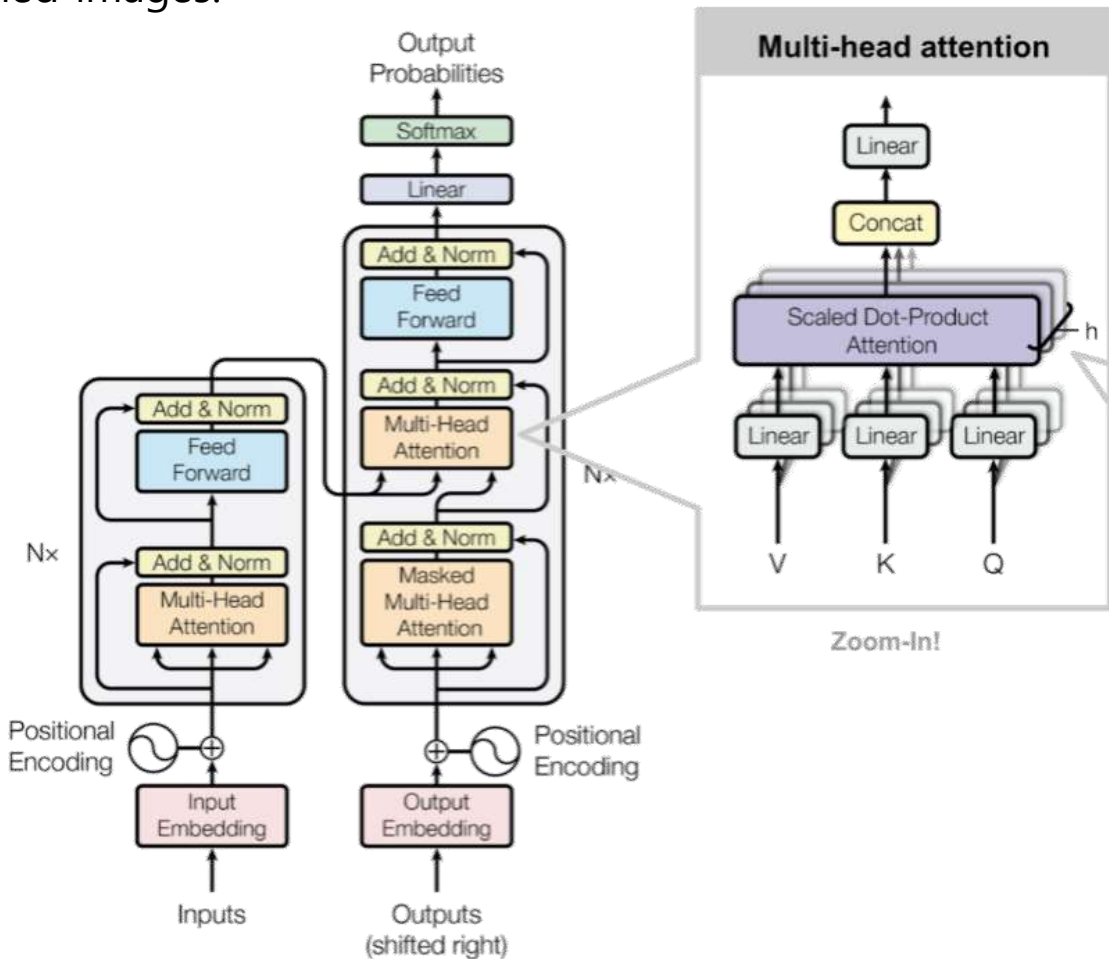
Illia Polosukhin*
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 13].



Link: 딥러닝 모델 트랜스포머 인코더 핵심 코드 구현을 통한 동작 메커니즘 이해하기

AI model Transformer Structure (Self-Attention, Positional Encoding)

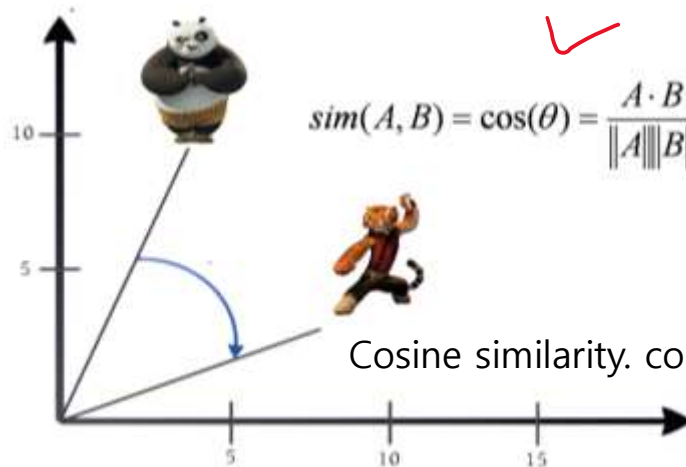
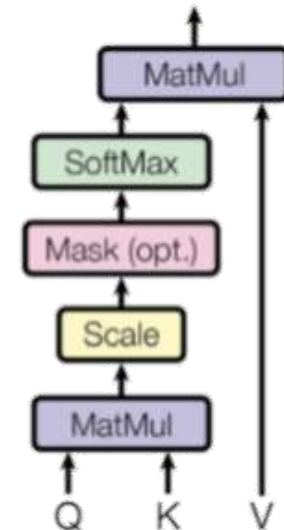
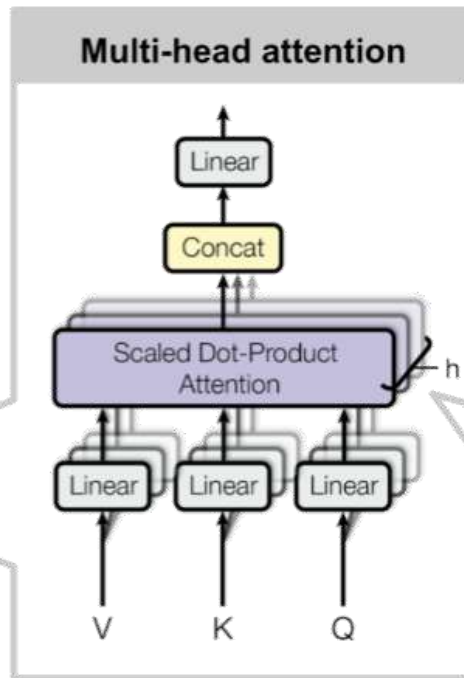
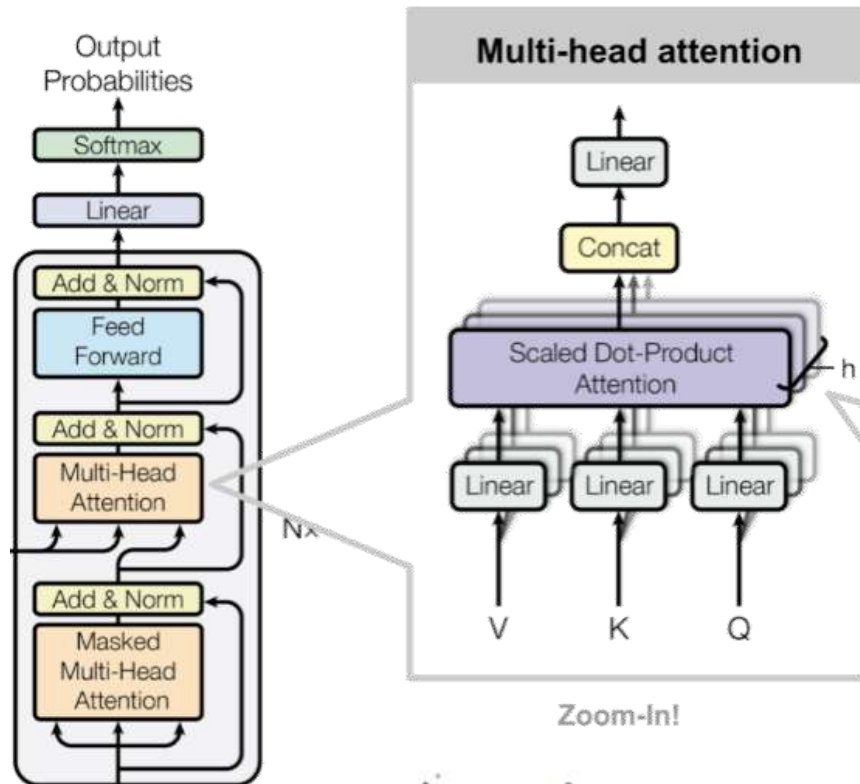
Attention

Each word in a sequence is weighted based on its relationship to all other elements, determining which word has the greatest influence on the prediction of the currently processed word.

The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .
The FBI is chasing a criminal on the run .

An example of attention in a given sentence (red indicates the current word token, blue indicates the token that is being activated by attention. Long Short-Term Memory-Networks for Machine Reading, 2016)

AI model Transformer Structure (Self-Attention, Positional Encoding)



The diagram illustrates the matrix multiplication steps for the attention mechanism. It shows the formula:

$$\text{softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V = Z$$

where Q is the Query matrix, K^T is the transposed Key matrix, V is the Value matrix, and Z is the resulting matrix. The diagram also shows the formula $V \times \text{Softmax}(QK^T)$.

[ex\) source code](#)

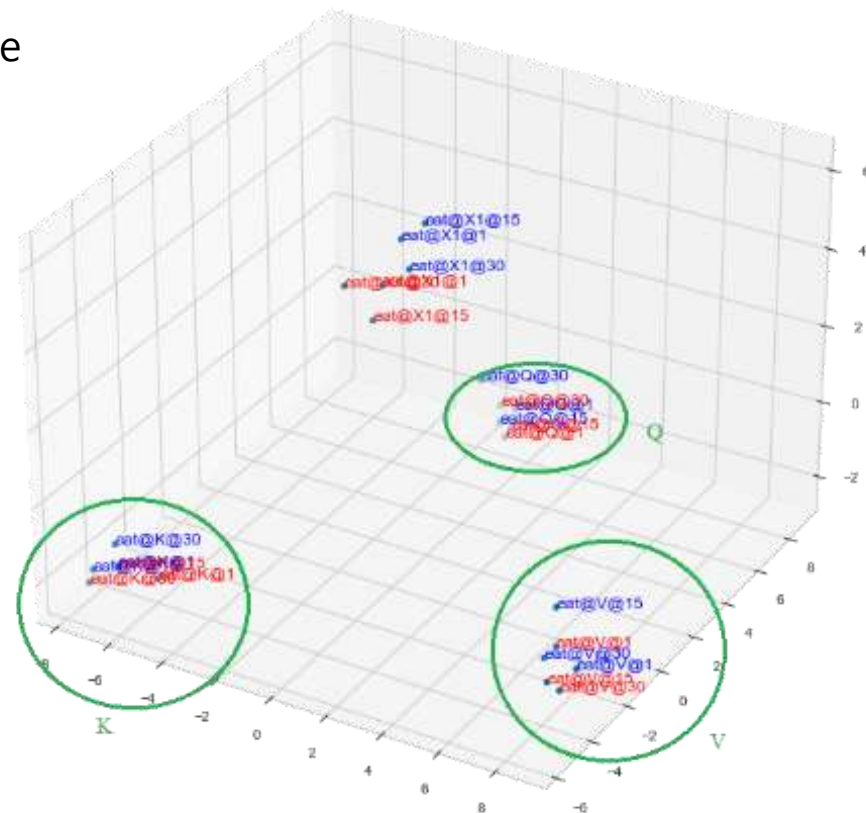
AI model Transformer Structure (Self-Attention, Positional Encoding)

Attention calculation process

1. Initially, QK^T computes a meaningless similarity $\rightarrow$ After softmax, it averages V and outputs it.
2. If this result deviates from the predicted label (e.g., the next word), the loss increases.
3. Backpropagation updates the weights W_Q , W_K , and W_V , which create Q , K , and V .
4. This process is repeated for tens of millions of sentences, and each QKV learns to play a different role in the context.

역할 = Query Key Value

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



AI model Transformer Structure (Self-Attention, Positional Encoding)

Contextual Embedding

Vector values that pass through self-attention have unique vector representations that reflect their meaning depending on the context (sentence) in which they are placed. For example, the embeddings for "fruit apple" and "action apple" have different vector values due to the influence of surrounding words.

"I am a student" → attention → out → FFN → logits (10000 dimensions) → softmax → "is" (prediction)

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$


out

Context-sensitive vector (dimensions: [seq_len, d_model])

Feed-Forward Network (FFN)

Nonlinearly transform each out vector (information enhancement)

Decoder Stack Last Output

Final output hidden state

Linear + Softmax

For each hidden state, create logits as large as the entire vocabulary size →

Convert to probability by applying softmax

Token Prediction

The word with the highest probability is predicted as the next word (e.g., "is" is predicted after "student").

AI model Transformer Structure (Self-Attention, Positional Encoding)

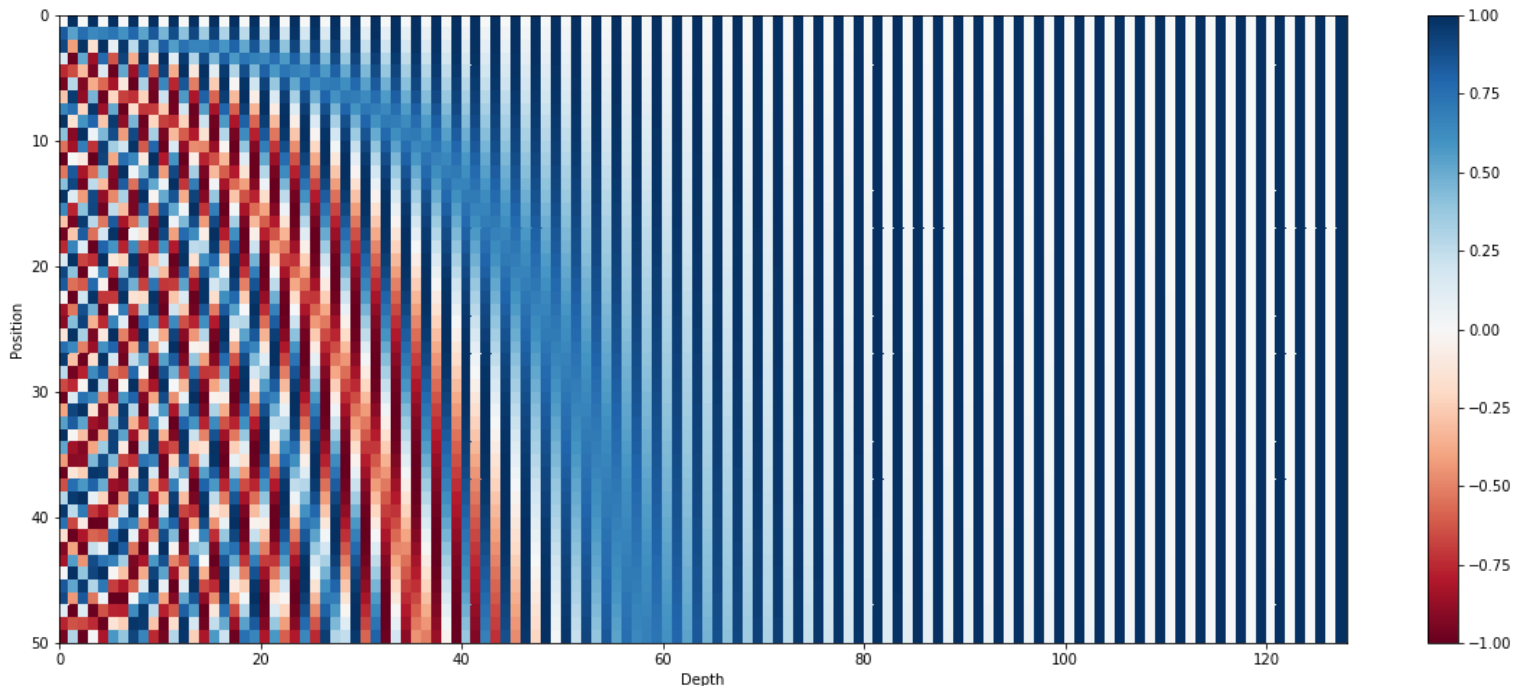
Positional Encoding

Because the transformer doesn't consider order, it encodes the positional information of each word. To achieve this, it generates a positional encoding vector based on the sine (sin) and cosine (cos) functions.

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$
$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Here, i = dimension index
 d_{model} = embedding length
 pos = word position within the sequence

```
class PositionalEncoding(nn.Module):  
    def forward(self, x):  
        # Calculate sin and cos vectors by location and  
        # add them to the input  
        return x + self.pe[:, :x.size(1)]
```



최대 길이 50인 문장에 대한 128차원 위치 인코딩 결과(각 행은 embedding vector를 나타냄. [Amirhossein Kazemnejad](#))

AI model Transformer Structure (Self-Attention, Positional Encoding)

Scaled Dot Product Attention

This module is the core component that performs the basic operations of attention. It performs the following operations using the input Q (Query), K (Key), and V (Value) vectors.

Multi-Head Attention

Calculating attention as a single vector results in significant information loss. Therefore, multiple attention "heads" are executed in parallel and the results are concatenated.

Positional Encoding

Because the transformer does not consider order, it encodes the positional information of each word. To achieve this, it generates a positional encoding vector based on the sine and cosine functions.

```
class ScaledDotProductAttention(nn.Module):
    def forward(self, Q, K, V):
        scores = torch.matmul(Q, K.transpose(-2, -1)) /
        math.sqrt(d_k)
        attn = F.softmax(scores, dim=-1)
        return torch.matmul(attn, V)
```

```
class MultiHeadAttention(nn.Module):
    def forward(self, Q, K, V):
        # Q, K, V 선형 변환 및 split
        # 각 헤드별 attention 계산
        # concat 후 출력 선형 변환
        return output
```

```
class PositionalEncoding(nn.Module):
    def forward(self, x):
        # 위치별 sin, cos vector 계산 후 입력에 더함
        return x + self.pe[:, :x.size(1)]
```

```
class EncoderLayer(nn.Module):
    def forward(self, x):
        x = x + self_attn(x, x, x) # Residual
        x = LayerNorm(x)
        x = x + FFN(x) # Residual
        x = LayerNorm(x)
        return x
```

AI model Transformer Structure (Self-Attention, Positional Encoding)

1. Input

```
input_sentence = ["I", "am", "a", "student"]    # english
target_sentence = ["<sos>", "저는", "학생", "입니다"] # Korean output (decoder in)
target_labels = ["저는", "학생", "입니다", "<eos>"] # Label (decoder out)
```

2. Encoder

```
X = embed(input_sentence)                # sequence embedding → [x_l, x_am, x_a,
x_student]
```

```
Q_enc = linear_Q(X)                     # Q from encoder input
K_enc = linear_K(X)                     # K from encoder input
V_enc = linear_V(X)                     # V from encoder input
```

```
encoder_outputs = self_attention(Q_enc, K_enc, V_enc) # encoder out
```


AI model

Transformer Structure (Self-Attention, Positional Encoding)

#3. decoder

```
Y = ["<sos>"]
logits_sequence = []

for t in range(len(target_labels)):

    Y_embed = embed(Y)

    #Decoder Self-Attention (with Masking)
    Q_dec = linear_Q_dec(Y_embed)
    K_dec = linear_K_dec(Y_embed)
    V_dec = linear_V_dec(Y_embed)
    dec_out = masked_self_attention(Q_dec, K_dec, V_dec)

    # Cross-Attention: Decoder → Encoder input reference
    Q_cross = linear_Q_cross(dec_out)
    K_cross = linear_K_cross(encoder_outputs)
    V_cross = linear_V_cross(encoder_outputs)
    cross_out = attention(Q_cross, K_cross, V_cross)
```

Initial input to the decoder (starting from <sos>)

Save the output logit at each point in time

Decoder input embedding generated so far

Q from decoder input

K from decoder input

V from decoder input

Decoder internal context calculation (future ma
(including skins)

Q from decoder context

K from encoder output

V from encoder output

Translation vector based on input sentence information

AI model

Transformer Structure (Self-Attention, Positional Encoding)

```
for t in range(len(target_labels)):
```

```
    ...
```

```
    # Generate output logit
```

```
    logits = linear_output(cross_out)
    logits_sequence.append(logits[-1])
```

```
    # Predicted logit of vocab size
```

```
    # Only store the last position logit (e.g. the most recent token)
```

```
side)
```

```
    # Insert the correct answer for the next decoder input (teacher forcing)
```

```
    next_token = target_sentence[t + 1]
```

```
    # Example: "I" then "student"
```

```
    Y.append(next_token)
```

```
    # Add to decoder input for next loop
```

#4. Loss calculation

```
loss = 0
```

```
for logit, label_token in zip(logits_sequence, target_labels):
```

```
    label_id = vocab_id(label_token)
```

```
    # Convert correct answer token to ID
```

```
    loss += cross_entropy_loss(logit, label_id)
```

```
    # CrossEntropy Loss between the predicted distribution and the correct ID
```

```
loss = loss / len(target_labels)
```

```
    #Average Loss value (divided by sequence length)
```

```
#5. Perform backpropagation and parameter update
```

AI model Transformer Structure (Self-Attention, Positional Encoding)

instruction string · lengths 	input string · lengths 	output string · lengths 
Find the most frequently used word in the given paragraph.	The cat is black, the dog is brown and the cow is white.	The: 3 times.
Generate a regex pattern to match the following string	String: "Hello123"	\w{5}[0-9]{3}
Write a code snippet to update the value of an element in an array.	Array: {3, 2, 5, 7, 8, 1} Index: 3 New value: 6	array[3] = 6;
Predict a label for the given sentence.	The phone is red .	Label: Color.
Print out the 5th element of an array.	arr = [55, 32, 10, 11, 8]	print(arr[4])
Classify the following text into two categories: 'Sports' and...	The critically acclaimed film 'Avatar' is about a paraplegic Marine who embark...	Entertainment
Create a formula in Excel to calculate the total cost given the price, tax,...	Price: 10 Tax: 0.08 Shipping: 6	=10+0.08*10+6
Classify the following given expression. Your response should be...	x = 0.5	float literal
Classify the following code as either a python function or code section.	x = "hello" z = x + " world"	Code section.
Generate a regex expression to match strings of length 8 with given...	Allowed letters: A-F and digits: 0-9	^[A-F0-9]{8}\$

AI model Transformer Structure (Self-Attention, Positional Encoding)

Datasets: junelee/ **sharegpt_deep1_ko** like 53

Dataset card Viewer Files and versions Community

Dataset Preview

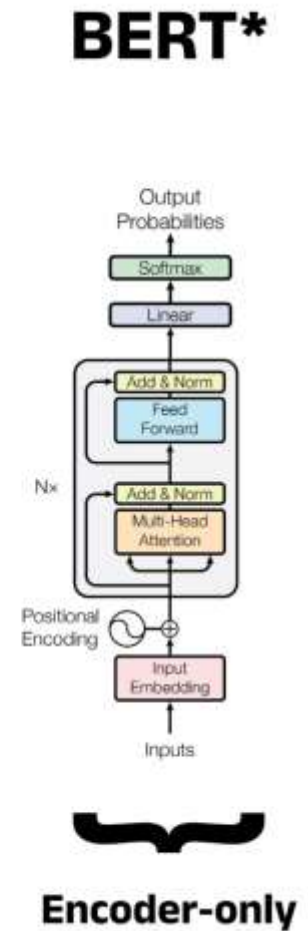
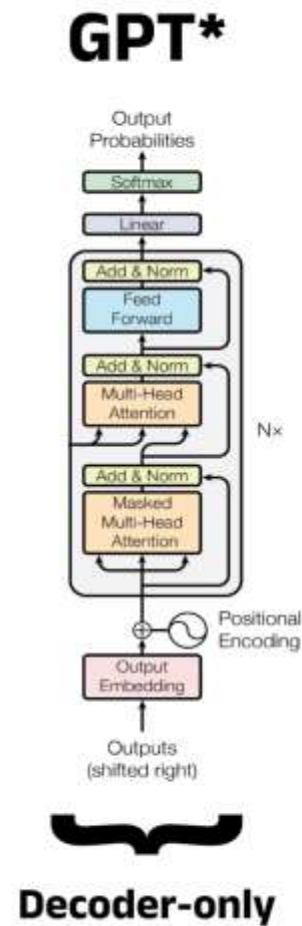
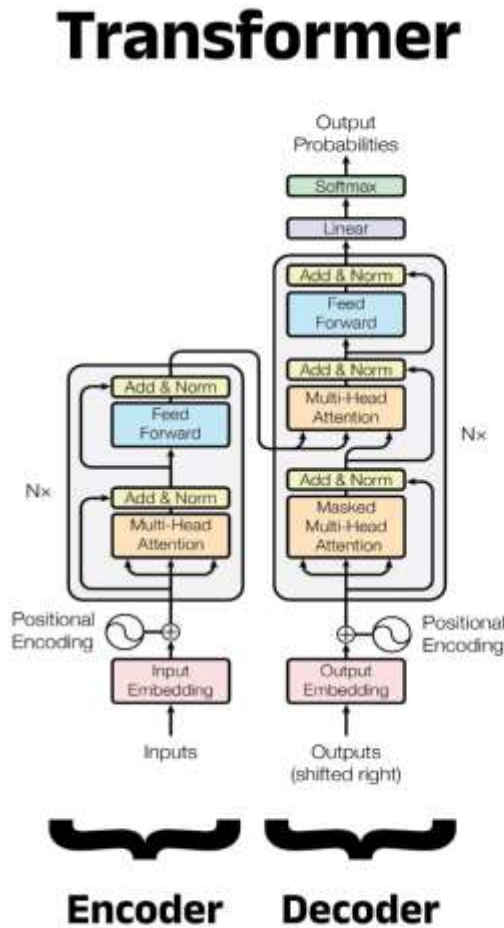
API View in Dataset Viewer

Split (1)
train

► The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

input string	instruction string	output string
	토니 로빈스가 설명하는 인간의 6가지 욕구를 요약해 보세요...	토니 로빈스는 인간의 행동과 동기를 이끄는 6가지 핵심 욕구를 설명합니다. 이 여섯 가지 욕구는 다음과 같습니다: 1. 확실성: 안전, 안정성, 예...
	고객 세그먼트가 잘 세분화되었는지 어떻게 알 수 있을까요? 세 가지로 요약할 수 있습니다.	1. 동질성: 세그먼트는 유사한 특성과 행동을 공유하는 고객으로 구성되어야 합니다. 2. 차별성: 세그먼트는 특성 및 행동 측면에서 다른 세그먼트...
	n Java에서 "이것은 {장소}에 있는 새로운 {객체}입니다"와 같은 문자열을 맵, {객체: "학생", "지점 3, 4"}로 바꾸고 "이 학생은 지점 3, 4...	을 사용하여 문자열의 자리 표시자를 맵의 값으로 바꿀 수 있습니다. 다음은 이를 수행하는 방법을 보여주는 코드 스니펫 예시입니다. ``java...
	전체 단락을 이와 같은 스타일로 작성하세요: 신들의 은총으로 은유적 언어의 신비롭고 수수께끼 같은 예술이 소환되어 우리 앞에 놓인 지침의 당혹...	보세요! 신성한 개입의 은총으로, 이해할 수 없고 불가사의한 은유적 언어의 예술이 우리 앞에 놓인 지침의 불가해한 전달 방식을 해명하기 위해 호...
	길이를 변경할 수 없습니다.	지식의 구도자 여러분, 잠시만요, 우리 앞에 제시된 지침의 복잡한 전달 방식을 설명하는 데 사용된 은유적 언어의 난해하고 수수께끼 같은 예술에 ...
	계속	은유적 언어의 사용은 단순히 보이는 이 지침에도 신비감과 초월성을 불어넣어 줍니다. 특히 '회상하다'라는 단어는 이 지침에 숨겨진 지식이나 비밀...
	다음과 같은 C++ 함수가 있습니다: void add_player(벡터& 플레이어)	주어진 C++ 함수의 'dummy' 변수는 'cin' 문을 사용하여

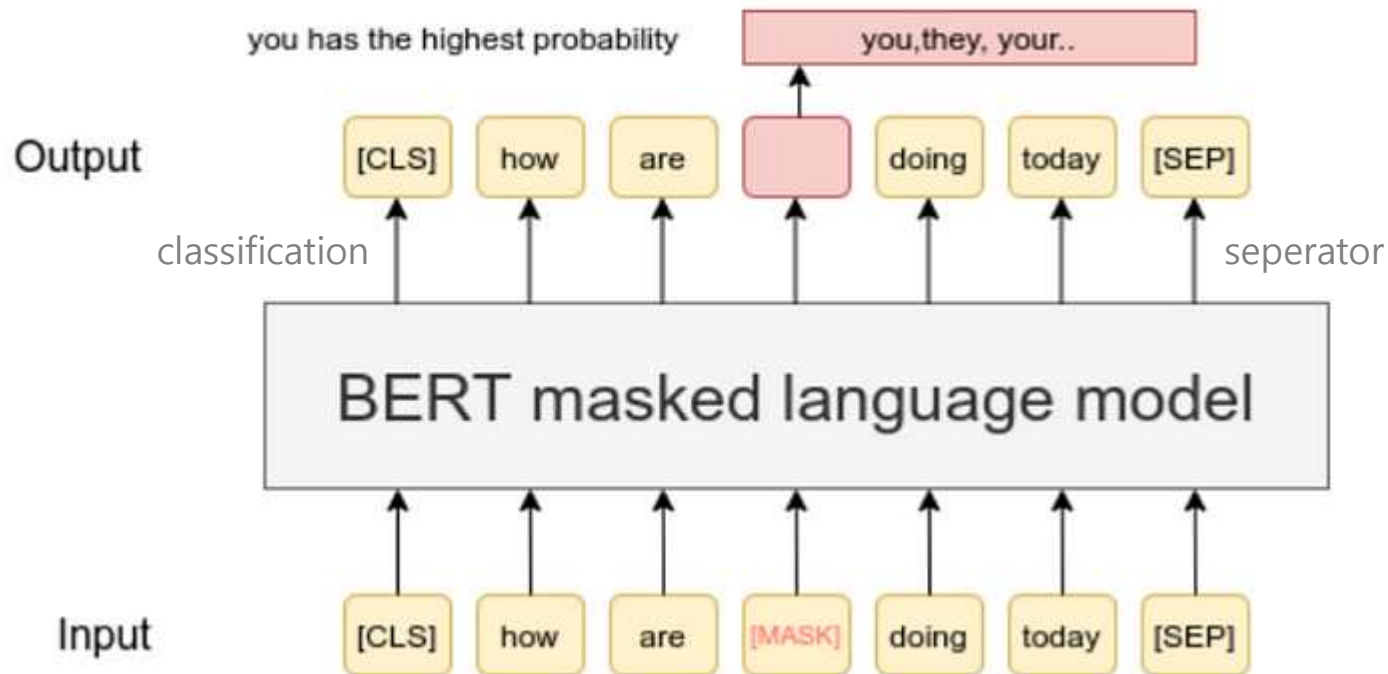
AI model BERT, GPT Mechanism



BERT

BERT (Bidirectional Encoder Representations from Transformers) is a technique for learning contextual relationships between words in a sentence. For this reason, it only uses the encoder of a Transformer. It calculates the importance of each word based on the preceding and following contexts of the sentence through a mechanism called self-attention.

BERT is best suited for a subset of NLP tasks, such as NER and sentiment analysis.



MLM(Masked Language Model)

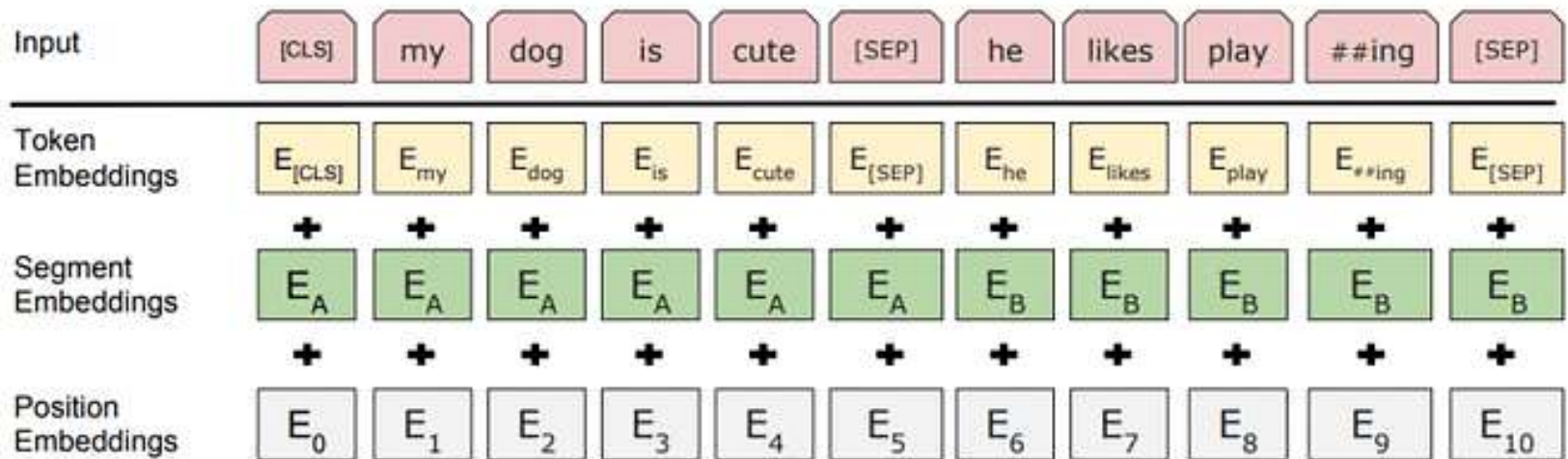
AI model BERT, GPT Mechanism

BERT

Token embedding: [CLS] tokens are added to the input word tokens at the beginning of the first sentence. [SEP] tokens are inserted at the end of each sentence.

Segment embedding: Markers indicating sentence A or sentence B are added to each token. This allows the encoder to distinguish sentences.

Position embedding: Position embeddings are added to each token to indicate its position in the sentence.

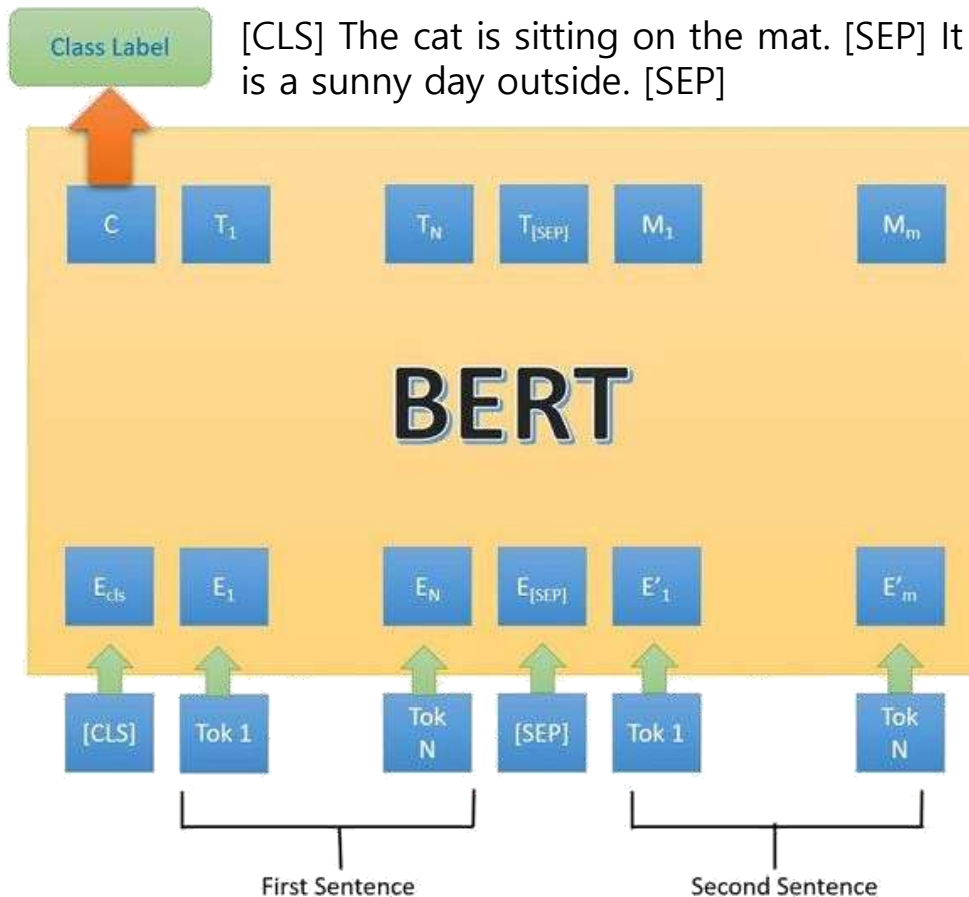


[google-research/bert](https://github.com/google-research/bert): TensorFlow code and pre-trained models for BERT

AI model BERT, GPT Mechanism

BERT

For Next Sentence Prediction (NSP), the model is trained to predict whether two sentences appear consecutively in the original text. A special delimiter token ([SEP]) is inserted between the two sentences. Additionally, a special classification token ([CLS]) is inserted at the beginning of the sentence and trained.



[PAD]

A padding token used to bring the total number of tokens in the input sequence up to 512.

[UNK]

A token not in BERT's vocabulary.

[CLS]

A classification token.

[SEP]

Used to distinguish two segments in a single input sequence. Up to two segments are allowed.

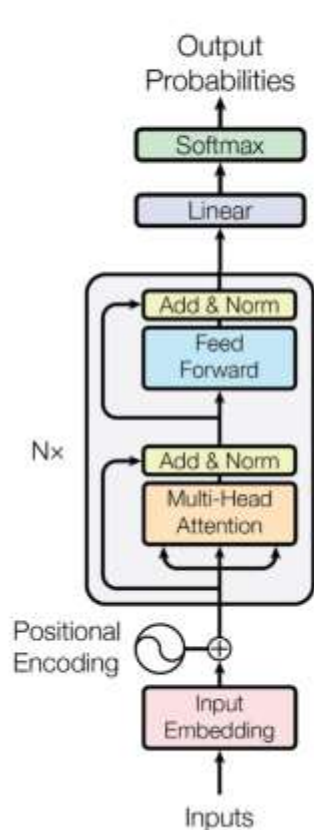
[MASK]

A token used to perform inference on the masked sequence.

AI model

BERT, GPT Mechanism

BERT



	BERT Base	BERT Large
Number of embedding dimensions, d_{model}	768	1024
Number of encoder blocks, N :	12	24
Number of attention heads per encoder block:	12	16
Size of hidden layer in feedforward network:	3072	4096
Total parameters:	110 million	340 million

AI model

BERT, GPT Mechanism

GPT

GPT was developed by OpenAI. Its goal is to predict new words one at a time and generate an output sequence. Each predicted word only utilizes the context provided by the previous word (the left context) because the subsequent word (the right context) has not yet been generated.

BIDIRECTIONAL CONTEXT

When predicting words within a sequence, all of the surrounding words can be used to gain contextual information.

Left context

Right context

← A man was [MASK] on a river bank. →

UNIDIRECTIONAL CONTEXT

When predicting future words, only the previous words can be used to gain contextual information.

Left context

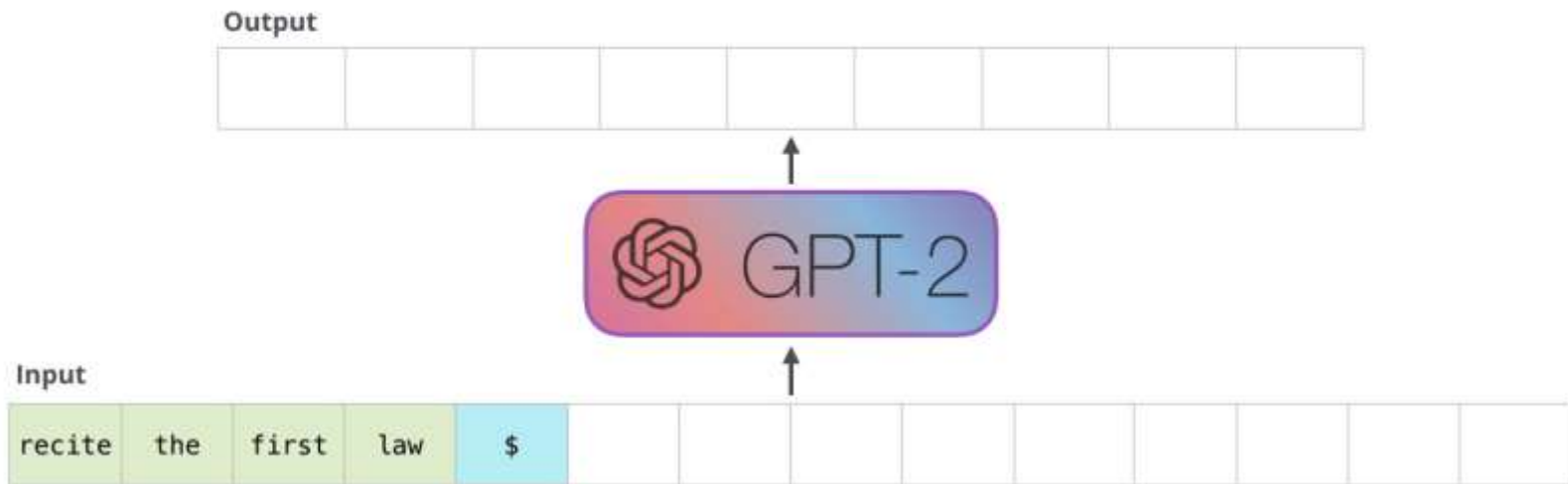
← Write a poem about a man fishing on a river bank. Upon a [NEXT TOKEN]

AI model

BERT, GPT Mechanism

GPT

GPT operates as an autoregressive model, predicting future values based on past values. After each token is generated, it is added to the input sequence.



AI model BERT, GPT Mechanism

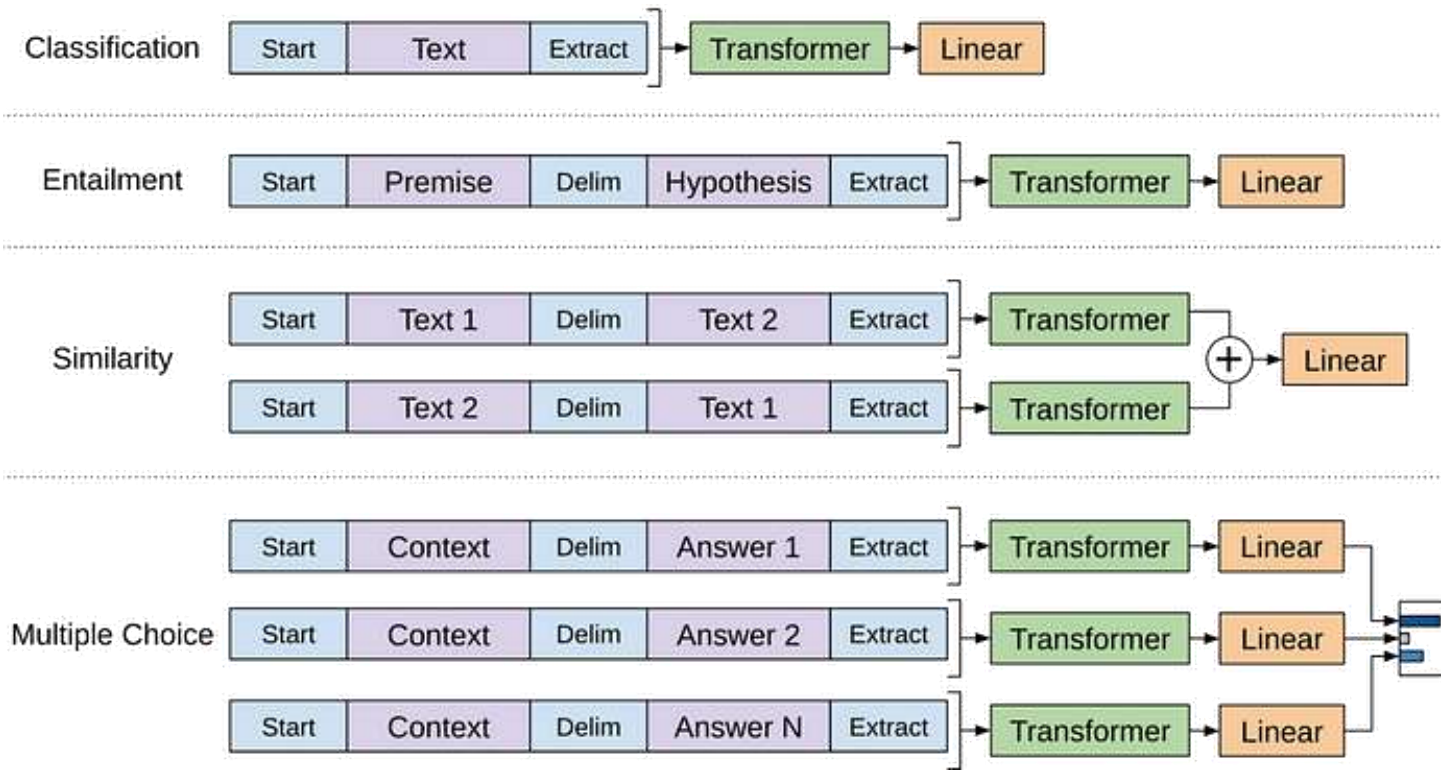
GPT

For tasks like text classification, the model can be fine-tuned directly by adding linear and softmax layers. Other tasks, such as question answering or text embedding, can accept structured inputs, such as aligned sentence pairs or documents, or triplet of questions and answers.

GPT uses a recurrent approach. This allows for minimal architecture changes across tasks.

All transformations include an initialized start token, a delimiter to separate multiple-sentence inputs, and an end token.

For example, it can be trained as a single sequence using premises and hypotheses surrounded by Start and Extract tokens. Multiple transformers can also be used and the outputs of the last layer can be merged for multiple-choice tasks.



[Step by Step into GPT. GPT stands for Generative Pre-Training... | by Yan Xu | Medium](#)

AI model BERT, GPT Mechanism

GPT

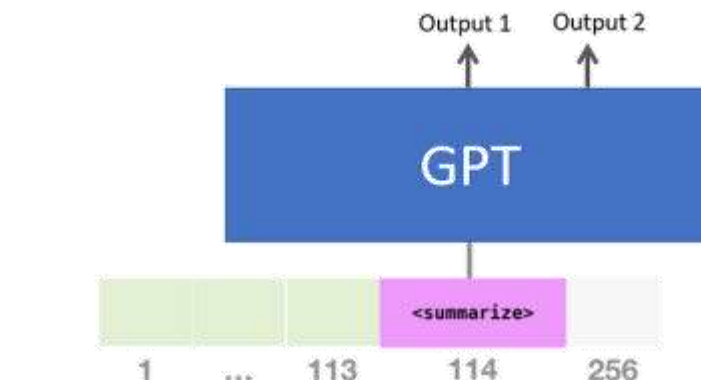
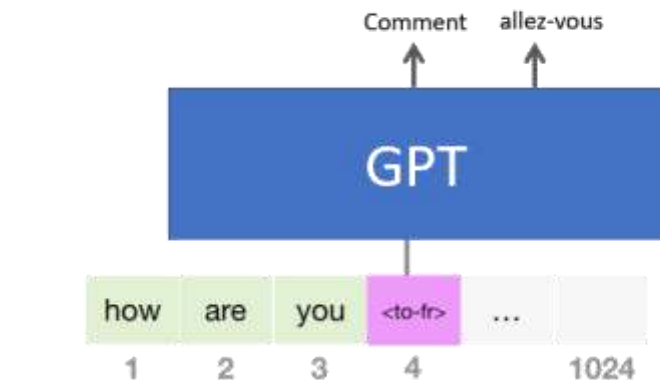
The GPT structure can be used for a variety of use cases without significant changes.

Training Dataset Eng to Fr

I	am	a	student	<to-fr>	je	suis	étudiant
let	them	eat	cake	<to-fr>	Qu'ils	mangent	de
good	morning	<to-fr>	Bonjour				

Training Dataset Summary

Article #1 tokens	<summarize>	Article #1 Summary
Article #2 tokens	<summarize>	Article #2 Summary padding
Article #3 tokens	<summarize>	Article #3 Summary



[Step by Step into GPT. GPT stands for Generative Pre-Training... | by Yan Xu | Medium](#)

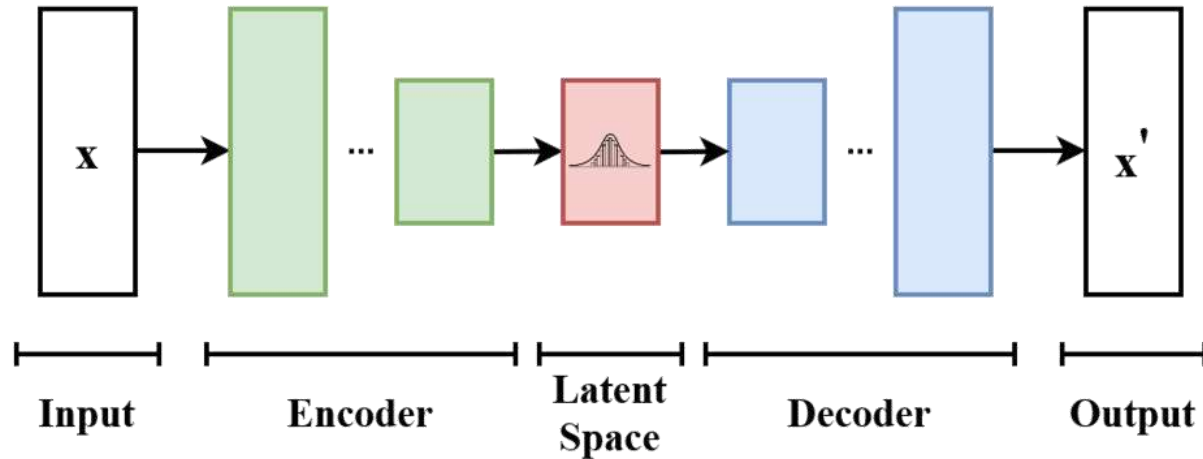
MuseNet

AI model Architecture Latent Space

Variational autoencoder (VAE)

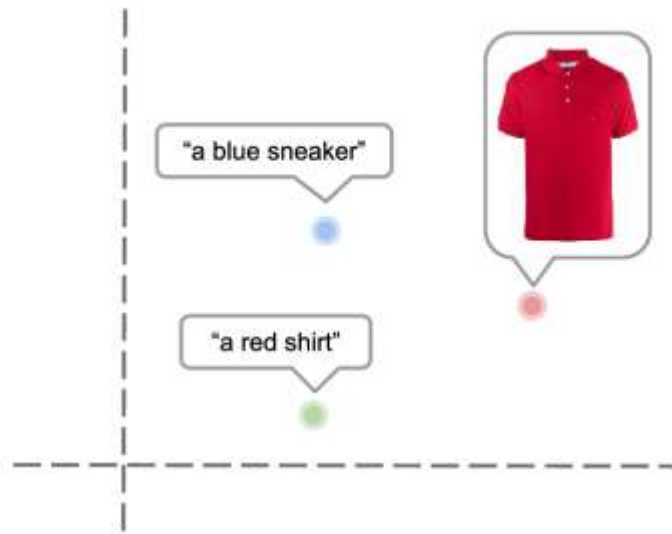


Villa savoye,
1929, Le Corusier

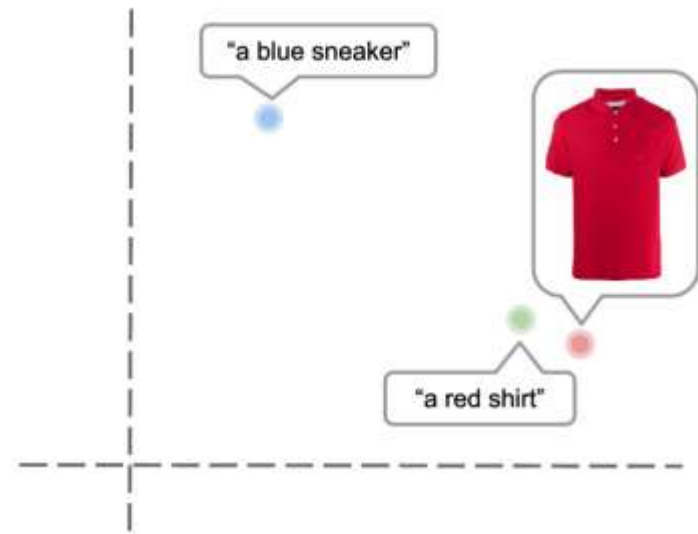


Variational autoencoder

Latent Space Before Training



Latent Space After Training



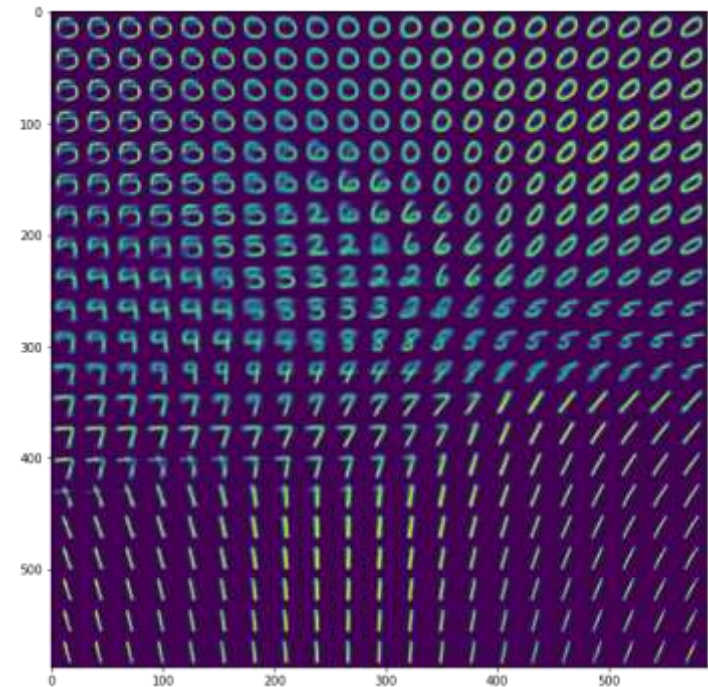
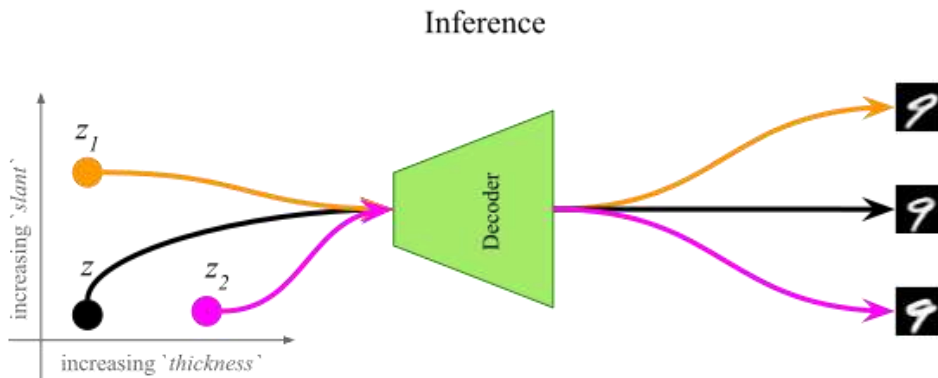
AI model Architecture VAE

VAE. Variational autoencoder

A VAE is a generative model that learns to encode data into a compressed latent space and then decode it back into data. This makes it particularly useful for generating variations of the input data. Instead of simply outputting a single latent vector for input data X , the encoder outputs two vectors representing the probability distribution of the latent space (mean μ and variance σ^2). The decoder directly samples the latent vector z from the mean and variance output by the encoder.

$$\text{Latent space vector } \mathbf{z} = \mu + \sigma \cdot \epsilon$$

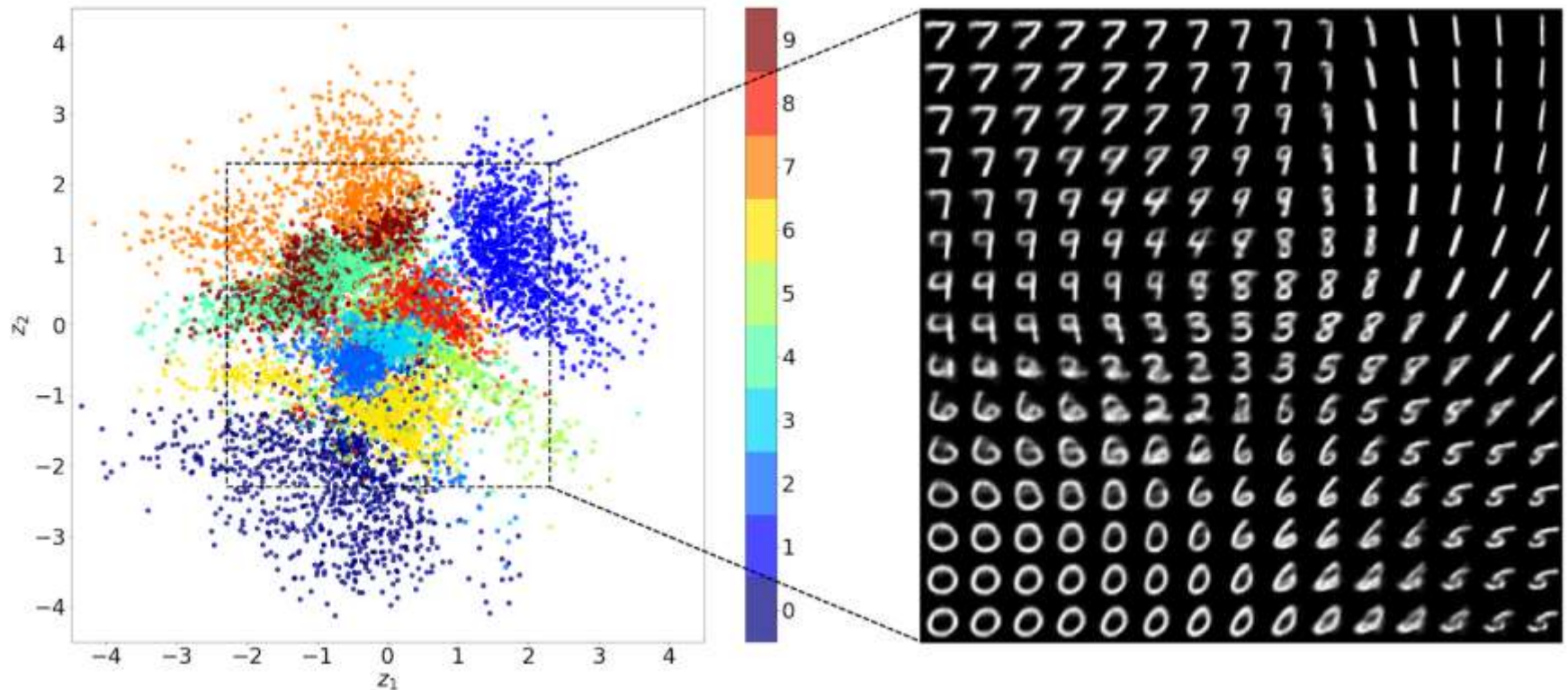
ϵ is noise sampled from a standard normal distribution ($N(0,1)$) so that this equation is differentiable. Note that $z = N(\mu, \sigma^2)$ is not differentiable.



AI model Architecture Latent Space

The smooth manifold

effect ($z = \mu + \sigma \cdot \epsilon$) injects noise into this latent space, allowing the encoder to learn a "distribution" beyond simply mapping data to specific points.



Data mapped (encoded) to latent space (Alexej Klushyn, December 2019, Learning Hierarchical Priors in VAEs)

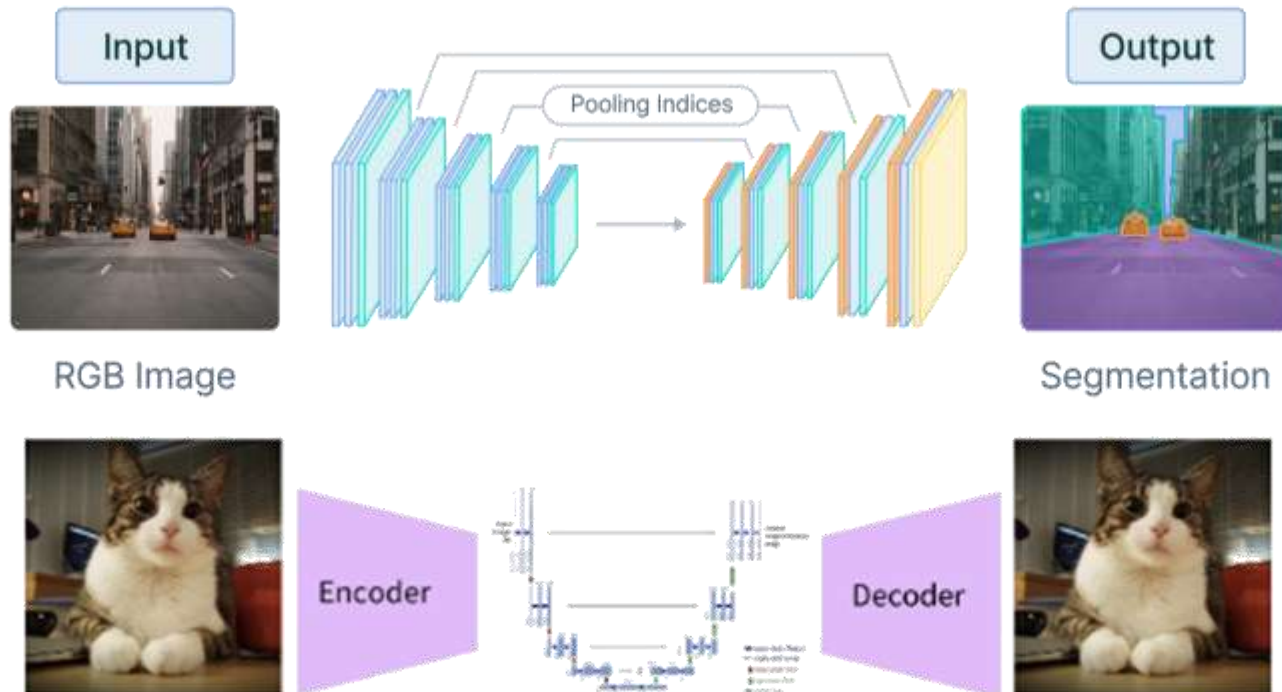
AI model Architecture U-Net

U-Net

Consists of an encoder that compresses the image and a decoder that expands it.

It preserves image details through skip connections that directly pass information from the encoder to the decoder, producing pixel-accurate images.

Convolutional encoder-decoder



AI model

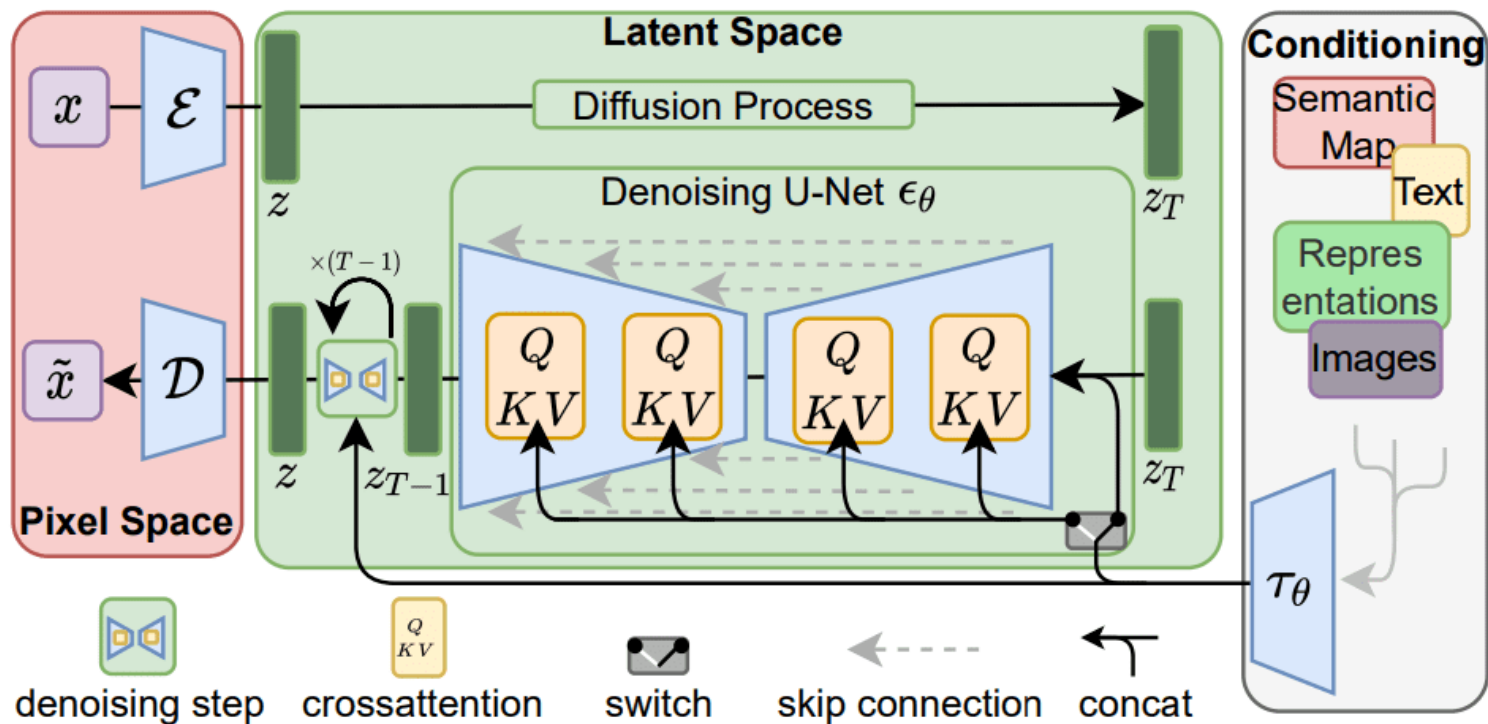
Why did Transformer become the standard?

Latent Diffusion Model

A latent diffusion model generates images tailored to text conditions. The input image is compressed and converted into a latent space, and then noise is gradually removed from the noisy representation z of the latent space.

External conditional information, such as text, is reflected through cross-attention within the U-Net to control the generation process.

Finally, the denoised latent representation is reconstructed into an image (decoder), generating an image that satisfies the given conditions.



AI model

Why did Transformer become the standard?

CLIP (Contrastive Language–Image Pre-training)

CLIP uses a contrastive learning approach to pre-train using a large-scale dataset of image-text pairs. The model's goal is to map image embeddings and text embeddings describing those images into a common latent space, ensuring that similar image-text pairs are close together and dissimilar pairs are far apart.



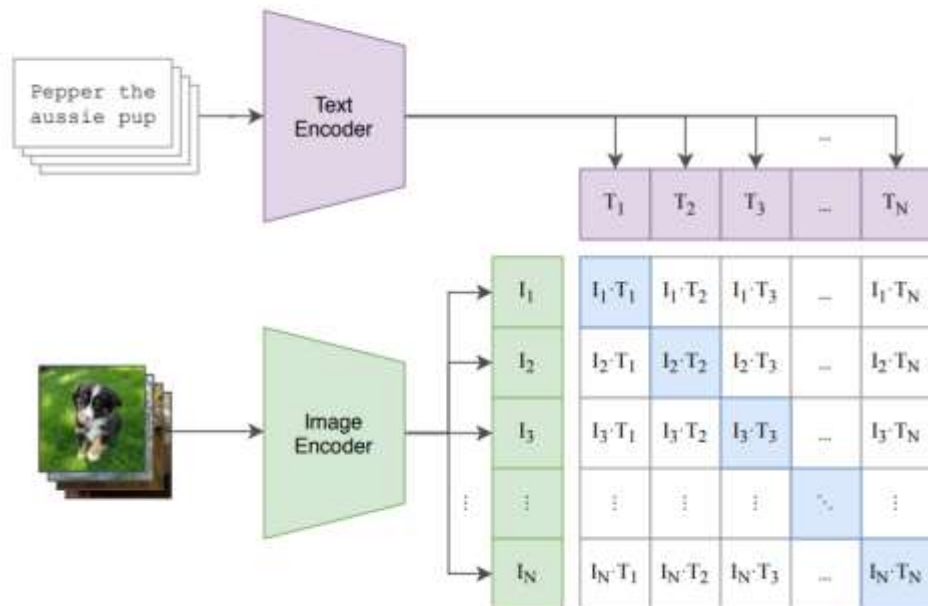
current label: King Charles Spaniel



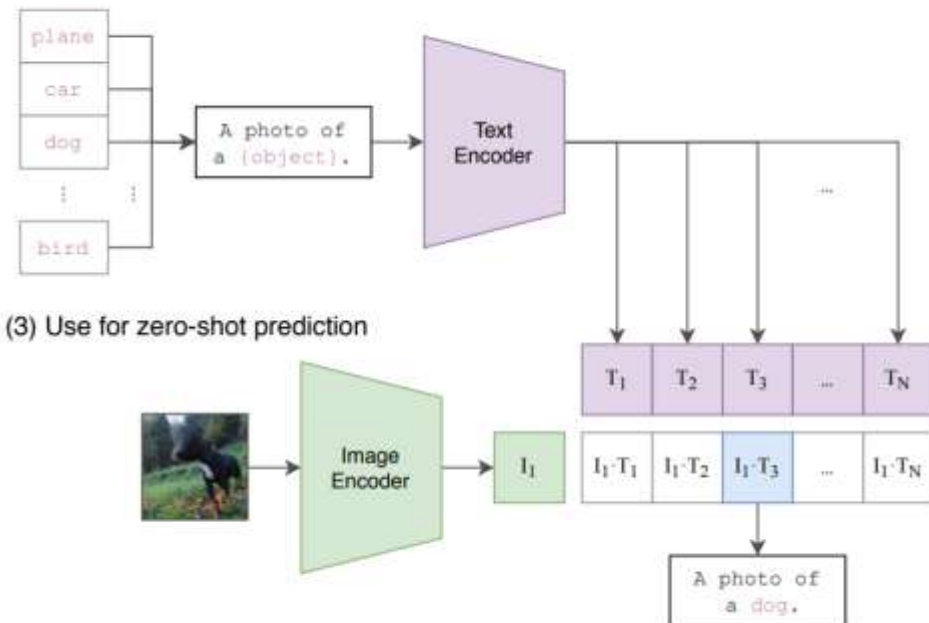
["Two young guys with shaggy hair look at their hands while hanging out in the yard.", "Two young, White males are outside near many bushes.", "Two men in green shirts are standing in a yard.", "A man in a blue shirt standing in a garden.", "Two friends enjoy time spent together."]

[CLIP](#)(Contrastive Language-Image Pre-Training. Open AI. 2021

(1) Contrastive pre-training



(2) Create dataset classifier from label text



AI model Geometry graph 기반 모델

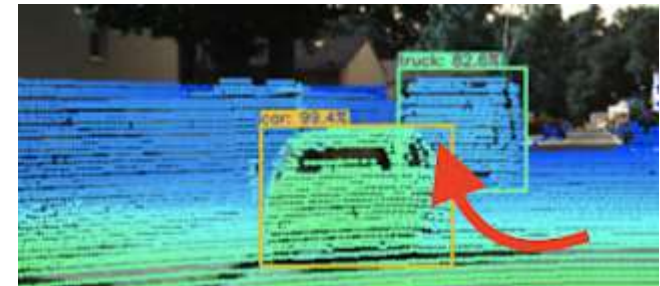
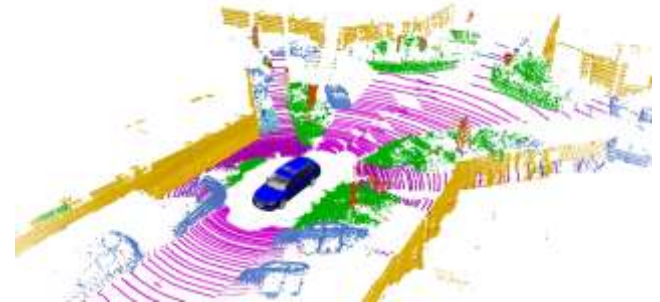
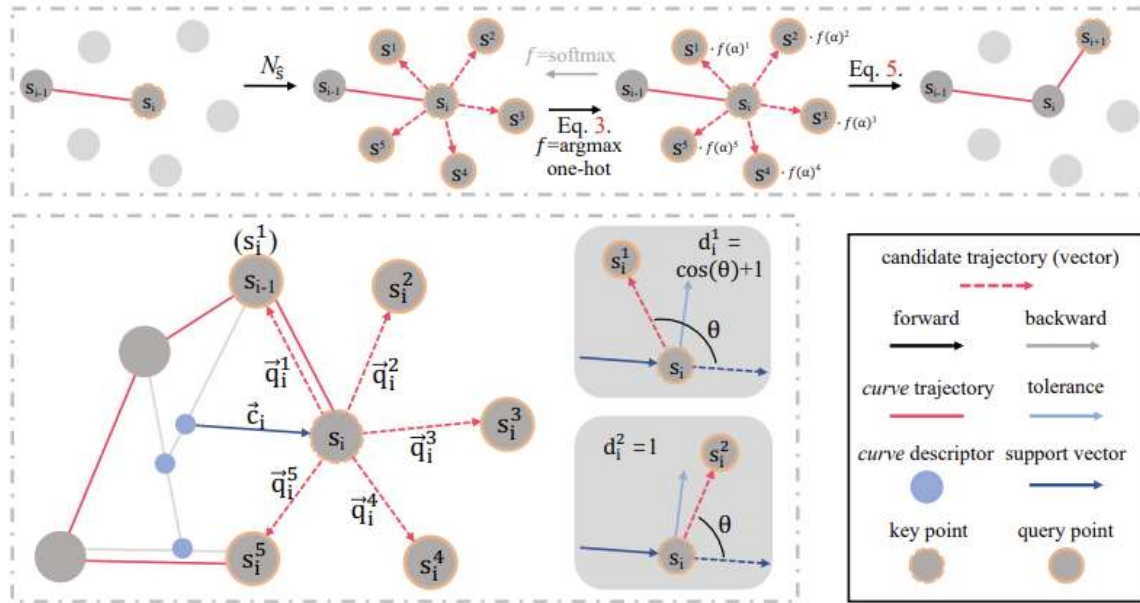
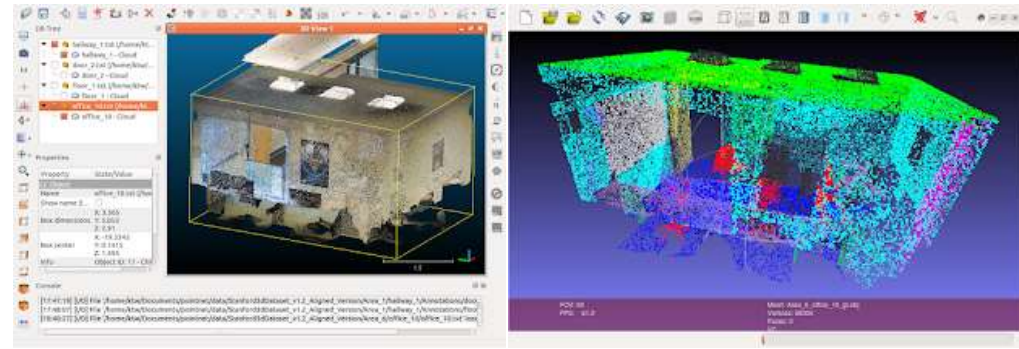
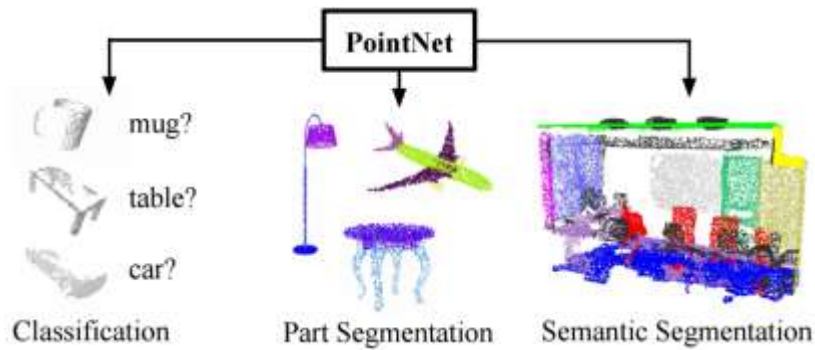


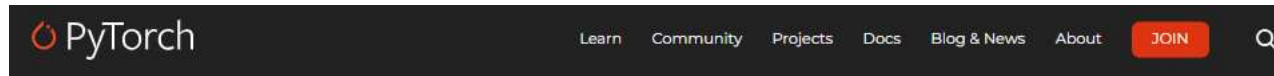
Figure 3. **Top:** An overview of our curve grouping process. **Bottom:** Visualization of the proposed dynamic momentum and the crossover suppression strategies.

Link: 딥러닝 기반 3차원 비전 객체 인식 PointNet 분석

Link: 최신 딥러닝 기반 3차원 스캔 데이터, 포인트 클라우드 학습, 3D 세그멘테이션 연구 논문 조사 분석

AI model Latest model

PyTorch Hub



PyTorch Hub For Researchers

PyTorch Hub For Researchers

Explore and extend models from the latest cutting edge research.

Discover and publish models to a pre-trained model repository designed for research exploration. Check out the models for [Researchers](#), or learn [How It Works](#). [Contribute Models](#).

*This is a beta release – we will be collecting feedback and improving the PyTorch Hub over the coming months.

PyTorch-Transformers

PyTorch implementations of popular NLP Transformers

🔗 147.5k



YOLOv5

Ultralytics YOLOv5 🚀 for object detection, instance segmentation and image classification.

🔗 54.8k

RoBERTa

A Robustly Optimized BERT Pretraining Approach

🔗 31.7k



FastPitch 2

The FastPitch model for generating mel spectrograms from text

🔗 14.4k



HiFi GAN

The HiFi GAN model for generating waveforms from mel spectrograms

🔗 14.4k



```
import torch
import matplotlib.pyplot as plt
from IPython.display import Audio
import warnings
warnings.filterwarnings('ignore')

device = torch.device("cuda") if torch.cuda.is_available() else torch.device("cpu")
print(f'Using {device} for inference')

Download and setup FastPitch generator model.

fastpitch_generator_train_setup = torch.hub.load('NVIDIA/DeepLearningExamples', 'fastpitch_generator_train_setup')

Download and setup vocoder and denoiser models.

hifigan_vocoder_train_setup, denoiser = torch.hub.load('NVIDIA/DeepLearningExamples', 'hifigan_vocoder_train_setup', denoiser)
```


AI model Latest model

TensorFlow Hub



TensorFlow Hub, 학습된 머신러닝 모델의 저장소

TensorFlow Hub는 어디서나 미세 조정 및 배포 가능한 학습된 머신러닝 모델의 저장소입니다. 몇 줄의 코드만으로 BERT 및 Faster R-CNN과 같은 학습된 모델을 재사용할 수 있습니다.

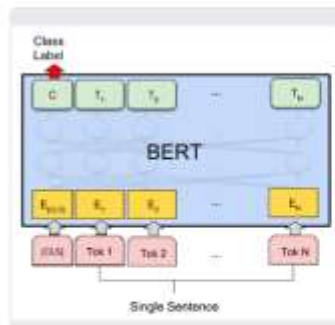
 [튜토리얼 보기](#)

```
!pip install --upgrade tensorflow_hub  
  
import tensorflow_hub as hub
```

(28/2")

모델

TFHub.dev에서 TensorFlow 커뮤니티의 학습된 모델 찾기



BERT

텍스트 분류 및 질문 답변 등 NLP 작업에서 BERT를 확실히 보세요.

[모델 보기](#)



객체 감지

이미지에서 객체를 감지하려면 Faster R-CNN Inception ResNet V2 640x640 모델을 사용하세요.

[모델 보기](#)



스타일 전이

이미지 하나의 스타일을 이미지 스타일 전이 모델을 사용하여 다른 이미지로 전이하세요.

[모델 보기](#)



기기 내 음식 분류기

이 TFLite 모델을 사용하여 휴대기기의 음식 사진을 분류하세요.

[모델 보기](#)

AI model Latest model

Ollama library



Library

Filter models

Popular ▾

deepseek-r1

DeepSeek-R1 is a family of open reasoning models with performance approaching that of leading models, such as O3 and Gemini 2.5 Pro.

[tools](#) [thinking](#) [1.5b](#) [7b](#) [8b](#) [14b](#) [32b](#) [70b](#) [671b](#)

↓ 55M Pulls ↻ 35 Tags ⌚ Updated 3 weeks ago

gemma3n

Gemma 3n models are designed for efficient execution on everyday devices such as laptops, tablets or phones.

[e2b](#) [e4b](#)

↓ 249.9K Pulls ↻ 9 Tags ⌚ Updated 1 month ago

gemma3

The current, most capable model that runs on a single GPU.

[vision](#) [1b](#) [4b](#) [12b](#) [27b](#)

↓ 10M Pulls ↻ 21 Tags ⌚ Updated 3 months ago



Gemma 3

This model requires Ollama 0.6 or later. [Download Ollama](#)

Gemma is a lightweight, family of models from Google built on Gemini technology. The Gemma 3 models are multimodal—processing text and images—and feature a 128K context window with support for over 140 languages. Available in 1B, 4B, 12B, and 27B parameter sizes, they excel in tasks like question answering, summarization, and reasoning, while their compact design allows deployment on resource-limited devices.

Models

Text

1B parameter model (32k context window)

```
ollama run gemma3:1b
```

Multimodal (Vision)

4B parameter model (128k context window)

```
ollama run gemma3:4b
```

12B parameter model (128k context window)

```
ollama run gemma3:12b
```

AI model Latest model

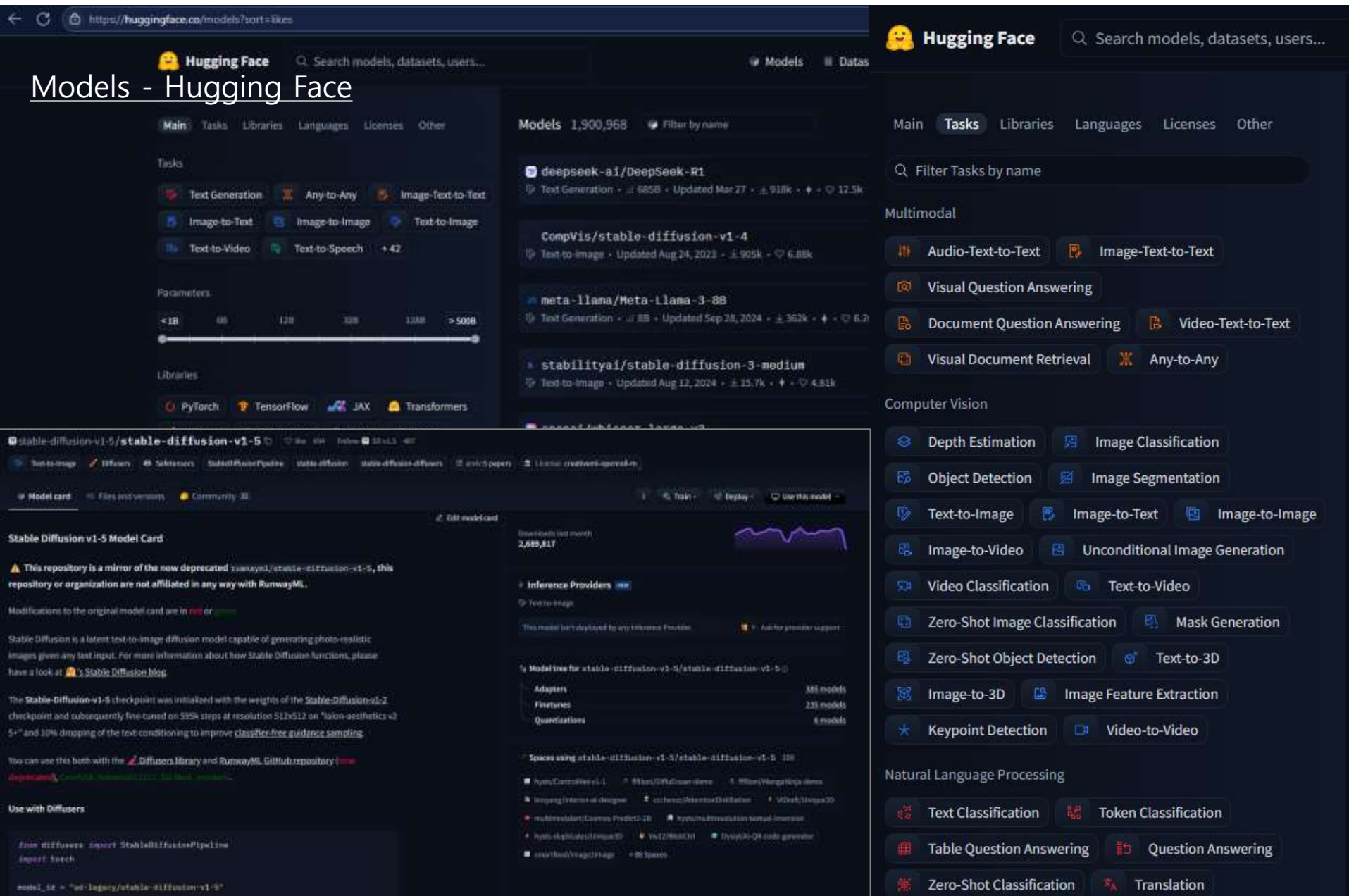
Open-LLM performances

Evaluating and comparing LLMs is hard. Our RLHF team realized this a year ago when they wanted to reproduce and compare results from several published models. It was a nearly impossible task: scores in papers or marketing releases were given without any reproducible

Of course, this ranking is only the beginning of the leaderboard, and we expect it to change soon, as more models get evaluated. You can check the queue status to see which models are currently running!

Rank	New Leaderboard Ranking	
★	Qwen/Qwen2-72B-Instruct	
2	meta-llama/Meta-Llama-3-70B-Instruct	
3	microsoft/Phi-3-medium-4k-instruct	
4	01-ai/Yi-1.5-34B-Chat	零一万物-AI2.0大模型技术和应用的全球公司 (01.AI)
5	CohereForAI/c4ai-command-r-plus	The Secure AI Platform for Enterprise Cohere
6	abacusai/Smaug-72B-v0.1	
7	Qwen/Qwen1.5-110B	
8	Qwen/Qwen1.5-110B-Chat	
9	microsoft/Phi-3-small-128k-instruct	
10	01-ai/Yi-1.5-9B-Chat	

AI model Latest model



AI model Latest model

LLM - Detect AI Generated Text | Kaggle

The screenshot shows the Kaggle website's 'Models' section. The left sidebar contains navigation links: Create, Home, Competitions, Datasets, Models (selected), Benchmarks, Code, Discussions, Learn, More, Your Work, VIEWED, LLM - Detect AI Gene..., Running Kaggle Kern..., Laptop Price, and PDF Q&A App by Lan... The main content area has a search bar and filters. The search results show 184 results (1455 Variations) for 'T5Gemma'. The top result is 'T5Gemma' by Google, with a description: 'T5Gemma is a family of encoder-decoder large language models. Google - T5 - 64 변형 - 노트북 1개'. Other results include 'DeepSeek R1 0528', 'Qwen 3', and 'Gemma'. The right sidebar shows the 'T5Gemma' model card, including a 'Model Card' tab, a 'Code (1)' tab, and a 'Discussion (1)' tab. The 'Model Card' tab is active, showing a warning: 'Access T5Gemma on Kaggle. To access T5Gemma on Kaggle, you need to first request access.' Below this, there is an 'Example Use' section with a code snippet for using the model.

Models

Search Models

Organization Community Hugging Face

All Filters Clear All X 9.00 or higher

184 Results (1455 Variations)

T5Gemma
T5Gemma is a family of encoder-decoder large language models. Google - T5 - 64 변형 - 노트북 1개

DeepSeek R1 0528
DeepSeek-R1-0523 model is unique for its "reasoning" DeepSeek - 2 Variations - 6 Notebooks

Qwen 3
Qwen3 is the latest generation of large language models. QwenLM - 39 Variations - 40 Notebooks

Gemma
Keras implementation of the Gemma model. This Keras implementation of the Gemma model. This Keras implementation of the Gemma model. Keras - 6 Variations - 422 Notebooks

T5Gemma

Model Card Code (1) Discussion (1) Competitions (0)

Access T5Gemma on Kaggle
To access T5Gemma on Kaggle, you need to first request access.

Example Use

Simple example usage:

```
# pip install accelerate
import kagglehub
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

T5GEMMA_PATH = kagglehub.model_download("google/t5gemma/transformers/t5gemma-2b-2b-prefixlm")

tokenizer = AutoTokenizer.from_pretrained(T5GEMMA_PATH)
model = AutoModelForSeq2SeqLM.from_pretrained(T5GEMMA_PATH)

input_text = "Write me a poem about Machine Learning. Answer:"
input_ids = tokenizer(input_text, return_tensors="pt")

outputs = model.generate(**input_ids, max_new_tokens=32)
print(tokenizer.decode(outputs[0]))
```

AI model Latest model

arXiv.org e-Print archive

Mathematics

- Mathematics (**math** new, recent, search)

includes: (see detailed description): Algebraic Geometry; Algebraic Topology; Analysis of PDEs; Category Theory; Classical Analysis and ODEs; Combinatorics; Commutative Algebra; Complex Variables; General Topology; Geometric Topology; Group Theory; History and Overview; Information Theory; K-Theory and Homology; Logic; Mathematical Physics; Metric Geometry; Number Theory; Numeric Representation Theory; Rings and Algebras; Spectral Theory; Statistics Theory; Symplectic Geometry

Computer Science

- [Computing Research Repository \(CoRR new, recent, s](#)

includes: (see detailed description); Artificial Intelligence; Society; Cryptography and Security; Data Structures and Algorithms; Human-Computer Interaction; Information Retrieval; Internet Systems; Other Computer Science; Performance; Programming Languages


Cornell University

[arXiv](#) > [cs.LG](#)

Machine Learning

Authors and titles for recent submissions

- Mon, 28 Jul 2025
- Fri, 25 Jul 2025
- Thu, 24 Jul 2025
- Wed, 23 Jul 2025
- Tue, 22 Jul 2025

See today's new changes

Total of 726 entries: 1-50 51-100 101-150 151-200 ... 701-726

Showing up to 50 entries per page: fewer | more | all

Mon, 28 Jul 2025 (showing first 50 of 103 entries)

[1] [arXiv:2507.19477 \[pdf, html, other\]](#)

Advancing Event Forecasting through Massive Training of Large Language Models: Challenges, Solutions, and Broader Impacts
 Sang-Woo Lee, Sohee Yang, Donghyun Kwak, Noah Y. Siegel
 Subject: Machine Learning (cs.LG); Artificial Intelligence (cs.AI); Computation and Language (cs.CL)

[2] [arXiv:2507.15455 \[pdf, html, other\]](#)

Forest-Guided Clustering – Shedding Light into the Random Forest Black Box
 Usa Barros de Andrade e Sousa, Gregor Miller, Ronan Le Gref, Dominik Thalmeier, Helena Petin, Marie Pilsud
 Subject: Machine Learning (cs.LG)

[3] [arXiv:2507.19437 \[pdf, html, other\]](#)

Observations Meet Actions: Learning Control-Sufficient Representations for Robust Policy Generalization
 Yuliang Gu, Hongpeng Cao, Marco Caccamo, Naira Hovakimyan
 Subject: Machine Learning (cs.LG)

[4] [arXiv:2507.19477 \[pdf, html, other\]](#)

Step-3 is Large yet Affordable: Model-system Co-design for Cost-effective Decoding
 StepFun, Bin Wang, Boyun Wang, Changyu Wan, Guanzhe Huang, Hanpeng Hu, Haonan Jia, Hao Ni, Mingtang Li, Nuo Chen, Sayu Chen, Song Yuan, Wuxun Xie, Xiaohu Song, Xing Lu, Houyi Li, Jingcheng Hu, Ka Man Lo, Ailin Huang, Bingxiao Jiao, Bo Li, Boyu Chen, Changlong Mao, Chang Lou, Chen Hu, Chen Xu, Chenfeng Yu, Chengyuan Yao, Daokuan Lv, Dapei Guin Yan, Han Zhang, Han Zhou, Hanghao Wu, Hangyu Guo, Hanqi Chen, Hanshan Zhang, Hao Wu, Haocheng Zhang, Haolong Yan, Haoran Lv, Haoran Wei, Hebin Zhou, Heng Wan Xie, Jian Zhou, Jianjiao Sun, Jiaoren Wu, Jiaran Zhang, Jiayu Liu, Jie Cheng, Jie Luo, Jie Yan, Jie Yang, Jieyi Hou, Jinguang Zhang, Jinlan Cao, Jisheng Yin, Junfeng Liu, Junhao Huang, Langyu Chen, Linyu Shi, Liguao Tan, Lin Lin, Lin Zhang, Lina Chen, Liwen Huang, Liying Shi, Longlong Gu, Mei Chen
 Subject: Machine Learning (cs.LG); Artificial Intelligence (cs.AI)

[5] [arXiv:2507.19411 \[pdf, html, other\]](#)

SILS: Strategic Influence on Liquidity Stability and Whale Detection in Concentrated-Liquidity DEXs
 Ali RajabiNekoo, Laleh Rasouli, Amirhamad Farhadi, Azadeh Zamanifar

- Control

Reddit/Machine Learning



LLM and AI Agent

LLM and RAG

Ollama and Open Source Models

AI Agent

Generative AI and Vibe Coding

[LLM_agent_vibe](#)

Key Question

What impact did Transformers have on LLM implementation?

What impact did Transformers have on generative AI?

What are the limitations of LLM?

Why did RAG emerge?

What is the difference between AI agents and RAGs?

What is a function call?

What are multi-AI agents?

How do Vibe coding work and what are its limitations?

What are the attitudes of researchers and developers toward the emergence of various AI agent frameworks?

LLM and AI Agent LLM



Introduction of Transformer Models
Transformer Models are introduced through papers like Google's Transformer: A Novel Neural Network Architecture for Language Understanding and Attention Is All You Need, Vaswani et al., 2017.

Tensor Flow Framework
November: Google releases its TensorFlow framework, which allows developers to build and train their own AI models.



2017

2018

Open AI Generative Pre-Trained Transformer
June: OpenAI publishes paper on Generative Pre-Trained Transformer (GPT) model for natural language processing with 110 million parameters, setting the foundation for future generative AI models



OpenAI Chat GPT
November 30: OpenAI debuts generative AI chatbot ChatGPT to explosive interest.

2023



Meta releases Llama 2
July 18th: Meta releases Llama 2 with open-source commercial license and 40% better performance than predecessor

$$y=f(w \cdot x+b)$$

Model	Parameters	Context Length
GPT-1	117M	512 tokens
GPT-4	175B - 1T (approx)	Up to 32,000 tokens
LLaMA 2	7B, 13B, 70B	4,096 tokens

$$\text{Memory (GB)} = \frac{\text{Parameters} \times \text{Bytes per Parameter}}{1024^3}$$

Substitute values:

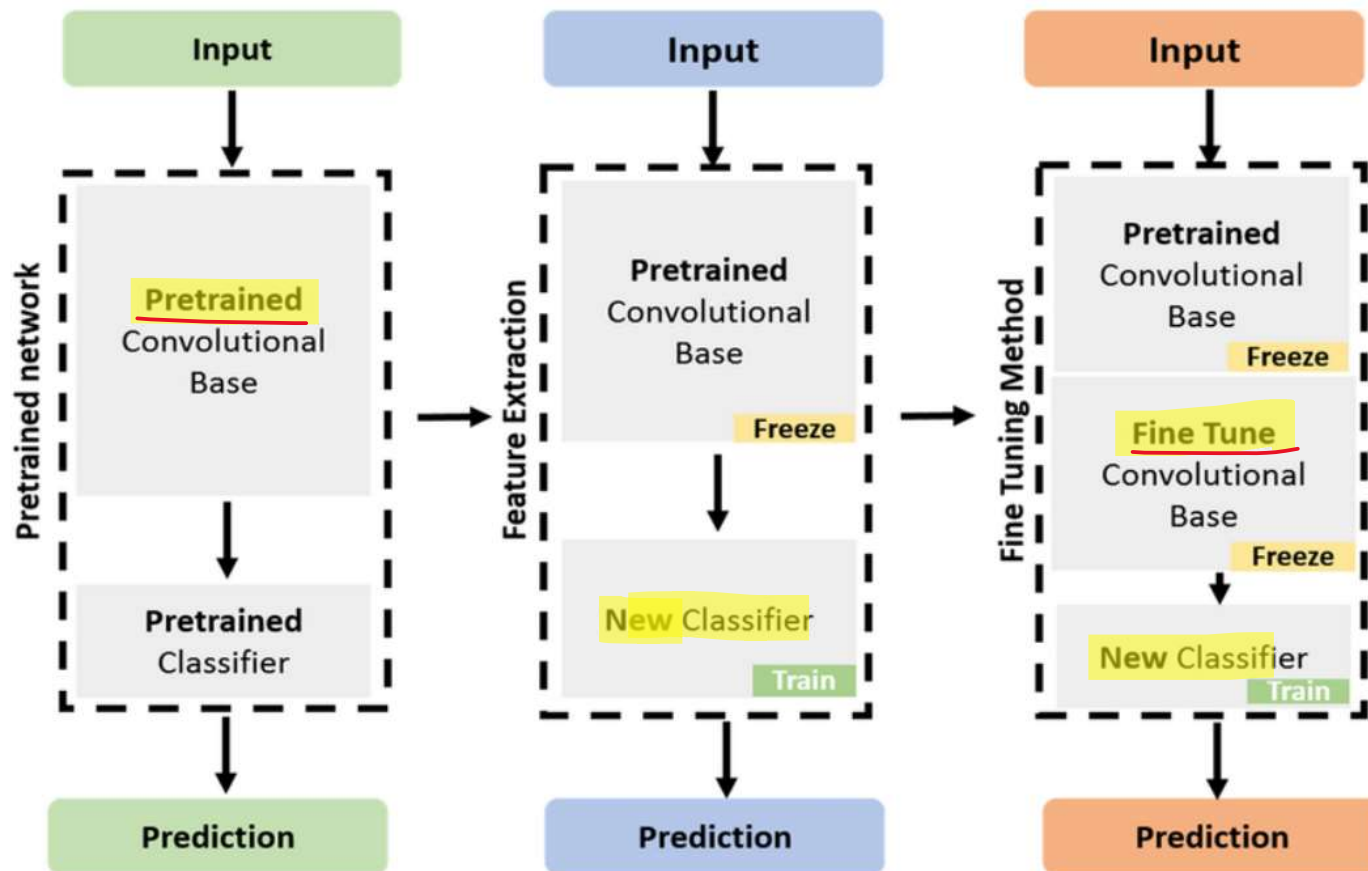
$$\text{Memory (GB)} = \frac{175 \times 10^9 \times 2}{1024^3}$$

$$\text{Memory (GB)} = \frac{350 \times 10^9}{1,073,741,824} \approx 327 \text{ GB}$$

A Timeline of Large Language Model Innovation

LLM and AI Agent LLM fine turning

Large language models (LLMs) require massive computational resources, and fine-tuning all parameters incurs significant memory usage and computational costs. Low-Rank Adaptation (LoRA) and Parameter-Efficient Fine-Tuning (PEFT) methods, such as Parameter-Efficient Fine-Tuning (PEFT), are being used to efficiently perform domain-specific fine-tuning.



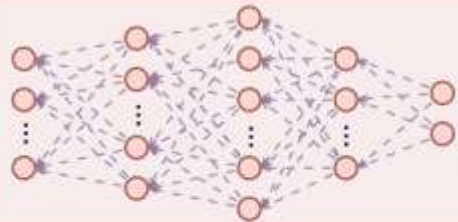
LLM and AI Agent LLM fine turning

Full Fine Tuning, LoRA and RAG



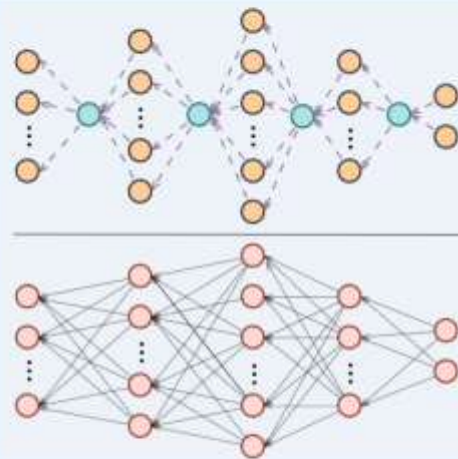
blog.DailyDoseofDS.com

Full Fine Tuning



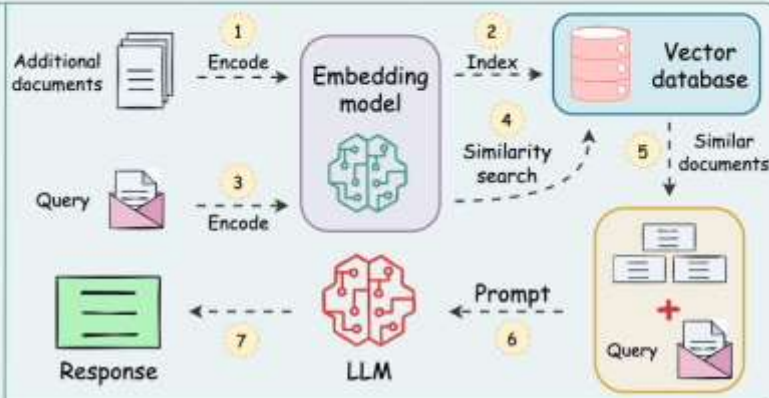
--- Gradient flow
○ Full pre-trained network

LoRA Fine Tuning



--- Gradient flow
○ Additional LoRA layers
○ Full pre-trained network
← No gradient flow

RAG



[finetune-bert-cot-train.py](#)

$$z = f(W \cdot x + b)$$

$$W' = W + \Delta W$$

$$\Delta W = A \cdot B$$

[finetune-gemma.py](#)

Retrieval-Augmented Generation

Full-model Fine-tuning vs.
LoRA vs. RAG - by Avi
Chawla

LLM and AI Agent

LLM fine turning

No fine-tuning

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.

The first thing you need to do is to keep your face clean. Wash your face twice a day with a mild soap. Do not use any oil base make up. Use only water base make up. Do not use any oil base cream. Use only water base cream. Do not use any oil base perfume. Use only water base perfume. Do not use any oil base toothpaste. Use only water base toothpaste. Do not use any oil base shaving cream. Use only water base shaving cream. Do not use any oil base hair oil. Use only water base hair oil. Do not use any oil base hair cream

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

What are the profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length.

The first thing you need to do is to keep your face clean. Wash your face twice a day with a mild soap. Do not use m. Do not use any oil base perfume. Use only water base perfume. Do not use any oil base toothpaste. Use only water oil base hair oil. Use only water base hair oil. Do not use any oil base hair cream

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

What are the profile properties of IfcStructuralCurveMemberVarying described by? So...

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The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying

LLM and AI Agent

LLM fine turning

After fine-tuning

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Hello. Acne is a universal issue in the age group of 15 to 25 years due to hormonal changes in adolescents. However, there are multiple effective treatment options depending on the grade and severity of the problem. You can visit a dermatologist for a clinical evaluation and further treatment. If you are looking for home remedies, you can apply tea tree oil. It has antibacterial properties that help in reducing acne. You can apply 3 to 4 teaspoons of tea tree oil in 8 oz of water twice a day for three weeks. Additionally, you can apply honey with lemon juice twice a day. Lemon juice

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

A start profile and an end profile (e.g. tapers) and the corresponding IfcStructuralCurveMemberConstants for linear structural elements. For non-linear structural elements, the profile properties are described by IfcStructuralCurveMemberHistory with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Hello. Acne is a universal issue in the age group of 15 to 25 years due to hormonal changes in adolescents. However, there are multiple effective treatment options depending on the grade and severity of the problem. You can visit a dermatologist for a clinical evaluation and further treatment. If you are looking for home remedies, you can apply tea tree oil. It has antibacterial properties that help in reducing acne. You can apply 3 to 4 teaspoons of tea tree oil in 8 oz of water twice a day for three weeks. Additionally, you can apply honey with lemon juice twice a day. Lemon juice

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

A start profile and an end profile (e.g. tapers) and the corresponding IfcStructuralCurveMemberConstants for linear structural elements. For non-linear structural elements, the profile properties are described by IfcStructuralCurveMemberHistory with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with

LLM and AI Agent LLM fine turning

fine-tuning

Epoch = 10 (over 5 hours)

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Asking to truncate to `max_length` but no `maximum length` is provided and the model has no predefined maximum length.

Hello. Acne is a universal issue in the age group of 15 to 25 years due to hormonal changes in adolescents. However, the severity of the problem. Grade 1 to grade 3 can be treated conservatively, while grade 4 often requires oral antibiotics. Skin care products like Cetaphil and Clinique can be used at home. However, I suggest you consult a dermatologist. I don't know if I can assist you further. Regards, For more information consult a dermatologist online --> <https://bit.ly/3k8w8w8> Family Physician, Dermatology Specialist, Skin Care Specialist, Acne Specialist, Cosmetic Dermatology, Anti-Aging Specialist, and Cosmetic injection specialist. For more information consult a dermatologist online --> <https://bit.ly/3k8w8w8> provided me. Please do not hesitate to ask further clarifications. Regards, For more information consult a dermatologist online --> <https://bit.ly/3k8w8w8>

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

A start profile and an end profile (e.g. tapers) and the corresponding `IfcStructuralCurveMemberEndProfile` and `IfcStructuralCurveMemberStartProfile` Type. In addition, a sweep operation is described by `IfcStructuralCurveMemberSweptAreaSolid`. For `IfcStructuralCurveMemberSweptAreaSolid` and `IfcStructuralCurveMemberVaryingSurface` pressures have to be defined as well as the tolerance for curve. Finally, the local axis system has to be defined too. For more information, see Löwner, M.O.; Gröger, G. In: *Computational Geometry*; 1313-0499. [CrossRef] [Web of Science (WoS)] [SCI] [e-EL] [Export to PDF] [Metadata] [BIMcert Handbuch] [BIMcert

LLM and AI Agent LLM fine turning

PEFT(Parameter-Efficient Fine-Tuning)

An approach that enables efficient fine-tuning by targeting only specific modules or layers for learning without adjusting all the weights of a pre-trained large language model.

LoRA: Complement linear operations by inserting low-rank matrices.

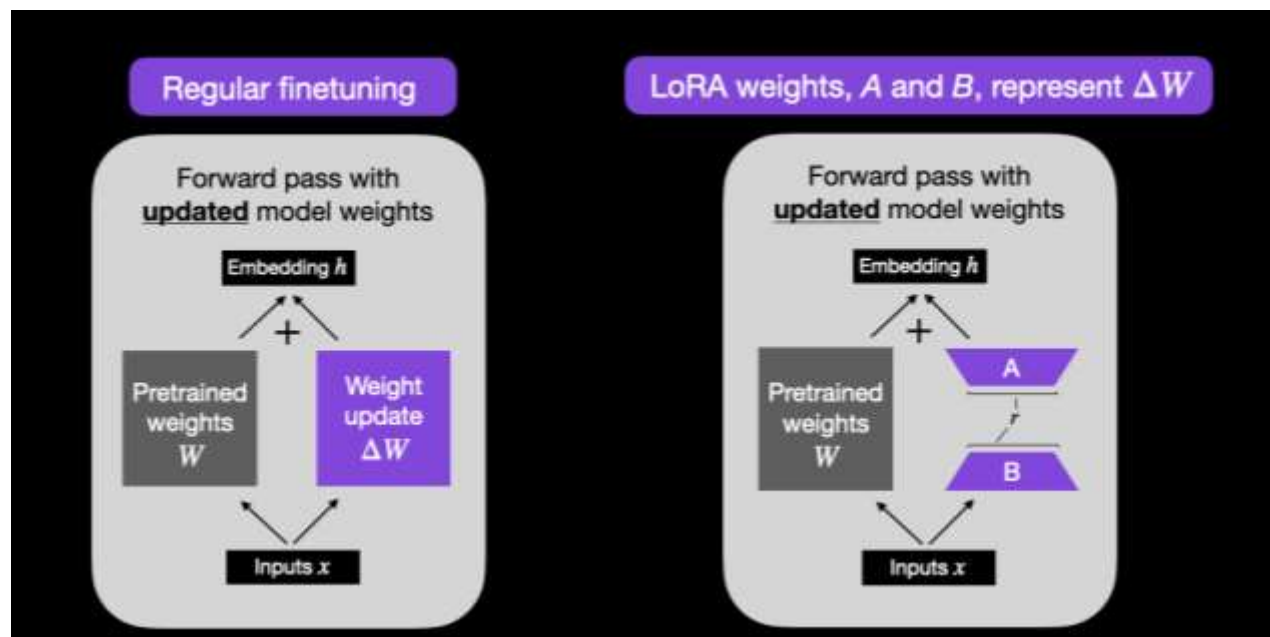
Adapter Tuning: Trains by adding a small network to the Transformer block.

Prefix/Prompt Tuning: Inserts domain-specific tokens into the input sequence.

LoRA(Low-Rank Adaptation)

By adding low-rank matrices to the fixed original weights for the linear layers of self-attention or feedforward blocks, the number of parameters can be reduced and efficient learning can be achieved.

$$y = Wx \quad y = (W + BA)x$$



LLM and AI Agent LLM fine turning

$$y = Wx \quad y = (W + BA)x$$

```
LoraConfig(  
    r=64,  
    lora_alpha=32,  
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"],  
    lora_dropout=0.05,  
    bias="none",  
    task_type="CAUSAL_LM"  
)
```

- `r`: The dimension of the low-rank matrix. Smaller values reduce computational complexity, while larger values increase expressiveness.
- `lora_alpha`: This serves as a scaling factor during training and is related to training stability.
- `target_modules`: The name of the linear layer within the Transformer into which LoRA will be inserted.
- `lora_dropout`: The dropout probability applied during training.
- `bias`: Whether to train the bias term alongside LoRA (none, all, lora_only, etc.).
- `task_type`: The type of fine-tuning task (CAUSAL_LM, SEQ_CLS, etc.). CAUSAL_LM: A language generation model that sequentially predicts the next token (e.g., GPT).
- SEQ_CLS: A classification model that predicts a single class for the entire input (e.g., sentiment analysis, sentence classification).

`q_proj`, `k_proj`, `v_proj`, `o_proj`: Correspond to the query, key, value, and output projections of Self-Attention.

`gate_proj`, `up_proj`, `down_proj`: These are the three core linear layers used in the Feedforward Network (FFN) of the Transformer block.

`gate_proj`: Processes the input before applying the gating mechanism.

`up_proj`: Expands the hidden dimension (e.g., from 4096 to 11008).

`down_proj`: Reduces the dimension again and restores the original output.

LLM and AI Agent LLM fine turning

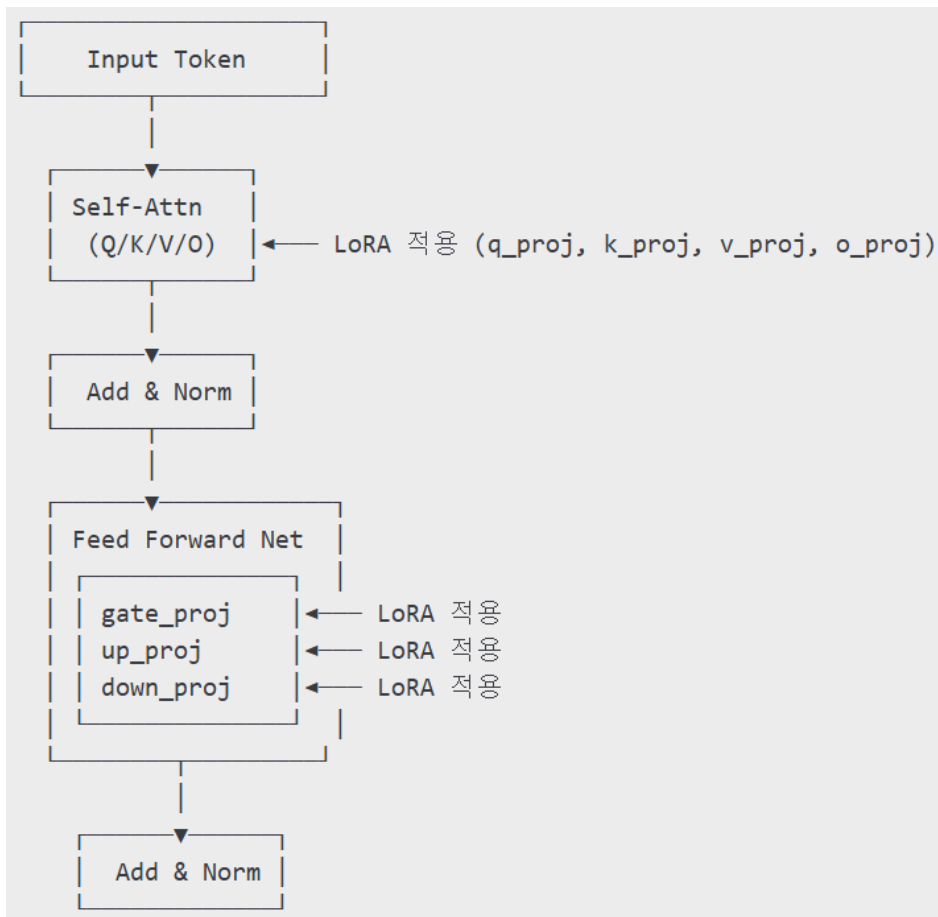
q_proj, k_proj, v_proj, o_proj: Correspond to the query, key, value, and output projections of Self-Attention.

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gate_proj: Processes the input before applying the gating mechanism.

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down_proj: Reduces the dimension again and restores the original output.



LLM and AI Agent 양자화 (Quantization)

Quantization

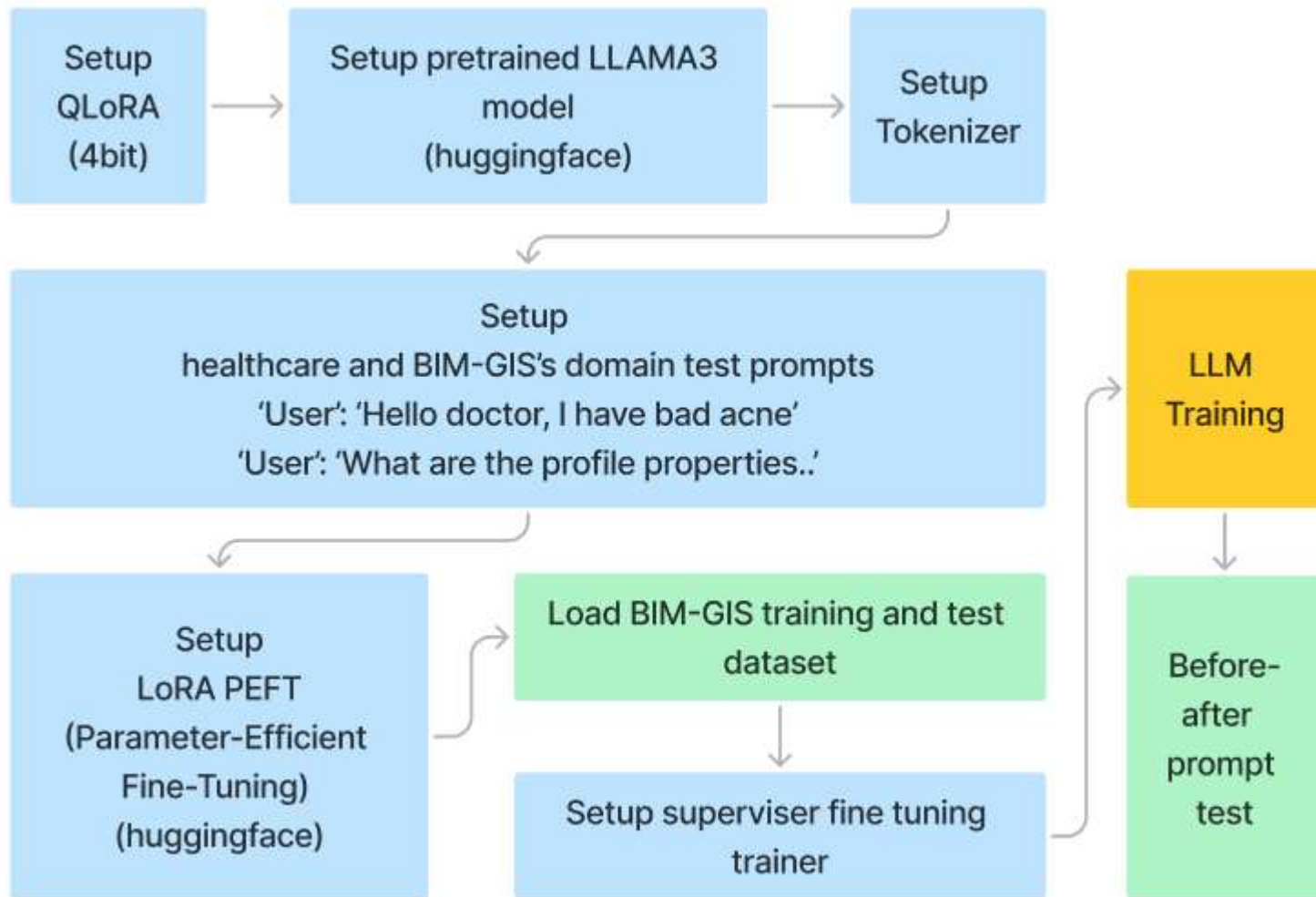
A technique that reduces resource usage and improves computational speed by expressing model parameters as int4/8 instead of float. bitsandbytes is the quantization library adopted by Hugging Face and is particularly effective in 4-bit quantization.

```
BitsAndBytesConfig(  
    load_in_4bit=True,  
    bnb_4bit_use_double_quant=True,  
    bnb_4bit_quant_type="nf4",  
    bnb_4bit_compute_dtype=torch.bfloat16  
)
```

- `load_in_4bit`: Whether to load the model in 4-bit.
- `bnb_4bit_use_double_quant`: Perform double quantization to improve expressiveness.
- `bnb_4bit_quant_type`:
 - `nf4`: NormalFloat4, normalized 4-bit floating point.
 - `fp4`: Basic floating point 4-bit quantization, with lower precision.
- `bnb_4bit_compute_dtype`: Specifies the compute dtype (bfloat16, float16, float32). Typically, bfloat16 is used.

LLM and AI Agent

LLM fine turning



```
{'loss': 2.6513, 'grad_norm': 1.0972543954849243, 'learning_rate': 0.00019908787541713017, 'epoch': 0.11}
{'loss': 2.3663, 'grad_norm': 1.041242241859436, 'learning_rate': 0.00019906562847608455, 'epoch': 0.12}
{'loss': 2.4729, 'grad_norm': 1.1358354091644287, 'learning_rate': 0.00019904338153503895, 'epoch': 0.12}
{'loss': 2.3378, 'grad_norm': 1.1035994291305542, 'learning_rate': 0.00019902113459399333, 'epoch': 0.12}
{'loss': 2.3338, 'grad_norm': 1.300874948501587, 'learning_rate': 0.0001989988876529477, 'epoch': 0.12} 1{'
```

LLM and AI Agent

LLM fine turning

Pretraining

► Dataset:

- Raw text
- Few trillions of tokens

► Task:

- Next word prediction

► Compute:

- O(1-10K) GPUs
- Weeks/months of training

Instruction-Tuning

► Dataset:

- Paired: (Prompt, Response)
- O(10-100K instructions)

► Task:

- Next word prediction (Masked)

► Compute:

- O(1-100) GPUs
- Few hrs/days

Learning from Human Feedback

► Dataset:

- Human Preference Data
- O(10-100K)

► Task:

- RLHF/DPO

► Compute:

- O(1-100) GPUs
- Few hrs/days

LLM and AI Agent

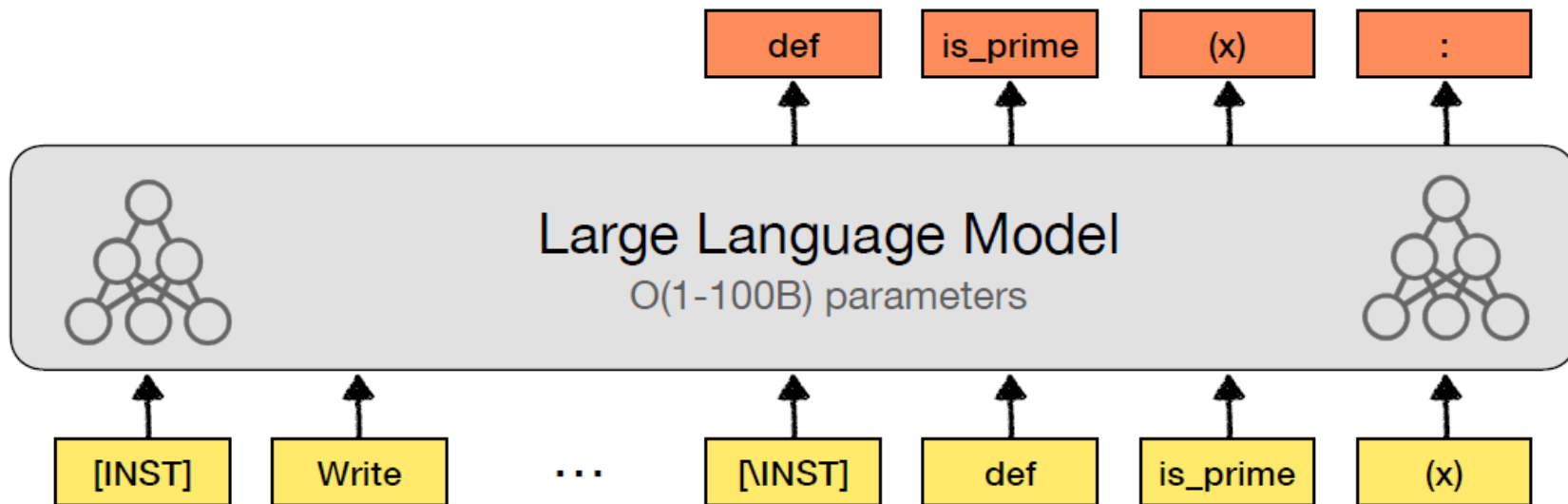
LLM fine turning

Prompt:

[INST] Write a python function to find whether the input number is prime. [INST]

Response:

```
def is_prime(x: int):  
    """  
    takes as input an integer x. Returns True if x is prime and False otherwise  
    """  
    if x <= 1:  
        return False  
    for i in range(2, int(x ** 0.5) + 1):  
        if x % i == 0:  
            return False  
    return True
```



LLM and AI Agent

LLM fine turning

Expand to see the descriptions of 34 models

Model A

Write a python function to find whether the input number is prime.

Here is a Python function to check if a given number is prime:

```
def is_prime(num):  
    if num < 2:  
        return False  
    for i in range(2, int(num**0.5) + 1):  
        if num % i == 0:  
            return False  
    return True  
  
# Test the function  
number = int(input("Enter a number: "))  
if is_prime(number):  
    print(f"{number} is a prime number.")  
else:  
    print(f"{number} is not a prime number.")
```

You can use this function to determine whether a given number is prime or not.

Model B

Write a python function to find whether the input number is prime.

Sure, let's break down the problem into steps and then I'll write the Python function for you.

A number is considered prime if it has exactly two factors: 1 and itself. All numbers greater than 1 have at least three factors, one of which is 1, so we can start our range from 2.

Here is a Python function that checks whether a number is prime:

```
def is_prime(number):  
    # We start from 2 because 1 and the number itself are always factors  
    for i in range(2, int(number**0.5) + 1):  
        # If we find a factor, the number is not prime  
        if number % i == 0:  
            return False  
  
    # If no factors are found, the number is prime  
    return True
```

This function takes an integer as input and returns a boolean value. It starts by checking if the number is less than or equal to 1. If so, it immediately returns `True` for 2 and `False` for all others because these are the only prime numbers below 12. If the number is greater than 1, the function then checks for factors from 2 up to the square root

A is better

B is better

Tie

Both are bad

Enter your prompt and press ENTER

Send

New Round

Regenerate

Share

LLM and AI Agent

AI Agent

hallucination problems



Why wasn't that answer 35 as you predicted?



I apologize for my mistake earlier. The correct set of operations to arrive at 35 is to double the number 8, which is 16, subtract 3 from it to get 13, multiply it by 5 to get 65, subtract 30 to get 35, and then divide it by 2 to get the final answer of 17.

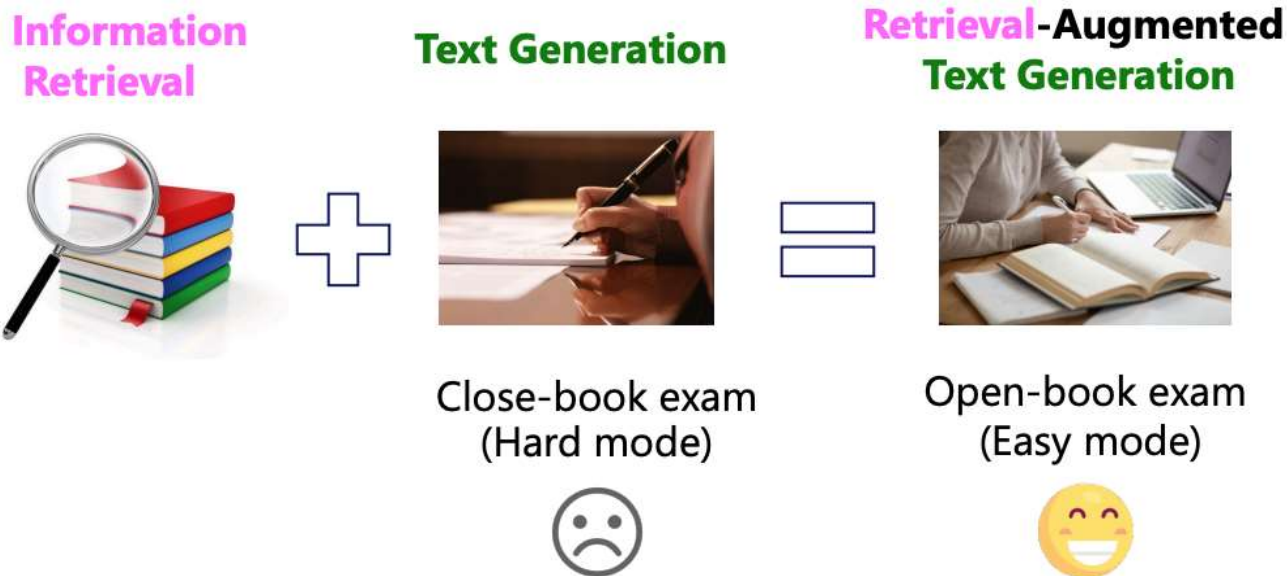
LLM and AI Agent RAG

Retrieval-Augmented Generation (RAG)

To overcome the limitations of large language models (LLMs), this method retrieves external knowledge bases and incorporates them into the generation process.

RAG consists of three stages: retrieving relevant information from external documents or databases, augmenting the input with this information, and ultimately generating text.

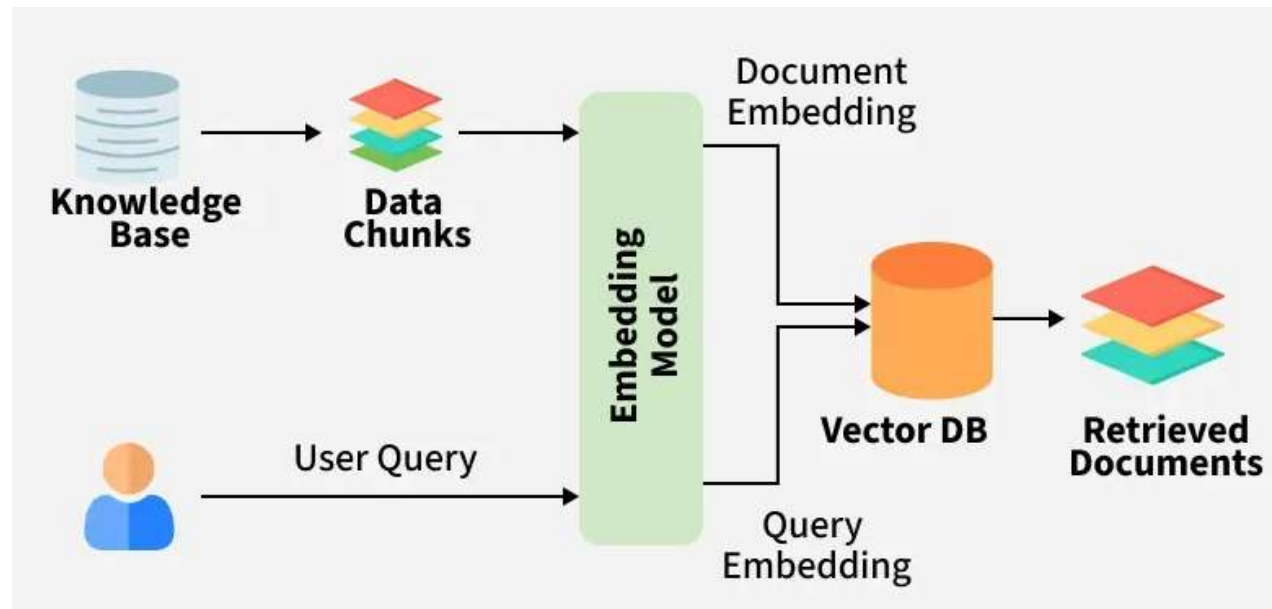
RAG was first proposed by Facebook AI Research (FAIR) in 2020.



LLM and AI Agent RAG

Search

Retrieves documents related to the query by calculating the similarity between the query sentence and the chunked document embedding vector stored in the vector database.



LLM and AI Agent RAG

The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries

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Jade Goldstein
Language Technologies Institute
Carnegie Mellon University
jade@cs.cmu.edu

MMR (Maximal Marginal Relevance) for Re-rank

This approach was introduced to address the redundancy and lack of diversity issues of Cosine Top-K.

Example problem: When a user asks, "What are the limitations of autonomous driving?", using only cosine similarity results in a continuous selection of documents related to "sensor accuracy issues." This results in missing other perspectives (legal, ethical, and technical).

Documents are selected by maintaining relevance to the question while reducing redundancy (diversity) with previously selected documents. The importance of relevance and redundancy is controlled through the λ value ($0 \leq \lambda \leq 1$). A higher λ value leads to a search based solely on relevance.

Input: query_embedding, candidate_docs, λ ($0 \leq \lambda \leq 1$), K

Initialize selected_docs = []

While len(selected_docs) < K:

 best_score = -inf

 best_doc = None

 For doc in candidate_docs:

relevance = cosine_similarity(query_embedding, doc.embedding)

diversity = max([cosine_similarity(doc.embedding, s.embedding) for s in selected_docs], default=0)

score = $\lambda * \text{relevance} - (1 - \lambda) * \text{diversity}$

 If score > best_score:

 best_score = score

 best_doc = doc

 Add best_doc to selected_docs

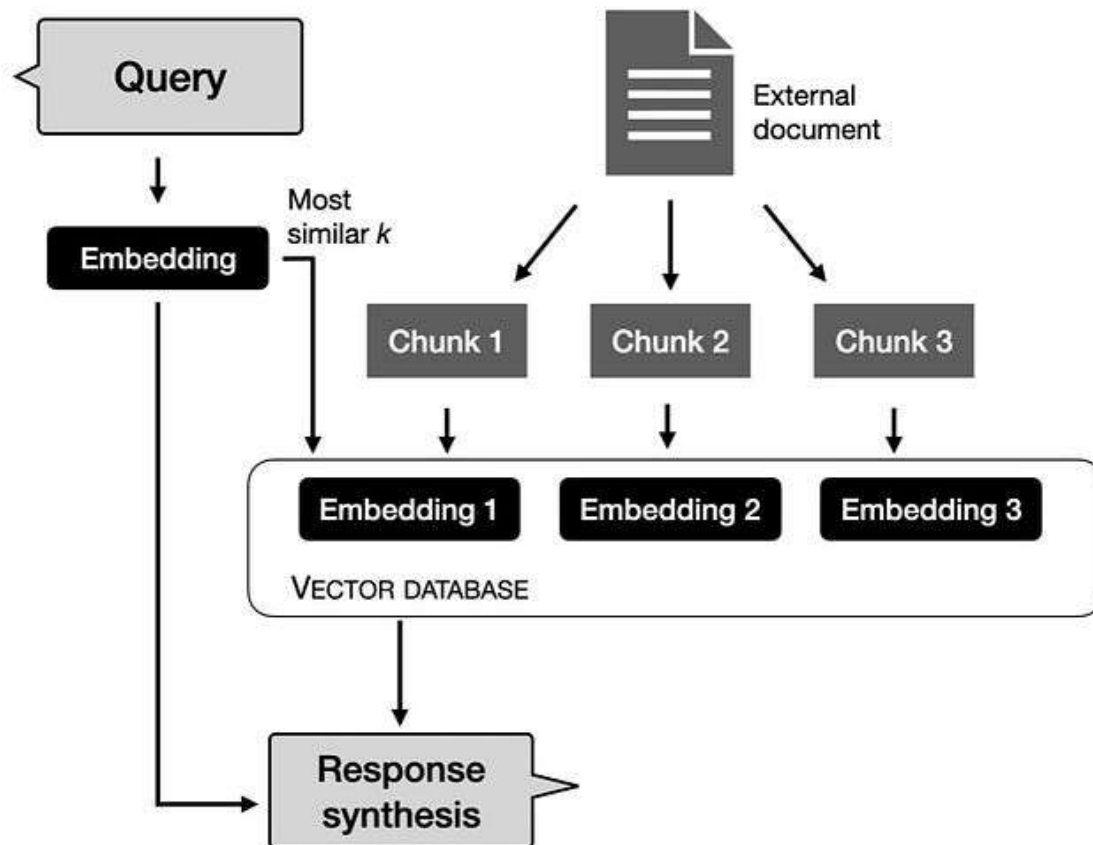
 Remove best_doc from candidate_docs

Return selected_docs

LLM and AI Agent RAG

Chunking

A preprocessing process that divides large-scale text data (e.g., long documents, books, or collections of web pages) into "chunks," independent units that are semantically organized and summarized and appropriately sized within a context window.

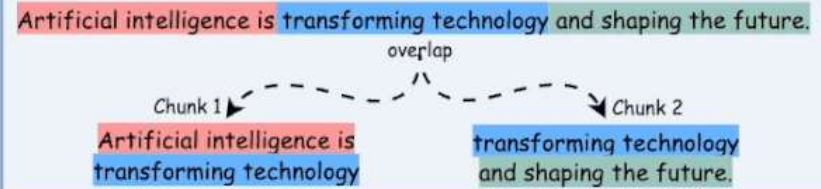


LLM and AI Agen

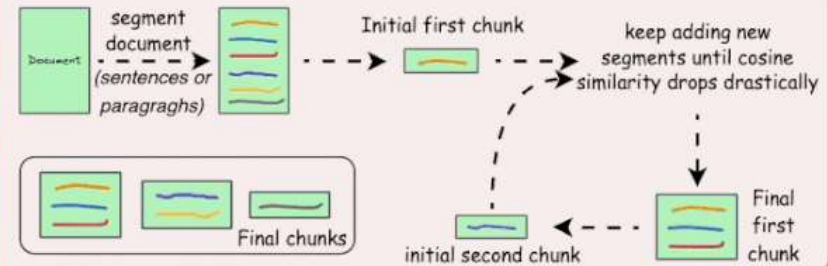
chunking

5 Chunking Strategies for RAG

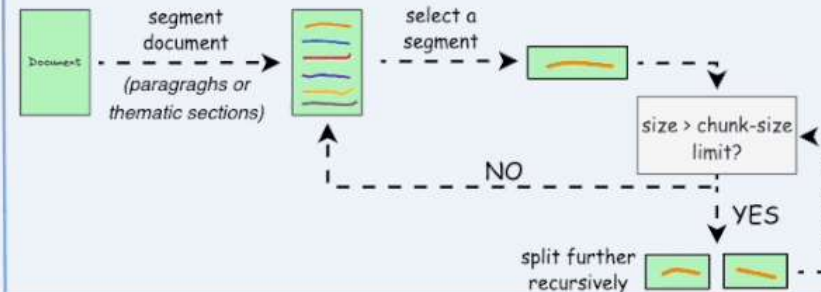
1) Fixed-size chunking



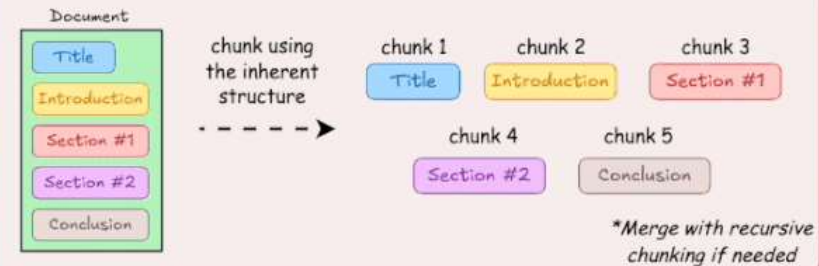
2) Semantic chunking



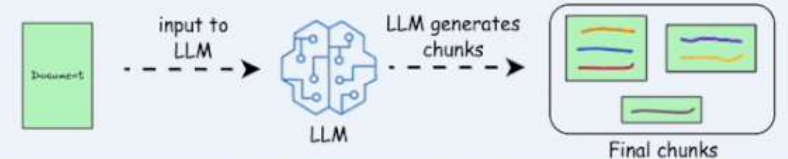
3) Recursive chunking



4) Document structure-based chunking



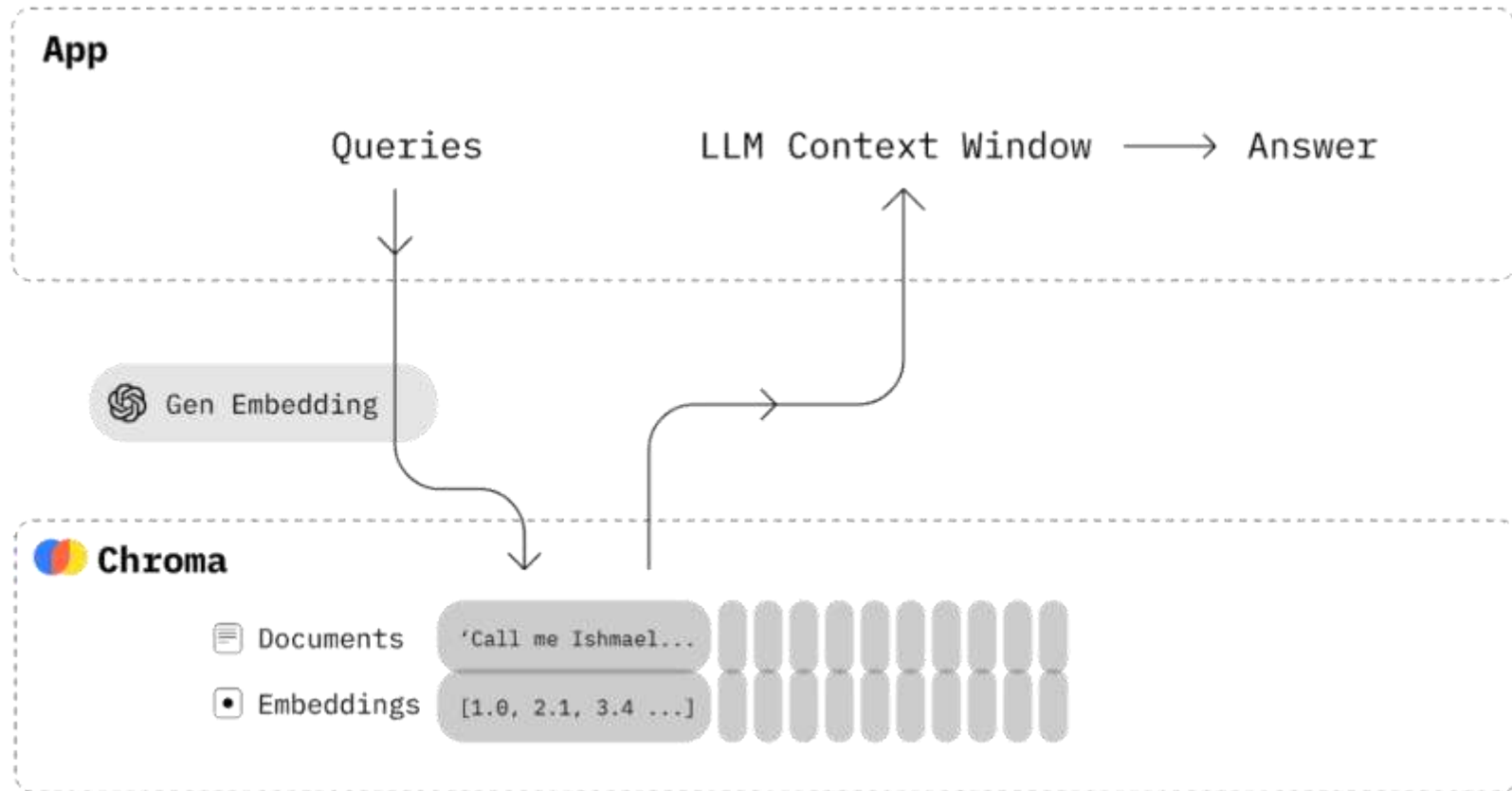
5) LLM-based chunking



LLM and AI Agent RAG

Augmented

Combine the retrieved documents with the user's original question to construct an LLM input prompt.



LLM and AI Agent RAG

Augmentation

To make a document searchable in a RAG system, the document must first be divided into manageable units, then each unit converted into a vector.

LLM and embedding models have a limited number of input tokens and cannot process long documents simultaneously.

Dividing into meaningful paragraphs or sentences prevents information loss and improves search quality. Diffing into detailed units (chunks) during the search phase allows for identifying the most relevant context for the query.

Excessive input of information during augmentation generation can result in noise, so dividing the data into appropriate units contributes to improved performance.

```
# 청킹
```

```
chunks = split_document(doc, chunk_size=500,  
overlap=100)
```

```
# embedding
```

```
for chunk in chunks:
```

```
    v = embed_model(chunk)
```

```
    vector_store.append(v)
```

LLM and AI Agent RAG

Augmented Generation

The prompt is passed to an LLM (OpenAI GPT-4, Claude, LLaMA, etc.) to output a response.

An augmented prompt is formed (user question + retrieved document).

The prompt is passed as input to the LLM.

The LLM generates a response by repeatedly generating the next token using pretrained probabilistic language modeling.

If necessary, sampling parameters such as temperature, top-k, and top-p are adjusted to control the generation diversity and quality.

You are a helpful assistant. Use the following context to answer the question.

Context:

{retrieved_context}

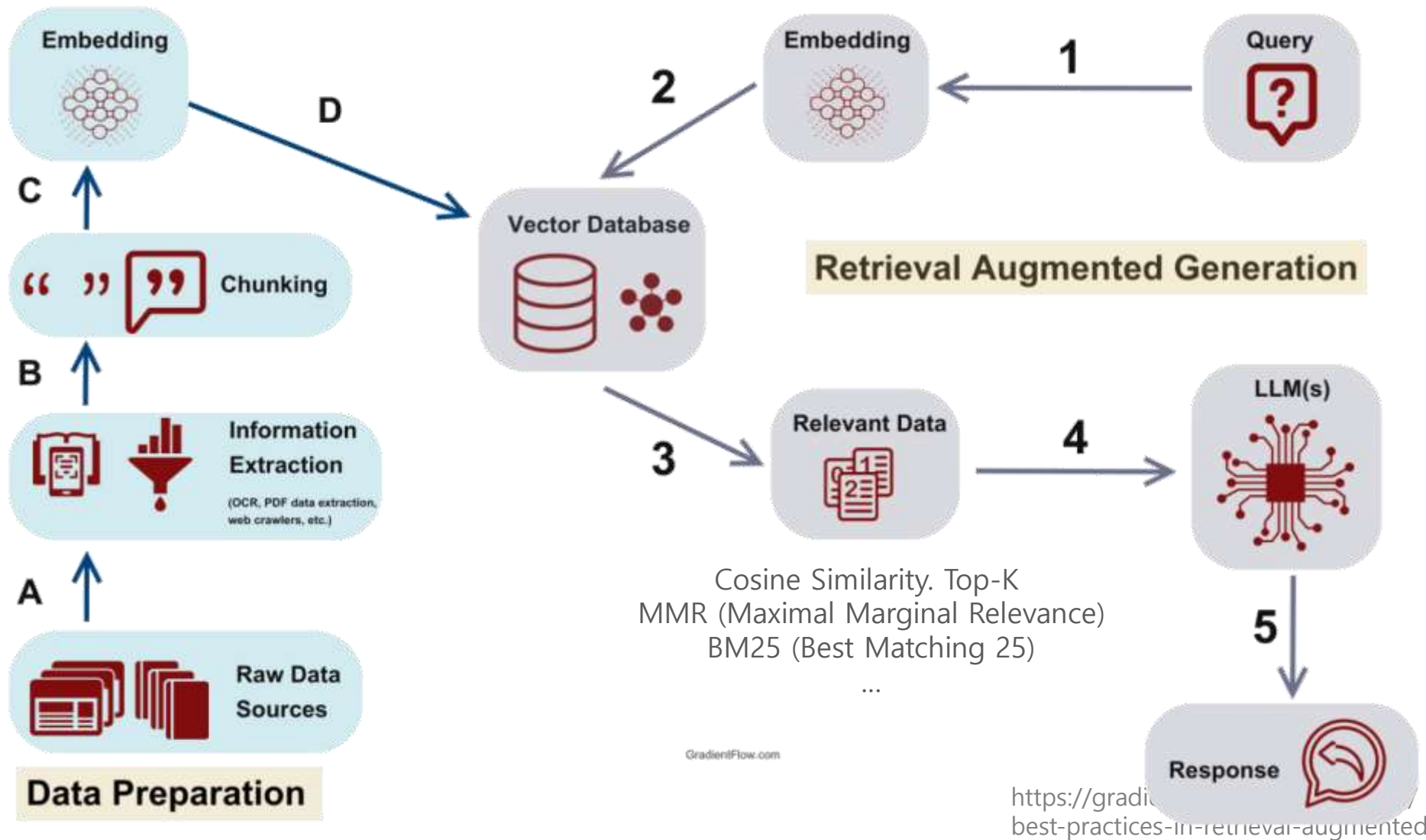
Question:

{user_question}

Answer:

LLM and AI Agent RAG

Full RAG flow



LLM and AI Agent RAG

Agent Development Tools

tool name	explanation	link
AutoGPT	An AI agent that automatically performs multiple tasks using text-based input. Designed to automate various tasks by leveraging GPT models.	AutoGPT GitHub
Pydantic	A data validation and configuration tool for building AI agents and models. Pydantic allows you to structure and validate the inputs and outputs of AI models.	PydanticAI
LlamaIndex	A tool that leverages the Llama model to implement efficient data retrieval and agent functionality. It is particularly strong in processing and retrieving large-scale text data.	LlamaIndex GitHub
CrewAI	A tool for managing AI agents, focusing on collaboration. Useful for connecting multiple AI agents to solve complex problems.	CrewAI Website
LangGraph	An AI agent management tool based on a graph database that helps effectively process and explore complex data relationships.	LangGraph GitHub

LLM and AI Agent RAG

PDF 읽기 및 청킹

```
from langchain.document_loaders import PyPDFLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
loader = PyPDFLoader("my_doc.pdf")
docs = loader.load()
splitter = RecursiveCharacterTextSplitter(chunk_size=500, chunk_overlap=100)
chunks = splitter.split_documents(docs)
```

embedding 및 vector 저장

```
from langchain.vectorstores import FAISS
from langchain.embeddings import OpenAIEmbeddings
embedding_model = OpenAIEmbeddings()
vectorstore = FAISS.from_documents(chunks, embedding_model)
```

질문-응답 체인 구성

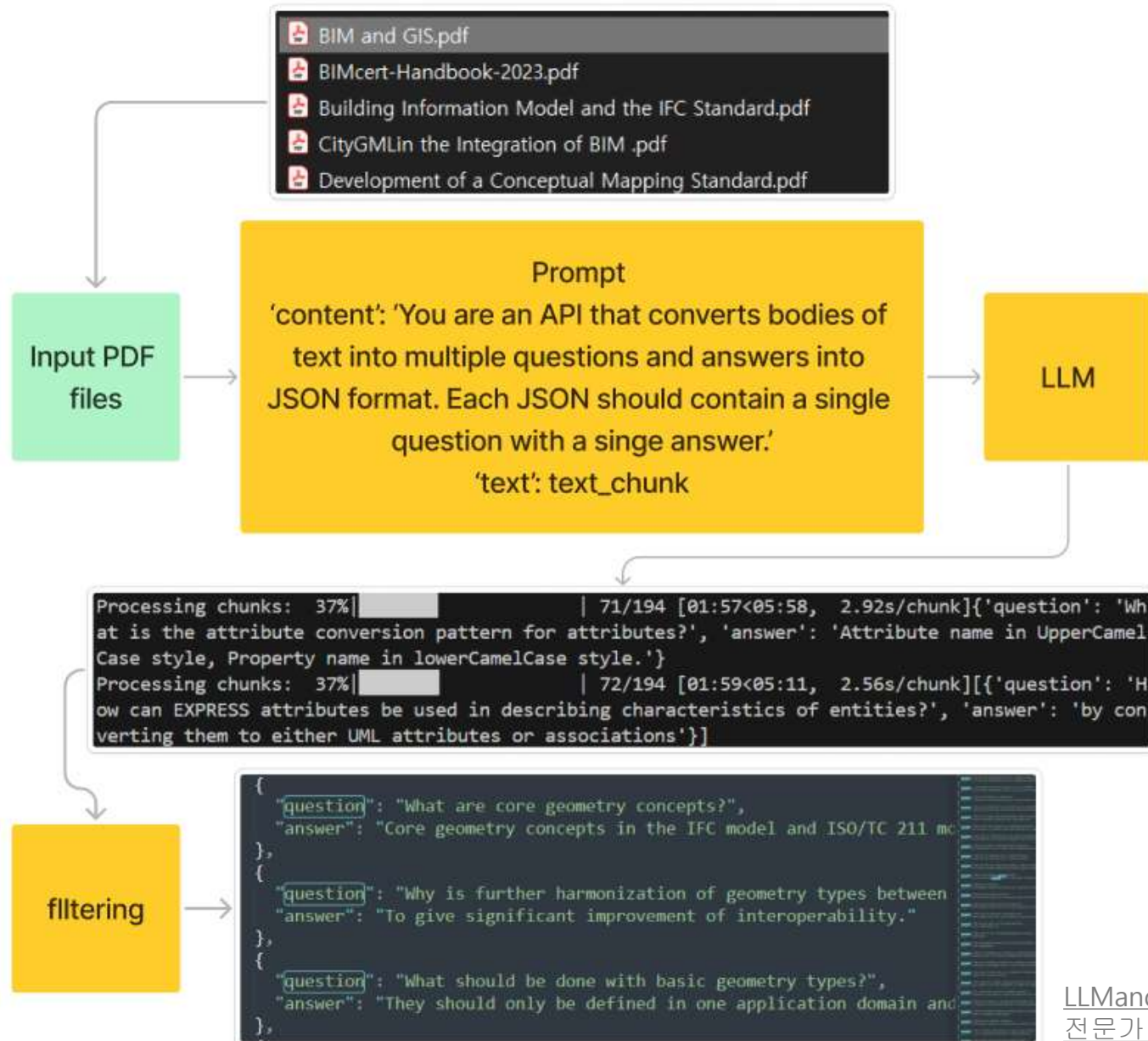
```
from langchain.chains import RetrievalQA
from langchain.chat_models import ChatOpenAI
retriever = vectorstore.as_retriever(search_type="mmr")
qa_chain = RetrievalQA.from_chain_type(
    llm=ChatOpenAI(),
    chain_type="stuff",
    retriever=retriever
)
```

질문에 대한 응답 Generative

```
query = "이 문서에서 저자의 주요 주장은 무엇인가요?"
result = qa_chain.run(query)
print(result)
```

LLM and AI Agent RAG

Example



LLM and AI Agent RAG

Example

```
client = OpenAI(  
    base_url = 'http://localhost:11434/v1',  
    api_key='ollama',  
)  
  
template = {  
    "question": " ",  
    "answer": " "  
}
```

```
def generate_questions_answers(text_chunk):  
    messages = [  
        {'role': 'system', 'content': 'You are an API that converts bodies of text  
            into multiple questions and answers into a JSON format. Each JSON  
            should contain a single question with a single answer. Only respond  
            with the JSON and no additional text. \n.'},  
        {'role': 'user', 'content': 'Text: ' + text_chunk}  
    ]  
  
    response = client.chat.completions.create(  
        model='llama3',  
        messages=messages,  
        max_tokens=2048,  
        n=1,  
        stop=None,  
        temperature=0.7,  
    )
```

LLM and AI Agent RAG

```
{
  "question": "What is the URL to access information on jetgeo?",
  "answer": ".com/jetgeo/IFC2GML"
},
{
  "question": "When was the ISO TC 211 introduction to UML accessed?",
  "answer": "20 March 2020"
},
{
  "question": "Who wrote Learning UML 2.0 and what is its edition?",
  "answer": "Miles, R.R.; Hamilton, K., 1st ed."
},
},
```

```
1555 {
1556   "question": "What problems were handled?",
1557   "answer": "Problems surrounding the integration of BIM and GIS heterogeneous m
1558 },
1559 {
1560   "question": "What is a method used for model integration?",
1561   "answer": "Schema integration, extension, linkage."
1562 },
1563 {
1564   "question": "What are limitations of model integration methods?",
1565   "answer": "Each method has its own respective limitation."
1566 },
1567 {
1568   "question": "What is BIM-GIS conceptual mapping?",
1569   "answer": "BIM-GIS conceptual mapping (B2GM) attempts to standardize the BIM-G
1570 },
1571 {
1572   "question": "How does B2GM divide heterogeneous processes?",
1573   "answer": "B2GM divides heterogeneous process into PD, EM, and LM, and links a
1574 },
```


LLM and AI Agent RAG

Open AccessArticle

Development of a Conceptual Mapping Standard to Link Building and Geospatial Information

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ISPRS Int. J. Geo-Inf. 2018, 7(5), 162; <https://doi.org/10.3390/ijgi7050162>

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Abstract

This study introduces the BIM (building information modeling)-GIS (geographic information system) conceptual mapping (B2GM) standard (ISO 19166) and proposes a mapping mechanism. In addition, the major issues concerning BIM-GIS integration, and the considerations that it requires, are discussed. The B2GM is currently being standardized by the spatial-information international standardization organization TC211. Previous studies on BIM-GIS integration seem to focus on the integration of different types of model schemas and on the implementation of service interfaces. B2GM concerns the clear definition of the conditions and methods for mapping the object information required from the user's use-case viewpoint for city-scale mapping. The benefits of the B2GM approach are that the user is able directly control the BIM-GIS linkage and integration process in order to acquire the necessary object information. This can reveal cases of possibly unclear BIM-GIS integration outside the black box in an explicit and standard way, so that the user can effectively predict the final output obtainable from the BIM-GIS integration. This study examined B2GM in terms of its development background, components, and several utilization examples, as well as the levels and considerations of the integration of different BIM-GIS models.

Keywords: BIM; GIS; conceptual mapping; ISO 19166; BG-IL

Version: 551.61

CUDA Version: 12.4

Bus-Id	Disp.A	Memory-Usage	Volatile GPU-Util	Uncorr. Compute M.	ECC MIG M.
00000000:01:00.0	On				N/A
5074MiB /	8192MiB		86%	Default	N/A

<https://daddynkidsmakers.blogspot.com/2024/02/github-copilot-ai.html>

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GeoBIM.pdf 3.6MB

Chat

what is BIM?

BIM (Building Information Modeling) is a method used in the construction industry to create and manage digital representations of physical and functional characteristics of places. It includes defining object information such as runtime, geometry, property, and relationship for building components like walls, doors, and rooms. The geometry information uses boundary representation with support for vertices, edges, loops, and faces. Property information categorizes attributes and defines name, value type, and initial value pairs. System properties, including the BIM element name and GUID, are automatically created when making a model. Relationships between elements are also defined, following UML relationships.

what is use cases of BIM?

BIM-GIS integration is used for scaling building information modeling (BIM) to city level use cases. This method includes a BIM-GIS element, geometry, and property set mapping with external datasets. The goal is to enable flexible information mapping and integration, focusing on user's specific requirements. Previous studies suggest various methods for BIM-GIS integration, but this study proposes using the B2GM standard under ISO/TC211.

tell me about BIM-GIS standard?

The BIM-GIS standard, also known as Building Information Modeling-Geographic Information System, is a method for

LLM and AI Agent

Ollama & Open source



ollama run llama3.2-vision
what is this figure? ./zaha_ddp.jpg

```
>>> what is this figure? ./zaha_ddp.jpg
Added image './zaha_ddp.jpg'
The image appears to be an interior view of a large, modern building with white architecture seems futuristic and minimalist, with clean lines and curves.
Upon closer inspection, it's possible that the image depicts a public space or building, such as a lobby, atrium, or hallway. The overall aesthetic suggests design, possibly intended to evoke feelings of calmness and serenity in those
```

The image appears to show the interior of a large, modern building with white walls and floors. The architecture is futuristic and minimalist, with clean lines and curves.

A closer look reveals that the image may depict a public space or commercial area within the building, such as a lobby, atrium, or hallway. The overall aesthetic suggests a modern, sophisticated design, likely intended to evoke a sense of calm and tranquility for those entering.

The image shows a large, open space with white walls and floors. The ceiling is also white, with several rows of recessed lights along the ceiling edge. In the center of the room is a long, curved wall that divides the space into two sections. A series of windows run along the wall's length, allowing natural light to enter the room. Before the curved wall is a large, open space with a gleaming white floor. This appears to be some kind of public or commercial space within the building. While no people are visible in the image, it is likely a place where people gather or congregate.

The overall feel of the image is modern and sleek. The clean lines, curves, and minimalist aesthetic suggest a contemporary design style. Using white as a primary color creates a sense of brightness and openness, which may be intended to evoke a sense of calm and tranquility for those entering the space.

LLM and AI Agent

Ollama & Open source

Ollama

A lightweight tool that helps you run and utilize large-scale language models (LLMs) locally.

Designed to load open source models like GPT, LLaMA, and Mistral directly into your local environment and use them for prompt response and RAG search.

It works without Docker and integrates easily with LangChain.

- Go to <https://ollama.com>
- Select your operating system (Windows) and download the installer (.exe).
- After installation, open a command prompt (CMD) or PowerShell and verify that the ollama command is recognized.

```
F:\projects>ollama list
NAME                                ID                                SIZE  MODIFIED
tinyllama:latest                    2644915ede35                     637 MB 8 days ago
llama3:8b-instruct-q4_K_M            9b8f3f3385bf                     4.9 GB 8 days ago
qwen2.5-coder:7b                    dae161e27b0e                     4.7 GB 3 weeks ago
codegemma:7b                        0c96700aaada                     5.0 GB 3 weeks ago
gemma3:latest                       a2af6cc3eb7f                     3.3 GB 5 weeks ago
llama3.2-vision:latest               38107a0cd119                     7.9 GB 8 months ago

F:\projects>ollama run gemma3
>>> 안녕
안녕하세요! 무엇을 도와드릴까요? 😊

>>> 너는 누구?
저는 Google에서 개발한 대규모 언어 모델입니다.

쉽게 말하면, 방대한 양의 텍스트 데이터를 학습하여 인간처럼 대화하고, 질문에 답하고, 텍스트를 생성할 수 있는 인공지능입니다.
```

LLM and AI Agent Ollama & Open source

Running a web-based Ollama service

Install Ollama from the following link

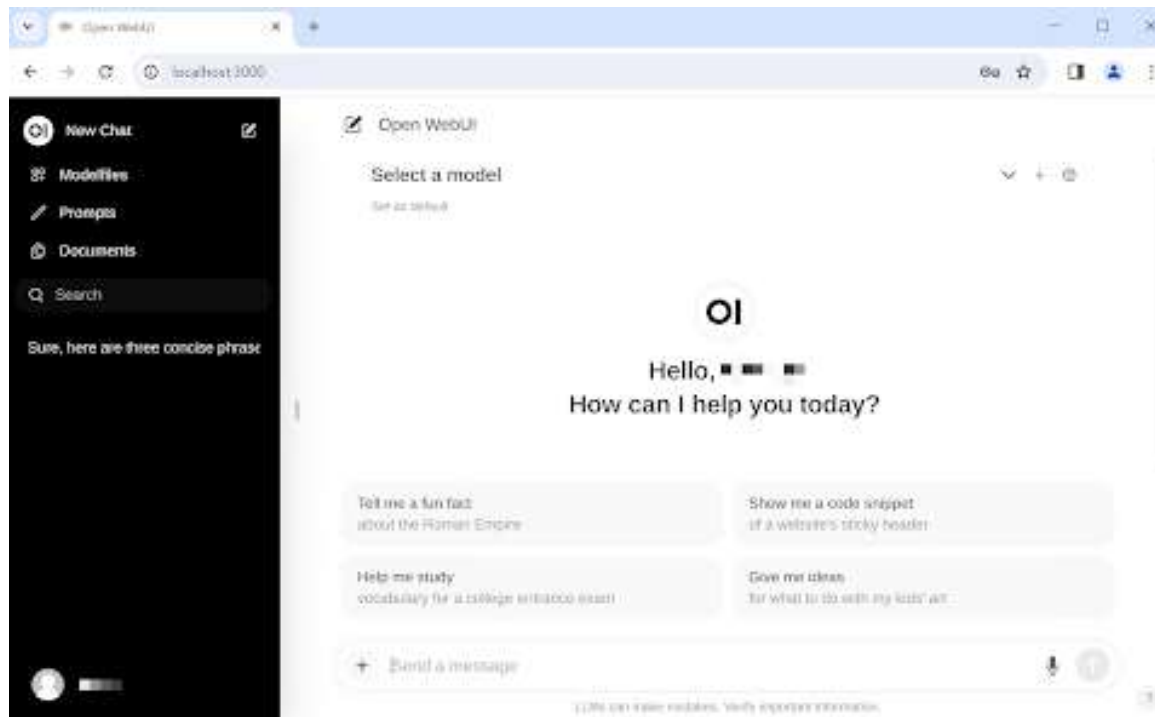
[Getting Started](#) | [Open WebUI](#)

Install Docker from the web using the following link. Reboot.

[Windows](#) | [Docker Docs](#)

Run the following command in the terminal command window

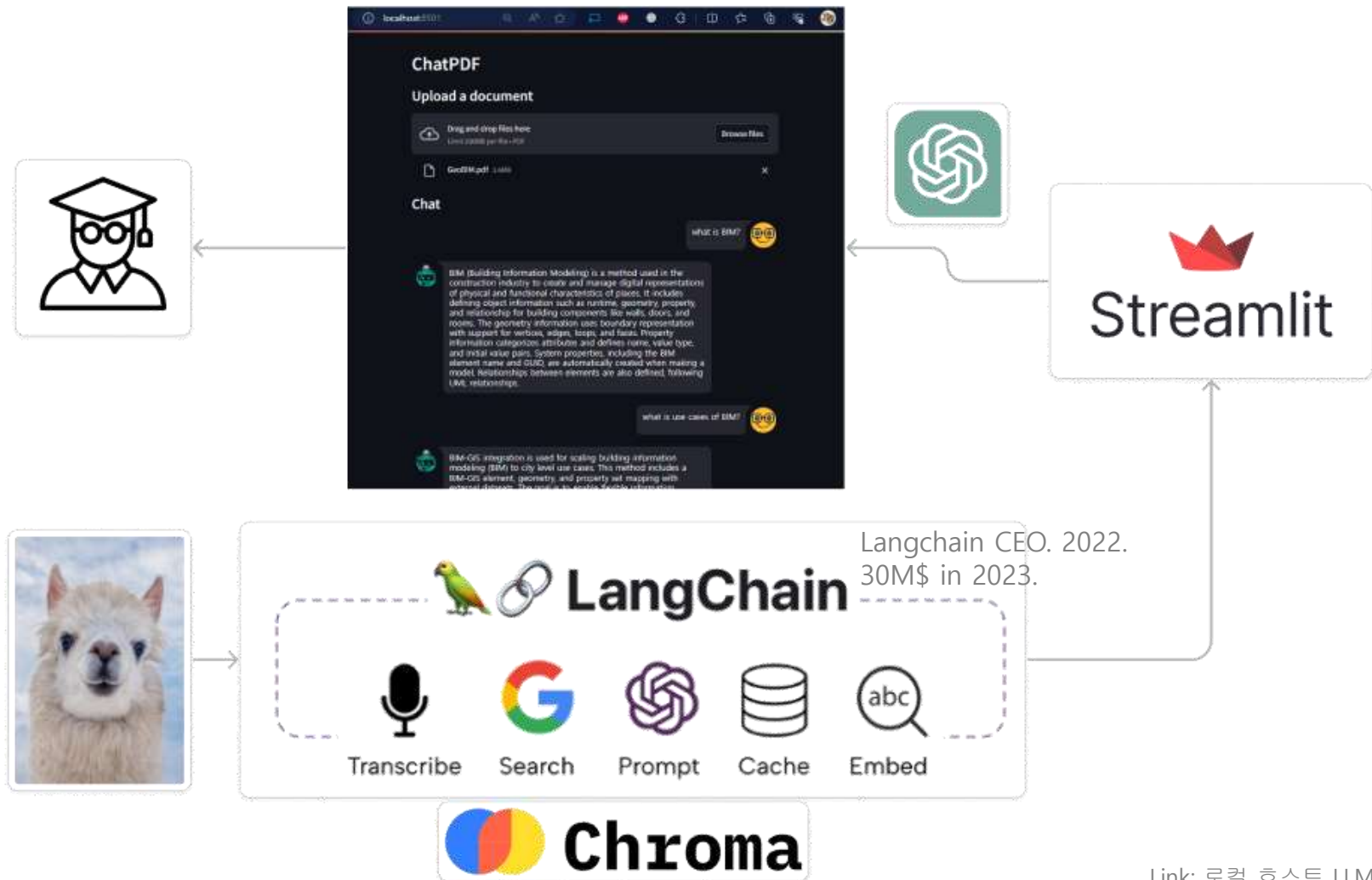
```
docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway -v open-webui:/app/backend/data --name open-webui --restart always ghcr.io/open-webui/open-webui:main
```



Link: [Web 기반 Ollama 서비스 구동 방법](#)

LLM and AI Agent Ollama & Open source

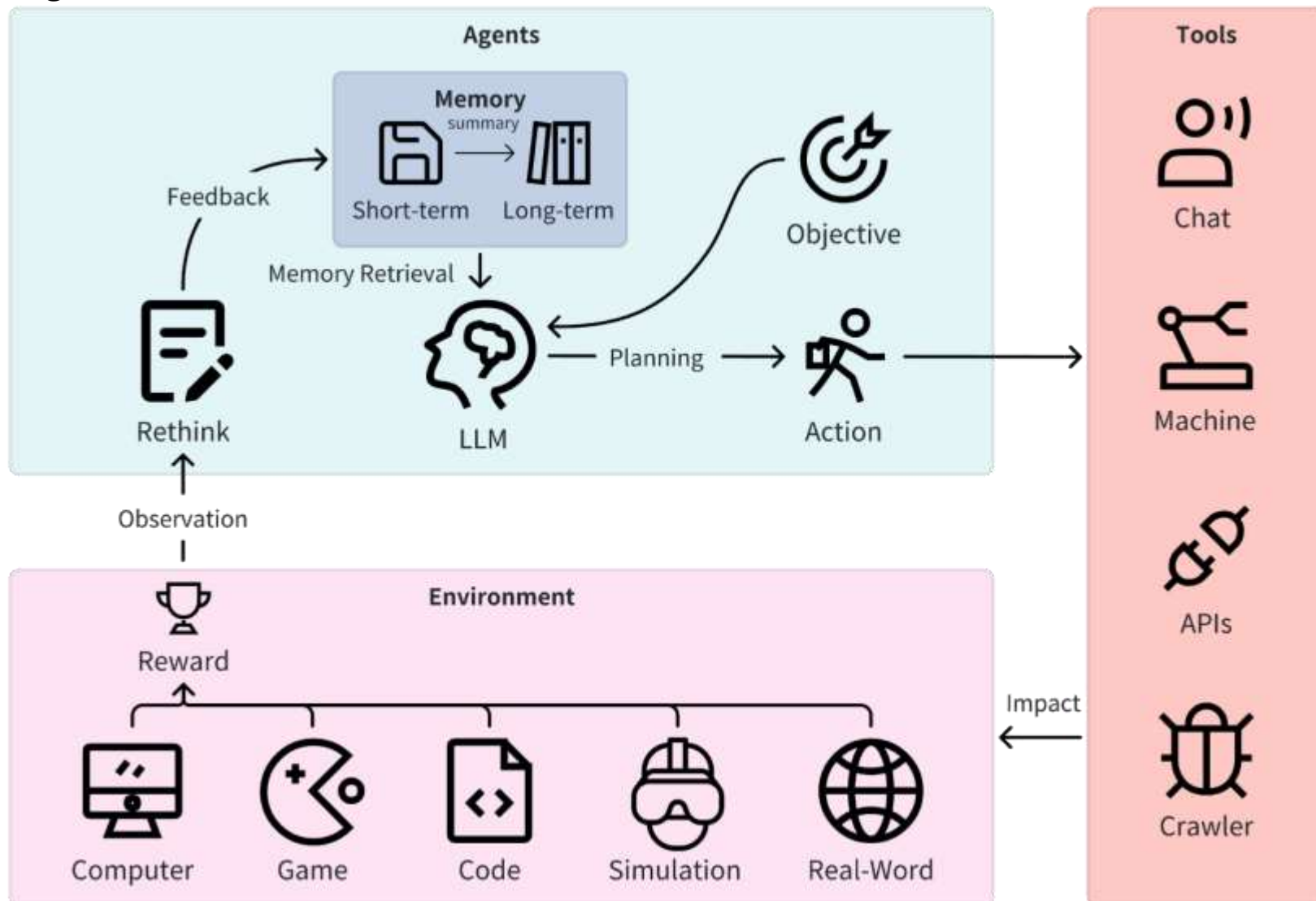
Example



[Link: 로컬 호스트 LLM open source OLLAMA 기반 PDF 지식 챗봇 서비스 간단히 만들어보기](#)

LLM and AI Agent

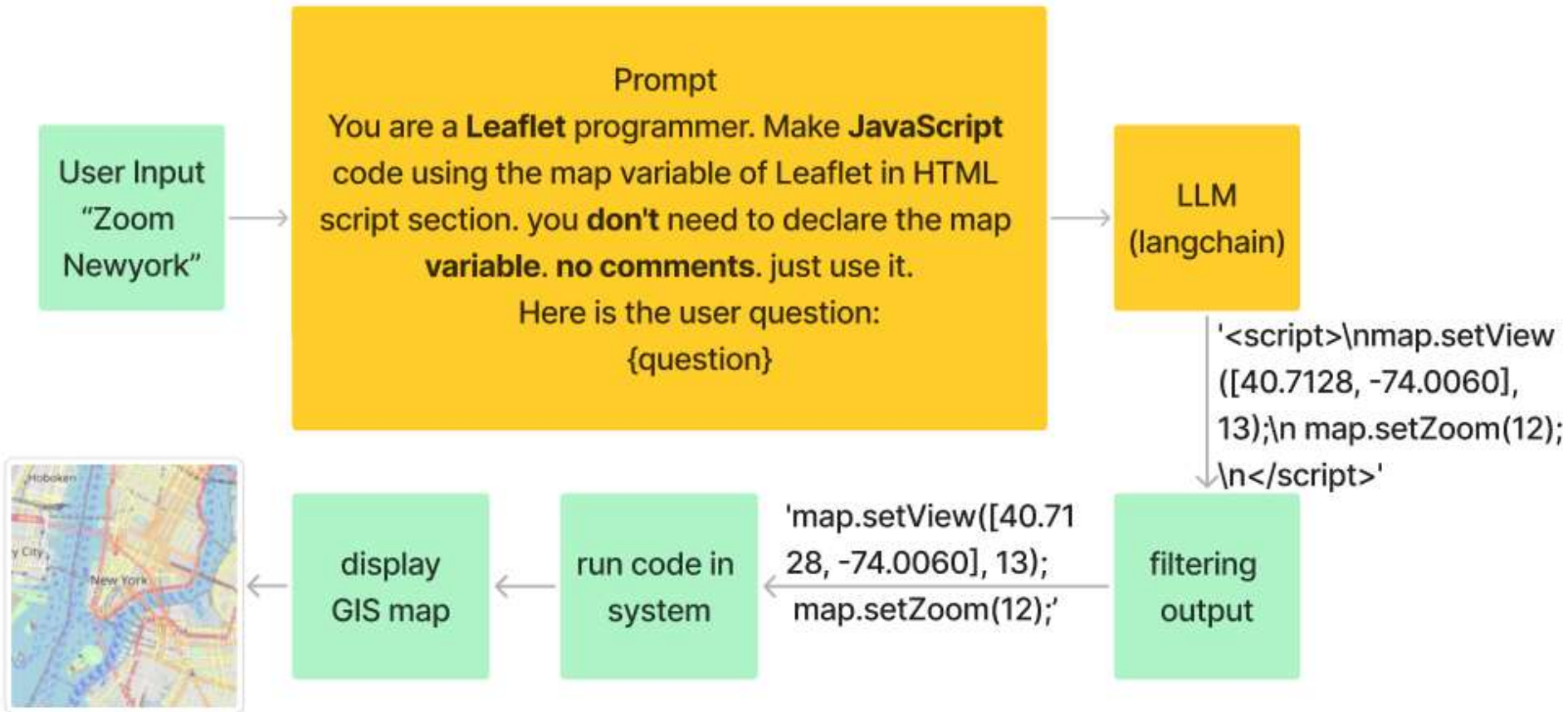
multi-agent model



LLM and AI Agent

AI Agent

Example



LLM and AI Agent

AI Agent

```
local_llm = 'llama3'
llm = ChatOllama(model=local_llm, temperature=0) # format="json",

def text_to_geo_code(query_text):
    prompt = " " + query_text

    prompt = PromptTemplate(
        template="""You are a Leaflet programmer. Make JavaScript code using the map variable
of Leaflet in HTML script section. I need script section. you don't need to declare
the map variable. no comments. just use it.
Here is the user question:
{question}
""",
        input_variables=["question"],
    )

    text_to_geo_chain = prompt | llm | StrOutputParser()
    generated_code = text_to_geo_chain.invoke({"question": query_text})

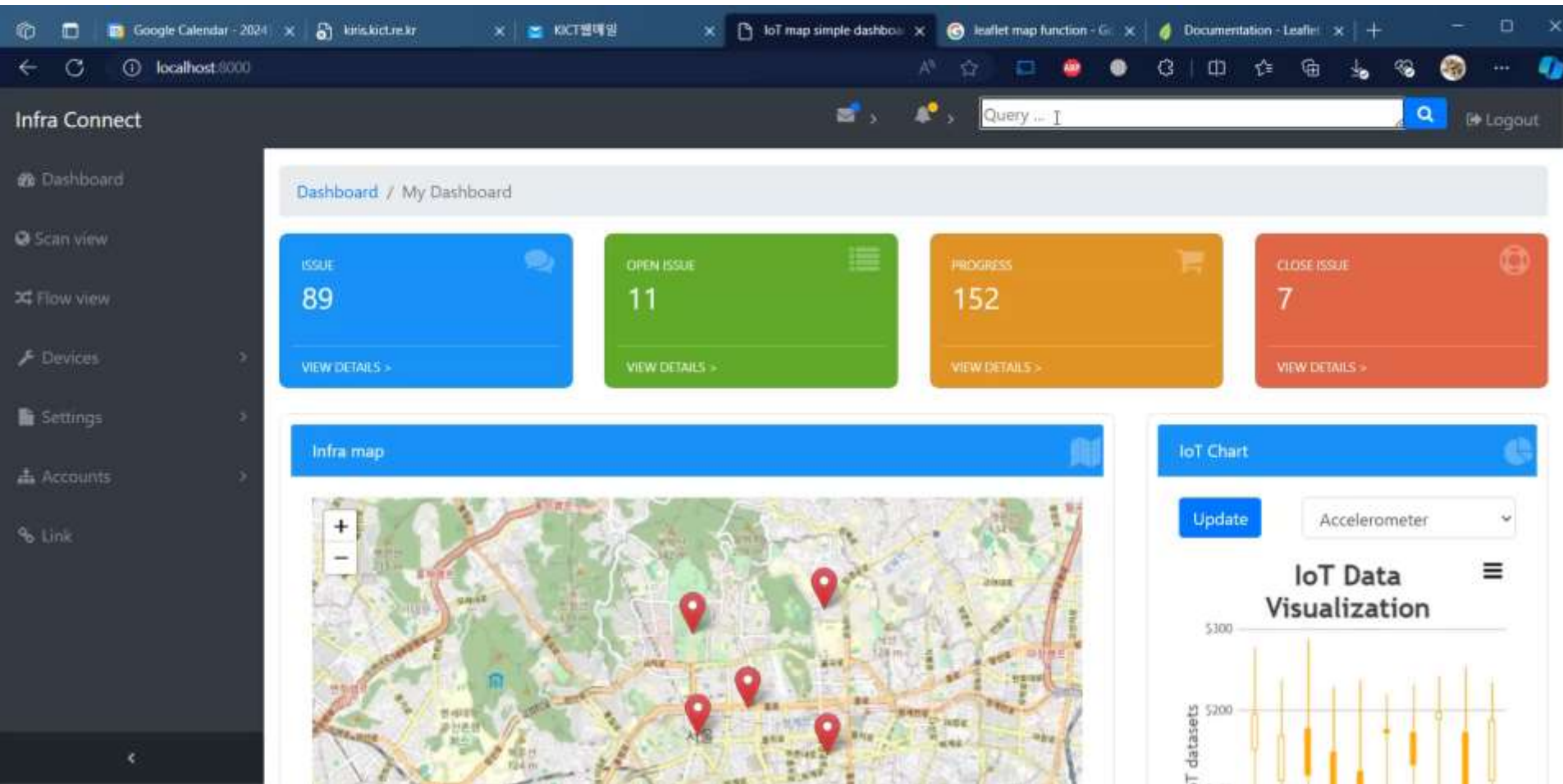
    script_regex = r"<script>(.*?)</script>"
    match = re.search(script_regex, generated_code, re.DOTALL)

    code = ''
    if match:
        code = match.group(1).strip() # Return the extracted code

    return code
```

LLM and AI Agent AI Agent

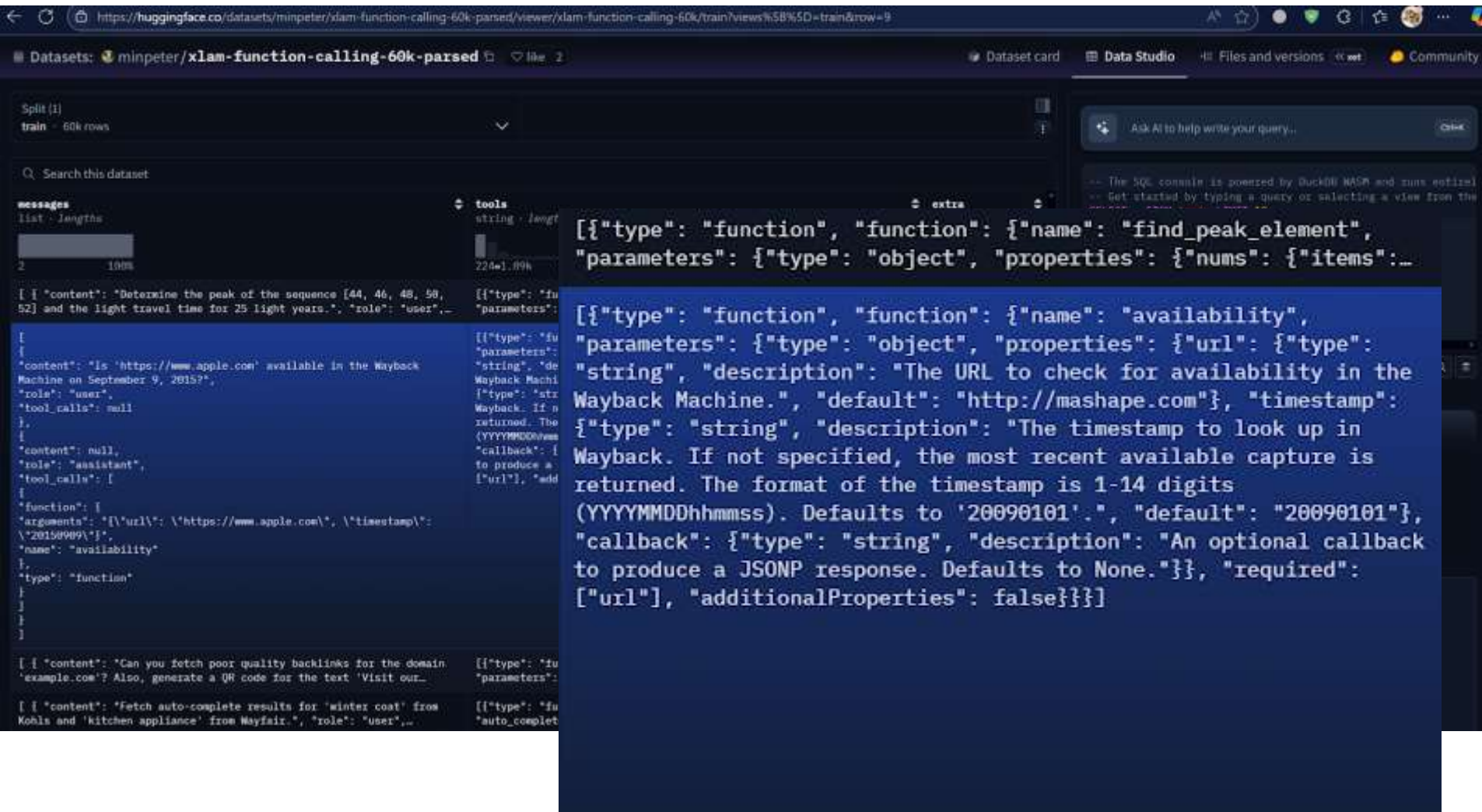
Example



LLM and AI Agent

AI Agent Design Pattern

Tool and Function Call



The screenshot displays the Hugging Face dataset viewer for the dataset 'minpeter/xlam-function-calling-60k-parsed'. The interface shows a table of data rows with columns for 'messages' (containing a list of messages), 'tools' (containing a list of tool names), and 'extra' (containing additional information). The 'messages' column is expanded, showing a list of messages. The 'tools' column is also expanded, showing a list of tool names. The 'extra' column is expanded, showing a list of JSON objects representing function calls. The JSON objects are structured as follows:

```
[{"type": "function", "function": {"name": "find_peak_element", "parameters": {"type": "object", "properties": {"nums": {"items": ...
```

```
[{"type": "function", "function": {"name": "availability", "parameters": {"type": "object", "properties": {"url": {"type": "string", "description": "The URL to check for availability in the Wayback Machine.", "default": "http://mashape.com"}, "timestamp": {"type": "string", "description": "The timestamp to look up in Wayback. If not specified, the most recent available capture is returned. The format of the timestamp is 1-14 digits (YYYYMMDDhhmmss). Defaults to '20090101'.", "default": "20090101"}, "callback": {"type": "string", "description": "An optional callback to produce a JSONP response. Defaults to None."}, "required": [{"url"}, {"additionalProperties": false}]}}
```

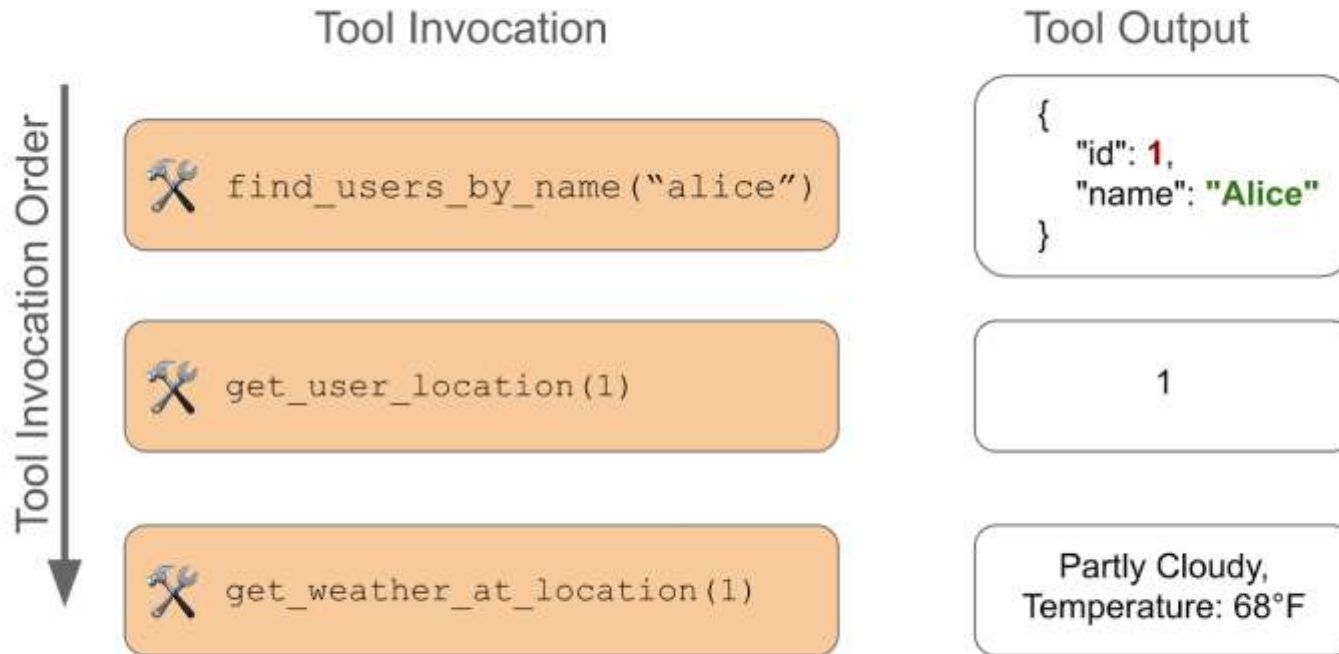
Link: [높은 성능의 AI 에이전트 구현을 위한 Gemma3 Function call 파인튜닝](#)

LLM and AI Agent

AI Agent Design Pattern

Tool and Function Call

? Is it likely that Alice needs an umbrella now?



! Based on the current weather, it is partly cloudy with a temperature of 68°F at Alice's location. Therefore, it is unlikely that Alice needs an umbrella right now.

LLM and AI Agent

AI Agent Design Pattern

Tooland Function Call

```
load_dotenv()
hf_token = os.getenv("HF_API_KEY")
login(token=hf_token)

model_name = "google/gemma-3-4b-it"
dataset_name = "Salesforce/xlam-function-calling-60k"
```

```
peft_config = LoraConfig(
    lora_alpha=16,
    lora_dropout=0.05,
    r=16,
    task_type="CAUSAL_LM", ...
)
```

```
trainer = SFTTrainer(
    model=model,
    args=training_arguments,
    train_dataset=dataset["train"],
    eval_dataset=dataset["test"],
    # tokenizer=tokenizer,
    peft_config=peft_config,
)
```

```
trainer.train()
trainer.save_model()
```

[Link: 높은 성능의 AI 에이전트 구현을 위한 Gemma3 Function call 파인튜닝](#)

LLM and AI Agent

AI Agent Design Pattern

Tool and Function Call

```
User query:
What are the roots of the quadratic equation  $2x^2 - 3x + 1 = 0$ ?<end_of_turn><eos>
<start_of_turn>assistant
<function_call>
{"name": "solve_quadratic", "arguments": {"a": 2, "b": -3, "c": 1}}
</function_call><end_of_turn><eos>
<start_of_turn>model

Converting train dataset to ChatML: 100%|          | 54000/54000 [00:04<00:00, 11761.83 examples/s]
Adding EOS to train dataset: 100%|          | 54000/54000 [00:03<00:00, 17786.93 examples/s]
Tokenizing train dataset: 100%|          | 54000/54000 [00:20<00:00, 2575.83 examples/s]
Truncating train dataset: 100%|          | 54000/54000 [00:00<00:00, 462257.72 examples/s]
Converting eval dataset to ChatML: 100%|          | 6000/6000 [00:00<00:00, 11820.43 examples/s]
Adding EOS to eval dataset: 100%|          | 6000/6000 [00:00<00:00, 18528.33 examples/s]
Tokenizing eval dataset: 100%|          | 6000/6000 [00:02<00:00, 2546.05 examples/s]
Truncating eval dataset: 100%|          | 6000/6000 [00:00<00:00, 317345.61 examples/s]
No label_names provided for model class `PeftModelForCausalLM`. Since `PeftModel` hides base models input
tomatically within `Trainer`. Note that empty label_names list will be used instead.
wandb: WARNING The `run_name` is currently set to the same value as `TrainingArguments.output_dir`. If t
the `TrainingArguments.run_name` parameter.
wandb: Using wandb-core as the SDK backend. Please refer to https://wandb.me/wandb-core for more inform
wandb: Currently logged in as: mac999 to https://api.wandb.ai. Use `wandb login --relogin` to force rel
wandb: Tracking run with wandb version 0.19.9
wandb: Run data is saved locally in F:\projects\func-call-gemma3-finetune\wandb\run-20250609_101702-jmwj
wandb: Run `wandb offline` to turn off syncing.
wandb: Syncing run gemma-3-4b-it-thinking-function_calling-V0
wandb: View project at https://wandb.ai/mac999/huggingface
wandb: View run at https://wandb.ai/mac999/huggingface/runs/jmwjveij
0%|          | 0/5061 [00:00<?, ?it/s]`use_ca
ache=False`.
```


LLM and AI Agent

AI Agent Design Pattern

Function Call caution

[Link: Gemma3 기반 Ollama 활용 AI 에이전트 개발 핵심
Function Call 구현해보기](#)

Clarity in Prompt Design

Clearly indicate the situation in which a function call is required, the method of calling (e.g., JSON format), and an example of the function call in the SYSTEM_MESSAGE.

"If a function call is determined to be necessary to answer a question, respond only in the JSON format below. Never include any other text or explanations."

Specificity of Function Definitions

Describe the function's purpose, parameters, return value, and usage examples in detail. Clarify the type, mandatory nature, and description of each parameter.

Example-Based Few-Shot Prompting

Include multiple examples of questions requiring a function call and the correct function call in the SYSTEM_MESSAGE or user message.

Retry Logic for Function Call Failures

If the model fails to call a function or responds with an incorrect format, the model internally re-prompts with an additional prompt, such as "A function call is required. Please respond only in JSON format."

Robustness of Output Parsing

The model can return text other than JSON. Extract only JSON portions from parsing logic, or parse incomplete JSON as much as possible.

Function Call Intent Reinforcement Prompt

Added the phrase "If a function call is required, always consider a function call first" to SYSTEM_MESSAGE. Eliminate unnecessary explanations, such as "If a function call is not necessary, do not explain why; just respond directly."

Model Version and Parameter Optimization

Use a model optimized for function calls. Reduce parameters like temperature and top_p to ensure consistent responses.

Collect and Improve Function Call Failure Cases

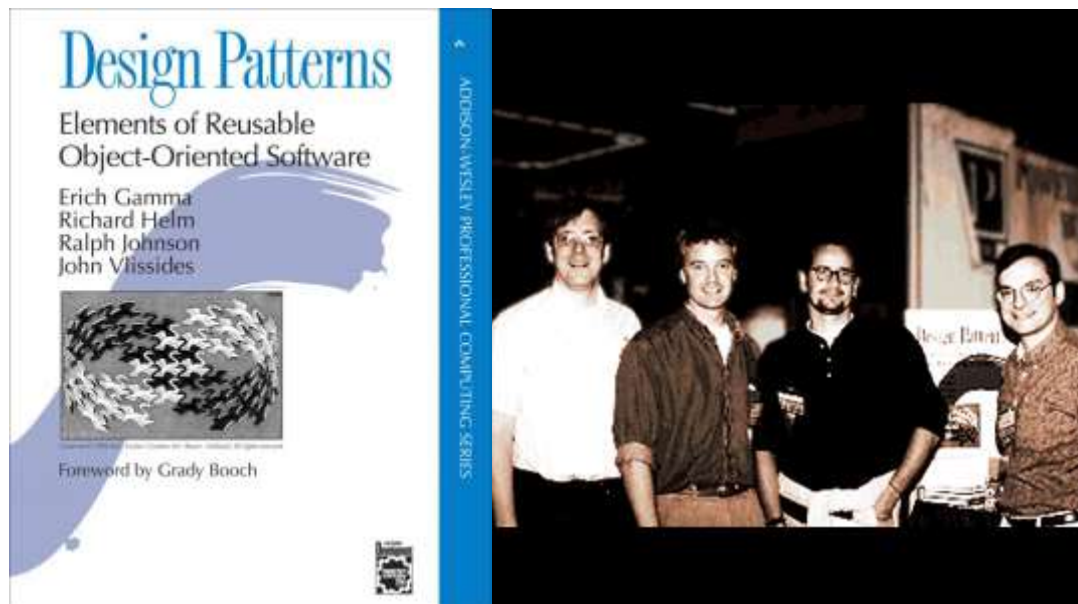
Continuously improve SYSTEM_MESSAGE and Example prompts by collecting cases where function calls were omitted during actual user input.

LLM and AI Agent

AI Agent Design Pattern

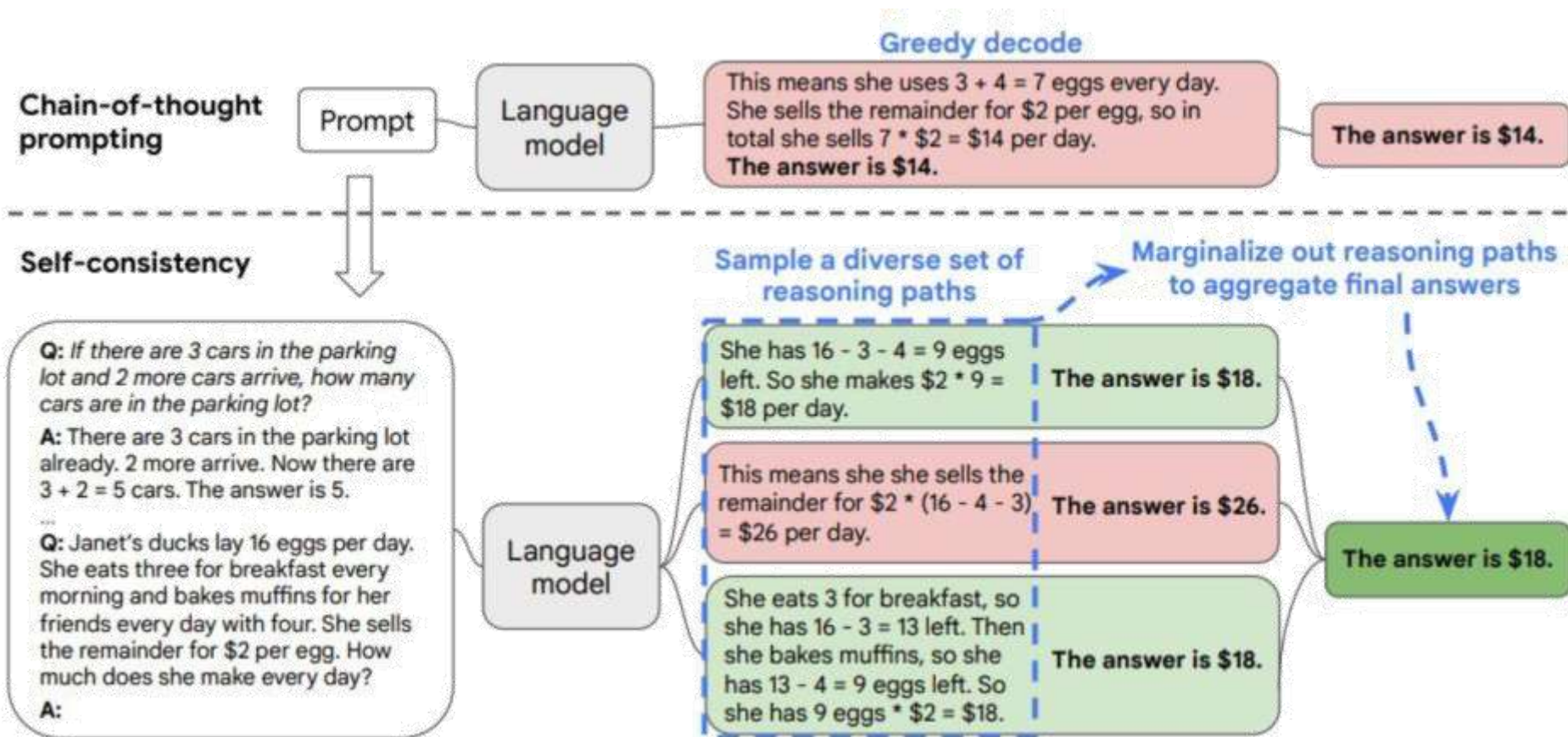
Design Patterns and Agent Patterns

First established in 1994 by the "Gang of Four" (GoF)—Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides—in their book "Design Patterns: Elements of Reusable Object-Oriented Software." They began as a patternized approach to solving recurring problems in object-oriented software design.



LLM and AI Agent

AI Agent Design Pattern



LLM and AI Agent

AI Agent Design Pattern

search_web

Question: What are the lens equations?
Thought: I should look at the database too see what I can query.
Action: rag_search
Action Input: <input to search the vector database using rag_search>: "What are the lens equations?"
Observation: No relevant results.
Thought: I should use search_web
Action: search_web
Action Input: <input to search the web search_web>: "What are the lens equations?"
Observation: The lens equation expresses the quantitative relationship between the object distance..., <https://www.physicsclassroom.com/class/refrn/Lesson-5/The-Mathematics-of-Lenses>
Thought: I have the final answer. No further tool usage required.
Final Answer: According to the website...

remember/checkpoint

Question: Checkpoint
Thought: I should checkpoint the current progress.
Action: checkpoint
Action Input: <input to checkpoint checkpoint>: current conversation, current topic in progress, function call stack etc.
Observation: The current conversation state, and the current tool call stack has been saved to long term memory.
Thought: I have the final answer. No further tool usage required.
Final Answer: Checkpoint successful

pop_quiz

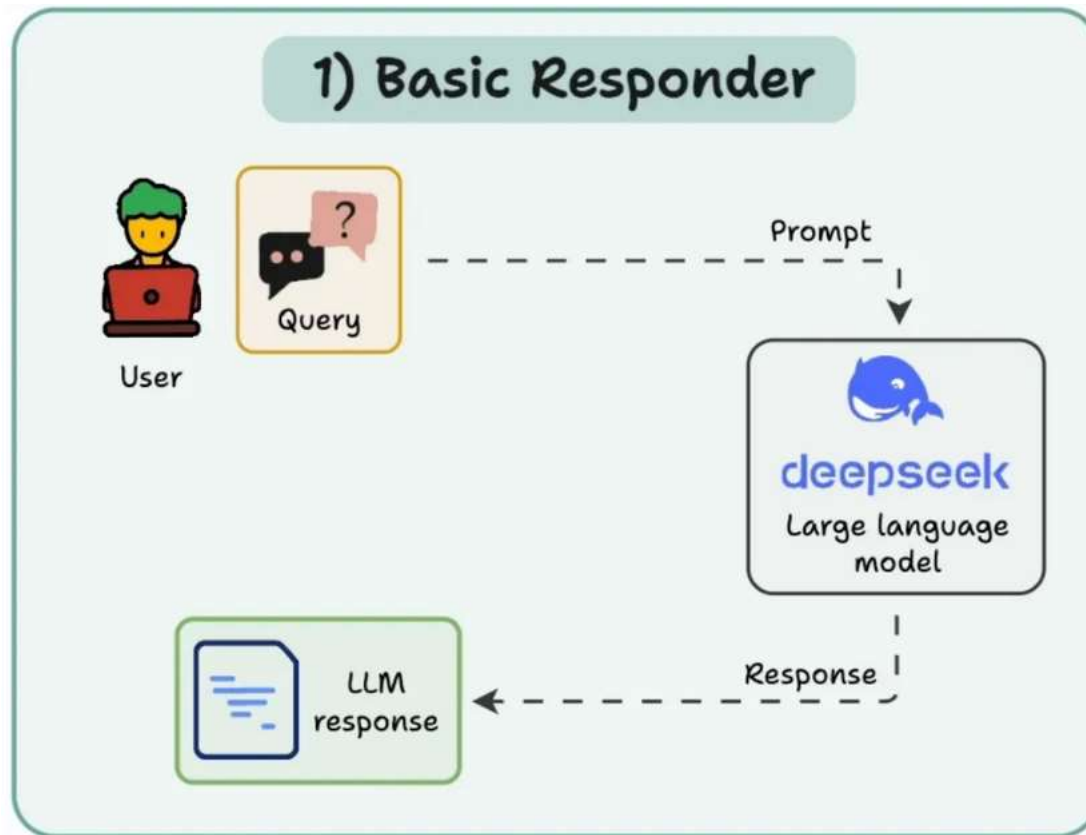
Question: Pop Quiz
Thought: I should generate a pop quiz using the template on the current topic of study
Action: pop_quiz
Action Input: <input to pop quiz pop_quiz>: current conversation, current topic in progress, previous topics completed
Observation: Pop quiz generated with topic A, B etc. with display UI and API backend.
Thought: I have the final answer. No further tool usage required.
Final Answer: Here is you pop quiz, please enter the answers for the respective questions and press submit at the end.

this is interesting since *pop_quiz* needs to interact with another API/system to generate the question answer UI and do something on submission

LLM and AI Agent

AI Agent Design Pattern

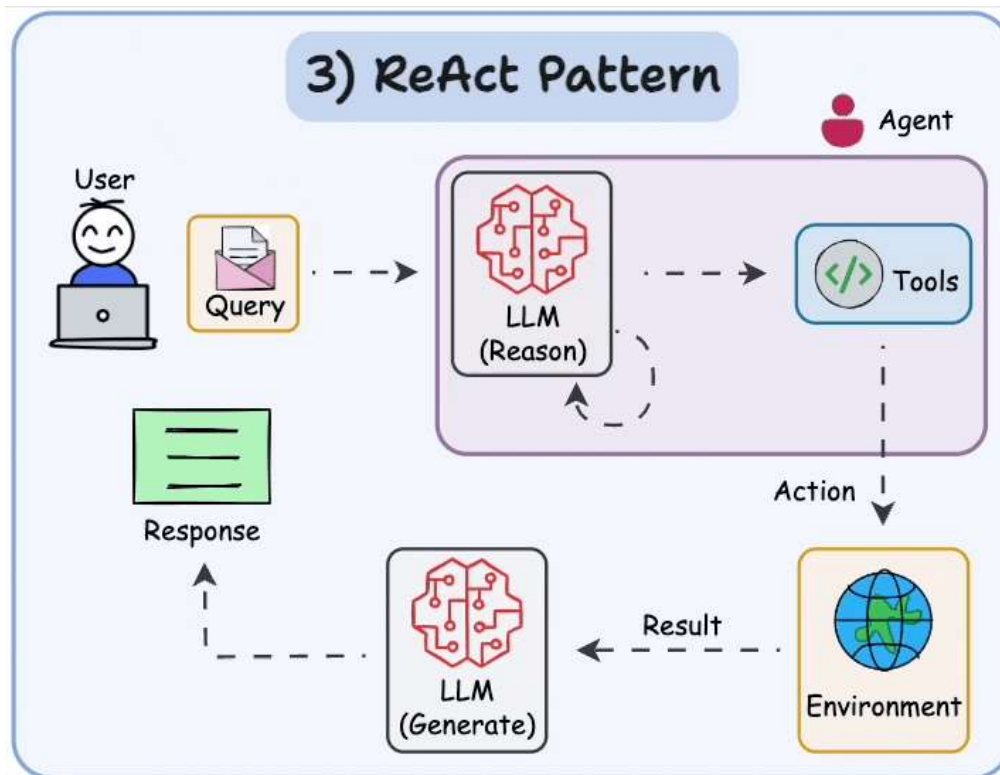
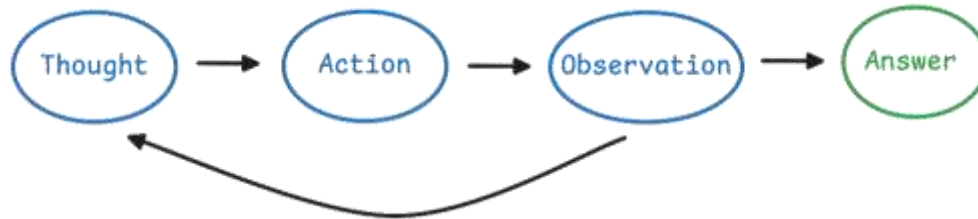
Basic Responder



LLM and AI Agent

AI Agent Design Pattern

ReAct



Question: What are the 3 Newton's laws on motion?

Thought: I should look at the database too see what I can query.

Action: rag_search

Action Input: <input to search the vector database using rag_search>: "Newton's three laws on motion"

Observation: Newton's three laws of motion applied to classical mechanics. They state....

Thought: I have the final answer. No further tool usage required.

Final Answer: Newton's three laws of motion applied to classical mechanics. They state....

LLM and AI Agent

AI Agent Design Pattern

Single Skill Agent

This pattern is used to build agents specialized for a single task. For example, it focuses on a single function, such as searching internal document-based FAQs, answering weather questions, or summarizing emails. It's built on a fine-tuned LLM or RAG-based architecture. The input is a simple query, and it consists of an embedding retriever and a single LLM. It's ideal for quickly deploying a single function as a service.

Multi-Tool Routing Agent

This pattern selectively executes various tools or skills based on input. For example, the command "calculate" invokes a Python executable, "tell me the weather" invokes an external API tool, and "summarize a document" invokes an LLM. It can be implemented in MCP-based tool systems, OpenAI Function/Tool Calling, LangChain ReAct, and others. The client embeds a selector (Classifier or Planner) that classifies the input.

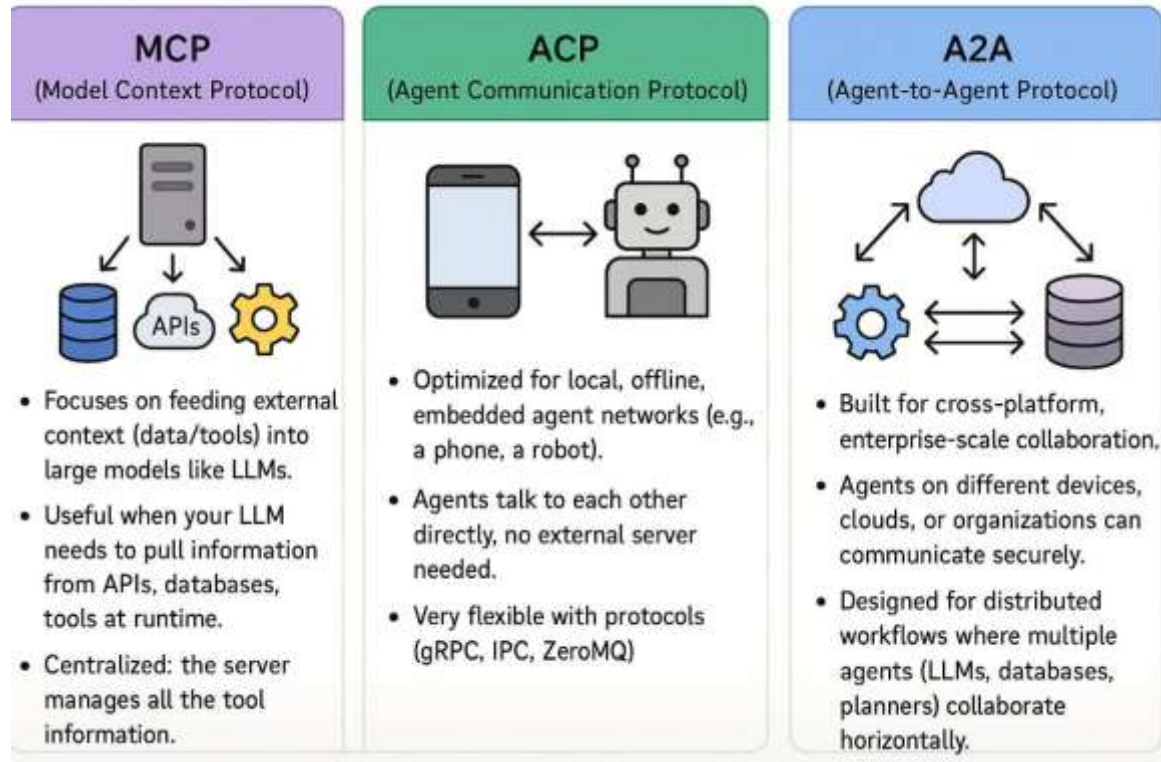
Expert Ensemble Agent

This pattern combines LLMs or agents specialized in different domains to generate collaborative responses. For example, medical questions might be routed to Med-PaLM, legal questions might be routed to GPT-Legal, or responses from all experts might be combined based on majority vote or score. This advanced strategy for combining multiple agents is ideal for expertise-based services.

Multimodal Retrieval Agent

This pattern simultaneously retrieves and analyzes data in various formats, including text, images, audio, video, PDFs, PPTs, and music. For this purpose, multimodal embedding models (e.g., CLIP, Whisper, MusicLM) are used, and documents are retrieved based on similarity in a unified vector search index.

LLM and AI Agent MCP



Model Context Protocol (MCP)

Proposed in mid-2023 by Anthropic, known as the developer of Claude.

MCP is a communication framework optimized for connecting external data sources or tools in an architecture centered on a Large Language Model (LLM). It is primarily designed for tool invocation and context injection.

Agent Communication Protocol (ACP)

Proposed by Anthropic and some community-based projects as a complementary protocol to MCP, it is specifically designed for efficient communication in local agent networks. Unlike MCP, ACP differs structurally in that it does not require a central server.

Agent-to-Agent Protocol (A2A)

An open standard that began discussions in 2024 among multiple companies and open source communities, including Microsoft, Hugging Face, and OpenAI. A2A aims to be a horizontal communication framework that enables multiple agents to collaborate in large-scale, distributed environments.

LLM and AI Agent MCP

MCP

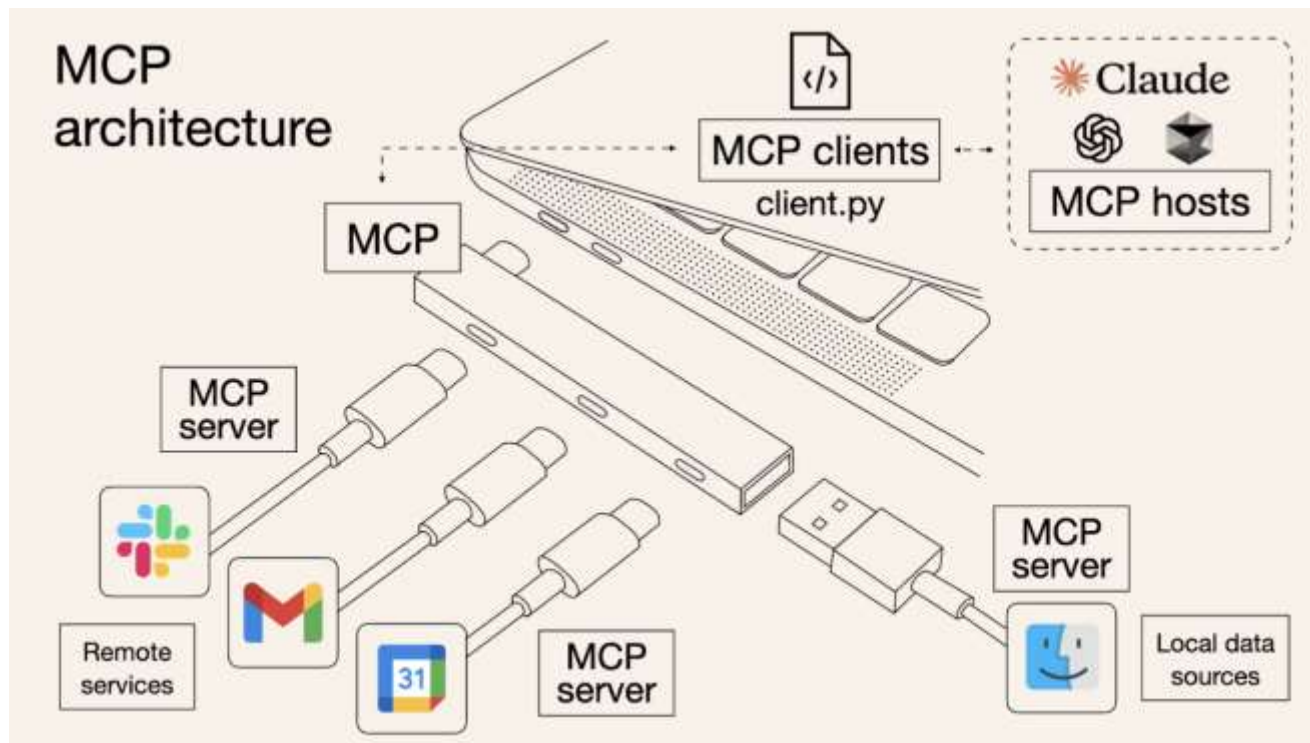
An open protocol that standardizes how applications provide context to LLM. Like a USB-C port, it serves as a unified interface for connecting various tools and data to LLM.

Claude Desktop: Download - Claude. Hosting. Currently supports Claude, Github Copilot, Cursor, OpenAI, and more.

Node.js — Run JavaScript Everywhere. Runs Node.JS-based server tools.

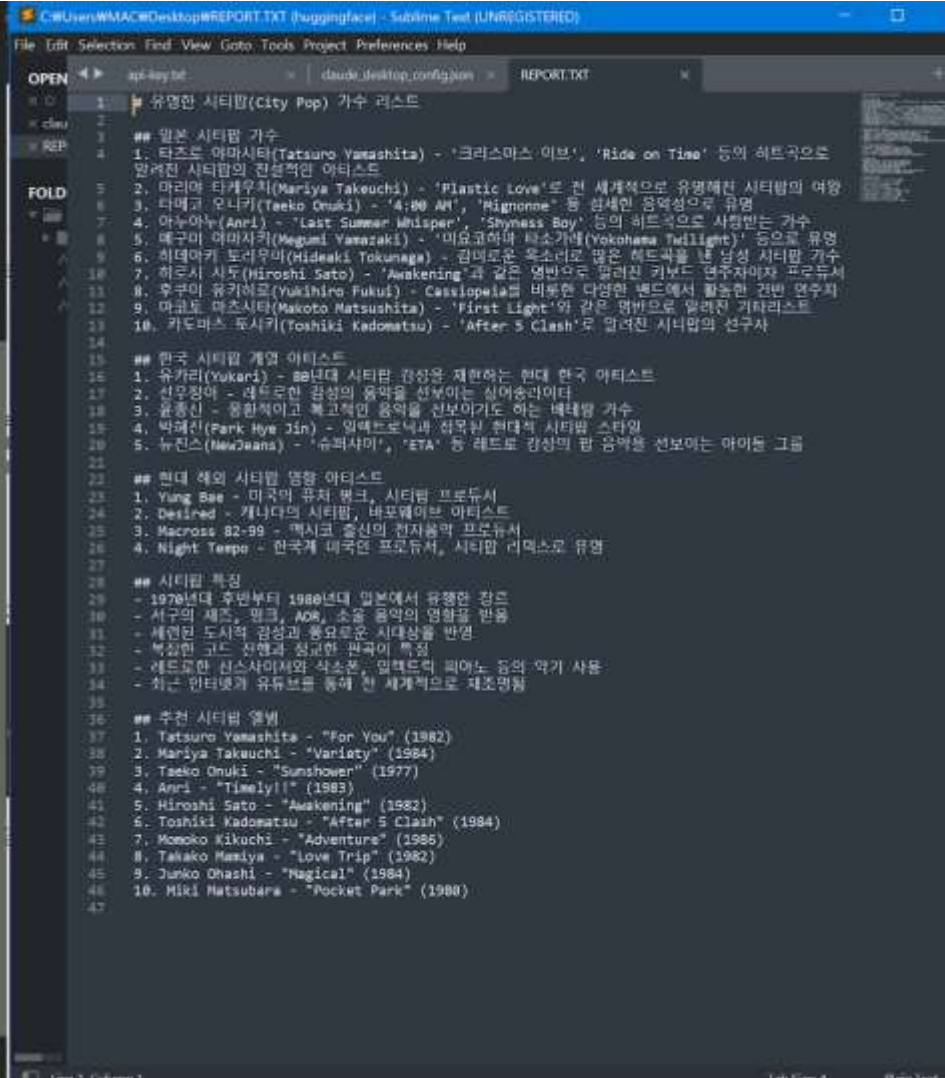
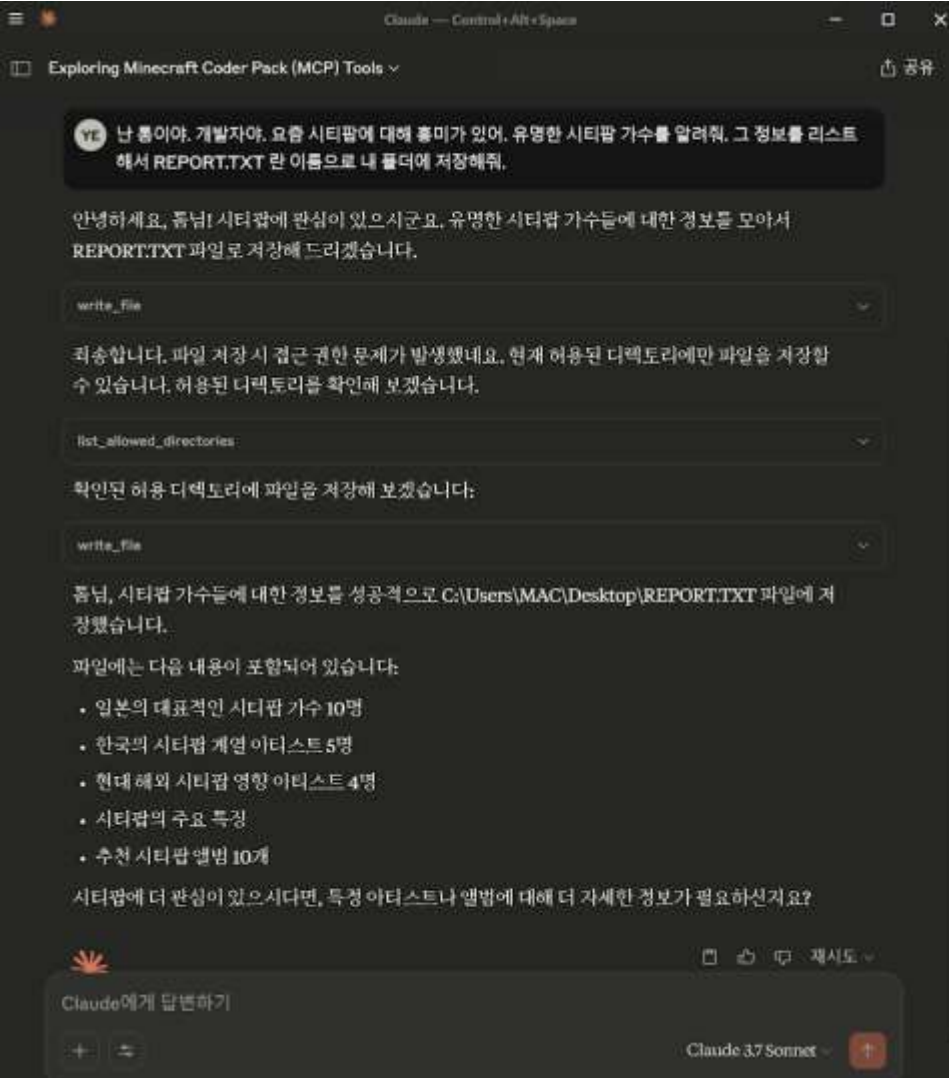
Python: Install a stable version (e.g., currently 3.11 or later, 3.12 or earlier recommended).

Install the latest Python packages. UV Tool: UV. An extremely fast Python package and project manager, written in Rust.



Link: [인공지능 AI 에이전트 표준 프로토콜 MCP 개념, 사용, 개발 및 동작 구조 분석하기](#)

LLM and AI Agent MCP



Link: 인공지능 AI 에이전트 표준 프로토콜 MCP 개념, 사용, 개발 및 동작 구조 분석하기

LLM and AI Agent MCP

The screenshot displays a development environment with the following components:

- Code Editor:** Shows the `mcp.json` file with the following configuration:

```
1 {
2   "inputs": {
3     // The "inputs" section defines the input
4     "type": "promptString"
5   },
6   "servers": {
7     // The "servers" section defines the MCP
8     // Start 1 cached tools
9     "fetch": {
10      "command": "uvx",
11      "args": ["mcp-server-fetch"]
12    },
13    "blender": {
14      // Running | Stop | Restart | 13 tools
15      "command": "uvx",
16      "args": [
17        "blender-mcp"
18      ]
19    }
20  }
21 }
```
- Chat Window:** Displays a conversation with the LLM (GitHub Copilot). The prompt is: "블렌더에 1층 집을 만들어줘. 각 벽에 창문이 있어야 해. 벽 하나는 문을 만들어줘." (Create a 1-story house in Blender. Each wall must have a window. One wall must have a door). The response is: "1층 집이 Blender에 성공적으로 생성되었습니다. 각 벽에는 창문이 있으며, 한 벽에는 문이 포함되어 있습니다. Blender에서 결과를 확인해 보세요! 추가 요청이 있으면 알려주세요." (A 1-story house has been successfully generated in Blender. Each wall has a window, and one wall includes a door. Check the results in Blender! Let me know if you have any further requests). The user's follow-up is: "지붕도 만들어줘." (Create the roof too).
- 3D Viewport:** Shows a 3D model of a house with a roof, generated in Blender.
- Terminal:** Displays the output of the MCP server, showing a successful response from the Blender MCP server.

Link: 인공지능 AI 에이전트 표준 프로토콜 MCP 개념, 사용, 개발 및 동작 구조 분석하기

LLM and AI Agent MCP

https://n8n.io

n8n Product Use cases Docs Community Enterprise Pricing GitHub 87

IT Ops can

- ⚡ On-board new employees and provision accounts

Sec Ops can

- ⚡ Enrich security incident tickets

Dev Ops can

- ⚡ Convert natural language commands into API calls

Sales can

- ⚡ Generate customer insights from grouped reviews

AI Workflow Automation Platform & Tools - n8n

The diagram illustrates an AI Agent workflow within the n8n platform. The workflow begins with a trigger node labeled 'On 'Create User' form submission'. This triggers an 'AI Agent Tools Agent' node. Below this agent node, four tools are listed: 'AI Anthropic Chat Model', 'Postgres Chat Memory', 'Microsoft Entra ID', and 'Jira Software'. Dashed lines connect these tools to the 'AI Agent Tools Agent' node. The workflow then proceeds to a decision node labeled 'is manager?'. From this node, two paths emerge: one labeled 'true' leading to an 'Add to channel' node (with the action 'invite: channel'), and another labeled 'false' leading to an 'Update profile' node (with the action 'updateProfile: user'). Both final nodes have a plus sign icon, indicating they can be expanded or configured further.

On 'Create User' form submission

AI Agent Tools Agent

AI Anthropic Chat Model

Postgres Chat Memory

Microsoft Entra ID
getAll: user

Jira Software
create: user

is manager?

Add to channel
invite: channel

Update profile
updateProfile: user

Link: 인공지능 AI 에이전트 표준 프로토콜 MCP 개념, 사용, 개발 및 동작 구조 분석하기

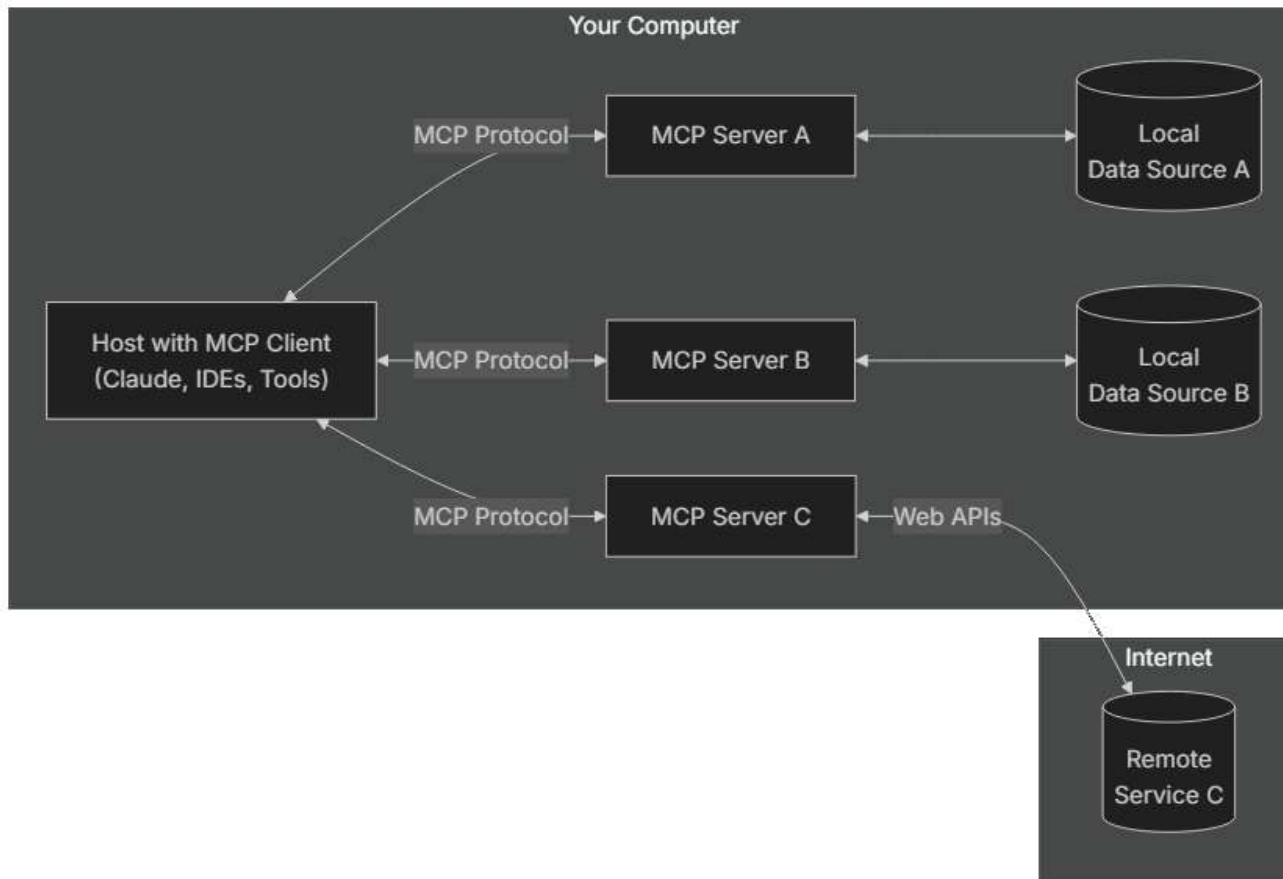
LLM and AI Agent MCP

MCP Host: An integrated environment for MCP clients, such as Claude Desktop, AI Agent, and Cursor IDE.

MCP Server: A program that exposes functionality and handles tool registration and execution. It operates in various runtime environments, including Python and Node.js.

Local Data Sources: Local resources such as the file system and database.

Remote Services: API-based external services such as GitHub and Brave Search.



LLM and AI Agent Gen AI& Vibe Coding

Generative AI (Gen AI)



Parent



Prompt

A painting for a house designed by Frank Lloyd Wright and the painting is made by Van Gogh using oil painting rough strokes warm colors dominant color is yellow --ar 16:9 --v 6.1

Description

This painting showcases a modernist house with bold yellow autumn trees in the background, emphasizing the blend of architecture and nature.

Details

Promptalot

LLM and AI Agent

Gen AI& Vibe Coding

OpenArt

Description: Provides the ability to generate images from text, style transfer, train custom models, and convert photos to videos.

Link: <https://openart.ai/>

Runway ML

Description: An AI content creation platform that supports text-based video generation and editing.

Link: <https://app.runwayml.com/>

Sora (OpenAI)

Description: OpenAI's experimental model that generates high-quality videos based on text descriptions.

Link: <https://sora.chatgpt.com/explore/videos>

Suno AI (formerly Mureka AI)

Description: A generative AI service that generates music from text prompts.

Link: <https://www.mureka.ai/>

Kling AI

Description: A high-performance video generation AI capable of character movement and scene simulation.

Link: <https://klingai.com/global/>

LLM and AI Agent Gen AI& Vibe Coding

Vibe Coding

A coding method focused on creating or reflecting a specific mood or feeling (vibe).

How to complete a program by copying and pasting Python-like code generated by requesting coding from ChatGPT.

How to create project files and source code using Gemini CLI, Claude Code CLI, and Codex CLI tools.

How to code in Vibe using tools like GitHub Copilot, Cursor, and Windsurf that are integrated with a development IDE like vscode.

How to create, build, and run an app in the cloud by entering requirements directly in Vibe Coding web services like Bubble.io or Canva.



Get started with GitHub Copilot in VS Code



Cursor - The AI Code Editor



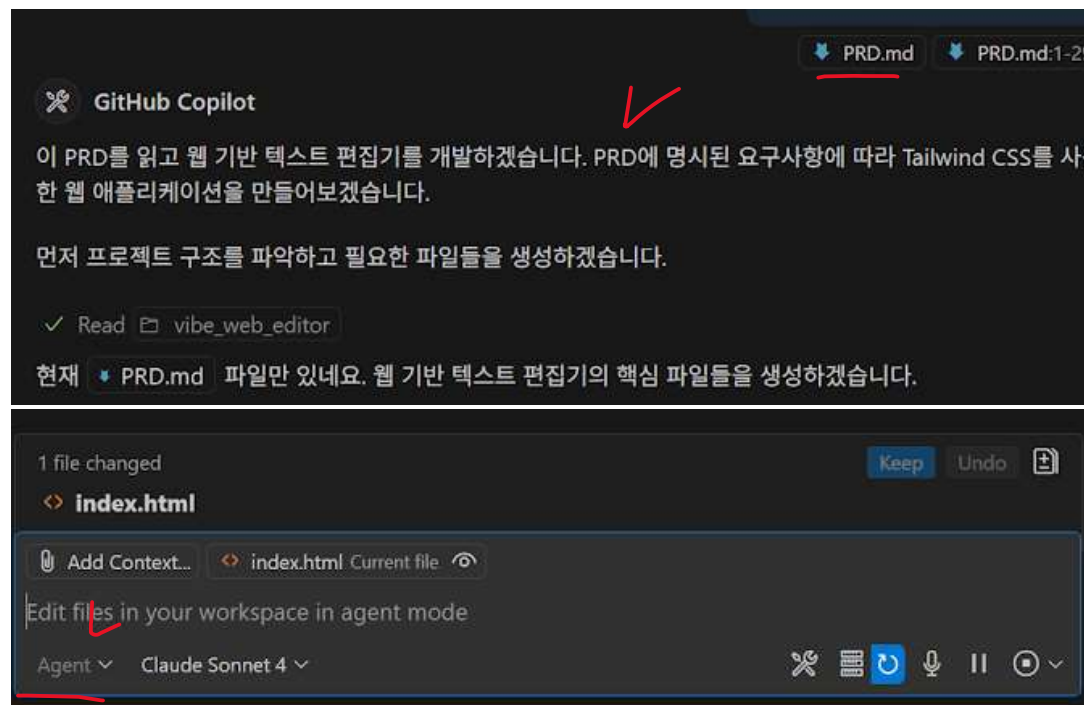
Claude Code: Deep coding at terminal velocity W Anthropic

LLM and AI Agent

Gen AI& Vibe Coding

I'm developing a GUI-based text editor. I'll use Python and Tkinter libraries. The menu will consist of the File menu, the Edit menu, the Theme selection menu (including a Dark Mode submenu), and the Find menu. For this program, I'll write a PRD in md format, including UI sketches.

It's important to specify the technology stack, such as the libraries you'll be using. Use trial and error to ensure a clear PRD is created. Include accurate information in the prompts. Use the md format for more clear documentation. The md format is a Markdown file format, and is the standard format for writing README files on GitHub.



LLM and AI Agent

Gen AI& Vibe Coding

Python/Tkinter-based GUI Text Editor Product Requirements Specification (PRD)

1. Overview

This document defines the requirements for a desktop text editor developed using Python and Tkinter, a standard GUI library. This application aims to provide basic text editing functionality, along with features such as a dark mode and search functionality for the user's convenience.

2. Target Users

- * **Beginner Developers and Students**: Users who want to write and edit simple code, scripts, notes, assignments, etc.
- * **General Users**: Users who need a lightweight and fast text editor similar to Windows Notepad.

3. Key Features

**3.1. File Menu

Function	Shortcut Keys	Description
---	---	---
New File	`Ctrl+N`	Opens a new, blank editing window.
Open	`Ctrl+O`	Loads an existing text file (e.g., `.txt`).
Save	`Ctrl+S`	Saves, overwriting the current file.
Save As	`Ctrl+Shift+S`	Saves the file with a new name or path.
Exit	`Alt+F4`	Exits the program. (Warning if unsaved data exists.)

**3.2. Edit Menu

Function	Shortcuts	Description
---	---	---
Undo	`Ctrl+Z`	Reverses the last action.
Redo	`Ctrl+Y`	Redoes the undone action.
Cut	`Ctrl+X`	Cuts the selected text and saves it to the clipboard.

**5. UI Sketch

This is a sketch of the overall layout of the application.

```
...
+-----+
| My Text Editor                                     - □ X |
+-----+
| File (F) | Edit (E) | Theme (T) | Find (S)          |
+-----+
| | ...
| |
#####
| | # ..... * **Text Area**: The main scrollable text editing area.
* **Status Bar**: At the bottom, displays simple status information, such as the
cursor's current position (line, column) and encoding.
```


LLM and AI Agent

Gen AI& Vibe Coding

The screenshot shows the Gemini AI web interface. At the top, the URL is <https://gemini.google.com/app/2fd4aa8c43d746b8>. The Gemini logo and '2.5 Pro' model name are visible. A prompt in a blue box asks for a GUI-based text editor using Python and Tkinter, including a PRD in md format. The response, marked with a red star icon, confirms the request and provides a detailed PRD for a 'Python/Tkinter 기반 GUI 텍스트 편집기' (Python/Tkinter-based GUI Text Editor). The PRD includes a '1. 개요' (Overview) section explaining the document's purpose and the application's features. At the bottom, there is a search bar 'Gemini에게 물어보기' and navigation links for '+', '동영상', 'Deep Research', and 'Canvas'.

GUI 기반 텍스트 편집기를 개발할꺼야. 파이썬과 Tkinter 라이브러릴 이용할꺼야. 메뉴는 파일 메뉴, 편집 메뉴, 테마 선택 메뉴(다크모드 하위 메뉴 포함), 찾기 메뉴로 구성됨. 이 프로그램 개발을 위한 PRD를 UI 스케치를 포함해 md 포맷으로 작성해.

생각하는 과정 표시

네, 파이썬과 Tkinter를 사용한 GUI 텍스트 편집기 개발을 위한 제품 요구사항 명세서(PRD)를 UI 스케치를 포함하여 작성해 드릴게요.

Python/Tkinter 기반 GUI 텍스트 편집기 제품 요구사항 명세서 (PRD)

1. 개요

본 문서는 파이썬(Python)과 표준 GUI 라이브러리인 티케이인터(Tkinter)를 사용하여 개발될 데스크톱 텍스트 편집기의 요구사항을 정의합니다. 이 애플리케이션은 기본적인 텍스트 편집 기능과 함께 사용자의 편의를

Gemini에게 물어보기

+ 동영상 Deep Research Canvas

Gemini는 인물 등에 관한 정보 제공 시 실수를 할 수 있으니 다시 한번 확인하세요. 개인 정보 보호 및 Gemini

LLM and AI Agent Gen AI& Vibe Coding

Gemini CLI

Install: <https://nodejs.org/en/download> <https://github.com/google-gemini/gemini-cli>

Key: <https://aistudio.google.com/apikey>



LLM and AI Agent

Gen AI& Vibe Coding

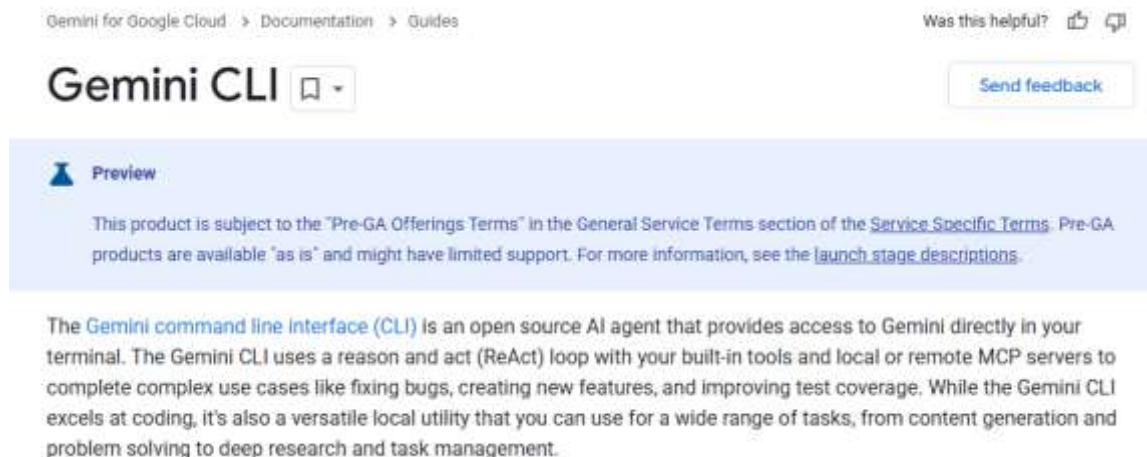
Gemini CLI

Commands: <https://github.com/google-gemini/gemini-cli/blob/main/docs/cli/commands.md>

MCP Extension: <https://github.com/google-gemini/gemini-cli/blob/main/docs/extension.md>

Base Tools: <https://github.com/google-gemini/gemini-cli/blob/main/docs/tools/index.md>

Pair Programming Method: <https://cloud.google.com/gemini/docs/codeassist/use-agentic-chat-pair-programmer>



gemini --help

-m: gemini model 지정. gemini -m "gemini-1.5-flash"

-p: 단일 프롬프트 실행. gemini -p "three.js 의 주요 예제 코드를 알려줘 "

-y: 사용자 확인 없이 자동 실행(yolo 모드)

-d: 디버그 옵션. 작업폴더, GEMINI.md 파일 유무, 모델 호출 상세 등 확인 가능

--sandbox: 격리된 환경에서 실행

--show_memory_usage: CLI 메모리 사용량 표시

gemini --help

-m: Specify the gemini model. gemini -m "gemini-1.5-flash"

-p: Run a single prompt. gemini -p "Show me the main example code for three.js"

-y: Run silently without user confirmation (yolo mode)

-d: Debug option. Check the working folder, the presence of the GEMINI.md file, model call details, etc.

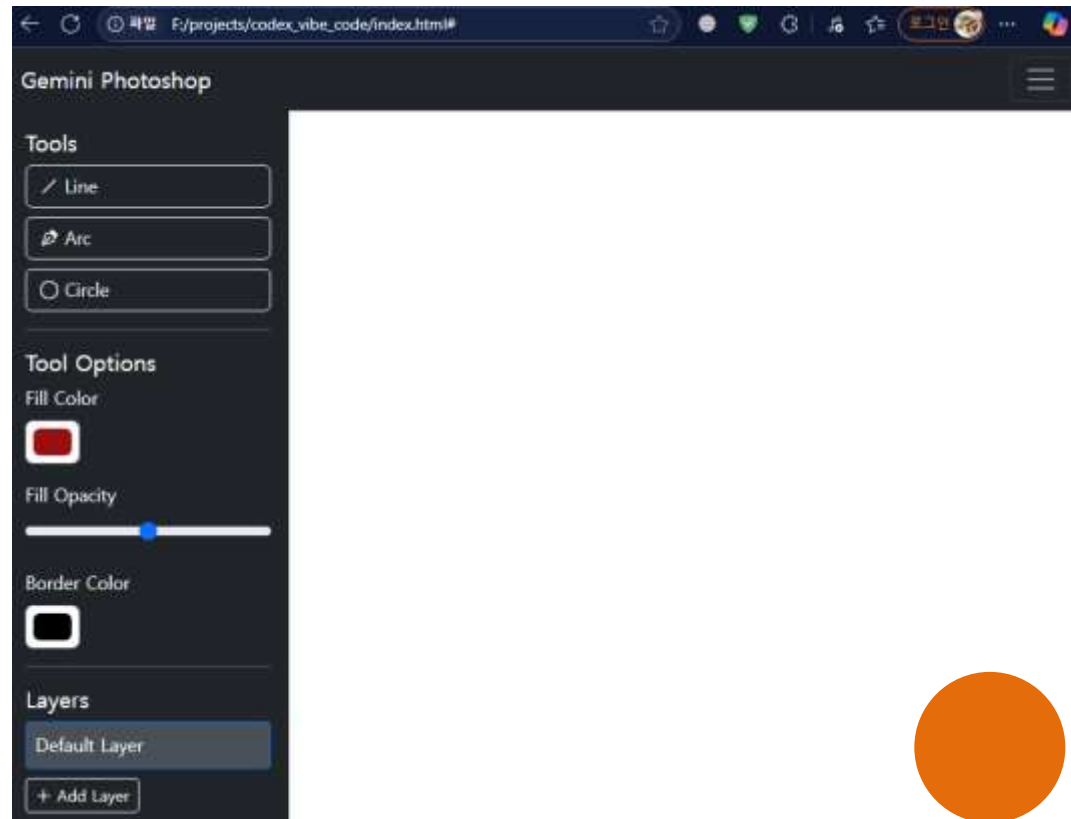
--sandbox: Run in an isolated environment

--show_memory_usage: Shows CLI memory usage

LLM and AI Agent

Gen AI& Vibe Coding

> make photoshop web app using three.js, bootstrap. Menus includes layer, line, arc, circle, fill color with transparent, border color, zoom in/out, pan, download file as JPG



LLM and AI Agent Gen AI& Vibe Coding

Codex

Vibe Coding Tool developed by OpenAI (released April 17th). Supports multi-AI agent functionality. Supports local CLI mode.

It goes beyond simple code generation and automates complex development tasks such as bug fixes, test execution, and code review suggestions.

Installation: openai/codex: Lightweight coding agent that runs in your terminal.

OpenAI o3 & o4-mini

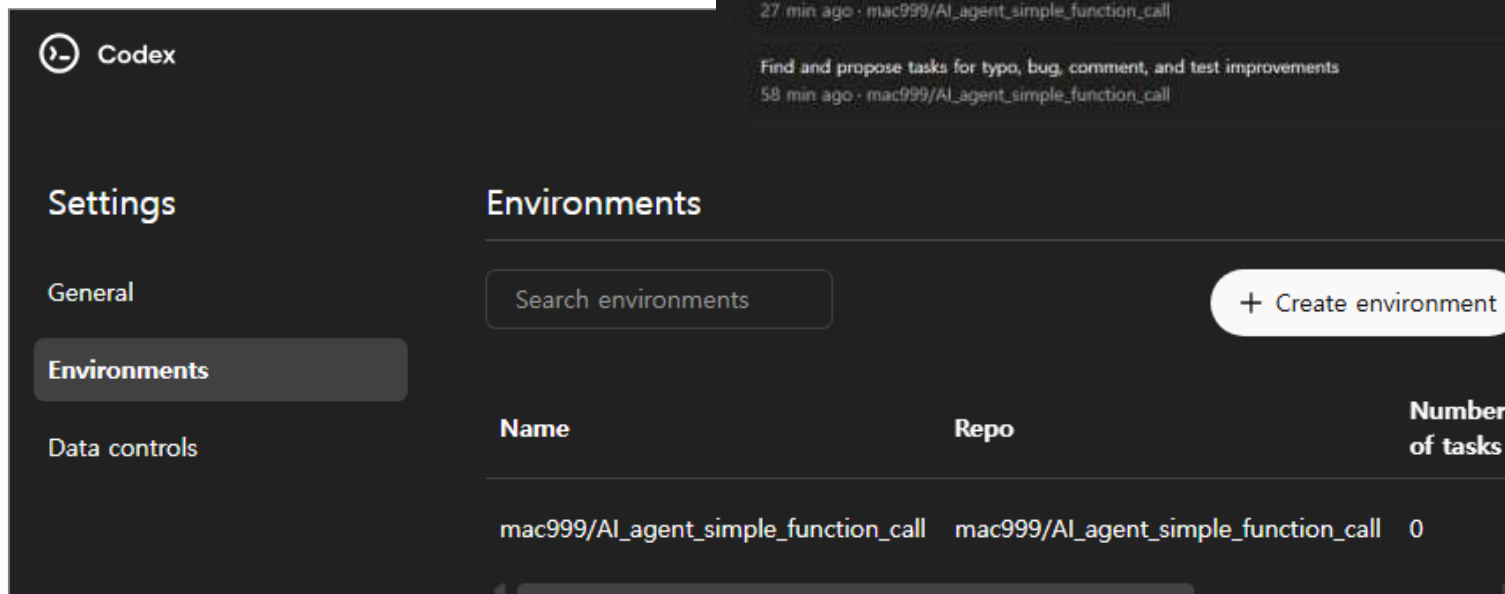
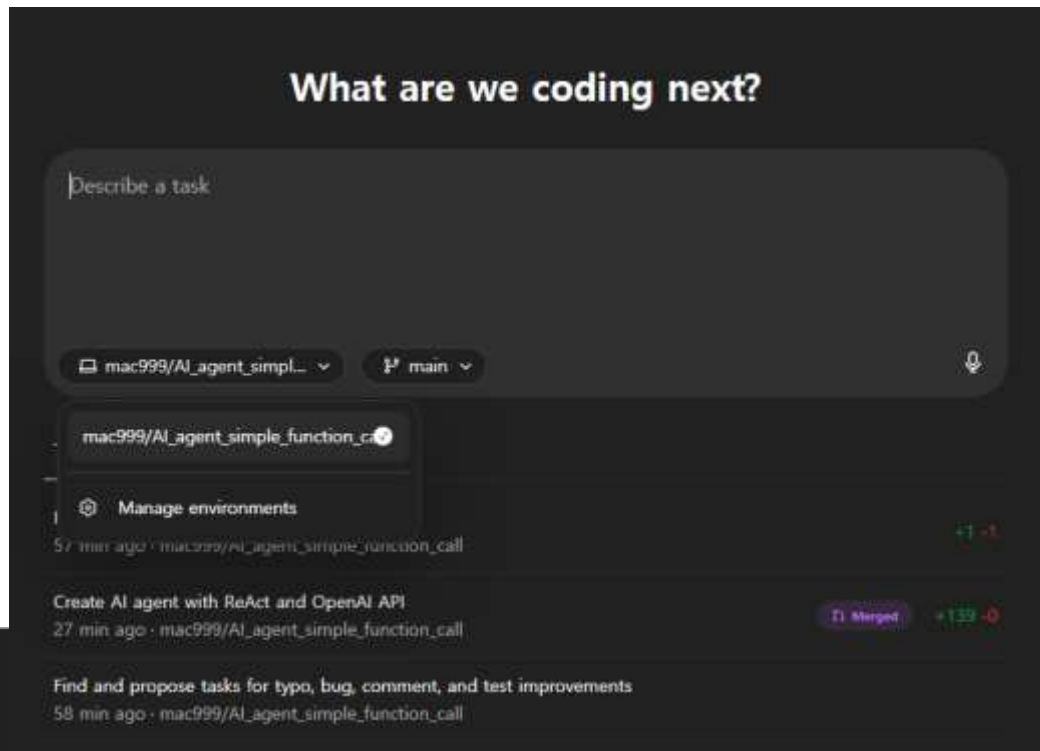
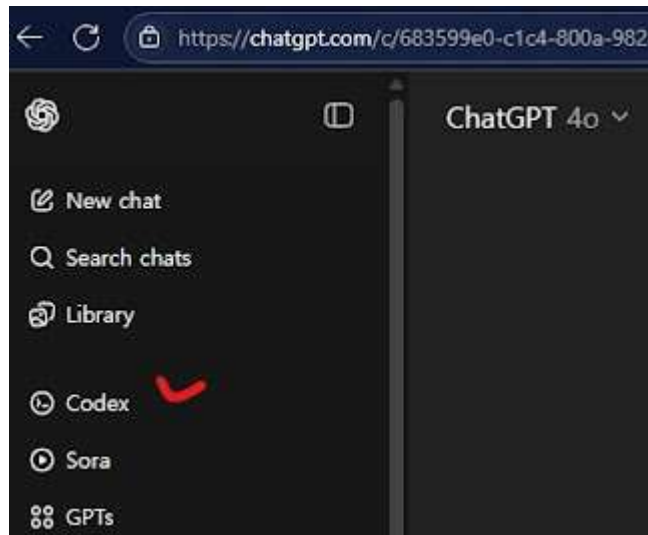


[Link: OpenAI 라이브 코딩 지원 멀티 에이전트 Codex 도구 사용법](#)

LLM and AI Agent

Gen AI& Vibe Coding

Codex



Link: [OpenAI 바이브 코딩 지원 멀티 에이전트 Codex 도구 사용법](#)



make AI agent simple source code using python to implement ReAct, with openai api. it supports CLI for singeline query mode, chatbot mode.

Worked for 1m 34s >

Notes

- Added `react_agent.py` containing a minimal ReAct-style loop that uses OpenAI function calling for searching and calculations.
- Updated `README.md` with instructions on how to run the new CLI tool.

Summary

- Documented the usage of the new ReAct CLI in the README, including single-query and chatbot modes ⁽ⁱ⁾
- Implemented `react_agent.py` to provide search and calculator tools and handle interactive or one-shot prompts ⁽ⁱ⁾

Testing

-  `pytest -q` (no tests were found) >

Files (2)

README.md

+19 -0

react_agent.py

New



Request changes or ask a question



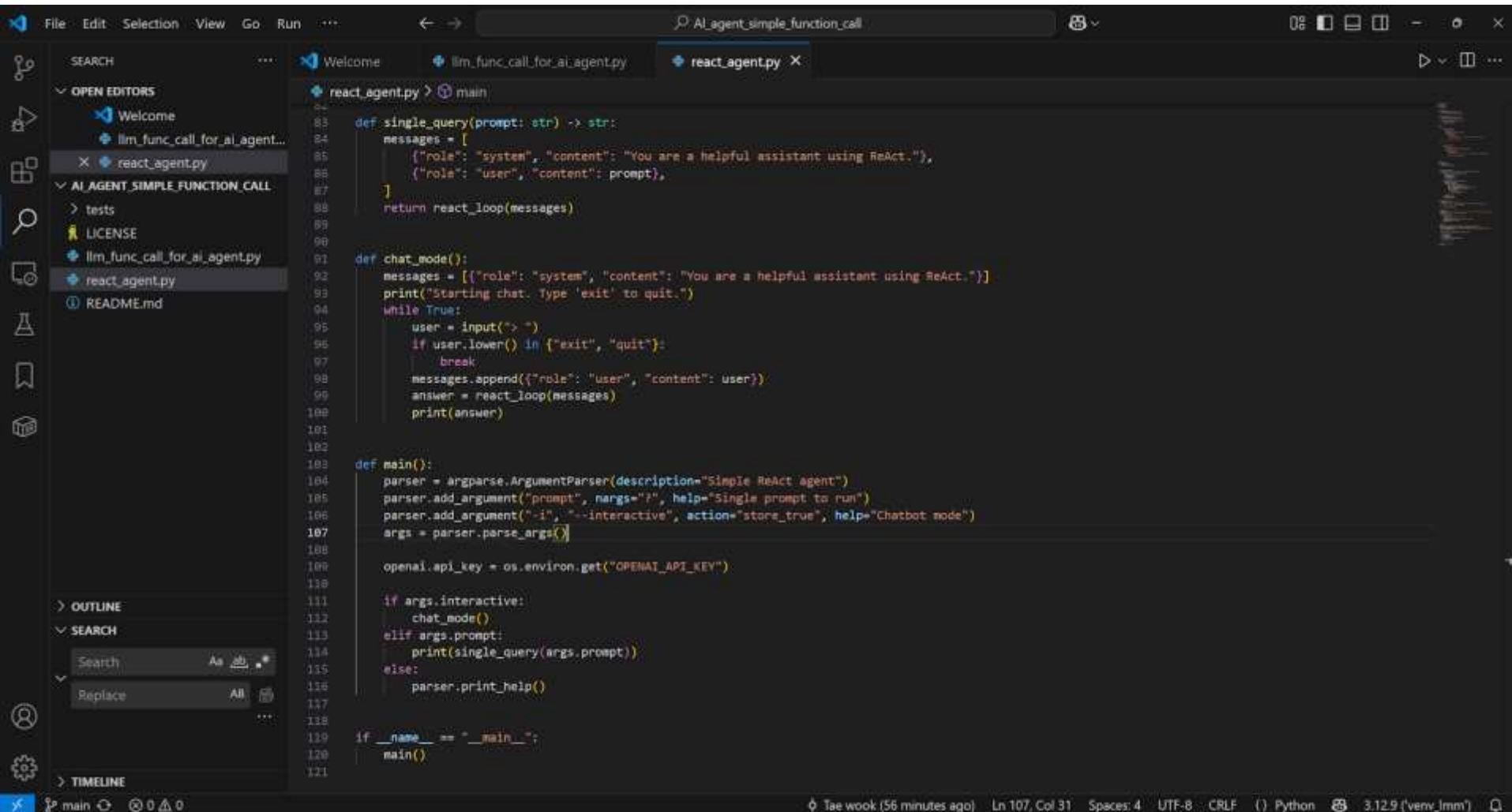
Diff Logs

▼ react_agent.py New

```
54 + }
55 +
56 +
57 + def react_loop(messages):
58 +     """Run the ReAct reasoning loop until a final answer is produced."""
59 +     while True:
60 +         response = openai.ChatCompletion.create(
61 +             model="gpt-4o",
62 +             messages=messages,
63 +             tools=tools,
64 +             tool_choice="auto",
65 +         )
66 +         msg = response["choices"][0]["message"]
67 +         if msg.get("tool_calls"):
68 +             messages.append(msg)
69 +             for call in msg["tool_calls"]:
70 +                 name = call["function"]["name"]
71 +                 args = json.loads(call["function"]["arguments"])
72 +                 result = FUNCTION_MAP[name](**args)
73 +                 messages.append({
74 +                     "role": "tool",
75 +                     "tool_call_id": call["id"],
76 +                     "content": result,
77 +                 })
78 +                 continue
79 +             messages.append(msg)
80 +             return msg["content"]
81 +
82 +
83 + def single_query(prompt: str) -> str:
84 +     messages = [
85 +         {"role": "system", "content": "You are a helpful assistant using ReAct."},
86 +         {"role": "user", "content": prompt},
87 +     ]
88 +     return react_loop(messages)
89 +
90 +
91 + def chat_mode():
92 +     messages = [{"role": "system", "content": "You are a helpful assistant using ReAct."}]
93 +     print("Starting chat. Type 'exit' to quit.")
94 +     while True:
95 +         user = input("> ")
96 +         if user.lower() in {"exit", "quit"}:
```

LLM and AI Agent Gen AI& Vibe Coding

Codex



```
def single_query(prompt: str) -> str:
    messages = [
        {"role": "system", "content": "You are a helpful assistant using ReAct."},
        {"role": "user", "content": prompt},
    ]
    return react_loop(messages)

def chat_mode():
    messages = [{"role": "system", "content": "You are a helpful assistant using ReAct."}]
    print("Starting chat. Type 'exit' to quit.")
    while True:
        user = input("> ")
        if user.lower() in {"exit", "quit"}:
            break
        messages.append({"role": "user", "content": user})
        answer = react_loop(messages)
        print(answer)

def main():
    parser = argparse.ArgumentParser(description="Simple ReAct agent")
    parser.add_argument("prompt", nargs="?", help="Single prompt to run")
    parser.add_argument("-i", "--interactive", action="store_true", help="Chatbot mode")
    args = parser.parse_args()

    openai.api_key = os.environ.get("OPENAI_API_KEY")

    if args.interactive:
        chat_mode()
    elif args.prompt:
        print(single_query(args.prompt))
    else:
        parser.print_help()

if __name__ == "__main__":
    main()
```

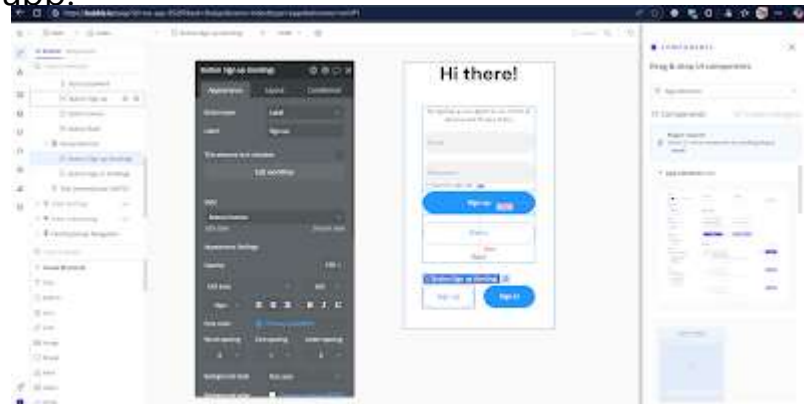
[Link: OpenAI 바이브 코딩 지원 멀티 에이전트 Codex 도구 사용법](#)

LLM and AI Agent

Gen AI& Vibe Coding

No-code

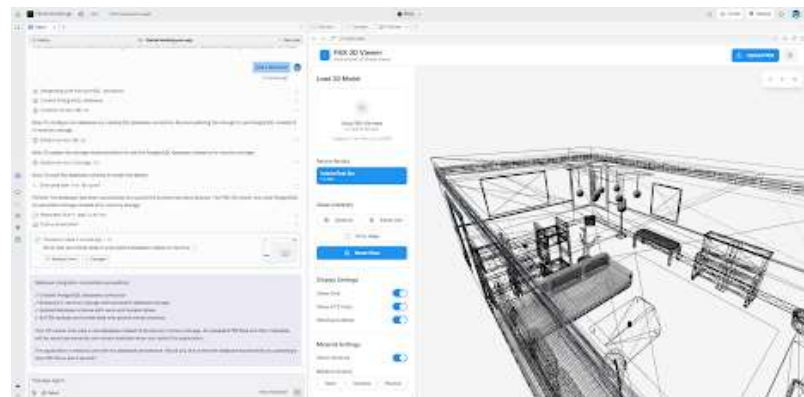
Bubble.io: Provides high flexibility and functionality to create almost any type of web app.



Nocodefusion: A tool focused on rapid development of AI-based apps by combining generative AI technology and a no-code platform.



Replit: An IDE that allows you to conveniently write, run, and share code on the web.



Prompt: create FBX 3D viewer web app using django, bootstrap, threejs. the app includes view menu such as zoom in/out, extent, rotate. the app's canvas renders grid, xyz mark defaultly

LLM and AI Agent

Example LLM agent vibe

1 finetuning/finetune-bert-cot-train.py
1 finetuning/finetune-bert-train-dialog.py
1 finetuning/finetune-gemma.ipynb

2 langchain/lang1-prompt-template.ipynb
2 langchain/lang4-func.ipynb
2 langchain/lang5-agent.ipynb
2 langchain/lang6-chain.ipynb

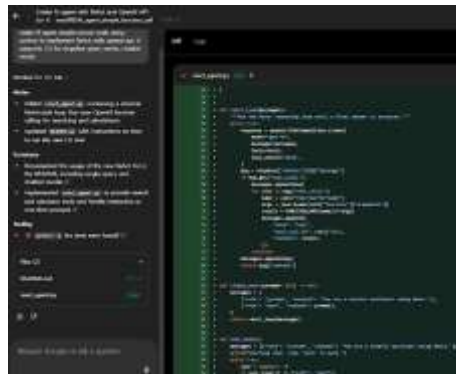
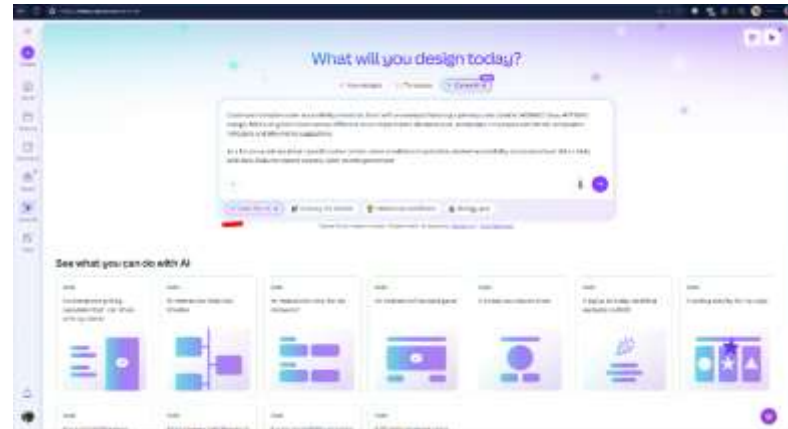
3 db/rag_db_pdf.ipynb
4 chatbot/gradio-pdf-web-rag-chatbot.ipynb
4 chatbot/streamlit-pdf-web-rag-chatbot.py
6 agent/lang-call-tools-news-python.py



LLM and AI Agent

Example LLM agent vibe

8 Gen AI
9 vibe-coding
vibe coding with Gemini CLI, codex



Overview of AI Model Servicing

Model Training vs. Serving

Introduction to API-Based AI Service Architecture

Overview of Lightweighting and Deployment Strategies,
Including ONNX, FastAPI, and Flask

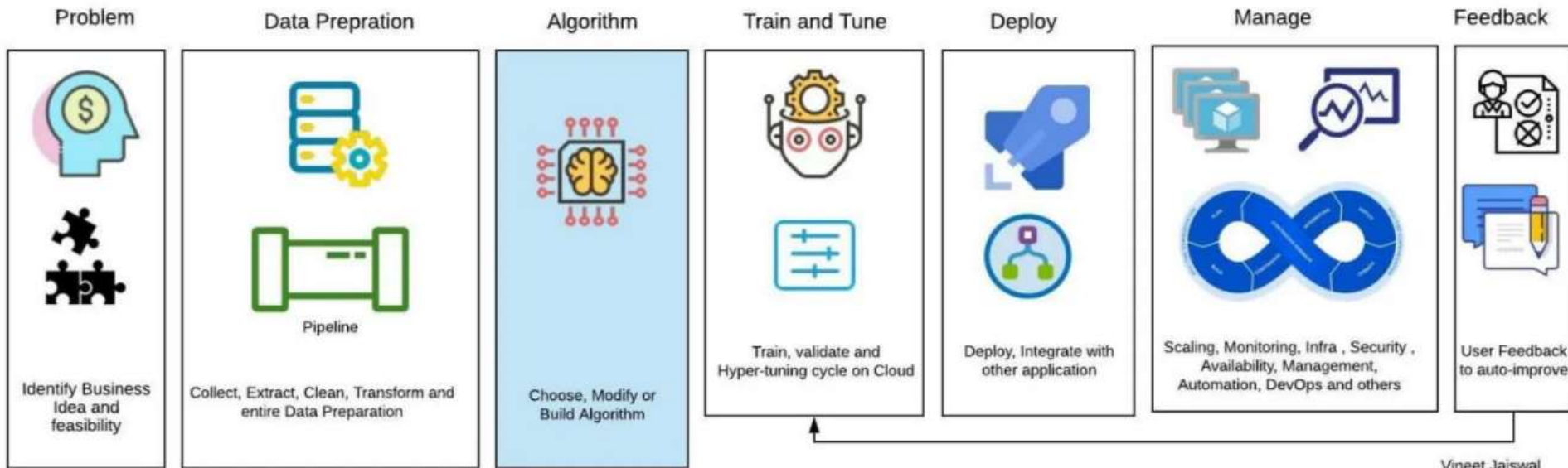
Service

Key Question

How do we serve learning models?
How do we deploy learning models effectively?
What is MLOps?

AI model service Train vs Serving

Machine Learning Lifecycle



Vineet Jaiswal

AI model service Train vs Serving

Model Serving Runtimes



BentoML



Torch Serve



TensorFlow Serving



NVIDIA Triton
Inference Server

Serving Platforms



Amazon
SageMaker



Vertex AI



Azure ML



BentoCloud

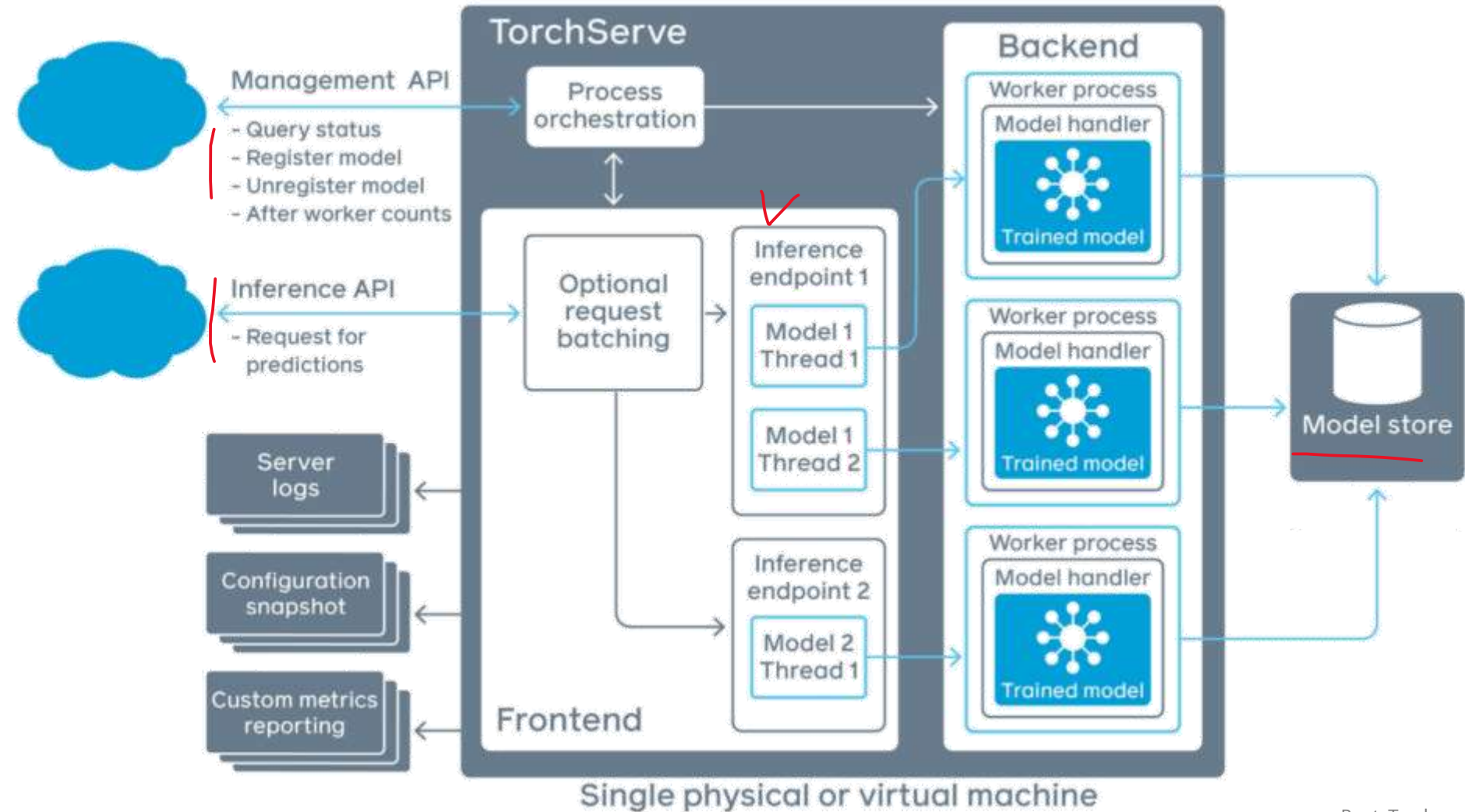


Seldon Core

Titan^{ML}

AI model service Train vs Serving

ML Serving Example



AI model service

AI service Architecture

LLM engineer toolkit

llm-engineer-toolkit Public

Watch 68

Fork 702

Star 4.2k

main

1 Branch

0 Tags

Go to file

Add file

Code

KalyanKS-NLP

Merge pull request #11 from dylanbouchard/main

2538144 · last month

17 Commits

LICENSE

Initial commit

4 months ago

README.md

ADD UQLM

2 months ago

README

Apache-2.0 license

LLM Engineer Toolkit

This repository contains a curated list of 120+ LLM libraries category wise.

Kalyan KS

Kalyan KS

Kalyan KS

Quick links

[LLM Training](#)

[LLM Application Development](#)

[LLM RA](#)

[LLM Inference](#)

[LLM Serving](#)

[LLM D](#)

[LLM Data Generation](#)

[LLM Agents](#)

[LLM E](#)

[LLM Monitoring](#)

[LLM Prompts](#)

[LLM Structured Outputs](#)

[LLM Safety and Security](#)

[LLM Embedding Models](#)

[Others](#)

About

A curated list of 120+ LLM libraries category wise.

[www.linkedin.com/in/kalyanksnlp/](#)

ai-engineer

large-language-models

llms

generative-ai

llm-engineer

Readme

Apache-2.0 license

Activity

4.2k stars

68 watching

702 forks

Report repository

LLM Training and Fine-Tuning

Library	Description	Link
unsloth	Fine-tune LLMs faster with less memory.	Link
PEFT	State-of-the-art Parameter-Efficient Fine-Tuning library.	Link
TRL	Train transformer language models with reinforcement learning.	Link
Transformers	Transformers provides thousands of pretrained models to perform tasks on different modalities such as text, vision, and audio.	Link

Contributors 6

AI model service ONNX, FastAPI, Flask Lightweight Deployment

TensorFlow Lite

A suite of tools that enables on-device machine learning by enabling developers to run models on mobile, embedded, and IoT devices.

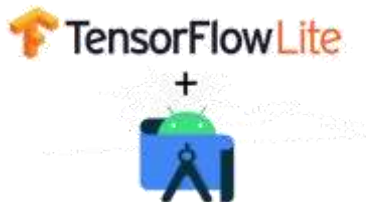


ARDUINO NANO 33 BLE SENSE



- Color, brightness, proximity and gesture sensor
- Digital microphone
- Motion, vibration and orientation sensor (9-axis IMU)
- Temperature, humidity and pressure sensor
- Arm Cortex-M4 microcontroller and BLE module

[Arduino Official Store](#) | [Boards Shields Kits Accessories](#)

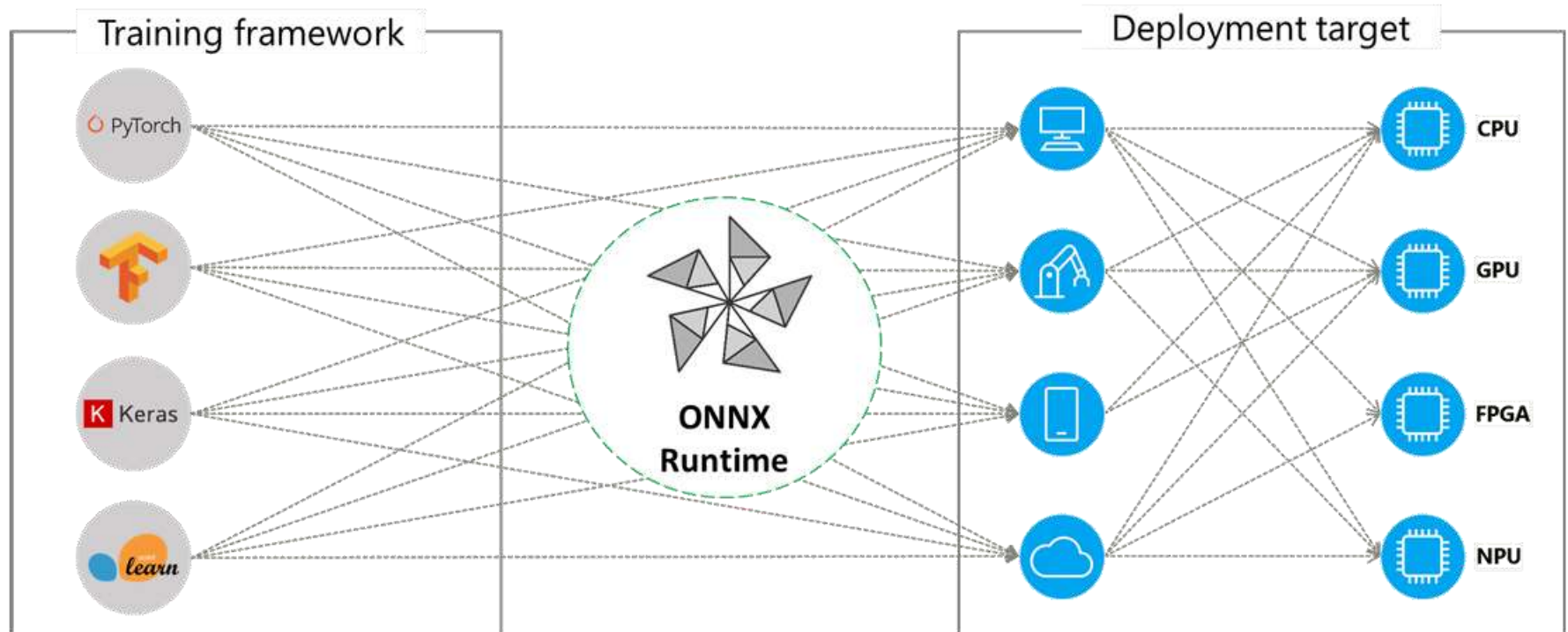


```
import tensorflow as tf
TFLITE_PATH = "./lite_model_cartoongan.tflite"
interpreter = tf.compat.v1.lite.Interpreter(model_path=TFLITE_PATH)
```


AI model service ONNX, FastAPI, Flask Lightweight Deployment

ONNX(Open Neural Network Exchange)

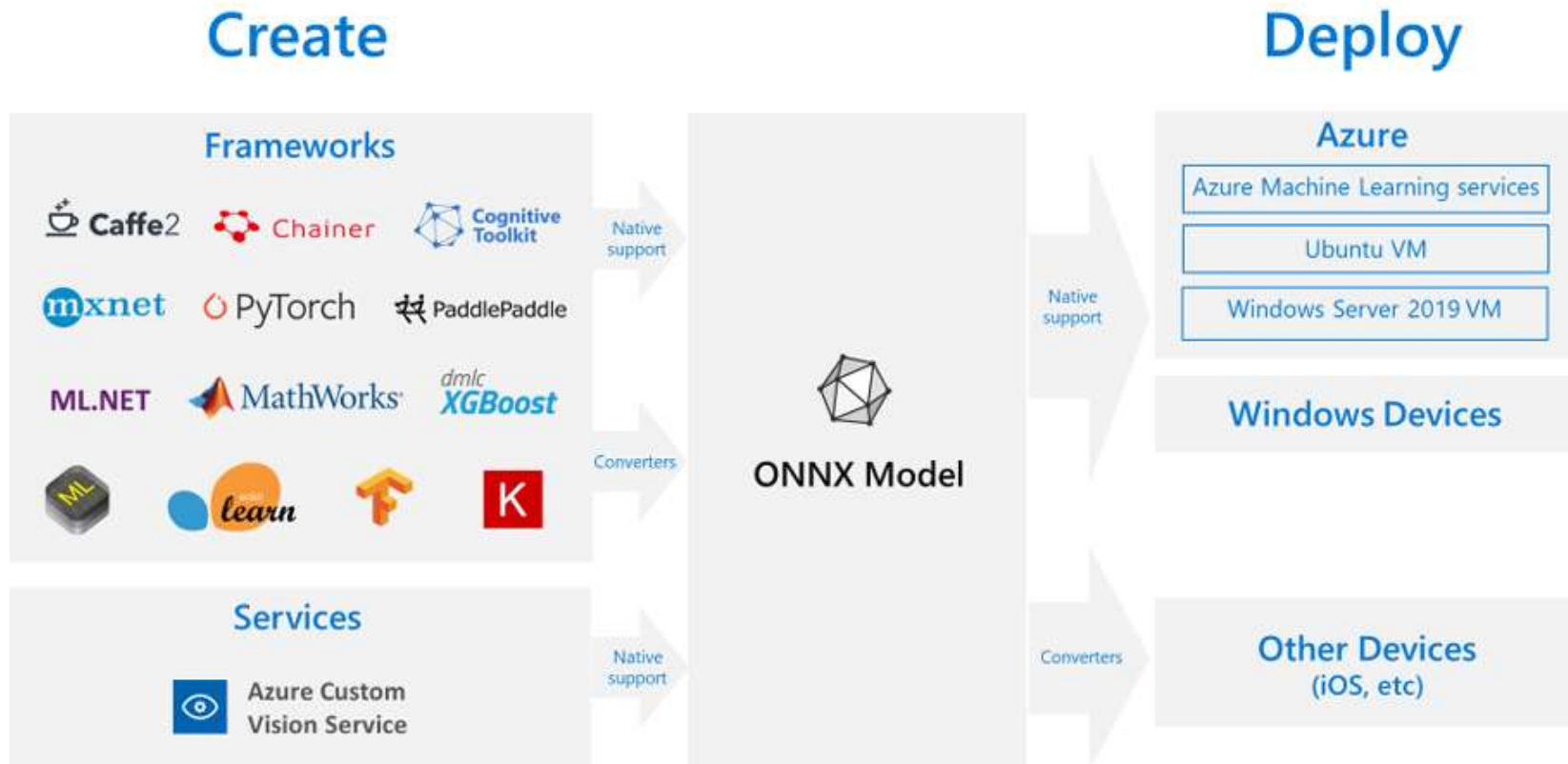
ONNX is a standardized deep learning model storage and management format. It provides standardized deep learning graph computation, input/output, and tensor formats. It provides model input/output compatibility across deep learning frameworks and various languages. It also offers compatibility with various TPUs as well as GPUs.



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ONNX(Open Neural Network Exchange)

pip install onnx onnxruntime-gpu



AI model service ONNX, FastAPI, Flask Lightweight Deployment

```
import os, torch, onnxruntime, numpy as np

# Defining PyTorch model
class MyModel(torch.nn.Module):
    def __init__(self):
        super(MyModel, self).__init__()
        self.fc1 = torch.nn.Linear(10, 512)
        self.fc2 = torch.nn.Linear(512, 1024)
        self.fc3 = torch.nn.Linear(1024, 512)
        self.fc4 = torch.nn.Linear(512, 256)
        self.fc5 = torch.nn.Linear(256, 10)

    def forward(self, x):
        x = torch.relu(self.fc1(x))
        x = torch.relu(self.fc2(x))
        x = torch.relu(self.fc3(x))
        x = torch.relu(self.fc4(x))
        x = self.fc5(x)
        return x

# Creating an instance
model = MyModel()

# Training data
X_train = torch.randn(1000, 10)
y_train = torch.randn(1000, 10)

# Training loop
optimizer = torch.optim.Adam(model.parameters(),
                               lr=0.01)
criterion = torch.nn.MSELoss()
```

```
for epoch in range(200):
    optimizer.zero_grad()
    outputs = model(X_train)
    loss = criterion(outputs, y_train)
    loss.backward()
    optimizer.step()

    if epoch % 20 == 0:
        print(f'Epoch {epoch}, Loss: {loss.item():.4f}')

# Defining input example
example_input = torch.randn(1, 10)

# Exporting to ONNX format
module_path =
os.path.dirname(os.path.abspath(__file__))
model_path = os.path.join(module_path, "model.onnx")
torch.onnx.export(model, example_input, model_path)

# Using ONNX model
session = onnxruntime.InferenceSession(model_path)
input_name = session.get_inputs()[0].name
onnx_output = session.run(None, {input_name:
example_input.numpy()})
```

```
Epoch 120, Loss: 0.9953
Epoch 140, Loss: 0.9953
-0.14417903 -0.27374357 0.04801298 -0.05633638]]
ONNX Output: [[-0.02958158 -0.07041439 0.16391063 -
-0.14417902 -0.27374354 0.04801297 -0.05633637]]
```

AI model service

ONNX, FastAPI, Flask Lightweight Deployment

```
# Quantizing the ONNX model
from onnxruntime.quantization import quantize_dynamic, QuantType

quantized_model_path = os.path.join(module_path, "model_quantized.onnx")
quantize_dynamic(model_path, quantized_model_path, weight_type=QuantType.QUInt8)

# Load the quantized model
quantized_session = onnxruntime.InferenceSession(quantized_model_path)
quantized_output = quantized_session.run(None, {input_name: example_input.numpy()})
print("Quantized ONNX Output:", quantized_output[0])

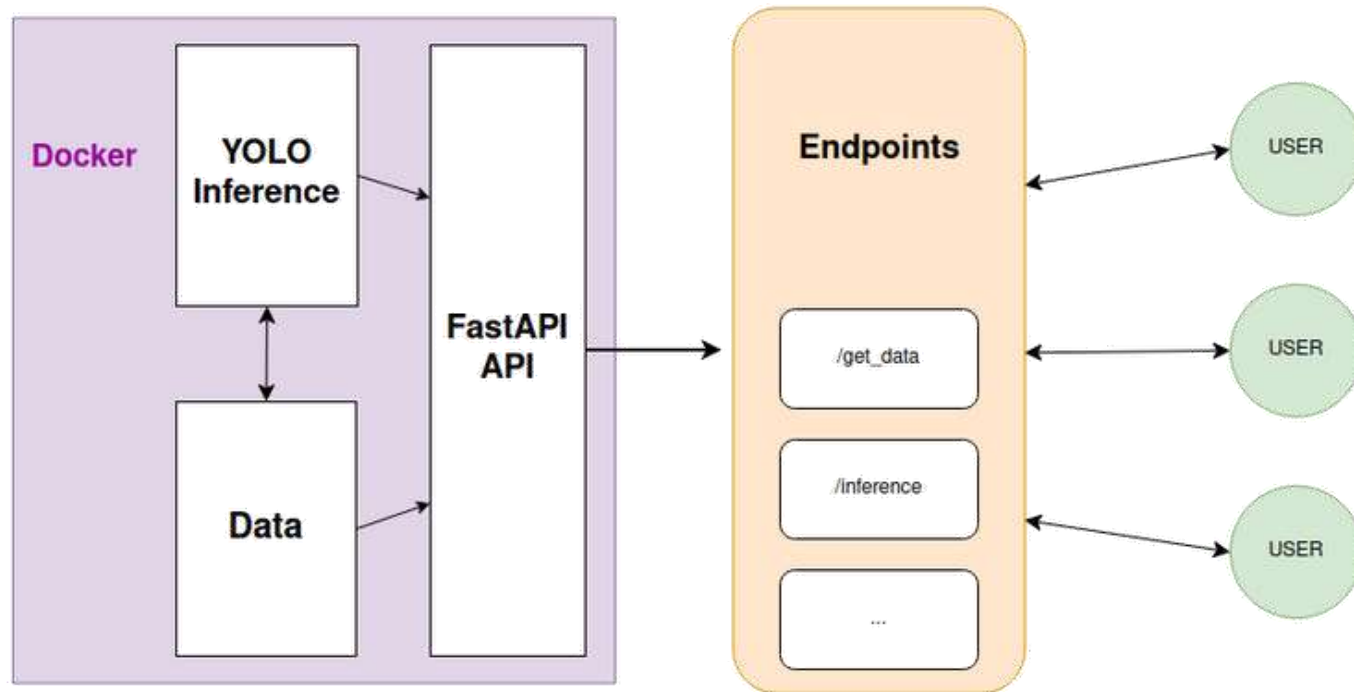
# Compare file sizes
original_size = os.path.getsize(model_path)
quantized_size = os.path.getsize(quantized_model_path)
print(f"Original model size: {original_size / 1024:.2f} KB")
print(f"Quantized model size: {quantized_size / 1024:.2f} KB")
print(f"Size reduction: {(1 - quantized_size/original_size)*100:.1f}%")
```

```
Quantized ONNX Output: [[ 0.7099464
 -0.5664326 -0.25450945 -1.1977834
Original model size: 4648.18 KB
Quantized model size: 1174.10 KB
Size reduction: 74.7%
```

AI model service ONNX, FastAPI, Flask Lightweight Deployment

FastAPI server with YOLO

FastAPI is a modern, fast (high-performance), web framework for building APIs with Python based on standard Python type hints.



Link: [FastAPI, Uvicorn, Websocket 기반 Open API 서버 개발](#)

[YOLOv8-docker/README.md at main · JavierMt5/YOLOv8-docker](#)

[YOLO inference with Docker via API | Towards Data Science](#)

AI model service

ONNX, FastAPI, Flask Lightweight Deployment

```
import os, json
from ultralytics import YOLO
from fastapi import FastAPI, UploadFile, File, Query
from typing import List, Dict, Any, Union
from PIL import Image
from io import BytesIO

app = FastAPI()

model_path = os.path.join(os.path.dirname(__file__), "yolov8m.pt")
model = YOLO(model_path)

def inference_on_path(imgs_path: str) -> List[Dict[str, Any]]:
    results: ultralytics.engine.results.Results = model(source=imgs_path, show=False, conf=0.45)
    results_data = []
    for result in results:
        result_metadata = {
            'shape': result.orig_shape,
            'path': result.path,
            'detections': json.loads(result.tojson())
        }
        results_data.append(result_metadata)

    return results_data

def inference_on_img(img: Image) -> List[Dict[str, Any]]:
    results: ultralytics.engine.results.Results = model(source=img, show=False, conf=0.45)
    result_data = json.loads(results[0].tojson())

    return result_data
```


AI model service

ONNX, FastAPI, Flask Lightweight Deployment

```
import os, json
from ultralytics import YOLO
from fastapi import FastAPI, UploadFile, File, Query
from typing import List, Dict, Any, Union
from PIL import Image
from io import BytesIO

app = FastAPI()

...

@app.post('/detect_img')
async def detect_img(
    img: UploadFile = File(...)
) -> Dict[str, Union[int, Dict[str, Any]]]:
    if not img:
        return {'status_code': 400, 'data': {'message': 'No upload file sent'}}

    img_content = await img.read()
    image_stream = BytesIO(img_content)
    image = Image.open(image_stream)

    inference_results_data = inference_on_img(img=image)
    return {'status_code': 200,
            'data': {
                'image_name': img.filename,
                'image_size': img.size,
                'inference_results': inference_results_data
            }
    }
```

AI model service

ONNX, FastAPI, Flask Lightweight Deployment

FastAPI server with YOLO

FastAPI

0.1.0

OAS 3.1

/openapi.json

default



GET

/ Home



GET

/detect Detect



POST

/detect_img Detect Img



GET

/images Get Available Images



Schemas

Body_detect_img_detect_img_post > Expand

HTTPValidationError > Expand all object

ValidationError > Expand all object

GET

/detect Detect



Runs YOLO inference for all the images available in the path received as query parameter

Parameters

Try it out

Name

Description

path ★ required
string
(query)

Path to the images directory

path

AI model service ONNX, FastAPI, Flask Lightweight Deployment

FastAPI server with YOLO

```
echo 3. Path-based Image Analysis (Current Folder)
curl -X GET "%API_BASE%/detect?path=cat.jpg" -H "Content-Type: application/json"

echo.
if exist "cat.jpg" (
    echo 4. Image Upload Test (cat.jpg)
    curl -X POST "%API_BASE%/detect_img" -F "img=@cat.jpg"
```

```
F:\projects\AI_foundation_tutorial\8_serving\fastapi_model_serving>test_api.bat
Object Detection Server API Test Start (curl)

1. Health Check
{"status_code":200,"data":{"api_health_status":"OK"}}

2. List Images in Current Folder
{"status_code":200,"data":{"available_images":["cat.jpg"],"size":1}}

3. Path-based Image Analysis (Current Folder)
{"status_code":200,"data":{"inference_results":[{"shape":[1536,2048],"path":"F:\\projects\\AI_foundation_tutorial\\8_serving\\fastapi_model_serving\\cat.jpg","detections":[{"name":"cat","class":15,"confidence":0.96055,"box":{"x1":3.84907,"y1":348.77158,"x2":1634.66895,"y2":1534.54602}}]}]}}

4. Image Upload Test (cat.jpg)
{"status_code":200,"data":{"image_name":"cat.jpg","image_size":322451,"inference_results":[{"name":"cat","class":15,"confidence":0.96055,"box":{"x1":3.84907,"y1":348.77158,"x2":1634.66895,"y2":1534.54602}}]}}
Test Completed!

F:\projects\AI_foundation_tutorial\8_serving\fastapi_model_serving>
```

AI model service ONNX, FastAPI, Flask Lightweight Deployment

Flask

Flask is a lightweight WSGI web application framework. It is designed to make getting started quick and easy, with the ability to scale up to complex applications.

pip install Flask



```
from markupsafe import escape
```

```
@app.route("/<name>")
```

```
def hello(name):
```

```
    return f"Hello, {escape(name)}!"
```

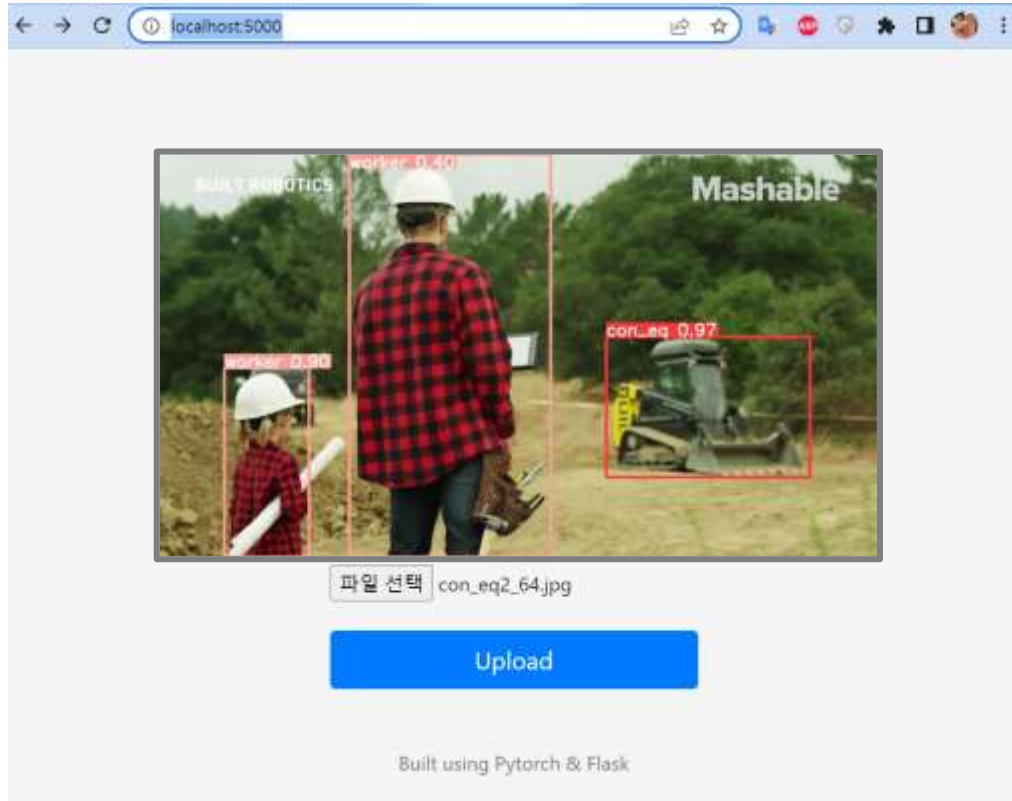
```
$ flask --app hello run
```

```
* Serving Flask app 'hello'
```

```
* Running on http://127.0.0.1:5000
```

AI model service ONNX, FastAPI, Flask Lightweight Deployment

Flask



projects > notebook > 6_yolo_practice > yolo5_flask

이름	수정한 날
----	-------

.git	2022-10-
.github	2022-10-
__pycache__	2022-10-
docs	
models	
static	
templates	
tests	
utils	
con_eq2_69.jpg	
README.md	
a_file.png	
yolov5_con.pt	
yolov5s.pt	
export.py	
get-pip.py	
hubconf.py	
restapi.py	
webapp.py	
requirements.txt	
openapi_test.bat	2022-10-
run.bat	2022-10-
dataset_construct_worker.yaml	2022-09-
.gitignore	2022-09-
Dockerfile	2022-09-



AI model service ONNX, FastAPI, Flask Lightweight Deployment

Flask

Curl utility installation <https://curl.se/download.html>

python restapi.py

curl -X POST -F image=@con_eq2_69.jpg "http://127.0.0.1:5000/v1/object_detection"

```
(venv_tf2) C:\projects\yolov5_flask>python restapi.py
YOLov5 1.3.2-gac18396 Python-3.9.12 torch-1.12.1+cu113 CUDA:0 (NVIDIA GeForce RTX 3080 Laptop GPU, 8192MiB)
```

```
Fusing layers...
Model summary: 213 layers, 7015519 parameters, 0 gradients, 15.8 GFLOPs
```

```
Adding AutoShape...
```

```
* Serving Flask app 'restapi'
```

```
* Debug mode: off
```

```
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
```

```
* Running on all addresses (0.0.0.0)
```

```
* Running on http://127.0.0.1:5000
```

```
* Running on http://172.30.1.88:5000
```

```
Press CTRL+C to quit
```

```
127.0.0.1 - - [16/Oct/2022 08:02:03] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
127.0.0.1 - - [16/Oct/2022 08:02:05] "GET / HTTP/1.1" 404 -
```

```
[{"xmin":24.6365852356,"ymin":561.6345214844,"xmax":129.339263916,"ymax":720.0,"confidence":0.9224505424,"class":1,"name":"worker"}, {"xmin":526.8735473633,"ymin":33.5504608154,"xmax":732.2012329102,"ymax":541.9304199219,"confidence":0.9142798185,"class":0,"name":"con_eq"}, {"xmin":1021.1021728516,"ymin":352.8305853711,"xmax":1228.3970947266,"ymax":514.4751586914,"confidence":0.910230875,"class":0,"name":"con_eq"}, {"xmin":908.7915039062,"ymin":256.1762390137,"xmax":964.9847412109,"ymax":316.9440612793,"confidence":0.8944447041,"class":0,"name":"con_eq"}, {"xmin":850.4646606445,"ymin":253.5209350586,"xmax":895.5513305664,"ymax":320.7725830078,"confidence":0.8598570824,"class":0,"name":"con_eq"}, {"xmin":983.1149291992,"ymin":266.1564941406,"xmax":1023.3728837695,"ymax":318.2346881758,"confidence":0.8123363853,"class":0,"name":"con_eq"}, {"xmin":691.6927490234,"ymin":109.5151672363,"xmax":793.2215576172,"ymax":394.5242614746,"confidence":0.7660834789,"class":0,"name":"con_eq"}, {"xmin":978.7466430664,"ymin":314.1014709473,"xmax":1081.8812255859,"ymax":403.5487365723,"confidence":0.7509086728,"class":0,"name":"con_eq"}, {"xmin":643.0079345703,"ymin":246.8774414062,"xmax":674.1931152344,"ymax":303.2550048828,"confidence":0.6875187159,"class":0,"name":"con_eq"}, {"xmin":964.1707763672,"ymin":258.2351989746,"xmax":1007.9593505859,"ymax":319.8886535645,"confidence":0.604162395,"class":0,"name":"con_eq"}, {"xmin":755.126159668,"ymin":161.3591308594,"xmax":803.8353881836,"ymax":321.1946411133,"confidence":0.5691390634,"class":0,"name":"con_eq"}, {"xmin":1243.7830810547,"ymin":410.7705993652,"confidence":0.428168267,"class":0,"name":"con_eq"}, {"xmin":867.5435791016,"ymin":246.4231872559,"xmax":906.6786542969,"ymax":315.4702453613,"confidence":0.40350898255,"class":0,"name":"con_eq"}, {"xmin":967.5684204102,"ymin":314.8999938965,"xmax":1001.7764282227,"confidence":0.3653090563,"class":0,"name":"con_eq"}, {"xmin":1242.9893798828,"ymin":337.2352294922,"xmax":1280.0,"ymax":457.3475952148,"confidence":0.3033725619,"class":0,"name":"con_eq"}]
```

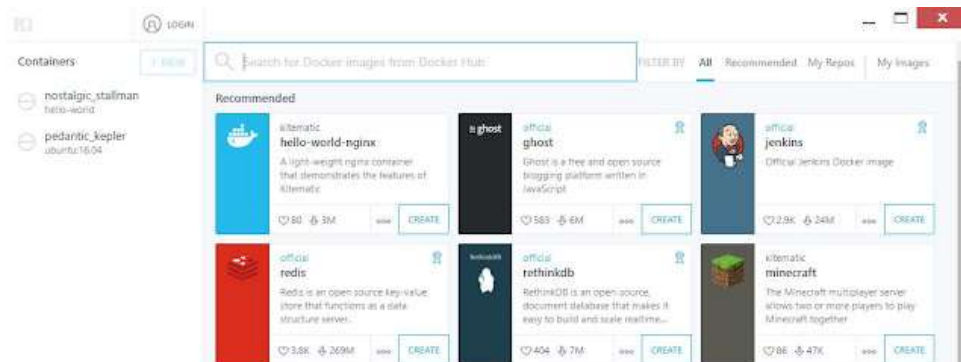
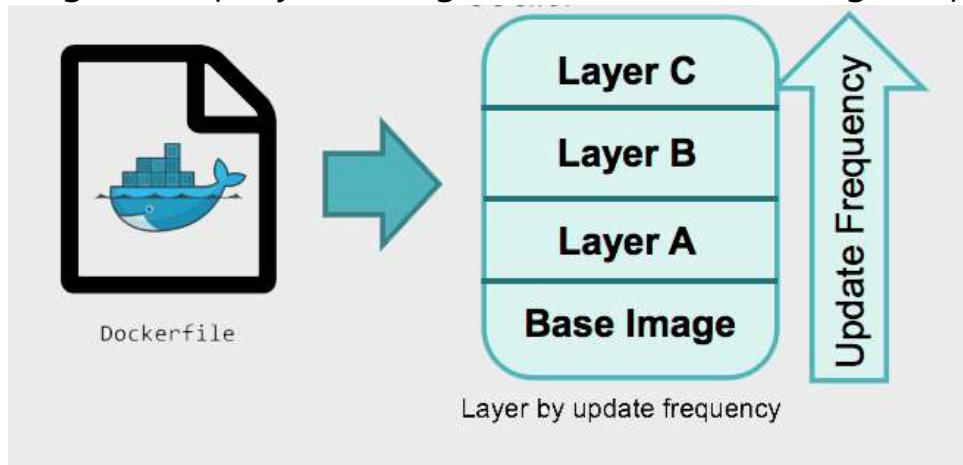
Link: FastAPI, Uvicorn, Websocket 기반 Open API 서버 개발

AI model service ONNX, FastAPI, Flask Lightweight Deployment

Docker: Accelerated Container Application Development

Docker is a container-based open-source virtualization platform. It allows users to connect to running containers, execute commands, and install packages using apt-get. It runs multiple processes in the background.

Docker first gained recognition in March 2013 at the Pycon conference in Santa Clara, where dotCloud founder Solomon Hykes presented a session titled "The Future of Linux Containers." Since then, Docker has grown rapidly, securing \$100 million in funding in April 2015.



```
FROM python:3.10-slim

# Set the working directory in the container
WORKDIR /app

# Copy the project files to the container
COPY ./src /app/src
COPY requirements.txt /app

# Install Python dependencies
RUN pip install --no-cache-dir -r requirements.txt

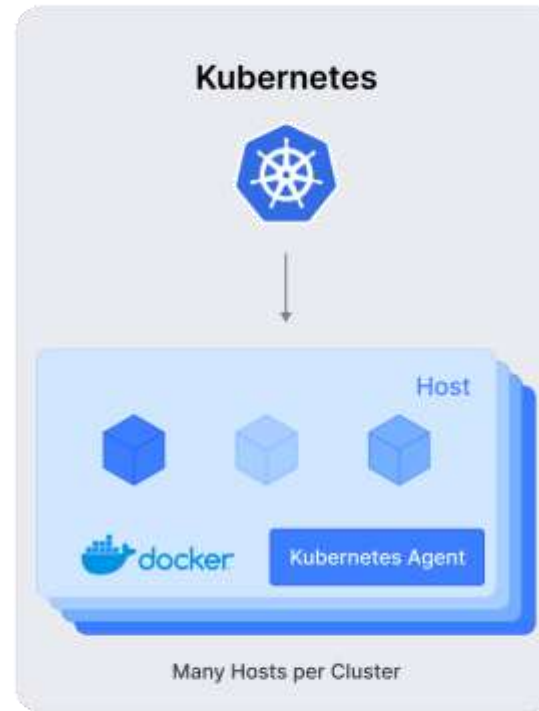
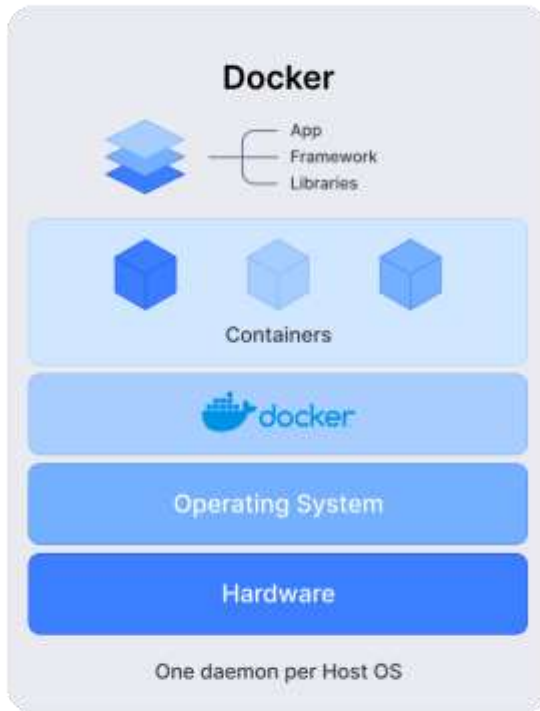
# Expose the port your app will run on (e.g., 7860 for Gradio)
EXPOSE 7860
ENV GRADIO_SERVER_NAME="0.0.0.0"

CMD ["python", "./src/app.py"]
```

Link: Docker 기반 우분투, 텐서플로우, PyTorch 설치, 개발 및 관련 명령

Link: 도커 기반 NVIDIA GPU 드라이버 설치 및 딥러닝 프레임워크 실행

AI model service ONNX, FastAPI, Flask Lightweight Deployment



kind: Deployment

spec:

replicas: 3

selector:

matchLabels:

app: onnx-inference

spec:

containers:

- name: onnx-inference

image: onnx-inference:latest

ports:

- containerPort: 5000

kind: Service

metadata:

name: onnx-inference-service

spec:

selector:

app: onnx-inference

ports:

- protocol: TCP
port: 80

Feature	Docker	Kubernetes
Type	Container runtime	Container orchestration tool
Scope	Operates on a single node	Manages clusters of nodes
Scalability	Limited built-in scaling features	Advanced scaling capabilities
Self-Healing	No native self-healing	Automatic rescheduling of failed containers
Networking	Basic networking capabilities	Advanced networking with service discovery

[Kubernetes and Docker: What's Best for Your Container Strategy?](#)

[Transform Your ML Ops: How I Built a Robust End-to-End Pipeline with PyTorch, ONNX & Kubernetes | by FAANG | Medium](#)

AI model service

ONNX, FastAPI, Flask Lightweight Deployment

```
# Use Python 3.11 slim image as base
FROM python:3.11-slim

# Set working directory and environment variables
WORKDIR /app
ENV PYTHONUNBUFFERED=1
ENV PYTHONDONTWRITEBYTECODE=1

# Install minimal system dependencies for OpenCV and YOLO
RUN apt-get update && \
    apt-get install -y --no-install-recommends \
    curl \
    libglib2.0-0 \
    ...

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .

# Create a non-root user
RUN useradd --create-home --shell /bin/bash app \
    && chown -R app:app /app
USER app

# Expose port 8000
EXPOSE 8000

# Command to run the application
CMD ["fastapi", "run", "server.py", "--host", "0.0.0.0", "--port", "8000"]
```



```
fastapi[standard]>=0.104.0
uvicorn[standard]>=0.24.0
ultralytics>=8.0.0
opencv-python>=4.8.0
Pillow>=10.0.0
python-multipart>=0.0.6
```

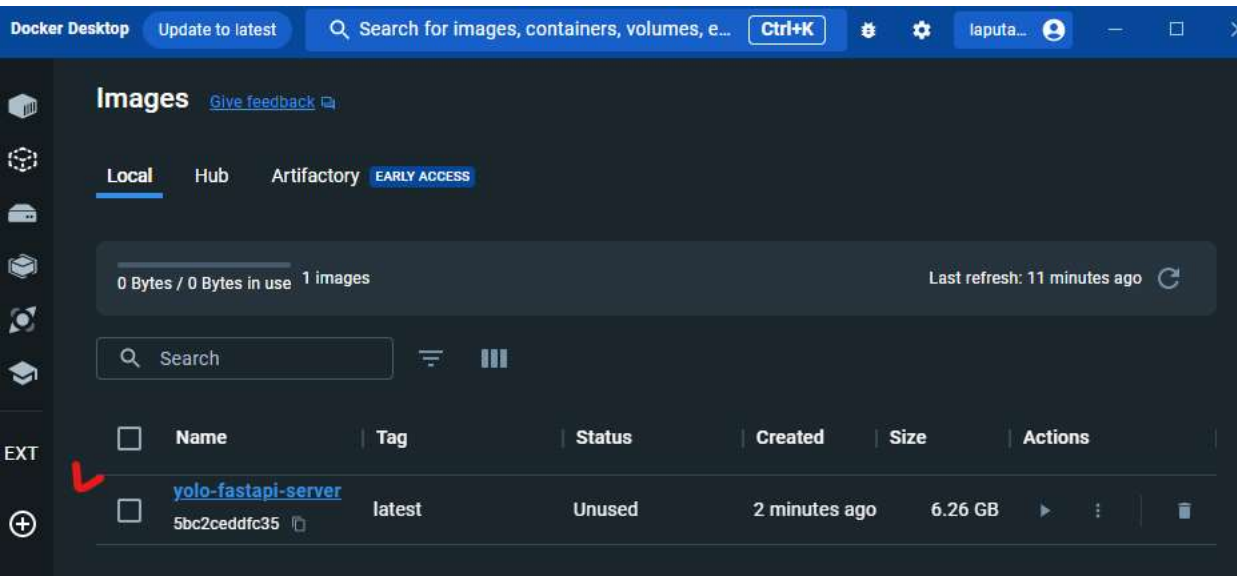
AI model service

ONNX, FastAPI, Flask Lightweight Deployment

```
echo Building Docker image...
docker build -t yolo-fastapi-server .
```

```
[+] Building 61.6s (9/12)                                docker:default
=> [internal] load build context                          0.0s
=> => transferring context: 1.29kB                        0.0s
=> CACHED [2/7] WORKDIR /app                             0.0s
=> [3/7] RUN apt-get update && apt-get install -y --no-install-recommends 4.9s
=> [4/7] COPY requirements.txt .                          0.0s
=> [5/7] RUN pip install --no-cache-dir -r requirements.
=> => # 0:00
=> => # Downloading torch-2.7.1-cp311-cp311-manylinux_2_
```

```
Container started successfully!
API is available at: http://localhost:8000
API Documentation: http://localhost:8000/docs
```

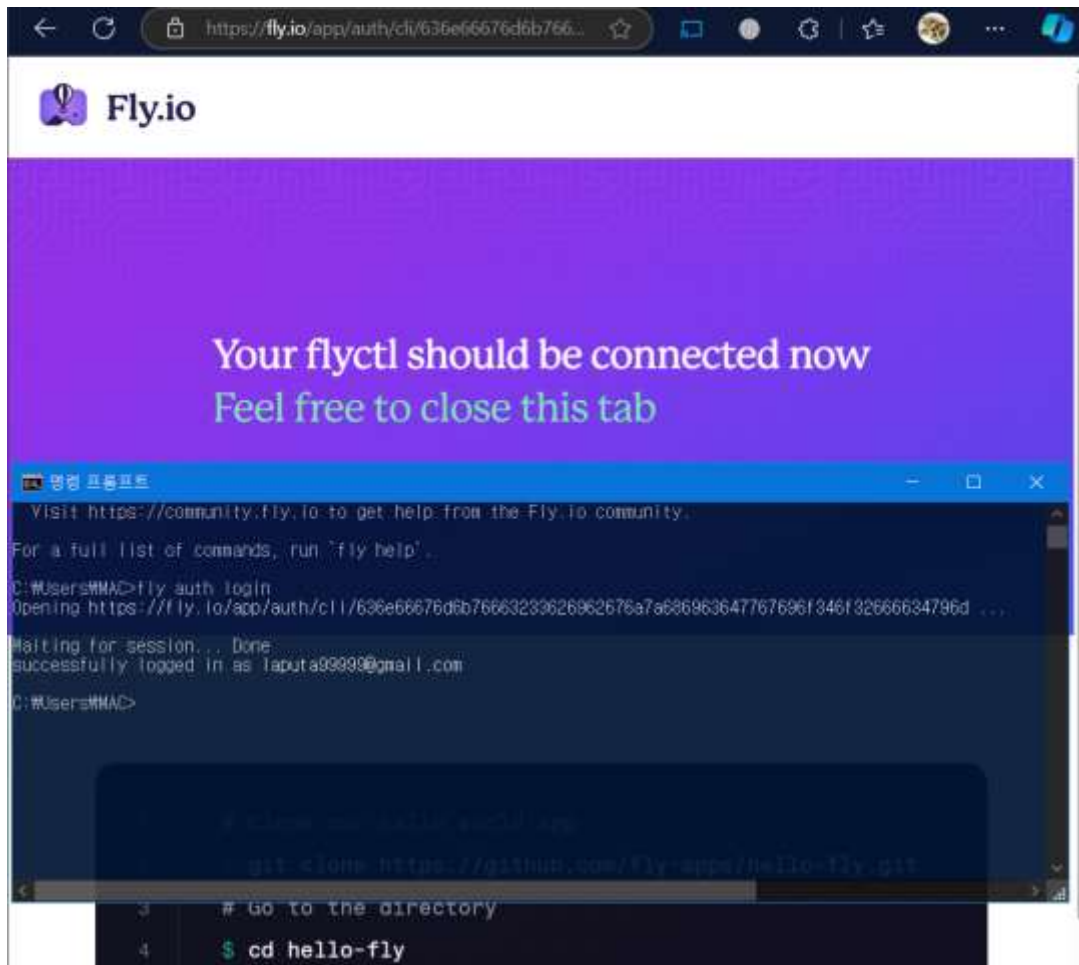


Serve your first model with Scikit-Learn +
Flask + Docker | by Carl W. Handlin |
Rappi Tech

AI model service ONNX, FastAPI, Flask Lightweight Deployment

FLY.IO

Co-founded and developed by Kurt Mackey, FLY.IO primarily provides a developer-friendly global application hosting service. Kurt previously co-founded Compose.io, a database hosting platform (sold to IBM).



Sign up and install

Go to the FLY.IO website (<https://fly.io>) and sign up.

Run the installation script in the terminal:
`pwsh -Command "iwr https://fly.io/install.ps1 -useb | iex"`

Run login in the terminal: `fly auth login`

`git clone https://github.com/fly-apps/hello-fly.git`
`cd hello-fly`
`fly launch --now`

Link: 가성비 있는 웹 서비스 호스팅 FLY.IO 사용하기

AI model service ONNX, FastAPI, Flask Lightweight Deployment

FLY.IO

The image displays four screenshots of the Fly.io platform, illustrating its use for deploying AI model services.

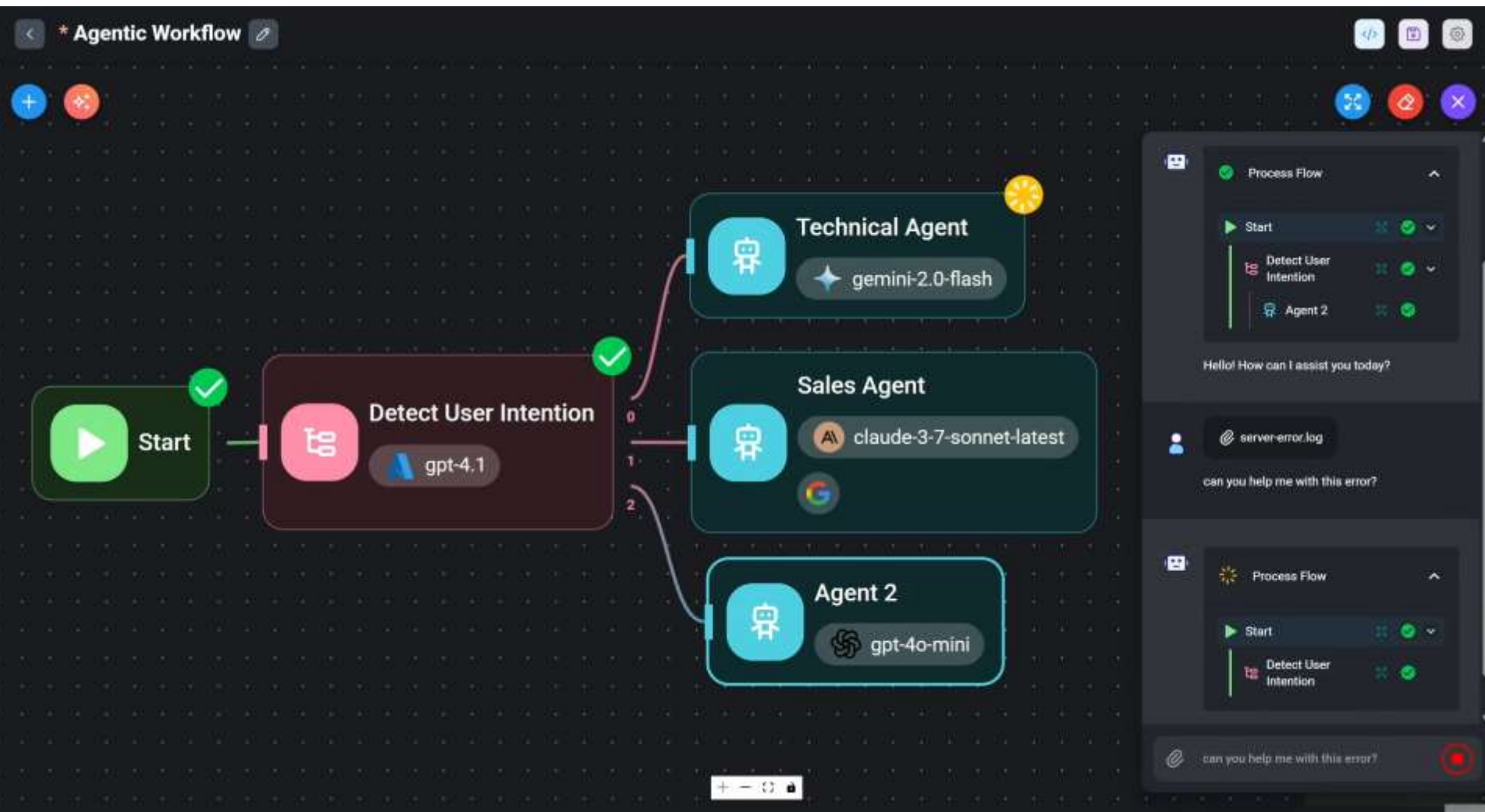
- Top Left:** The Fly.io Dashboard (https://fly.io/dashboard/taewook-kang) showing a list of deployed applications. The 'Apps' section includes filters for 'Deployed', 'Pending', and 'Suspended'. Applications listed include 'bim-data-quality-checker', 'scan-to-model-app', and 'ai-agent-simple-function-call'.
- Top Right:** The 'Scan to Model Pipeline (ver 0.2. prototype)' application interface (https://scan-to-model-app.fly.dev). It provides instructions for uploading a pipeline configuration file (JSON) and point cloud data file (LAS, LAZ) to process data and download results as a zip file. It includes upload buttons for 'Pipeline Configuration File' and 'Point Cloud Data File'.
- Bottom Left:** The 'Fly App - Dashboards - Grafana' interface (https://fly-metrics.net/id/fly-app/fly-app?orgId=6452368&var-app=...). It shows a dashboard with various metrics, including 'Data Out' (region, instance, net), 'Network I/O' (0 B/s, -2 kB/s, -4 kB/s, -6 kB/s), and 'Edge' (0 B/s, 100 B/s).
- Bottom Right:** The 'BIM Quality Checker (ver 0.46. prototype)' application interface (https://bim-data-quality-checker.fly.dev). It provides instructions for uploading a BIM Check RuleSet (JSON) Configuration File and BIM Files (COBie xls, csv, IFC, LandXML) to check BIM data quality and download the Quality Report. It includes upload buttons for 'BIM Check RuleSet (JSON) Configuration File' and 'BIM Files (COBie xls, csv, IFC, LandXML)'.

Link: 가성비 있는 웹 서비스 호스팅 Fly.io 사용하기

AI model service ONNX, FastAPI, Flask Lightweight Deployment

Flowise - Build AI Agents, Visually

Open source agentic systems development platform

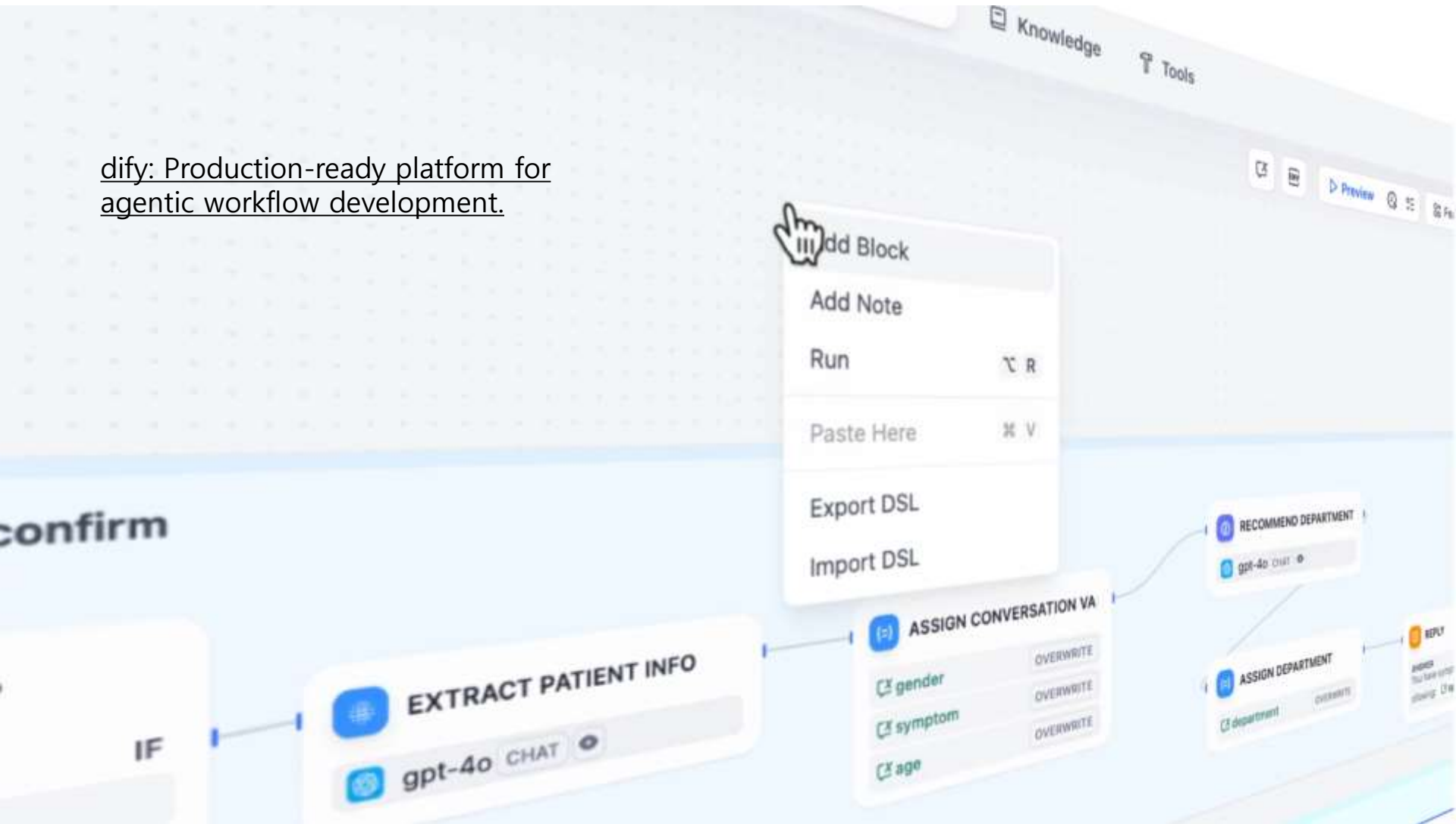


AI model service ONNX, FastAPI, Flask Lightweight Deployment

Dify: Leading Agentic AI Development Platform

Build Production – Ready Agentic AI Solutions

dify: Production-ready platform for agentic workflow development.



AX Strategy

Considerations for Enterprise AI Implementation

Questions & Answers and Recommended Learning Resources

[AX reference](#)

Key Question

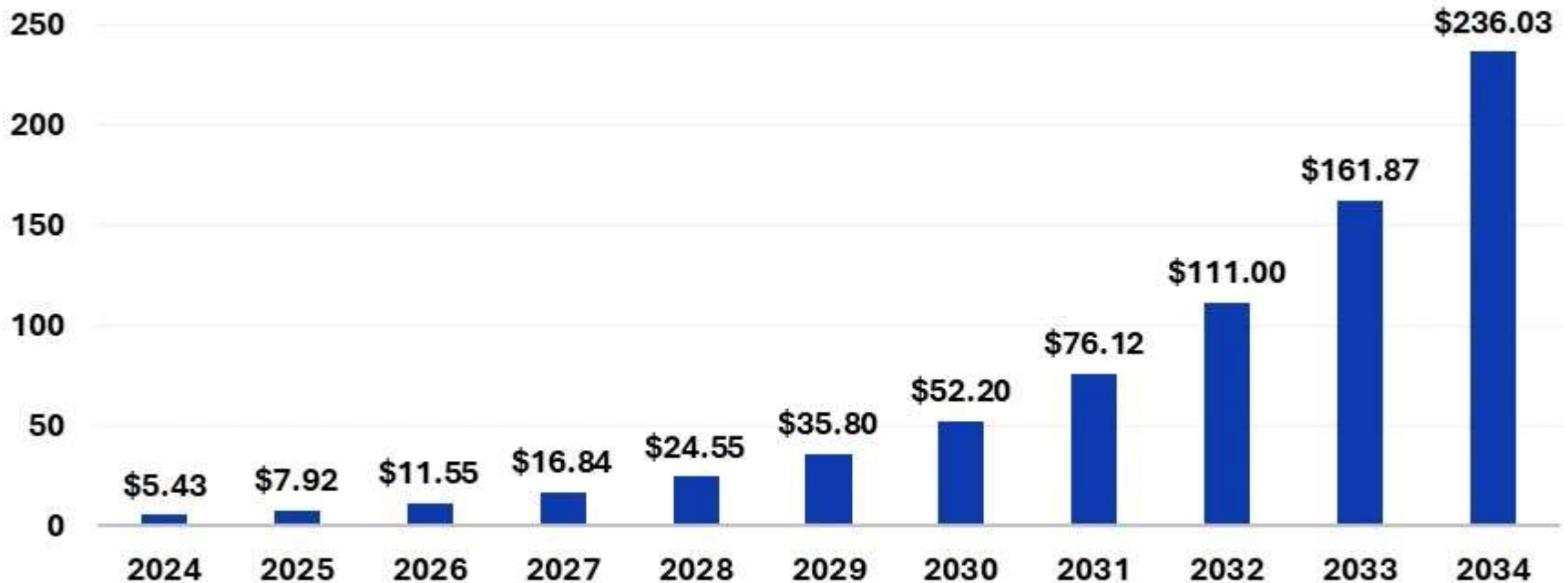
Why does AI fail?

What should you consider when developing an AX?

How can you brand yourself and your company in the AX revolution?

AX Strategy

AI Agents Market Size 2024 to 2034 (USD Billion)



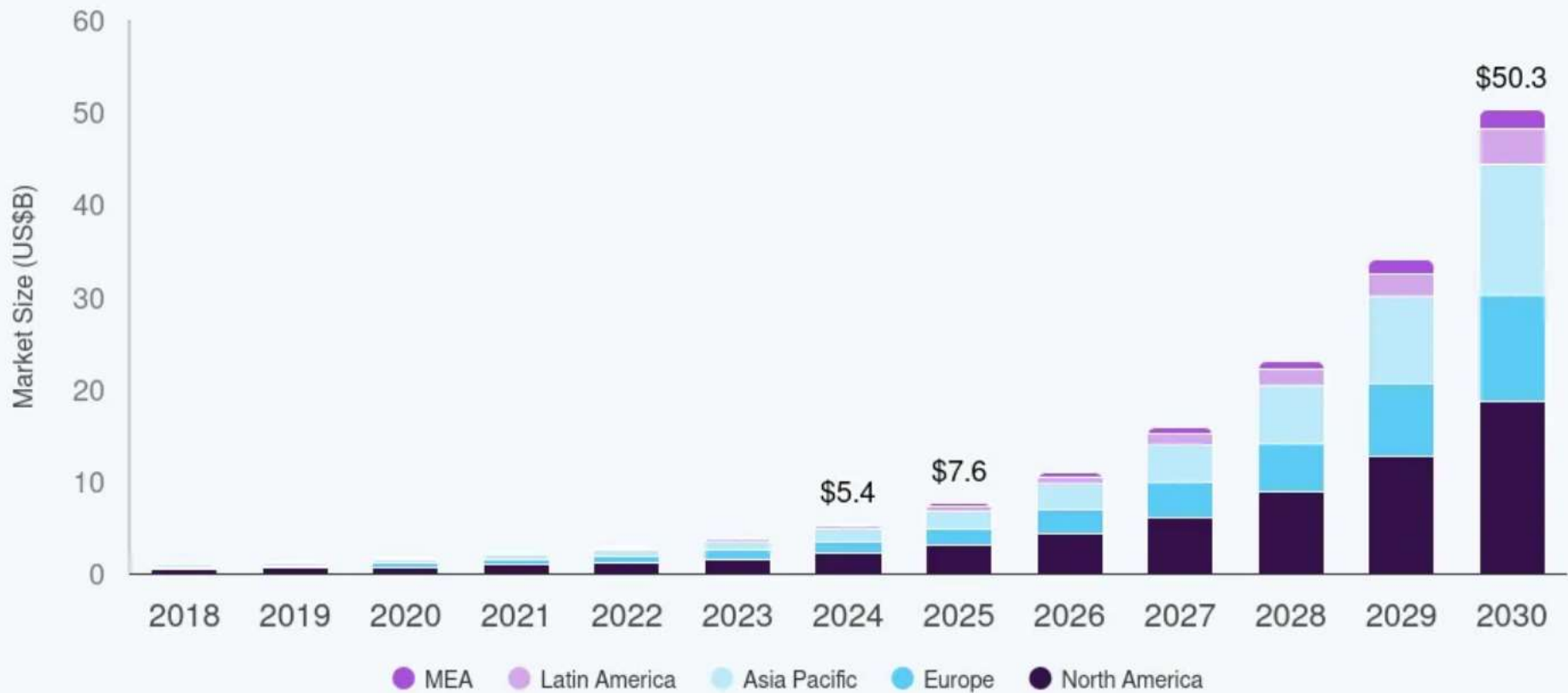
The global ai agents market size is predicted to increase from USD 5.43 billion in 2024 to approximately USD 236.06 billion by 2034, expanding at a CAGR of 45.82% from 2025 to 2034.

Source: <https://www.precedenceresearch.com/ai-agents-market>

AX Strategy

AI Agents Market

Size, by Region, 2018 - 2030



AX Strategy

Key AI Agents Companies:

The following are the leading companies in the **AI agents market** with the largest market share and dictate industry trends.

- Alibaba Group Holding Limited
- Amazon Web Services, Inc.
- Apple Inc.
- Baidu
- Google
- IBM Corporation
- Meta
- Microsoft
- NVIDIA Corporation
- Salesforce, inc.

- **Agent System Outlook (Revenue, USD Million, 2017 - 2030)**
 - Single Agent Systems
 - Multi Agent Systems
- **Type Outlook (Revenue, USD Million, 2017 - 2030)**
 - Ready-to-Deploy Agents
 - Build-Your-Own Agents
- **Application Outlook (Revenue, USD Million, 2017 - 2030)**
 - Customer Service and Virtual Assistants
 - Robotics and Automation
 - Healthcare
 - Financial Services
 - Security and Surveillance
 - Gaming and Entertainment
 - Marketing and Sales
 - Human Resources
 - Legal and Compliance
 - Others

AX Strategy

유스케이스

Deloitte Report

- Marketing content creation
- Coding development support
- Customer service support
- Product design support
- Research report writing
- Synthetic and training data generation
- Enterprise data retrieval, organization, analysis, and summarization
- Game content design and development
- Language translation
- Simulation and inference
- Advanced personalized education and training
- Field work support



About the Deloitte AI Institute

The Deloitte AI Institute™ helps organizations connect all the different dimensions of the robust, highly dynamic, and rapidly evolving Artificial Intelligence ecosystem. The AI Institute leads conversations on applied AI innovation across industries, with cutting-edge insights, to promote human-machine collaboration in the “Age of With.”

The Deloitte AI Institute aims to promote the dialogue and development of AI, stimulate innovation, and examine challenges to AI implementation and ways to address them. The AI Institute collaborates with an ecosystem composed of academic research groups, start-ups, entrepreneurs, innovators, mature AI product leaders, and AI visionaries to explore key areas of artificial intelligence including risks, policies, ethics, the future of work and talent, and applied AI use cases. Combined with Deloitte's deep knowledge and experience in artificial intelligence applications, the Institute helps make sense of this complex ecosystem, and as a result, delivers impactful perspectives to help organizations succeed by making informed AI decisions.

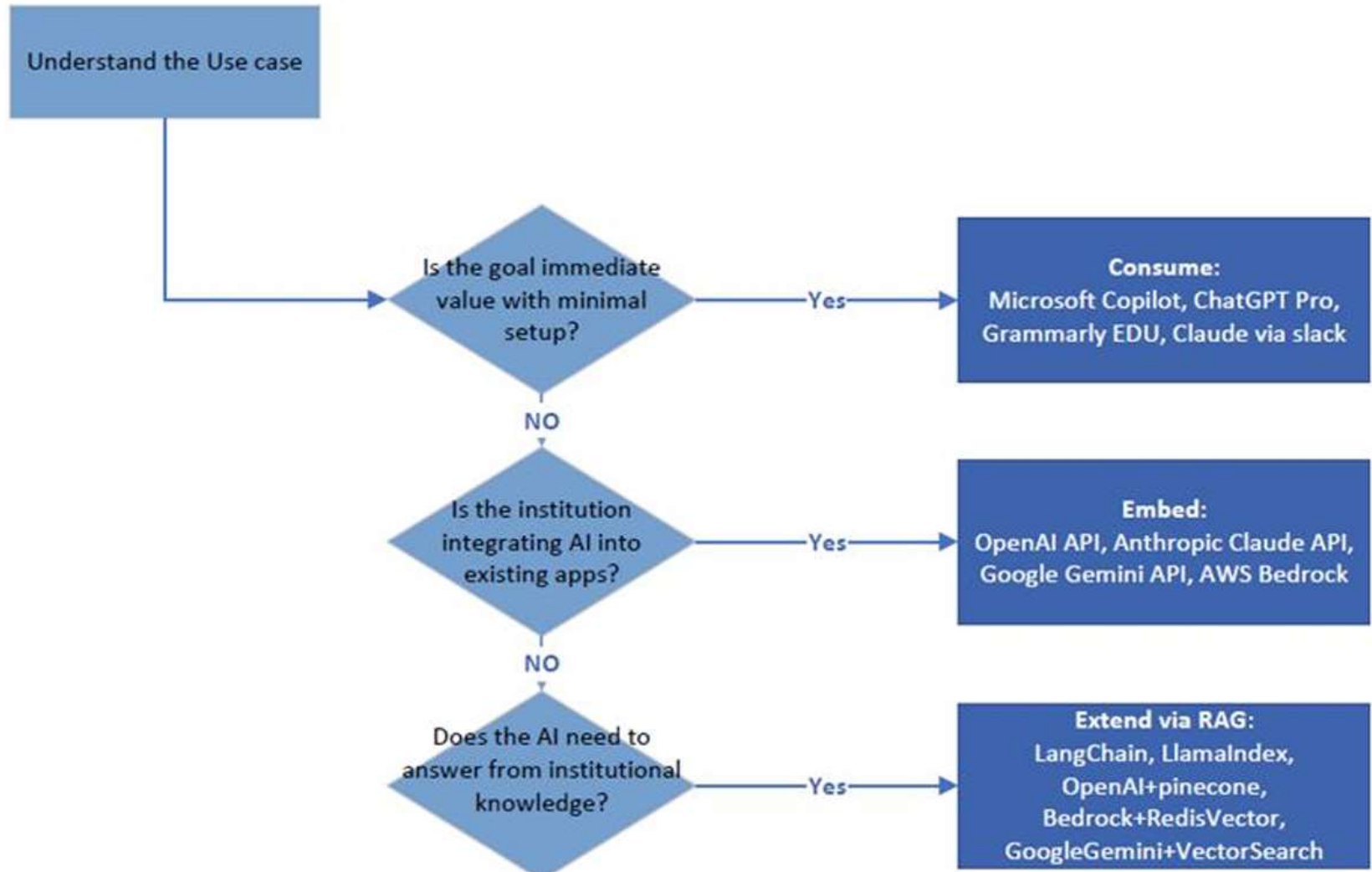
No matter what stage of the AI journey you are in: whether you are a board member or a C-Suite leader driving strategy for your organization—or a hands-on data scientist bringing an AI strategy to life—the Deloitte AI Institute can help you learn more about how enterprises across the world are leveraging AI for a competitive advantage. Visit us at the Deloitte AI Institute for a full body of our work, subscribe to our podcasts and newsletter, and join us at our meet-ups and live events. Let's explore the future of AI together.

www.deloitte.com/us/AIInstitute

AX Strategy

Considerations for Implementing Enterprise AI

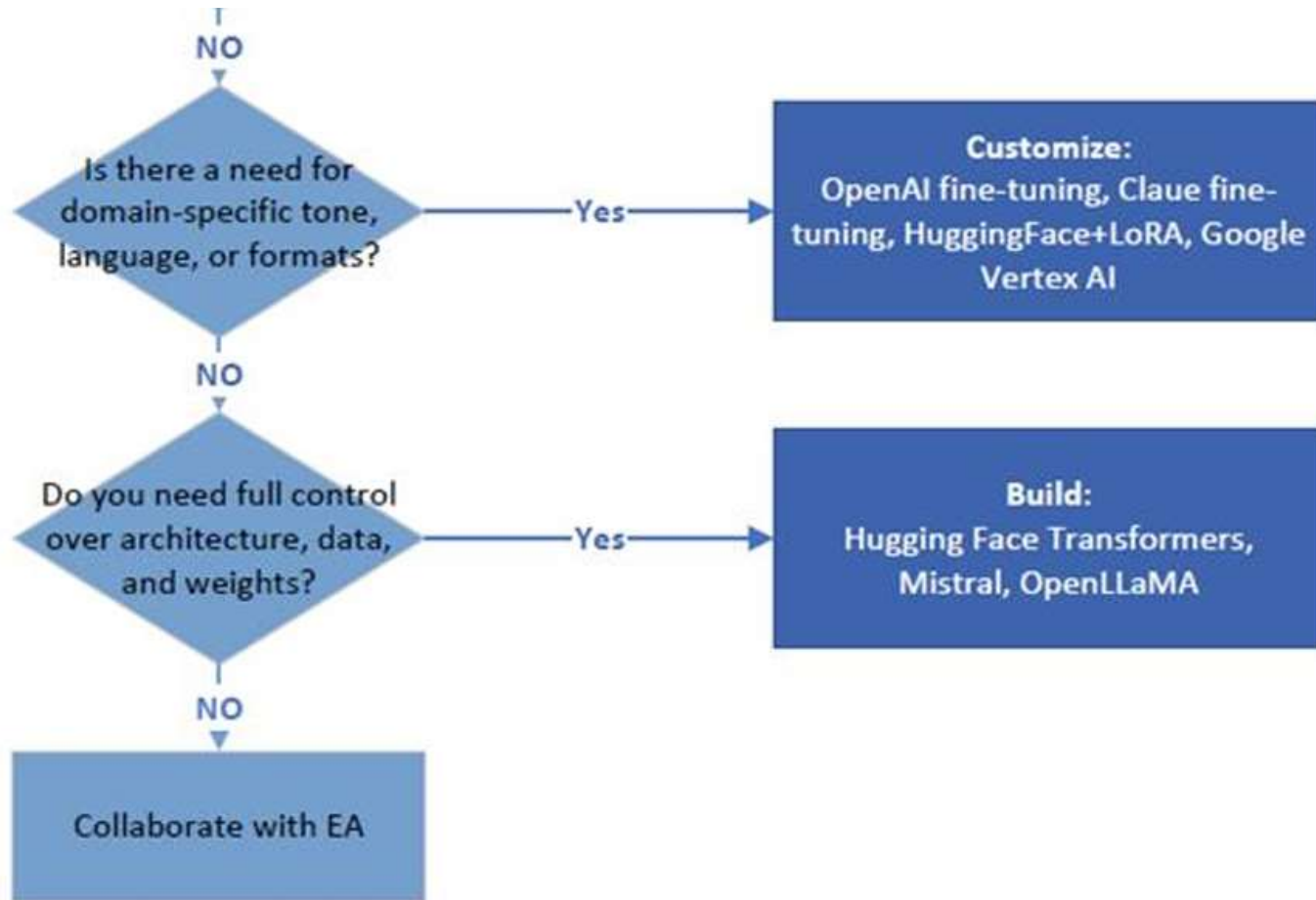
Example of AX service development decisions based on use cases



AX Strategy

Considerations for Implementing Enterprise AI

Example of AX service development decisions based on use cases



AX Strategy

Considerations for Implementing Enterprise AI

AI Tool Evaluation Methods Example

Appendix: AI Toolset Evaluation

To ensure objective and consistent selection of AI tools, Columbia proposes a scoring framework based on six key criteria. Each tool is scored on a scale of 1 to 10 using standardized rubrics, and then weighted to reflect institutional priorities.

Evaluation Criteria

Criterion	What It Measures
Performance	Speed, scalability, and accuracy under real-world workloads
Cost	Licensing, usage, infrastructure cost, and transparency
Customization	Ability to fine-tune or configure for domain-specific use
Compliance & Security	Support for regulatory/security standards (FERPA, HIPAA, etc.)
Integration & Tooling	Compatibility with systems, APIs, SDKs, and documentation
Community & Support	Docs, vendor responsiveness, and community activity

Criterion	Weight
Performance	25%
Cost	20%
Customization	15%
Compliance & Security	15%
Integration & Tooling	15%
Community & Support	10%

AX Strategy

Top 5 Causes of Failure

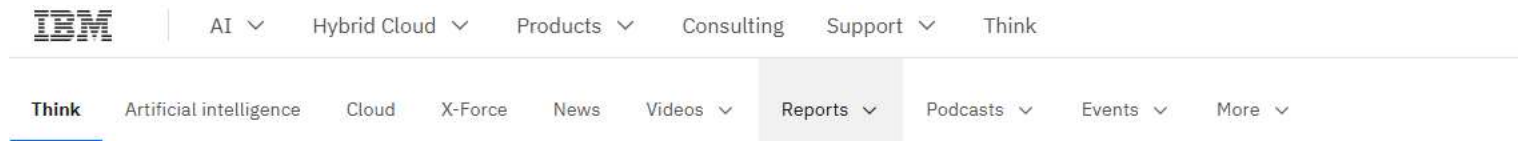
Concerns about data accuracy or bias (45%)

Insufficient proprietary data available to customize models (42%)

Inadequate generative AI expertise (42%)

Inadequate financial justification or business case (42%)

Concerns about privacy or confidentiality of data and information (40%)



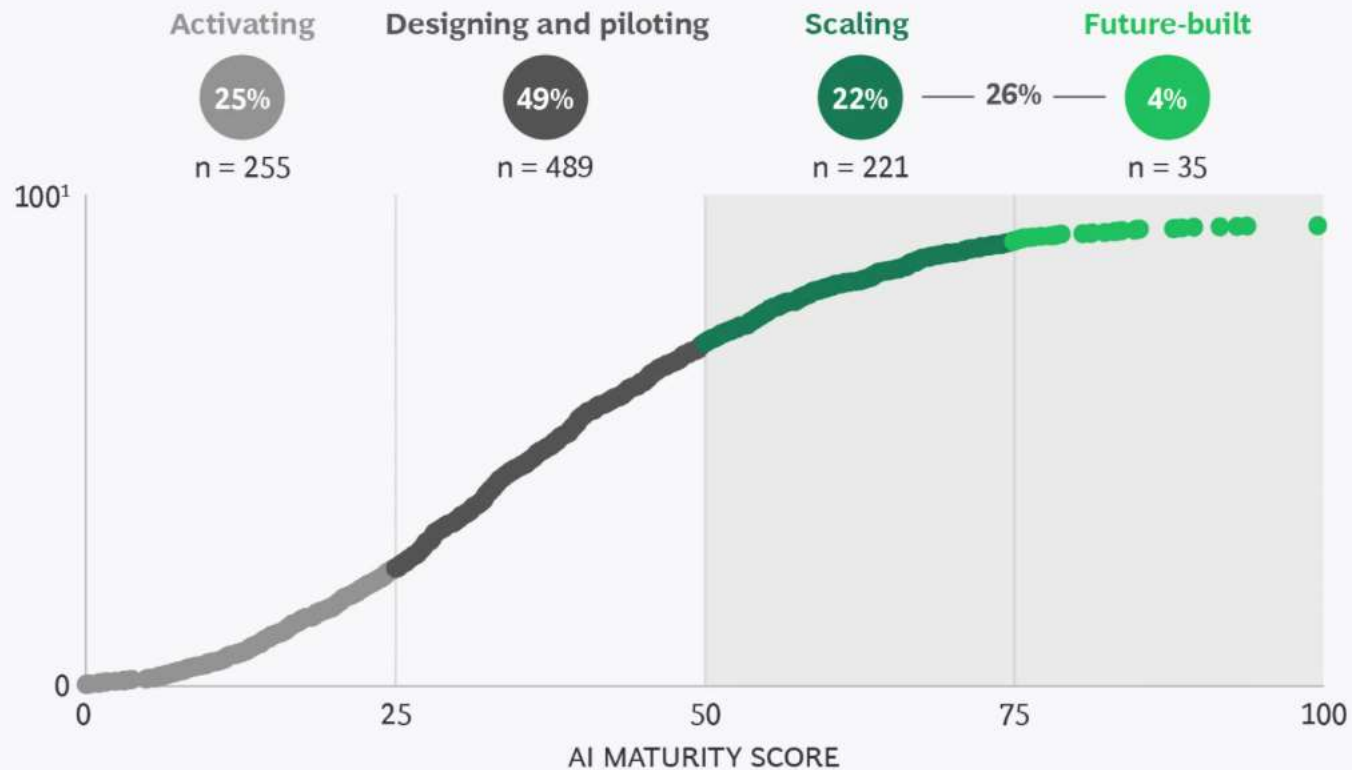
The 5 biggest AI adoption challenges for 2025



AX Strategy

AI Transformation Challenge

Only one in four companies have found real value from AI ...



Source: BCG Build for the Future 2024 global study, merged with data from BCG's Digital Acceleration Index; n = 1000.

Note: The AI maturity score is based on 30 AI-specific foundational capabilities, derived by regressing 53 "Build for the Future" capabilities against AI value generation and weighted by relative importance.

¹Y axis represents a company's percentile rank among those studied.

AX Strategy

AI transformation principle

(Gen)AI SETUP

Define **value pools**

- BCG (Gen)AI Maturity Assessment
- BCG Workforce Diagnostic

Articulate **vision** based on **strategy**

Select **priority opportunities** across
Deploy, Reshape, Invent PLAYS

Build the **business case**

DEPLOY *(Gen)AI in everyday tasks across the enterprise*

RESHAPE *critical functions*

In every business function:

Build **MVPs, re-design workflows** and pilot to
prove value...

...while designing **future state** operating model

Cascade changes and **scale E2E**

Transformation of function 1

Transformation of function 2

Transformation of function 3...

INVENT *new business models and products*

E2E CHANGE MANAGEMENT & DELIVERY

Enable **leaders and upskill**

Drive adoption, engagement & culture change, leveraging **behavioral science**

Steer through central **governance** and measure impact **via AI delivery office**

ENTERPRISE FOUNDATIONS

Make coordinated investments in core **tech & data, people, and responsible AI**



Read about BCG's perspective on Responsible AI

AX Strategy

OpenAI

Identifying and scaling AI use cases



How early adopters focus their AI efforts

AI transformation principle

Keep these three principles in mind. They're the backdrop to all the practical guidance you'll find in the pages ahead.

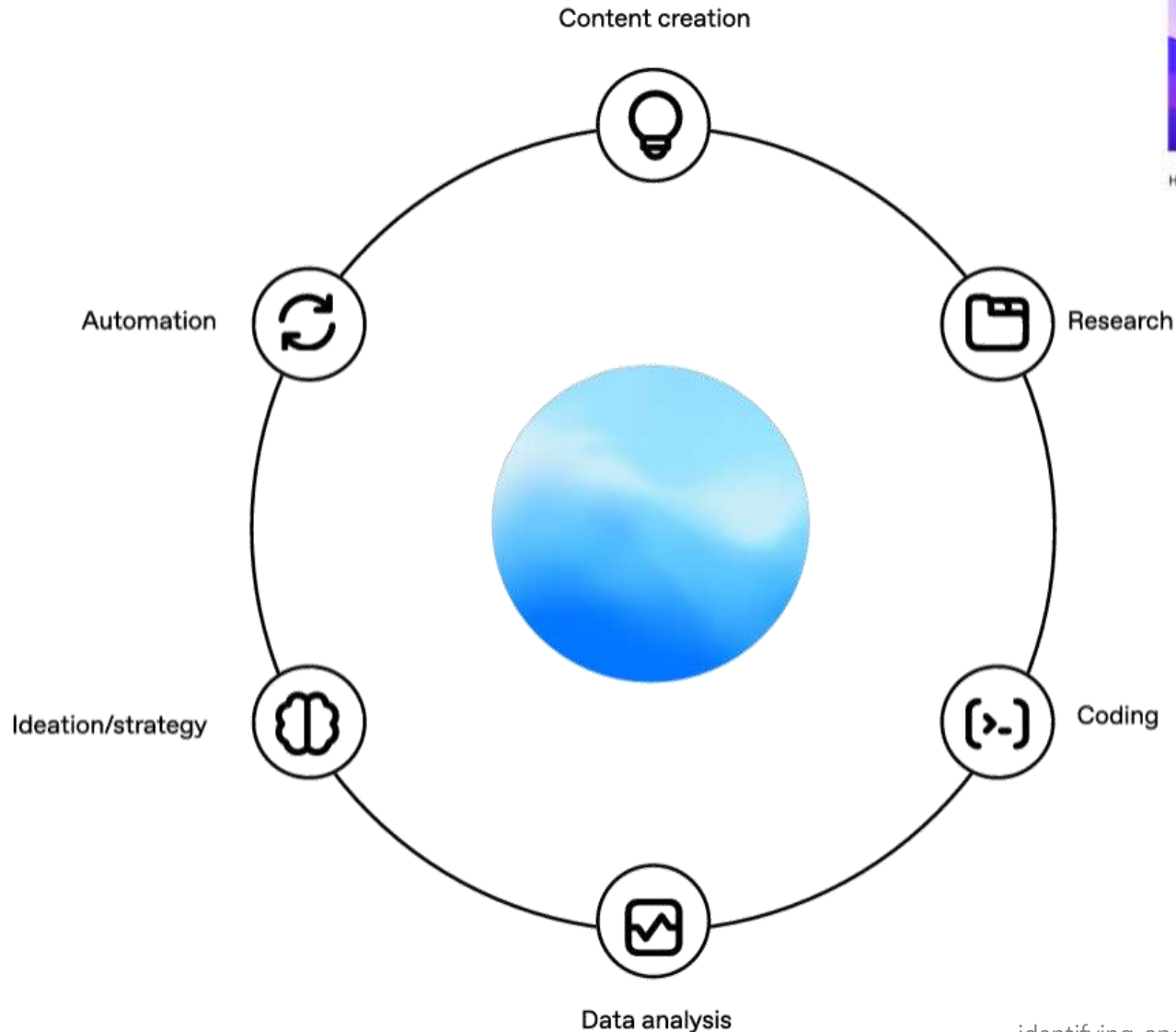
01 AI should be led and encouraged by leadership.

02 Complex use cases can feel impressive, but often slow you down. Instead, empowering employees to find use cases that work best for them, and your company, is often a faster path to success.

03 Encouraging adoption with hackathons, use case workshops, and peer-led learning sessions is a catalyst for many of our customers.

AX Strategy

AI transformation principle



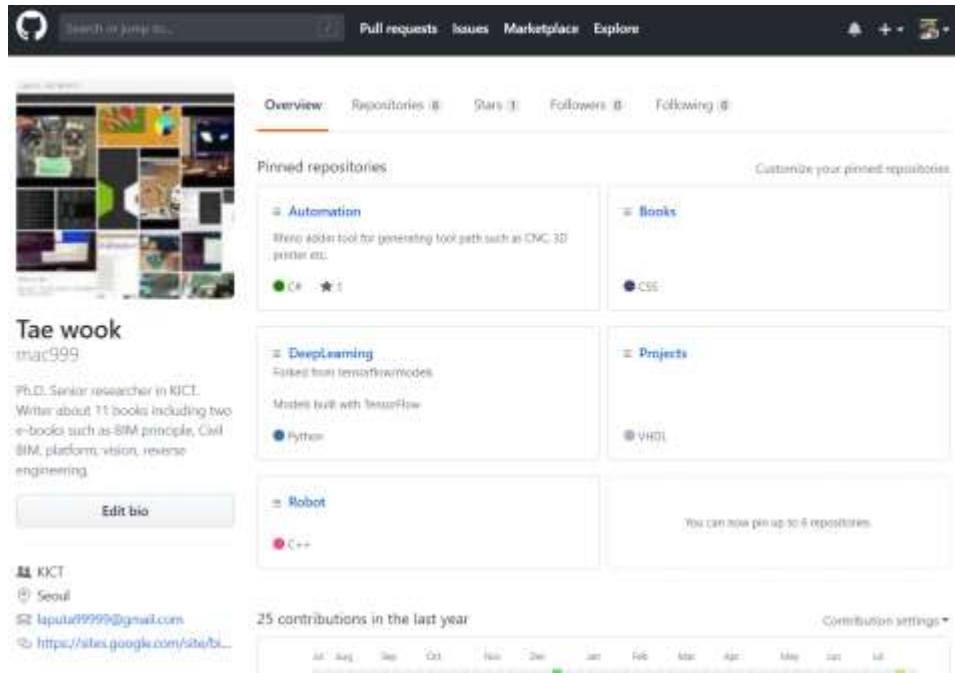
OpenAI

Identifying and scaling AI use cases



How early adopters focus their AI efforts

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Search or jump to... Pull requests Issues Marketplace Explore

Tae wook
mac999

Ph.D. Senior researcher in KICT.
Writer about 11 books including two e-books such as BIM principle, Civil BIM, platform, vision, reverse engineering.

Edit bio

KICT
Seoul
laputa99999@gmail.com
https://sites.google.com/site/bi...

Overview Repositories (8) Stars (3) Followers (0) Following (0)

Pinned repositories

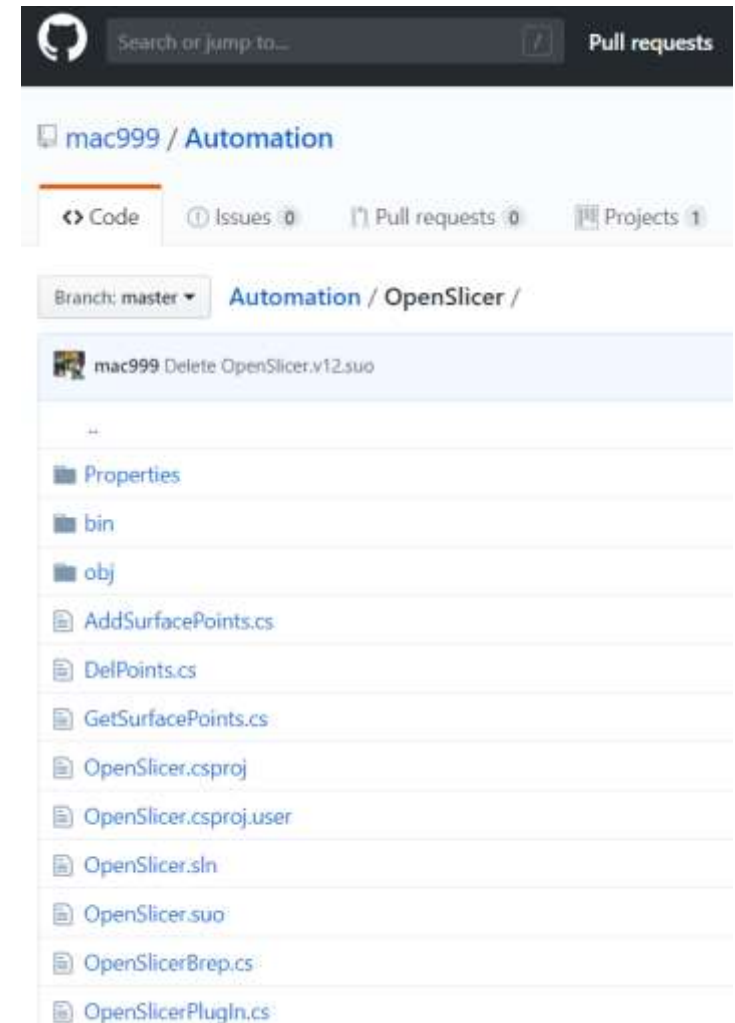
- Automation**
Brene addin tool for generating tool path such as CNC, 3D printer etc.
C# 1 star
- Books**
CSS
- DeepLearning**
Extract from tensorflowmodels
Models built with tensorflow
Python
- Projects**
VHDL
- Robot**
C++

You can now pin up to 6 repositories.

25 contributions in the last year

Contribution settings

Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul



Search or jump to... Pull requests

mac999 / Automation

Code Issues (0) Pull requests (0) Projects (1)

Branch: master Automation / OpenSlicer /

mac999 Delete OpenSlicer.v12.suo

Properties

- bin
- obj
- AddSurfacePoints.cs
- DelPoints.cs
- GetSurfacePoints.cs
- OpenSlicer.csproj
- OpenSlicer.csproj.user
- OpenSlicer.sln
- OpenSlicer.suo
- OpenSlicerBrep.cs
- OpenSlicerPlugin.cs

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라이선스 정책

GNU General Public License (GPL) 2.0

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- Copyright notice. LGPL must be specified. Modified library source code must be made public.

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- No obligation to disclose source code. Unlimited use of commercial software.

Apache License

- Similar to BSD. No obligation to disclose source code.

Mozilla Public License (MPL)

- No obligation to disclose source code. Modified code must be distributed under the MPL. No disclosure of combined program code is required.

MIT License

- Only the license/copyright must be specified.

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meta-llama / llama-prompt-ops

<> Code

Issues 7

Pull requests 4

Actions

Projects

Security

Insights

llama-prompt-ops

Public

Watch 10

Fork 57

Star 574

main 6 Branches 0 Tags

Go to file

Add file

Code

heyjustinal Merge pull request #35 from meta-llama/update-facility-eval

add env in ci last month

configs modify and update integration test last month

docs feat: Add pre-optimization summary display (issue #19) 3 weeks ago

About

An open-source tool for general prompt optimization.

Readme

MIT license

Code of conduct

Activity

Custom properties

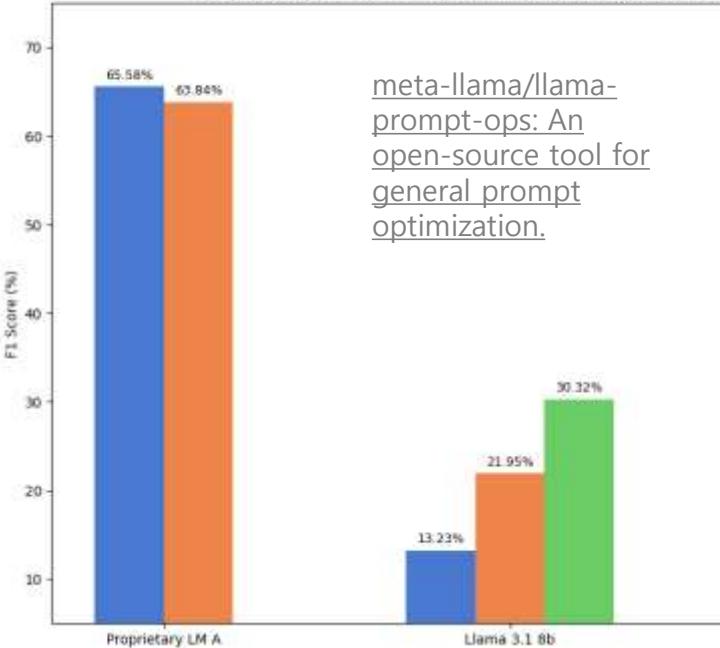
574 stars

10 watching

57 forks

Original Proprietary LM Prompt	Optimized Llama Prompt
You are a helpful assistant. Extract and return a JSON with the following keys and values: "urgency": one of <code>high</code>, <code>medium</code>, <code>low</code> "sentiment": one of <code>negative</code>, <code>neutral</code>, <code>positive</code> "categories": Create a dictionary with categories as keys and boolean values (True/False), where the value indicates whether the category matches tags like <code>emergency_repair_services</code>, <code>routine_maintenance_requests</code>, etc. Your complete message should be a valid JSON string that can be read directly.	You are an expert in analyzing customer service messages. Your task is to categorize the following message based on urgency, sentiment, and relevant categories. Analyze the message and return a JSON object with these fields: "urgency": Classify as "high", "medium", or "low" based on how quickly this needs attention "sentiment": Classify as "negative", "neutral", or "positive" based on the customer's tone "categories": Create a dictionary with facility management categories as keys and boolean values Only include these exact keys in your response. Return a valid JSON object

HotpotQA Benchmark: Explainable Multi-hop Question Answering



Model	Series 1 (Blue)	Series 2 (Orange)	Series 3 (Green)
Proprietary LM A	65.58%	63.84%	-
Llama 3.1 8B	13.23%	21.95%	30.32%

meta-llama/llama-prompt-ops: An open-source tool for general prompt optimization.

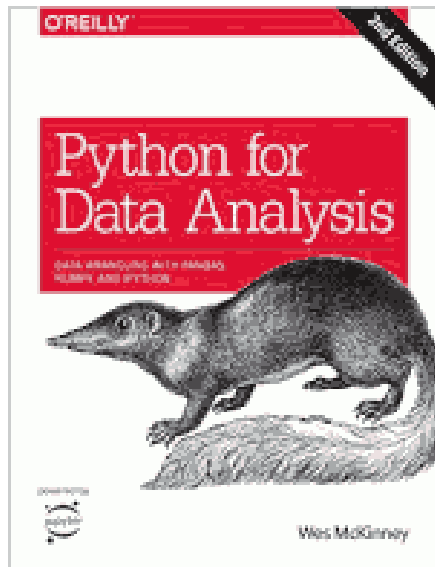
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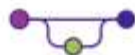
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NUMFOCUS
OPEN CODE • BETTER SCIENCE



VOLTRON DATA



Quansight Labs

TIDELIFT

bodo.ai

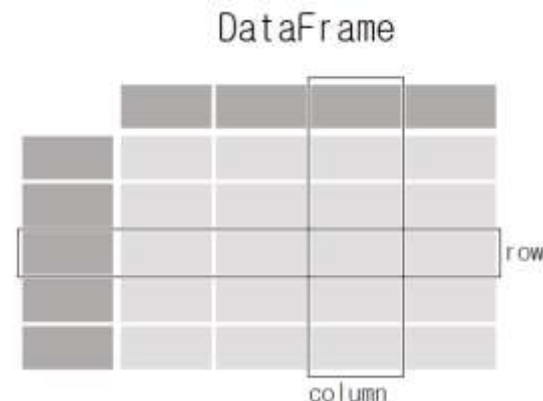
2σ TWO SIGMA

d-fine

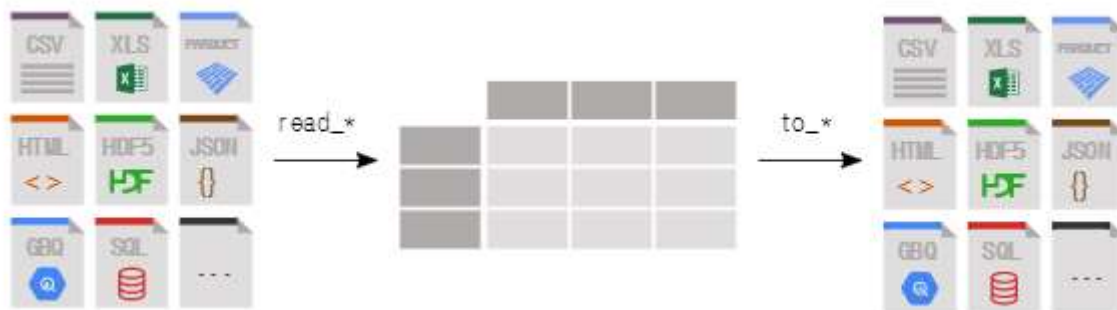


NVIDIA

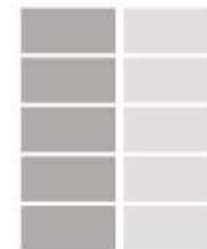
Chan
Zuckerberg
Initiative



Each column in a DataFrame is a Series



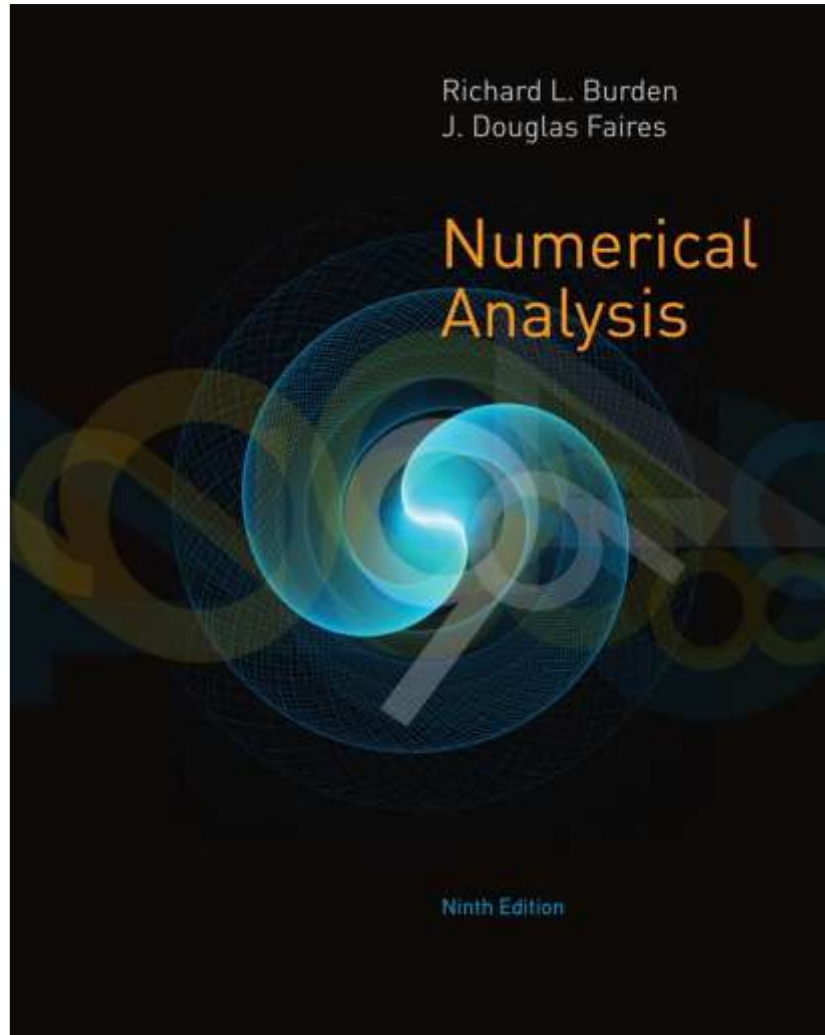
Series



<https://pandas.pydata.org/>

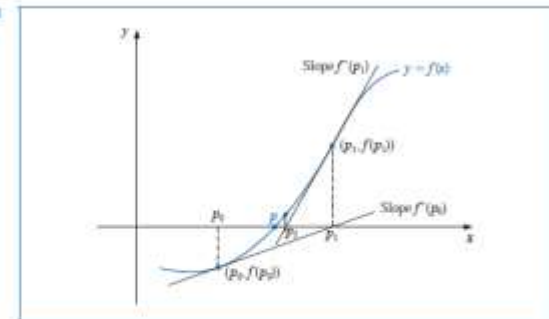
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68 CHAPTER 2 • Solutions of Equations in One Variable

Figure 2.8



ALGORITHM 2.3

Newton's

To find a solution to $f(x) = 0$ given an initial approximation p_0 :

INPUT initial approximation p_0 ; tolerance TOL ; maximum number of iterations N_0 .

OUTPUT approximate solution p or message of failure.

Step 1 Set $i = 1$.

Step 2 While $i \leq N_0$ do Steps 3–6.

Step 3 Set $p = p_0 - f(p_0)/f'(p_0)$. (Compute p_1 .)

Step 4 If $|p - p_0| < TOL$, then
OUTPUT (p); (The procedure was successful.)
STOP

Step 5 Set $i = i + 1$.

Step 6 Set $p_0 = p$. (Update p_0 .)

Step 7 OUTPUT ("The method failed after N_0 iterations, $N_0 = "$, N_0);
(The procedure was unsuccessful.)
STOP

The stopping-technique inequalities given with the Bisection method are applicable to Newton's method. That is, select a tolerance $\varepsilon > 0$, and construct $p_1, \dots, p_N$ until

$$|p_N - p_{N-1}| < \varepsilon, \quad (2.8)$$

$$\frac{|p_N - p_{N-1}|}{|p_N|} < \varepsilon, \quad p_N \neq 0, \quad (2.9)$$

or

$$|f(p_N)| < \varepsilon. \quad (2.10)$$

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Reference



목차(Contents)

<http://youtu.be/Mxple2Zzg-A>

Chapter 1. 벡터(Vectors)	*1.1 공학과 수학에서의 벡터: n -공간 *1.2 내적과 직교 1.3 직선과 평면의 벡터방정식 연습문제
Chapter 2. 선형연립방정식	2.1 선형연립방정식 2.2 Gauss 소거법과 Gauss-Jordan 소거법 연습문제
Chapter 3. 행렬과 행렬대수	3.1 행렬연산 3.2 역행렬 3.3 기본행렬 3.4 부분공간과 일차독립 3.5 선형연립방정식의 해집합과 행렬 3.6 특수행렬들(Special matrices) 연습문제
Chapter 4. 행렬식	4.1 행렬식의 정의와 기본정리 4.2 여인자 전개와 행렬식의 응용 4.3 크래머 공식 *4.4 행렬식의 응용 4.5 고유값과 고유벡터
Chapter 5. 행렬모델	*5.1 Blackout Game *5.2 Sage를 활용한 선형모델 *5.3 퀴즈 및 중간고사 예시 http://matrix.skku.ac.kr/2014-Album/MC.html http://matrix.skku.ac.kr/CLA-Exams-Sol.pdf
Chapter 6. 선형변환	6.1 함수(변환)로서의 행렬 6.2 선형변환의 기하학적 의미 6.3 핵과 치역 6.4 선형변환의 합성과 가역성 *6.5 Sage를 활용한 컴퓨터 그래픽 연습문제
Chapter 7. 차원과 부분공간	7.1 기저와 차원의 성질 7.2 행렬이 갖는 기본공간들 7.3 차원정리(Rank-Nullity 정리) 7.4 Rank 정리

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CS231n: Deep Learning for
Computer Vision

Lecture 1: Introduction



deeplearning.ai

Introduction to
Deep Learning

Welcome

AndrewNg

Fai-Fai U & Brian Adell

CS231n: Lecture 1 - 1

April 2, 2024



Lecture 1: Overview



Brief Introduction to Natural Language Processing

EE599 Deep Learning

Keith M. Chugg
Spring 2020



USC University of
Southern California

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[In-depth tutorials on LLMs, RAGs and real-world AI agent applications.](#)

[AI-agents-for-beginners: 11 Lessons to Get Started Building AI Agents](#)

[Agents-towards-production](#)

[LLM Engineering: Master AI, Large Language Models & Agents](#)

[Become an LLM Engineer - AI-Powered Learning for Developers](#)

The screenshot shows the GitHub repository page for 'ai-agents-for-beginners' by Microsoft. The repository is public and has 313 watches, 9.6k forks, and 32.5k stars. It is currently on the 'main' branch with 23 branches and 0 tags. The repository contains a list of files and folders, including .devcontainer, .github, 00-course-setup, 01-intro-to-ai-agents, 02-explore-agentic-frameworks, 03-agentic-design-patterns, 04-tool-use, 05-agentic-rag, 06-building-trustworthy-agents, 07-planning-design, and 08-multi-agent. The right sidebar shows the repository's description: '11 Lessons to Get Started Building AI Agents', a link to the repository URL, and a list of related repositories: ai-agents, autogen, generative-ai, semantic-kernel, ai-agents-framework, agentic-framework, agentic-rag, and agentic-ai. The repository also has a README, MIT license, Code of conduct, Security policy, Activity, Custom properties, 32.5k stars, 313 watching, 9.6k forks, and a Report repository button.

microsoft / ai-agents-for-beginners

Code Issues (5) Pull requests (3) Actions Projects Security Insights

ai-agents-for-beginners Public

Watch 313 Fork 9.6k Star 32.5k

main 23 Branches 0 Tags

Go to file Add file Code

koreyspace Merge pull request #246 from microsoft/update-translations 1e5ab0 · 2 days ago 541 Commits

.devcontainer	Update devcontainer configuration for improved compatability	2 months ago
.github	Add translation automation workflow	2 weeks ago
00-course-setup	Merge pull request #134 from indocoder/fga_token	3 days ago
01-intro-to-ai-agents	upgraded sk in requirements, improved setup steps	2 months ago
02-explore-agentic-frameworks	upgraded to new project endpoint	last month
03-agentic-design-patterns	docs: fix minor errors in lessons 01-03	3 months ago
04-tool-use	upgraded to new project endpoint	last month
05-agentic-rag	upgraded to new project endpoint	last month
06-building-trustworthy-agents	docs: fix errors found in lessons 04, 06, 09	3 months ago
07-planning-design	docs: small typo fix	3 months ago
08-multi-agent	docs: fix errors in lesson 08 multi-agent examples	3 months ago

About

11 Lessons to Get Started Building AI Agents

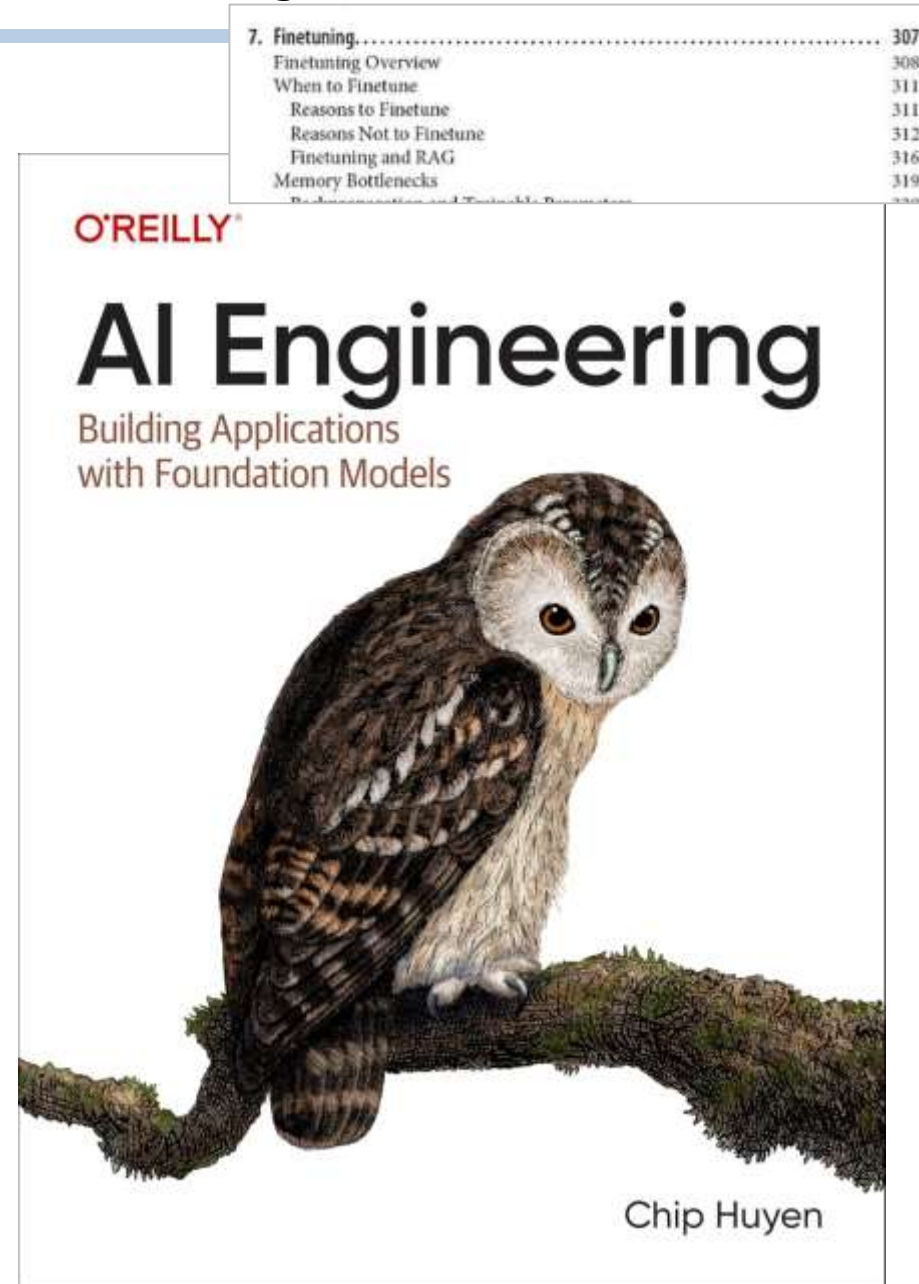
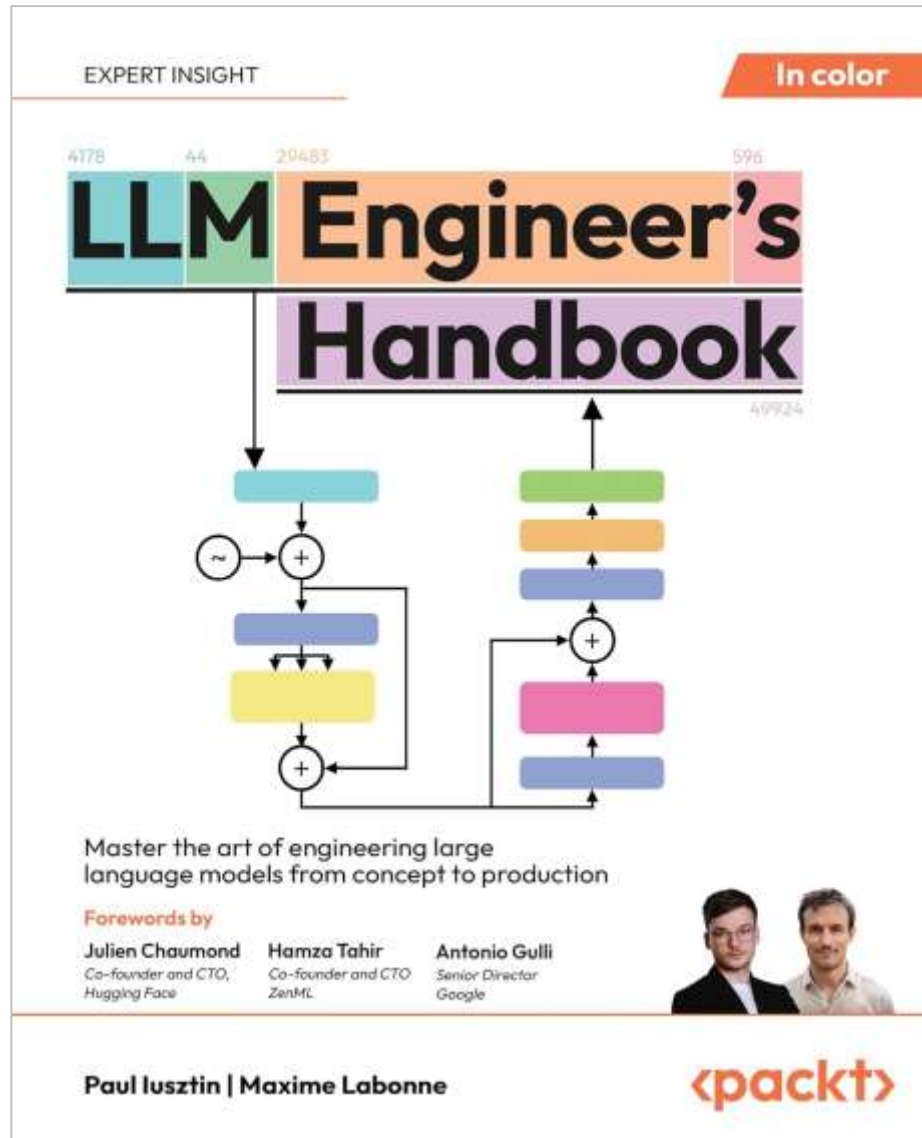
microsoft.github.io/ai-agents-for-beginners

ai-agents autogen generative-ai semantic-kernel ai-agents-framework agentic-framework agentic-rag agentic-ai

Readme MIT license Code of conduct Security policy Activity Custom properties 32.5k stars 313 watching 9.6k forks Report repository

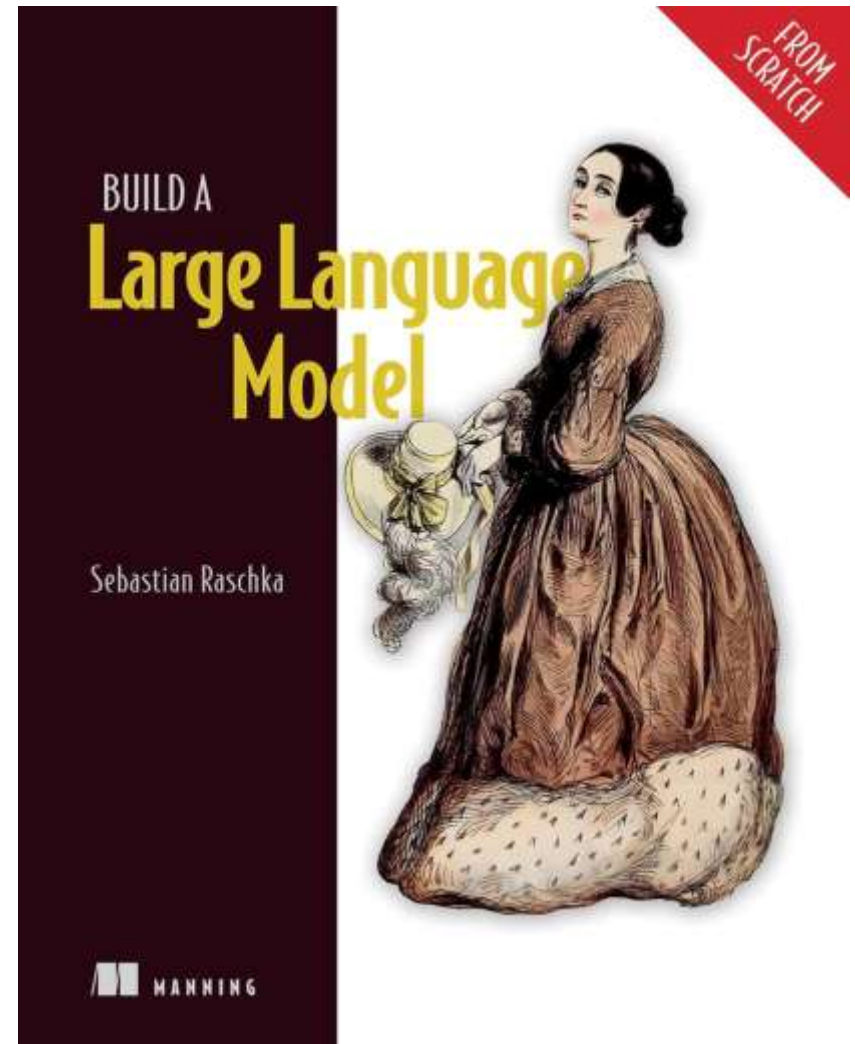
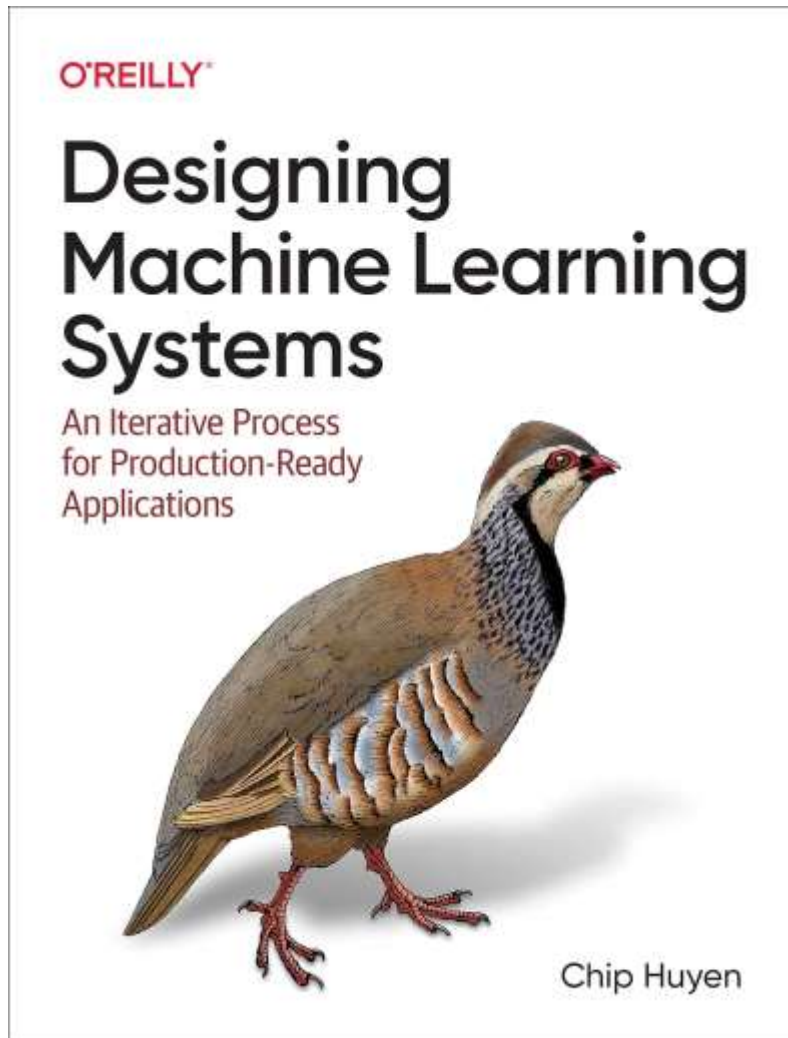
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Thanks

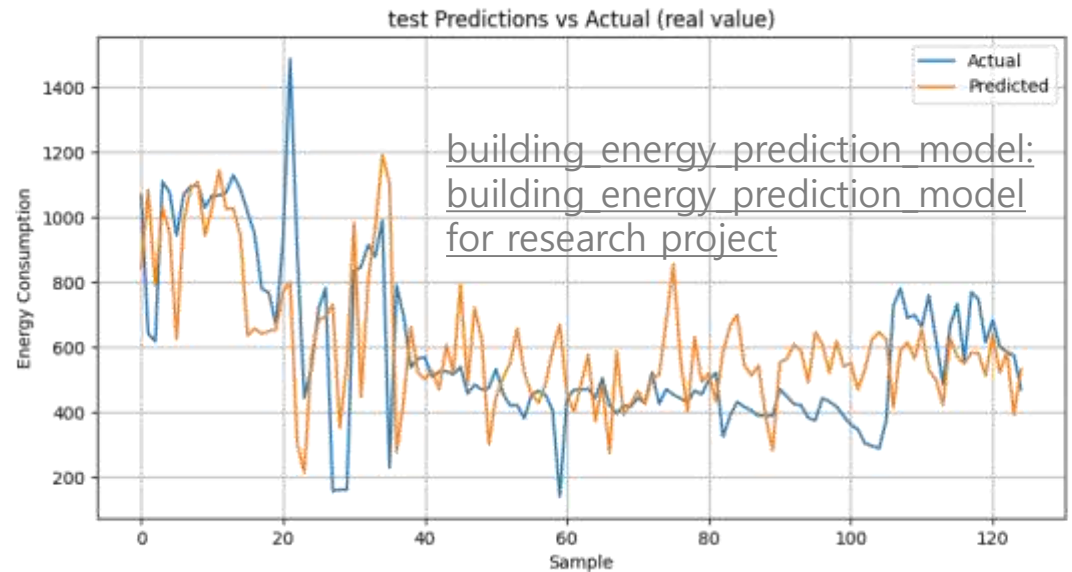
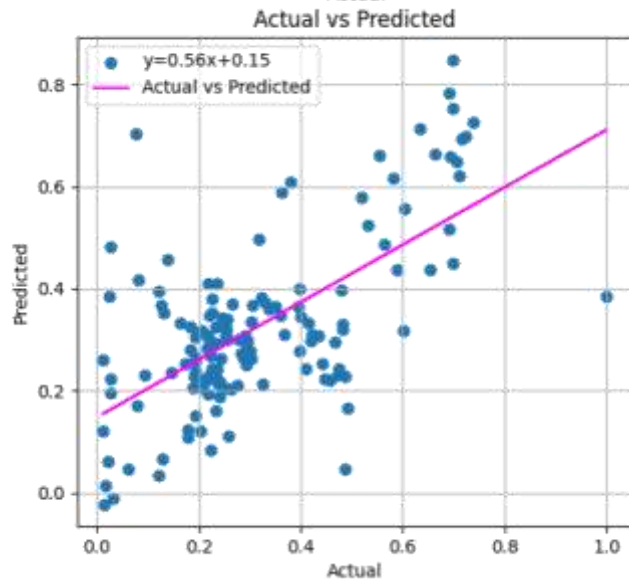
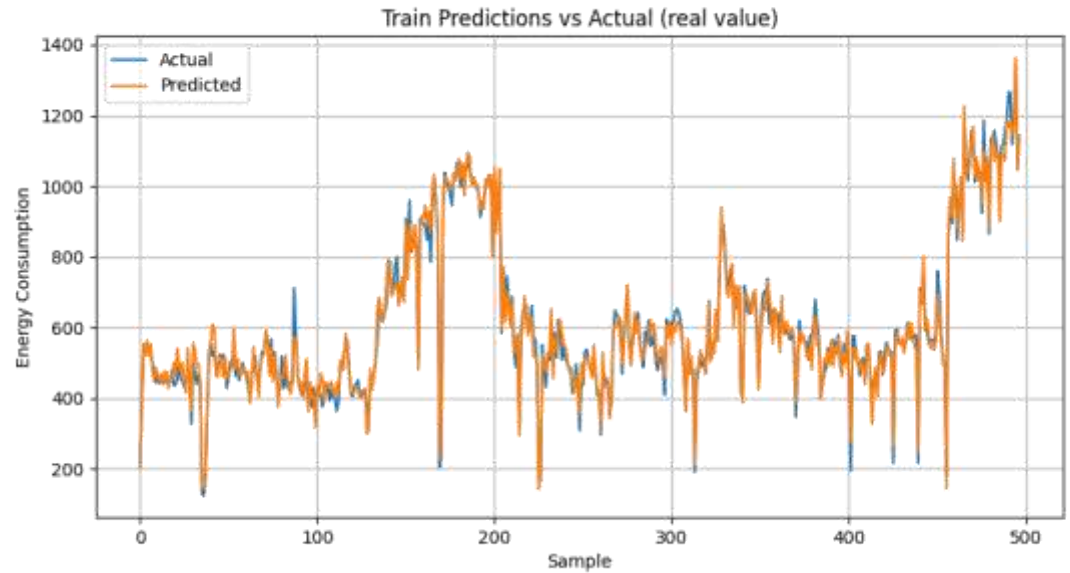
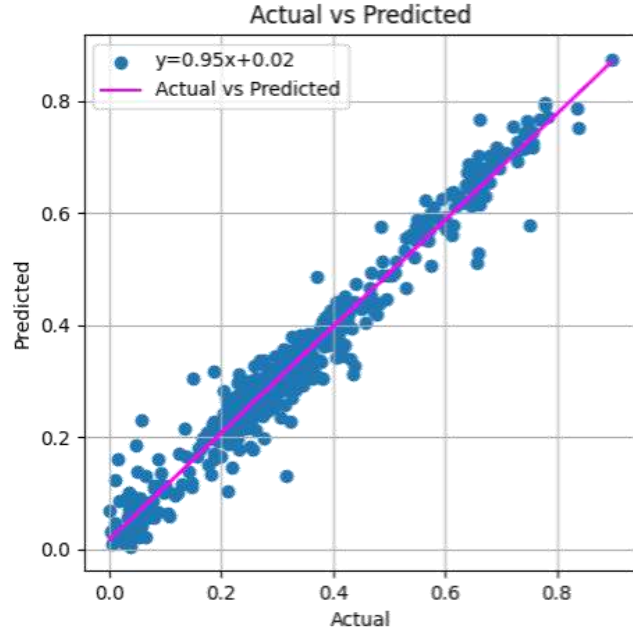
AX foundation and Trend 2025



laputa99999@gmail.com

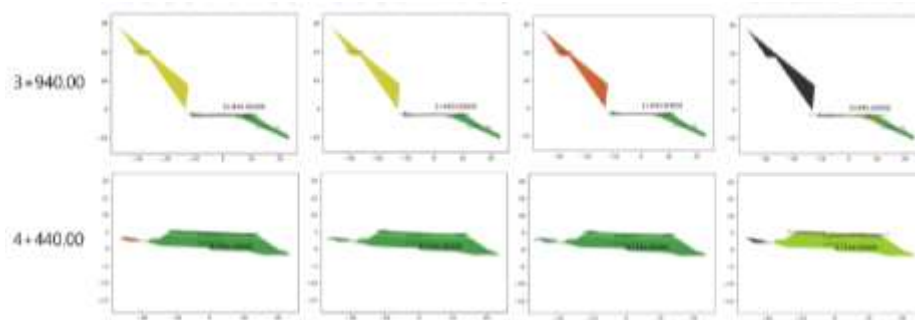
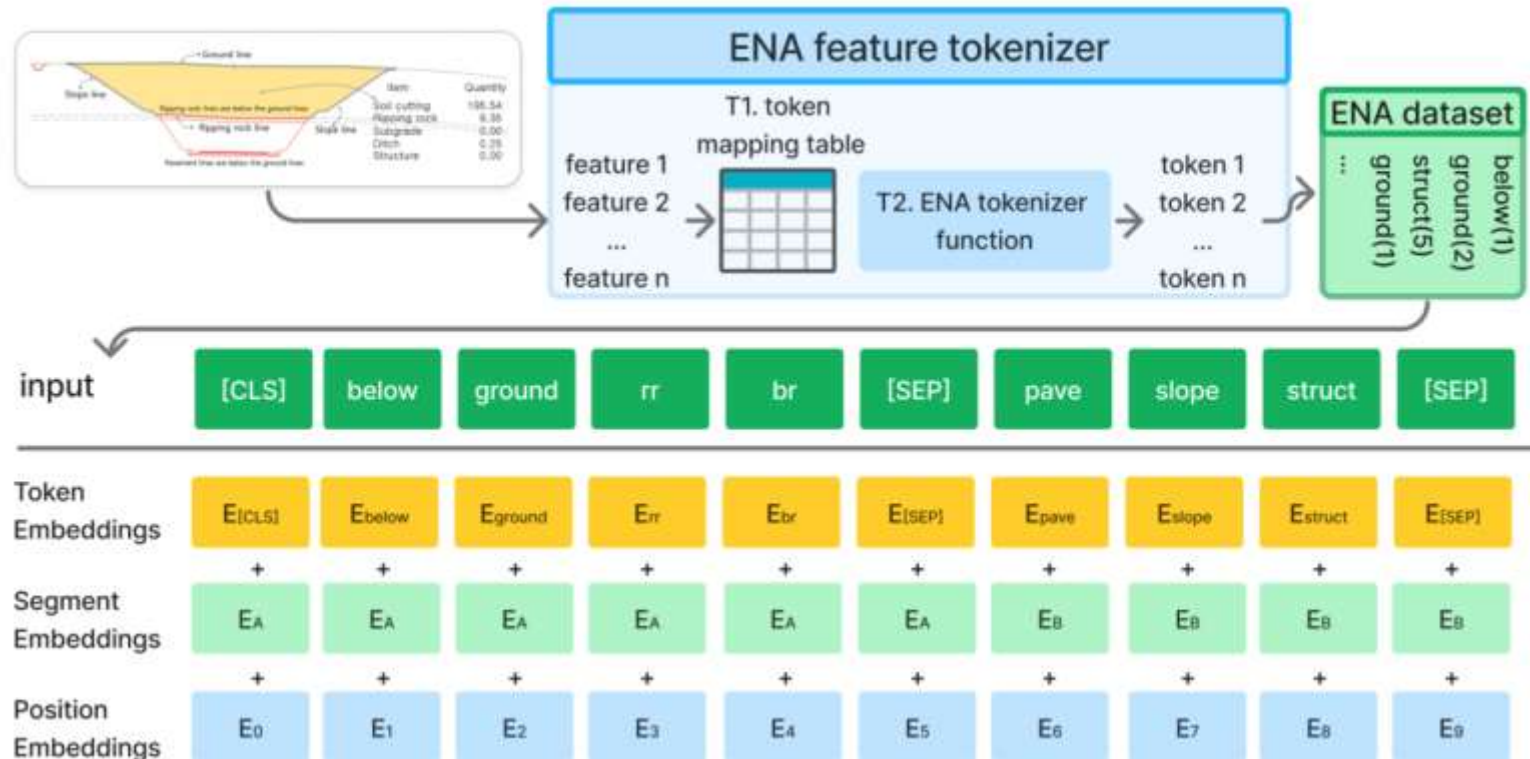
Usecase

Energy consumption forecast MLP, LSTM, TR, UCM



Loss=0.002. Train MAPE & Acc = (0.025, 95.30%). Test MAPE & Acc = (0.292, 70.76%)

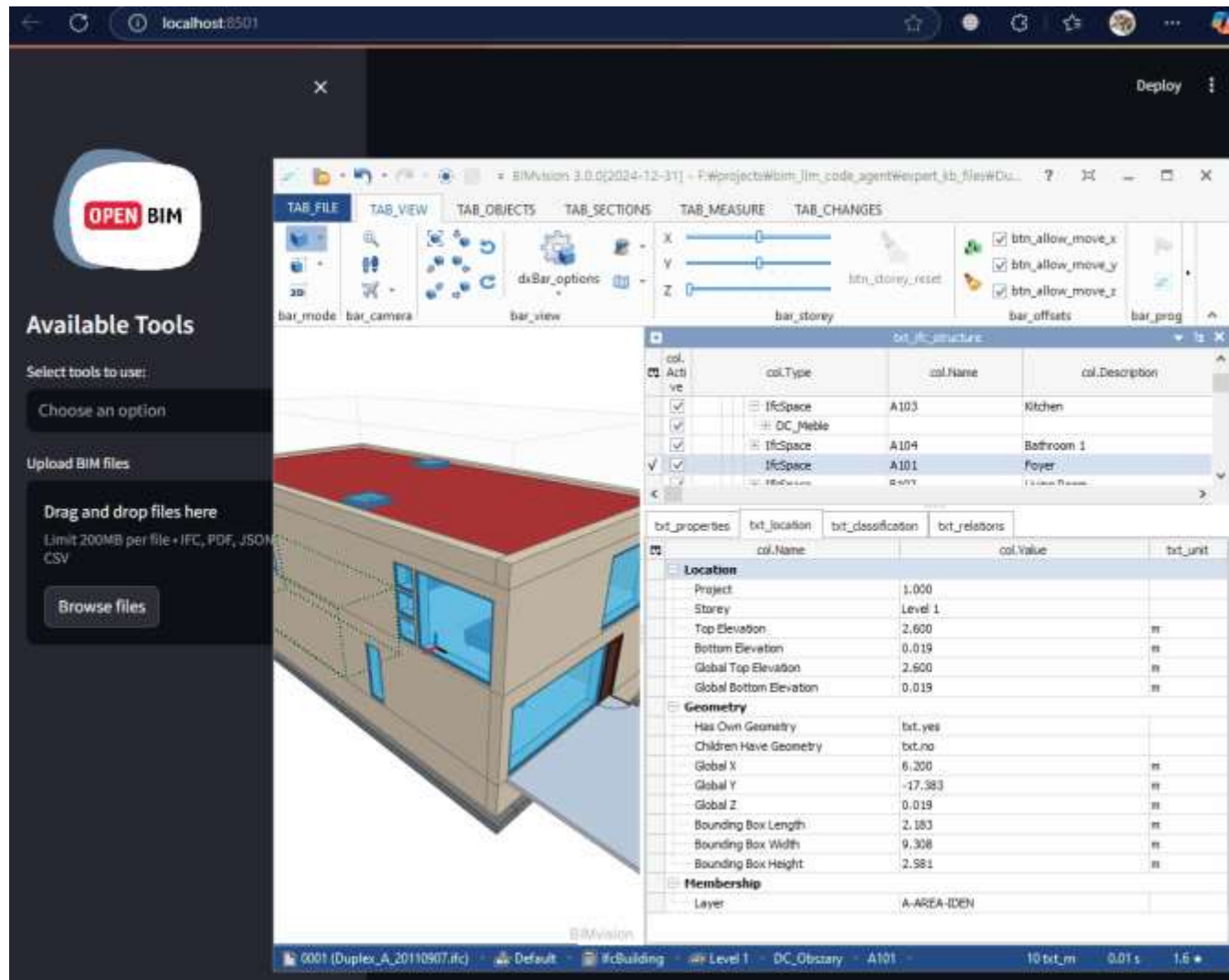
Usecase Predicting drawing quantity classification patterns MLP, LSTM, TR, BERT



[earthwork-net-model · Hugging Face](#)

[Earthwork Network Architecture \(ENA\):
Research for Earthwork Quantity Estimation
Method Improvement with Large Language
Model](#)

Usecase AI agent based on digital twin model LLM, OLLAMA, RAG, LANGCHAIN



BIM LLM code agent: BIM agent using RAG

MLP scratch code

Optimizer, Learning Rate, Batch

```
from math import exp
from random import seed
from random import random
```

Initialize a network

```
def initialize_network(n_inputs, n_hidden, n_outputs):
    network = list()
    hidden_layer = [{'weights':[random() for i in range(n_inputs + 1)]] for i in range(n_hidden)]
    network.append(hidden_layer)
    output_layer = [{'weights':[random() for i in range(n_hidden + 1)]] for i in range(n_outputs)]
    network.append(output_layer)
    return network
```

Calculate neuron activation for an input

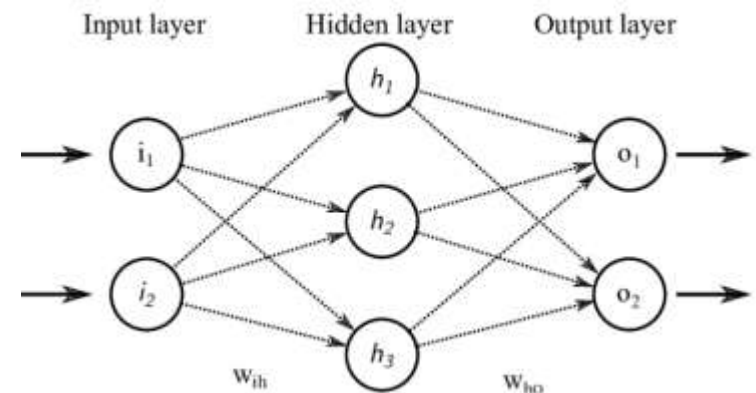
```
def activate(weights, inputs):
    activation = weights[-1]
    for i in range(len(weights)-1):
        activation += weights[i] * inputs[i]
    return activation
```

Transfer neuron activation

```
def transfer(activation):
    return 1.0 / (1.0 + exp(-activation))
```

Forward propagate input to a network output

```
def forward_propagate(network, row):
    inputs = row
    for layer in network:
        new_inputs = []
        for neuron in layer:
            activation = activate(neuron['weights'], inputs)
            neuron['output'] = transfer(activation)
            new_inputs.append(neuron['output'])
        inputs = new_inputs
    return inputs
```



INPUT = 2
HIDDEN LAYERS
OUTPUT = 2

MLP scratch code

Optimizer, Learning Rate, Batch

Calculate the derivative of an neuron output

def transfer_derivative(output):

return output * (1.0 - output)

$$\sigma(x)' = \sigma(x)(1 - \sigma(x))$$

$$\sigma(x) = 1 / (1 + e^{-x})$$

Backpropagate error and store in neurons

def backward_propagate_error(network, expected):

for i in reversed(range(len(network))):

layer = network[i]

errors = list()

if i != len(network)-1:

for j in range(len(layer)):

error = 0.0

for neuron in network[i + 1]:

error += (neuron['weights'][j] * neuron['delta'])

errors.append(error)

else:

for j in range(len(layer)):

neuron = layer[j]

errors.append(neuron['output'] - expected[j])

for j in range(len(layer)):

neuron = layer[j]

neuron['delta'] = errors[j] * transfer_derivative(neuron['output'])

Update network weights with error

def update_weights(network, row, l_rate):

for i in range(len(network)):

inputs = row[:-1]

if i != 0:

inputs = [neuron['output'] for neuron in network[i - 1]]

for neuron in network[i]:

for j in range(len(inputs)):

neuron['weights'][j] -= l_rate * neuron['delta'] * inputs[j]

neuron['weights'][-1] -= l_rate * neuron['delta']

MLP scratch code

Optimizer, Learning Rate, Batch

```
# Train a network for a fixed number of epochs
```

```
def train_network(network, train, l_rate, n_epoch, n_outputs):
```

```
    for epoch in range(n_epoch):
```

```
        sum_error = 0
```

```
        for row in train:
```

```
            outputs = forward_propagate(network, row)
```

```
            expected = [0 for i in range(n_outputs)]
```

```
            expected[row[-1]] = 1
```

```
            sum_error += sum([(expected[i]-outputs[i])**2 for i in range(len(expected))])
```

```
            backward_propagate_error(network, expected)
```

```
            update_weights(network, row, l_rate)
```

```
    print('>epoch=%d, lrate=%.3f, error=%.3f' % (epoch, l_rate, sum_error))
```

```
# Test training backprop algorithm
```

```
seed(1)
```

```
dataset = [[2.7810836,2.550537003,0],
```

```
            [1.465489372,2.362125076,0],
```

```
            [3.396561688,4.400293529,0],
```

```
            [1.38807019,1.850220317,0],
```

```
            [3.06407232,3.005305973,0],
```

```
            [7.627531214,2.759262235,1],
```

```
            [5.332441248,2.088626775,1],
```

```
            [6.922596716,1.77106367,1],
```

```
            [8.675418651,-0.242068655,1],
```

```
            [7.673756466,3.508563011,1]]
```

```
n_inputs = len(dataset[0]) - 1
```

```
n_outputs = len(set([row[-1] for row in dataset]))
```

```
network = initialize_network(n_inputs, 2, n_outputs)
```

```
train_network(network, dataset, 0.5, 20, n_outputs)
```

```
for layer in network:
```

```
    print(layer)
```

```
>epoch=0, lrate=0.500, error=6.350
```

```
>epoch=1, lrate=0.500, error=5.531
```

```
>epoch=2, lrate=0.500, error=5.221
```

```
>epoch=3, lrate=0.500, error=4.951
```

```
>epoch=4, lrate=0.500, error=4.519
```

```
>epoch=5, lrate=0.500, error=4.173
```

```
>epoch=6, lrate=0.500, error=3.835
```

```
>epoch=7, lrate=0.500, error=3.506
```

```
>epoch=8, lrate=0.500, error=3.192
```

```
>epoch=9, lrate=0.500, error=2.898
```

```
>epoch=10, lrate=0.500, error=2.626
```

```
>epoch=11, lrate=0.500, error=2.377
```

```
>epoch=12, lrate=0.500, error=2.153
```

```
>epoch=13, lrate=0.500, error=1.953
```

```
>epoch=14, lrate=0.500, error=1.774
```

```
>epoch=15, lrate=0.500, error=1.614
```

```
>epoch=16, lrate=0.500, error=1.472
```

```
>epoch=17, lrate=0.500, error=1.346
```

```
>epoch=18, lrate=0.500, error=1.233
```

```
>epoch=19, lrate=0.500, error=1.132
```

```
{ 'weights': [-1.4688375095432327, 1.850887325439514, 1.0858178629550297], 'output': 0.029963625}, { 'weights': [0.37711098142462157, -0.0625909894552989, 0.2765123702642716], 'output': 52850863837}}
```

```
{ 'weights': [2.515394649397849, -0.3391927502445985, -0.9671565426390275], 'output': 0.236487}, { 'weights': [-2.5584149848484263, 1.0036422106209202, 0.42383086467582715], 'output': 6437354}}
```